

30. Ethernet Message Forwarding Engine (MFWD)

30.1 Overview

Ethernet Message Forwarding Engine (MFWD) is part of ESWM and aims to filter, forward, and route the switch frames. MFWD snoops the frame information from fabric to redistribute them to the agent after filtering, forwarding, and routing mechanisms.

Table 30.1 lists the MFWD specifications and Figure 30.1 shows a block diagram.

Table 30.1 MFWD specifications

Function		Details
Overall function	Cut-through	Forwarding frames while receiving them depending on Layer 2
	Direct forwarding	Forwarding frames coming from GWCA's by bypassing the forwarding engine [GWCA]
	Layer 3 forwarding	Forwarding frames based on a stream ID
	Layer 2 forwarding	Forwarding frames based on destination MAC addresses and VLAN
	Port based forwarding	Forwarding frames depending on source port
	L23U	Update of Layer 2 and Layer 3 depending on Layer 3 forwarding
	PSFP	PSFP filtering [PSFP] depending on Layer 3 forwarding
	ATS	ATS traffic shaping [ATS]
	FRER	FRER [FRER] depending on Layer 3 forwarding
Data provision	Ethernet Frames	Corresponds to the IEEE 802.3, 802.1Q and 802.1CB standards [802.3] [802.1Q] [802.1CB].
	Descriptors	Local Ethernet Descriptor, Local Direct Descriptor (Agents to MFWD) Forwarding descriptor (MFWD to Agents)

Figure 30.1 shows MFWD block diagram.

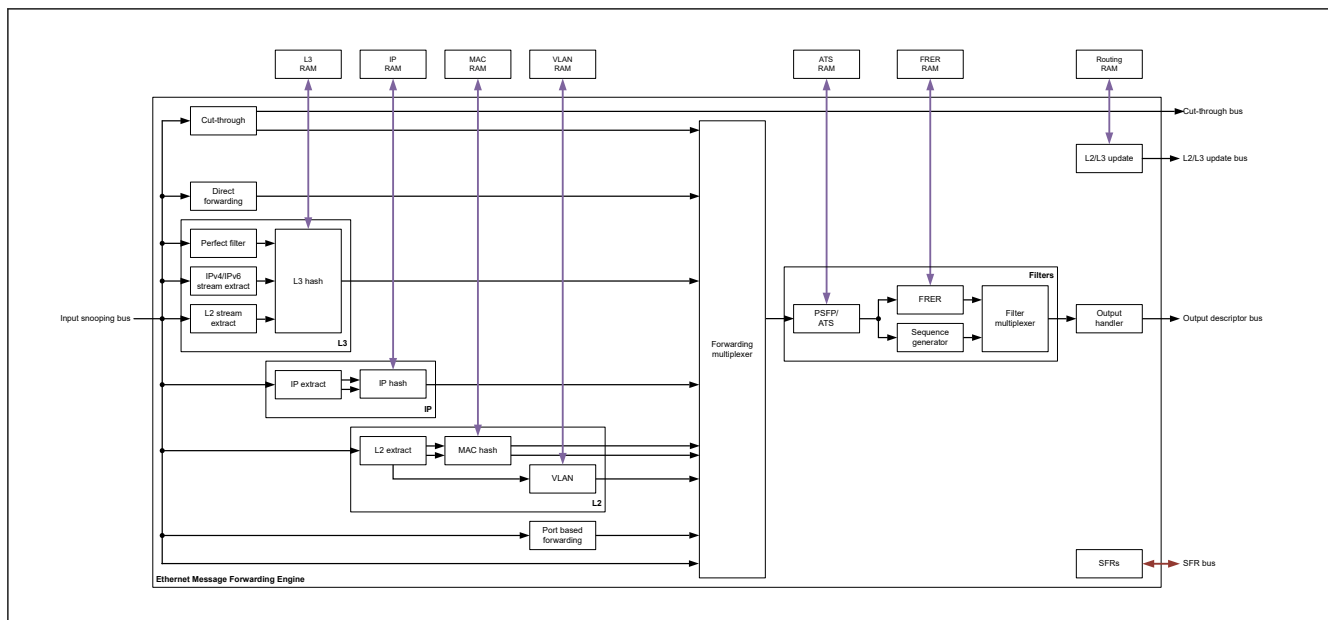


Figure 30.1 MFWD block diagram

Table 30.2 MFWD functional blocks (1 of 2)

Block name	Function
Cut-through	Filter frame L2 and send them to while receiving them.

Table 30.2 MFWD functional blocks (2 of 2)

Block name	Function
Direct forwarding	Forward frames coming from a CPU with the forwarding information already present.
L3	L2 to L4 Stream based forwarding/routing engine
L2	L2 switch
Port based forwarding	Source port-based forwarding
Forwarding multiplexer	Arbitrate between all the forwarding/routing mechanisms
PSFP/ATS	Per stream filtering and policing [802.1Qci] and asynchronous traffic shaping [802.1Qcr]
FRER	Frame Replication and Elimination for Reliability [802.1CB]
Sequence generator	Sequence number generation for FRER proxy functionality [802.1CB]
Filter multiplexer	Arbitrate between forwarding results and filtering results.
Output handler	Copy and send descriptors to targets. Manage overflow and security error recovery.

30.2 MFWD Register List

Table 30.3 MFWD registers (1 of 7)

Offset address	Register name	Register symbol
0x0000	General Configuration Register	FWGC
0x0010	TAG TPID Configuration Register 0	FWTTC0
0x0014	TAG TPID Configuration Register 1	FWTTC1
0x0020	CPU Exceptional Path Target Configuration Register	FWCEPTC
0x0024	CPU Exceptional Path Reason Configuration Register 0	FWCEPRC0
0x0028	CPU Exceptional Path Reason Configuration Register 1	FWCEPRC1
0x002C	CPU Exceptional Path Reason Configuration Register 2	FWCEPRC2
0x0030	CPU Learning Path Target Configuration Register	FWCLPTC
0x0034	CPU Learning Path Reason Configuration Register	FWCLPRC
0x0040	CPU Mirroring Path Target Configuration Register	FWCMPTC
0x0044	Ethernet Mirroring Path Target Configuration Register	FWEMPTC
0x0050	Source-Destination Mirroring Path Target Configuration Register	FWSDMPTC
0x0054	Source-Destination Mirroring Path Vector Configuration Register	FWSDMPVC
0x0080 + 0x04 × i	Port i Level Based Watermark Configuration Register (i = 0 to 2)	FWLBWMCi
0x0100 + 0x10 × i	Port i Configuration Register 0 (i = 0 to 2)	FWPCi0
0x0104 + 0x10 × i	Port i Configuration Register 1 (i = 0 to 2)	FWPCi1
0x0108 + 0x10 × i	Port i Configuration Register 2 (i = 0 to 2)	FWPCi2
0x0400 + 0x40 × i	Cut-Through General Configuration Register i0 (i = 0 to 7)	FWCTGCI0
0x0404 + 0x40 × i	Cut-Through General Configuration Register i1 (i = 0 to 7)	FWCTGCI1
0x0408 + 0x40 × i	Cut-Through Target Configuration Register i0 (i = 0 to 7)	FWCTTCi0
0x040C + 0x40 × i	Cut-Through Target Configuration Register i1 (i = 0 to 7)	FWCTTCi1
0x0410 + 0x40 × i	Cut-Through Target Configuration Register i20 (i = 0 to 7)	FWCTTCi20
0x0420 + 0x40 × i	Cut-Through Separation Configuration Register i0 (i = 0 to 7)	FWCTSCi0
0x0424 + 0x40 × i	Cut-Through Separation Configuration Register i1 (i = 0 to 7)	FWCTSCi1
0x0428 + 0x40 × i	Cut-Through Separation Configuration Register i2 (i = 0 to 7)	FWCTSCi2
0x042C + 0x40 × i	Cut-Through Separation Configuration Register i3 (i = 0 to 7)	FWCTSCi3
0x0430 + 0x40 × i	Cut-Through Separation Configuration Register i4 (i = 0 to 7)	FWCTSCi4

Table 30.3 MFWD registers (2 of 7)

Offset address	Register name	Register symbol
0x1000 + 0x10 × i	Two-Byte Filter Configuration Register i (i = 0 to 15)	FWTWBFCi
0x1004 + 0x10 × i	Two-Byte Filter Value Configuration Register i (i = 0 to 15)	FWTWBFVCI
0x1400 + 0x10 × i	Three-Byte Filter Configuration Register i (i = 0 to 15)	FWTHBFCi
0x1404 + 0x10 × i	Three-Byte Filter Value 0 Configuration Register i (i = 0 to 15)	FWTHBFV0Ci
0x1408 + 0x10 × i	Three-Byte Filter Value 1 Configuration Register i (i = 0 to 15)	FWTHBFV1Ci
0x1800 + 0x10 × i	Four-Byte Filter Configuration Register i (i = 0 to 15)	FWFOBFCi
0x1804 + 0x10 × i	Four-Byte Filter Value 0 Configuration Register i (i = 0 to 15)	FWFOBVF0Ci
0x1808 + 0x10 × i	Four-Byte Filter Value 1 Configuration Register i (i = 0 to 15)	FWFOBVF1Ci
0x1C00 + 0x10 × i	Range Filter Configuration Register i (i = 0, 1)	FWRFCi
0x1C04 + 0x10 × i	Range Filter Value Configuration Register i (i = 0, 1)	FWRFVCI
0x2000 + 0x40 × i	Cascade Filter Configuration Register i (i = 0 to 15)	FWCFCi
0x2004 + 0x40 × i + 0x04 × j	Cascade Filter Mapping Configuration Register ij (i = 0 to 15) (j = 0 to 6)	FWCFMCij
0x4008	IPv4 Stream Configuration Register	FWIP4SC
0x4018	IPv6 Stream Configuration Register	FWIP6SC
0x401C	IPv6 Offset Configuration Register	FWIP6OC
0x4020	Layer 2 Stream Configuration Register	FWL2SC
0x4030	Stream Filter Hash Equation Configuration Register	FWSFHEC
0x4040	Software Hash Calculation Request Register 0	FWSHCR0
0x4044	Software Hash Calculation Request Register 1	FWSHCR1
0x4048	Software Hash Calculation Request Register 2	FWSHCR2
0x404C	Software Hash Calculation Request Register 3	FWSHCR3
0x4050	Software Hash Calculation Request Register 4	FWSHCR4
0x4054	Software Hash Calculation Request Register 5	FWSHCR5
0x4058	Software Hash Calculation Request Register 6	FWSHCR6
0x405C	Software Hash Calculation Request Register 7	FWSHCR7
0x4060	Software Hash Calculation Request Register 8	FWSHCR8
0x4064	Software Hash Calculation Request Register 9	FWSHCR9
0x4068	Software Hash Calculation Request Register 10	FWSHCR10
0x406C	Software Hash Calculation Request Register 11	FWSHCR11
0x4070	Software Hash Calculation Request Register 12	FWSHCR12
0x4074	Software Hash Calculation Request Register 13	FWSHCR13
0x4078	Software Hash Calculation Request Result Register	FWSHCRR
0x4090	L3 Hash Entry Configuration Register	FWLTHHEC
0x4094	L3 Hash Configuration Register	FWLTHHC
0x40A0	L3 Table Learn Register 0	FWLTHTL0
0x40A4	L3 Table Learn Register 1	FWLTHTL1
0x40A8	L3 Table Learn Register 2	FWLTHTL2
0x40AC	L3 Table Learn Register 3	FWLTHTL3
0x40B0	L3 Table Learn Register 4	FWLTHTL4
0x40B4	L3 Table Learn Register 5	FWLTHTL5
0x40B8	L3 Table Learn Register 6	FWLTHTL6
0x40BC	L3 Table Learn Register 7	FWLTHTL7

Table 30.3 MFWD registers (3 of 7)

Offset address	Register name	Register symbol
0x40C0	L3 Table Learn Register 80	FWLTHTL80
0x40D0	L3 Table Learn Register 9	FWLTHTL9
0x40D4	L3 Table Learn Result Register	FWLTHTLR
0x40E0	L3 Table Initialization Monitoring Register	FWLTHTIM
0x40E4	L3 Table Entry Monitoring Register	FWLTHTEM
0x4100	L3 Table Search Register 0	FWLHTS0
0x4104	L3 Table Search Register 1	FWLHTS1
0x4108	L3 Table Search Register 2	FWLHTS2
0x410C	L3 Table Search Register 3	FWLHTS3
0x4110	L3 Table Search Register 4	FWLHTS4
0x4120	L3 Table Search Result Register 0	FWLHTSR0
0x4124	L3 Table Search Result Register 1	FWLHTSR1
0x4128	L3 Table Search Result Register 2	FWLHTSR2
0x412C	L3 Table Search Result Register 3	FWLHTSR3
0x4130	L3 Table Search Result Register 40	FWLHTSR40
0x4140	L3 Table Search Result Register 5	FWLHTSR5
0x4150	L3 Table Read Register	FWLHTR
0x4154	L3 Table Read Result Register 0	FWLHTRR0
0x4158	L3 Table Read Result Register 1	FWLHTRR1
0x415C	L3 Table Read Result Register 2	FWLHTRR2
0x4160	L3 Table Read Result Register 3	FWLHTRR3
0x4164	L3 Table Read Result Register 4	FWLHTRR4
0x4168	L3 Table Read Result Register 5	FWLHTRR5
0x416C	L3 Table Read Result Register 6	FWLHTRR6
0x4170	L3 Table Read Result Register 7	FWLHTRR7
0x4174	L3 Table Read Result Register 8	FWLHTRR8
0x4180	L3 Table Read Result Register 90	FWLHTRR90
0x4190	L3 Table Read Result Register 10	FWLHTRR10
0x4620	MAC Hash Entry Configuration Register	FWMACHEC
0x4624	MAC Hash Configuration Register	FWMACHC
0x4630	MAC Table Learn Register 0	FWMACTL0
0x4634	MAC Table Learn Register 1	FWMACTL1
0x4638	MAC Table Learn Register 2	FWMACTL2
0x463C	MAC Table Learn Register 3	FWMACTL3
0x4640	MAC Table Learn Register 40	FWMACTL40
0x4650	MAC Table Learn Register 5	FWMACTL5
0x4654	MAC Table Learn Result Register	FWMACTLR
0x4660	MAC Table Initialization Monitoring Register	FWMACTIM
0x4664	MAC Table Entry Monitoring Register	FWMACTEM
0x4670	MAC Table Search Register 0	FWMACTS0
0x4674	MAC Table Search Register 1	FWMACTS1
0x4678	MAC Table Search Result Register 0	FWMACTSR0

Table 30.3 MFWD registers (4 of 7)

Offset address	Register name	Register symbol
0x467C	MAC Table Search Result Register 1	FWMACTSR1
0x4680	MAC Table Search Result Register 20	FWMACTSR20
0x4690	MAC Table Search Result Register 3	FWMACTSR3
0x46A0	MAC Table Read Register	FWMACTR
0x46A4	MAC Table Read Result Register 0	FWMACTRR0
0x46A8	MAC Table Read Result Register 1	FWMACTRR1
0x46AC	MAC Table Read Result Register 2	FWMACTRR2
0x46B0	MAC Table Read Result Register 3	FWMACTRR3
0x46B4	MAC Table Read Result Register 4	FWMACTRR4
0x46C0	MAC Table Read Result Register 50	FWMACTRR50
0x46D0	MAC Table Read Result Register 6	FWMACTRR6
0x4880	MAC Aging US Prescaler Configuration Register	FWMACAGUSPC
0x4884	MAC Aging Configuration Register	FWMACAGC
0x4888	MAC Aging Monitoring Register 0	FWMACAGM0
0x488C	MAC Aging Monitoring Register 1	FWMACAGM1
0x4900	VLAN Table Entry Configuration Register	FWVLANTEC
0x4910	VLAN Table Learn Register 0	FWVLANTL0
0x4914	VLAN Table Learn Register 1	FWVLANTL1
0x4918	VLAN Table Learn Register 2	FWVLANTL2
0x4920	VLAN Table Learn Register 30	FWVLANTL30
0x4930	VLAN Table Learn Register 4	FWVLANTL4
0x4934	VLAN Table Learn Result Register	FWVLANTLR
0x4940	VLAN Table Initialization Monitoring Register	FWVLANTIM
0x4944	VLAN Table Entry Monitoring Register	FWVLANTEM
0x4950	VLAN Table Search Register	FWVLANTS
0x4954	VLAN Table Search Result Register 0	FWVLANTSRO
0x4958	VLAN Table Search Result Register 1	FWVLANTSRI
0x4960	VLAN Table Search Result Register 20	FWVLANTSRO20
0x4970	VLAN Table Search Result Register 3	FWVLANTSRI3
0x4A00 + 0x10 × i	Port i Port Based Forwarding Configuration Register (i = 0 to 2)	FWPBFCi
0x4A04 + 0x10 × i	Port i Port Based Forwarding CSD Configuration Register 0 (i = 0 to 2)	FWPBFCSDCi0
0x4E00	Layer 2/Layer 3 Update Rule Learn Register 0	FWL23URL0
0x4E04	Layer 2/Layer 3 Update Rule Learn Register 1	FWL23URL1
0x4E08	Layer 2/Layer 3 Update Rule Learn Register 2	FWL23URL2
0x4E0C	Layer 2/Layer 3 Update Rule Learn Register 3	FWL23URL3
0x4E10	Layer 2/Layer 3 Update Rule Learn Result Register	FWL23URLR
0x4E20	Layer 2/Layer 3 Update Table Initialization Monitoring Register	FWL23UTIM
0x4E30	Layer 2/Layer 3 Update Rule Read Register	FWL23URR
0x4E34	Layer 2/Layer 3 Update Rule Read Result Register 0	FWL23URRR0
0x4E38	Layer 2/Layer 3 Update Rule Read Result Register 1	FWL23URRR1
0x4E3C	Layer 2/Layer 3 Update Rule Read Result Register 2	FWL23URRR2
0x4E40	Layer 2/Layer 3 Update Rule Read Result Register 3	FWL23URRR3

Table 30.3 MFWD registers (5 of 7)

Offset address	Register name	Register symbol
0x4F00 + 0x04 × i	Layer 2/Layer 3 Update Remapping Configuration Register i (i = 0 to 31)	FWL23URMCi
0x5000 + 0x04 × i	PSFP MSDU Filter Global Configuration Register i (i = 0 to 15)	FWPMFGCi
0x5600 + 0x20 × i	PSFP Meter i Filter Configuration Register (i = 0 to 31)	FWPMTRFCi
0x5604 + 0x20 × i	PSFP Meter i CBS Configuration Register (i = 0 to 31)	FWPMTRCBSCi
0x5608 + 0x20 × i	PSFP Meter i CIR Configuration Register (i = 0 to 31)	FWPMTRCIRCi
0x560C + 0x20 × i	PSFP Meter i EBS Configuration Register (i = 0 to 7)	FWPMTREBSCi
0x5610 + 0x20 × i	PSFP Meter i EIR Configuration Register (i = 0 to 7)	FWPMTREIRCi
0x5614 + 0x20 × i	PSFP Meter i Filter Monitoring Register (i = 0 to 31)	FWPMTRFMI
0x6000	FRER Table Learn Register 0	FWFTL0
0x6004	FRER Table Learn Register 1	FWFTL1
0x6008	FRER Table Learn Result Register	FWFTLR
0x6010	FRER Timeout Configuration Register	FWFTOC
0x6014	FRER Timeout Prescaler Configuration Register 0	FWFTOPC
0x6020	FRER Table Initialization Monitoring Register	FWFTIM
0x6030	FRER Table Read Register	FWFTR
0x6034	FRER Table Read Result Register 0	FWFTRR0
0x6038	FRER Table Read Result Register 1	FWFTRR1
0x603C	FRER Table Read Result Register 2	FWFTRR2
0x6100 + 0x08 × i	Sequence Number Generation i Configuration Register (i = 0 to 31)	FWSEQNGCi
0x6104 + 0x08 × i	Sequence Number Generation i Monitoring Register (i = 0 to 31)	FWSEQNGMi
0x6200	Sequence Number Reset Configuration Register	FWSEQNRC
0x6300 + 0x20 × i E: 0x6E00 + 0x20 × i	Port i Cut-Through Forwarded Descriptor Counter Register (i = 0, 1)	FWCTFDCNi
0x6340 E: 0x6E40	Port 2 Direct Descriptor Forwarded Descriptor Counter Register	FWDDFDCN2
0x6304 + 0x20 × i E: 0x6E04 + 0x20 × i	Port i Layer 3 Forwarded Descriptor Counter Register (i = 0 to 2)	FWLTHFDCNi
0x630C + 0x20 × i E: 0x6E0C + 0x20 × i	Port i Layer 2 Forwarded Descriptor Counter Register (i = 0 to 2)	FWLTWFDCNi
0x6310 + 0x20 × i E: 0x6E10 + 0x20 × i	Port i Port Based Forwarded Descriptor Counter Register (i = 0 to 2)	FWPBFDCNi
0x6314 + 0x20 × i E: 0x6E14 + 0x20 × i	Port i MAC Hardware Learn Counter Register (i = 0 to 2)	FWMHLCNi
0x6504 + 0x20 × i E: 0x7004 + 0x20 × i	Port i Watermark Rejected Descriptor Counter Register (i = 0 to 2)	FWWMRDCNi
0x6508 + 0x20 × i E: 0x7008 + 0x20 × i	Port i Cut-Through Rejected Descriptor Counter Register (i = 0, 1)	FWCTRDCNi
0x6548 E: 0x7048	Port 2 Direct Descriptor Rejected Descriptor Counter Register	FWDDRDCN2
0x650C + 0x20 × i E: 0x700C + 0x20 × i	Port i Layer 3 Rejected Descriptor Counter Register (i = 0 to 2)	FWLTHRDCNi
0x6514 + 0x20 × i E: 0x7014 + 0x20 × i	Port i Layer 2 Rejected Descriptor Counter Register (i = 0 to 2)	FWLTWRDCNi
0x6518 + 0x20 × i E: 0x7018 + 0x20 × i	Port i Port Based Rejected Descriptor Counter Register (i = 0 to 2)	FWPBRDCNi
0x6700 + 0x04 × i E: 0x7200 + 0x04 × i	PSFP MSDU i Filtered Descriptor Counter Register (i = 0 to 15)	FWPMFDCNi

Table 30.3 MFWD registers (6 of 7)

Offset address	Register name	Register symbol
0x6800 + 0x10 × i E: 0x7300 + 0x10 × i	PSFP Meter i Green Descriptor Counter Register (i = 0 to 31)	FWPMGDCNi
0x6804 + 0x10 × i E: 0x7304 + 0x10 × i	PSFP Meter i Yellow Descriptor Counter Register (i = 0 to 7)	FWPMYDCNi
0x6808 + 0x10 × i E: 0x7308 + 0x10 × i	PSFP Meter i Red Descriptor Counter Register (i = 0 to 31)	FWPMRDCNi
0x6A00 + 0x08 × i E: 0x7500 + 0x08 × i	FRER i Passed Packet Counter Register (i = 0 to 127)	FWFRPPCni
0x6A04 + 0x08 × i E: 0x7504 + 0x08 × i	FRER i Discarded Packet Counter Register (i = 0 to 127)	FWFRDPCni
0x7900 + 0x10 × i	Port i Error Interrupt Status Register 0 (i = 0 to 2)	FWEIS0i
0x7904 + 0x10 × i	Port i Error Interrupt Enable Register 0 (i = 0 to 2)	FWEIE0i
0x7908 + 0x10 × i	Port i Error Interrupt Disable Register 0 (i = 0 to 2)	FWEID0i
0x7A00	Error Interrupt Status Register 1	FWEIS1
0x7A04	Error Interrupt Enable Register 1	FWEIE1
0x7A08	Error Interrupt Disable Register 1	FWEID1
0x7A10	Error Interrupt Status Register 2	FWEIS2
0x7A14	Error Interrupt Enable Register 2	FWEIE2
0x7A18	Error Interrupt Disable Register 2	FWEID2
0x7A40	Error Interrupt Status Register 5	FWEIS5
0x7A44	Error Interrupt Enable Register 5	FWEIE5
0x7A48	Error Interrupt Disable Register 5	FWEID5
0x7A50	Error Interrupt Status Register 60	FWEIS60
0x7A54	Error Interrupt Enable Register 60	FWEIE60
0x7A58	Error Interrupt Disable Register 60	FWEID60
0x7A60	Error Interrupt Status Register 61	FWEIS61
0x7A64	Error Interrupt Enable Register 61	FWEIE61
0x7A68	Error Interrupt Disable Register 61	FWEID61
0x7A70	Error Interrupt Status Register 62	FWEIS62
0x7A74	Error Interrupt Enable Register 62	FWEIE62
0x7A78	Error Interrupt Disable Register 62	FWEID62
0x7A80	Error Interrupt Status Register 63	FWEIS63
0x7A84	Error Interrupt Enable Register 63	FWEIE63
0x7A88	Error Interrupt Disable Register 63	FWEID63
0x7A90	Error Interrupt Status Register 70	FWEIS70
0x7A94	Error Interrupt Enable Register 70	FWEIE70
0x7A98	Error Interrupt Disable Register 70	FWEID70
0x7AA0	Error Interrupt Status Register 71	FWEIS71
0x7AA4	Error Interrupt Enable Register 71	FWEIE71
0x7AA8	Error Interrupt Disable Register 71	FWEID71
0x7AB0	Error Interrupt Status Register 72	FWEIS72
0x7AB4	Error Interrupt Enable Register 72	FWEIE72
0x7AB8	Error Interrupt Disable Register 72	FWEID72
0x7AC0	Error Interrupt Status Register 73	FWEIS73

Table 30.3 MFWD registers (7 of 7)

Offset address	Register name	Register symbol
0x7AC4	Error Interrupt Enable Register 73	FWEIE73
0x7AC8	Error Interrupt Disable Register 73	FWEID73
0x7AD0	Error Interrupt Status Register 80	FWEIS80
0x7AD4	Error Interrupt Enable Register 80	FWEIE80
0x7AD8	Error Interrupt Disable Register 80	FWEID80
0x7AE0	Error Interrupt Status Register 81	FWEIS81
0x7AE4	Error Interrupt Enable Register 81	FWEIE81
0x7AE8	Error Interrupt Disable Register 81	FWEID81
0x7AF0	Error Interrupt Status Register 82	FWEIS82
0x7AF4	Error Interrupt Enable Register 82	FWEIE82
0x7AF8	Error Interrupt Disable Register 82	FWEID82
0x7B00	Error Interrupt Status Register 83	FWEIS83
0x7B04	Error Interrupt Enable Register 83	FWEIE83
0x7B08	Error Interrupt Disable Register 83	FWEID83
0x7C00	Monitoring Interrupt Status Register 0	FWMIS0
0x7C04	Monitoring Interrupt Enable Register 0	FWMIE0
0x7C08	Monitoring Interrupt Disable Register 0	FWMID0

Note: Base address: MFWD = 0x403C_0000, MFWD_NS = 0x503C_0000

Note: The address preceded by "E:" correspond to an emulation address, which allows to read a register without modifying its content.

Note: If nothing specified, a register need an authorization from secured APB interface to be accessed by unsecured APB interface. This applies for Read and Write accesses. Any exception to this rule will be specified under "Security restrictions" or "Security no restrictions" labels.

30.3 MFWD Register Descriptions

30.3.1 MFWD Function Registers

This section describes MFWD function registers.

For MFWD function procedure details, see [section 30.4.1. Software Flows](#).

30.3.1.1 General Function Registers

30.3.1.1.1 FWGC : General Configuration Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x0000

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	SVM[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	SVM[1:0]	Switch VLAN Mode 0 0: No VLAN mode (VLAN not used for forwarding) 0 1: C-TAG mode (C-TAG used for forwarding) 1 0: SC-TAG mode (S-TAG used for forwarding) 1 1: Reserved	R/W
31:2	—	These bits are read as 0. The write value should be 0.	R/W

SVM[1:0] bits (Switch VLAN Mode)

HW: If FWGC.SVM is set to 00b, incoming packet PCP and DEI fields will be considered as 0 by MFWD.

For more details, see [section 30.5.1.2. ESWM Modes](#).

30.3.1.1.2 FW TTC0 : TAG TPID Configuration Register 0

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x0010

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	STT[15:0]															
Value after reset:	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	CTT[15:0]															
Value after reset:	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	CTT[15:0]	C-TAG TPID [801.2Q] Configure the C-TAG TPID.	R/W
31:16	STT[15:0]	S-TAG TPID [801.2Q] Configure the S-TAG TPID.	R/W

Note:

- SW: FW TTC0.CTT, FW TTC0.STT and FW TTC1.RTT registers should always be set to different values.
- SW: This register cannot be set to 0x0000 to 0x05FF, 0x0800 nor 0x86DD.

30.3.1.1.3 FW TTC1 : TAG TPID Configuration Register 1

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x0014

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	RTT[15:0]															
Value after reset:	1	1	1	1	0	0	0	1	1	1	0	0	0	0	0	1

Bit	Symbol	Function	R/W
15:0	RTT[15:0]	R-TAG TPID [801.2CB] Configure the R-TAG TPID.	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

- Note:
- SW: FWTTTC0.CTT, FWTTTC0.STT and FWTTTC1.RTT registers should always be set to different values.
 - SW: This register cannot be set to 0x0000 to 0x05FF, 0x0800 nor 0x86DD.

30.3.1.1.4 FWCEPTC : CPU Exceptional Path Target Configuration Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x0020

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	EPSL	—	—	—	—	—	—	—	EPCS
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	EPIPV[2:0]					—	—	—	—	EPCSD[5:0]					
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
5:0	EPCSD[5:0]	Exceptional Path CPU Sub Destination Configures the CPU sub destination to which exceptional frames will be sent.	R/W
11:6	—	These bits are read as 0. The write value should be 0.	R/W
14:12	EPIPV[2:0]	Exceptional Path Internal Priority Value Configures the priority to which exceptional frames will be sent.	R/W
15	—	This bit is read as 0. The write value should be 0.	R/W
16	EPCS	Exceptional Path CPU Select*1 Configures CPU to which exceptional frames will be sent. See Fabric specification for port mapping [MFAB].	R/W
23:17	—	These bits are read as 0. The write value should be 0.	R/W
24	EPSL	Exceptional Path Security Level Configures descriptor security level for exceptional path.	R/W
31:25	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. SW: These bits should not be set to a value greater than 1.

30.3.1.1.5 FWCEPRC0 : CPU Exceptional Path Reason Configuration Register 0

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x0024

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	DDFS FEF	—	DDEE F	—	—	GDNE EF	GTFE F	—	GSEQ EEF	GAXE EF	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	ETFEF	EOEE F	EUEE F	EBOE EF	ERPO OEF	ERFF EF	ECFF CEEF	ECFS EEF	EFFM EEF	EFCS EEF	ENIBE EF	EPCR CEEF	EPHY EEF
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	EPHYEEF	Ethernet PHY Error Exceptional Forwarding 0: Frames received with PHY Error are discarded. 1: Frames received with PHY Error are forwarded to exceptional path.	R/W

Bit	Symbol	Function	R/W
1	EPCRCEEF	Ethernet PCH CRC Error Exceptional Forwarding 0: Frames received with PCH CRC Error are discarded. 1: Frames received with PCH CRC Error are forwarded to exceptional path.	R/W
2	ENIBEEF	Ethernet Nibble Error Exceptional Forwarding 0: Frames received with Nibble Error are discarded. 1: Frames received with Nibble Error are forwarded to exceptional path.	R/W
3	EFCSEEF	Ethernet FCS Error Exceptional Forwarding 0: Frames received with FCS Error are discarded. 1: Frames received with FCS Error are forwarded to exceptional path.	R/W
4	EFFMEEF	Ethernet Final Fragment Missing Error Exceptional Forwarding 0: Frames received with Final Fragment Missing Error are discarded. 1: Frames received with Final Fragment Missing Error are forwarded to exceptional path.	R/W
5	ECFSEEF	Ethernet C-Fragment SMD Error Exceptional Forwarding 0: Frames received with C-Fragment SMD Error are discarded. 1: Frames received with C-Fragment SMD Error are forwarded to exceptional path.	R/W
6	ECFFCEEF	Ethernet C-Fragment FRAG_COUNT Error Exceptional Forwarding 0: Frames received with C-Fragment FRAG_COUNT Error are discarded. 1: Frames received with C-Fragment FRAG_COUNT Error are forwarded to exceptional path.	R/W
7	ERFFEF	Ethernet RMAC Frame Filtered Exceptional Forwarding 0: Frames received with RMAC Frame Filtered flag are discarded. 1: Frames received with RMAC Frame Filtered flag are forwarded to exceptional path.	R/W
8	ERPOOEF	Ethernet Reception Partially Out of Operation Exceptional Forwarding 0: Frames received with Reception Partially Out of Operation flag are discarded. 1: Frames received with Reception Partially Out of Operation flag are forwarded to exceptional path.	R/W
9	EBOEEF	Ethernet Buffer Overflow Error Exceptional Forwarding 0: Frames received with Buffer Overflow Error are discarded. 1: Frames received with Buffer Overflow Error are forwarded to exceptional path.	R/W
10	EUEEF	Ethernet Undersize Error Exceptional Forwarding 0: Frames received with Undersize Error are discarded. 1: Frames received with Undersize Error are forwarded to exceptional path.	R/W
11	EOEEF	Ethernet Oversize Error Exceptional Forwarding 0: Frames received with Oversize Error are discarded. 1: Frames received with Oversize Error are forwarded to exceptional path.	R/W
12	ETFEF	Ethernet TAG Filtering Exceptional Forwarding 0: Frames received with TAG Filtering flag are discarded. 1: Frames received with TAG Filtering flag are forwarded to exceptional path.	R/W
16:13	—	These bits are read as 0. The write value should be 0.	R/W
17	GAXEEF	GWCA AXI Error Exceptional Forwarding 0: Frames received with AXI Error are discarded. 1: Frames received with AXI Error are forwarded to exceptional path.	R/W
18	GSEQEEF	GWCA Sequence Error Exceptional Forwarding 0: Frames received with Sequence Error are discarded. 1: Frames received with Sequence Error are forwarded to exceptional path.	R/W
19	—	This bit is read as 0. The write value should be 0.	R/W
20	GTFEF	GWCA TAG Filtering Exceptional Forwarding 0: Frames received with TAG Filtering flag are discarded. 1: Frames received with TAG Filtering flag are forwarded to exceptional path.	R/W
21	GDNEEF	GWCA Descriptor Number Error Exceptional Forwarding 0: Frames received with Descriptor Number Error are discarded. 1: Frames received with Descriptor Number Error are forwarded to exceptional path.	R/W
23:22	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
24	DDEEF	Direct Descriptor Error Exceptional Forwarding 0: Frames with Direct Descriptor Error are discarded. 1: Frames with Direct Descriptor Error are forwarded to exceptional path.	R/W
25	—	This bit is read as 0. The write value should be 0.	R/W
26	DDFSFEF	Direct Descriptor Format Security Filtering Exceptional Forwarding 0: Frames with Direct Descriptor Format Security Error are discarded. 1: Frames with Direct Descriptor Format Security Error are forwarded to exceptional path.	R/W
31:27	—	These bits are read as 0. The write value should be 0.	R/W

Note:

- For ethernet error/flag explanations, refer to Ethernet agent specification document [ETHA].
- For GWCA error/flag explanations, refer to GWCA specification document [GWCA].

30.3.1.1.6 FWCEPRC1 : CPU Exceptional Path Reason Configuration Register 1

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x0028

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	FSFFE F	FIFFE F	—	—	—	—	—	FMTR FEF	—	FMSD UFEF
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	FMSDUFEF	MSDU Filtering Exceptional Forwarding 0: Frames filtered by PSFP MSDU filter are discarded. 1: Frames filtered by PSFP MSDU filter are forwarded to exceptional path.	R/W
1	—	This bit is read as 0. The write value should be 0.	R/W
2	FMTRFEF	Meter Filtering Exceptional Forwarding 0: Frames filtered by PSFP Meter filter are discarded. 1: Frames filtered by PSFP Meter filter are forwarded to exceptional path.	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W
8	FIFFEF	Individual FRER Filtering Exceptional Forwarding 0: Frames filtered by FRER individual recovery are discarded. 1: Frames filtered by FRER individual recovery are forwarded to exceptional path.	R/W
9	FSFFE F	Sequence FRER Filtering Exceptional Forwarding 0: Frames filtered by FRER Sequence recovery are discarded. 1: Frames filtered by FRER Sequence recovery are forwarded to exceptional path.	R/W
31:10	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.1.7 FWCEPRC2 : CPU Exceptional Path Reason Configuration Register 2

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x002C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	FWMF EF	—	—	—	—	FVLA NSLF EF	FSMA CSLF EF	FDMA CSLF EF	—	—	FLTHS LFEF
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	FPBN TFEF	FLTW NTFE F	—	FLTHN TFEF	FDDN TFEF	—	—	FVLA NUFE F	FSMA CUFE F	FDMA CUFE F	—	—	FLTHU FEF
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	FLTHUFEF	Layer 3 Unknown Filtering Exceptional Forwarding 0: Frames filtered because their stream ID is unknown are discarded. 1: Frames filtered because their stream ID is unknown are forwarded to exceptional path.	R/W
2:1	—	These bits are read as 0. The write value should be 0.	R/W
3	FDMACUFEF	Destination MAC Unknown Filtering Exceptional Forwarding 0: Frames filtered because their Destination MAC is unknown are discarded. 1: Frames filtered because their Destination MAC is unknown are forwarded to exceptional path.	R/W
4	FSMACUFEF	Source MAC Unknown Filtering Exceptional Forwarding 0: Frames filtered because their Source MAC is unknown are discarded. 1: Frames filtered because their Source MAC is unknown are forwarded to exceptional path.	R/W
5	FVLNUFEF	VLAN Unknown Filtering Exceptional Forwarding 0: Frames filtered because their VLAN is unknown are discarded. 1: Frames filtered because their VLAN is unknown are forwarded to exceptional path.	R/W
7:6	—	These bits are read as 0. The write value should be 0.	R/W
8	FDDNTFEF	Direct Descriptor No Target Filtering Exceptional Forwarding 0: Frames filtered because no target during direct forwarding is detected are discarded. 1: Frames filtered because no target during direct forwarding is detected are forwarded to exceptional path.	R/W
9	FLTHNTFEF	Layer 3 No Target Filtering Exceptional Forwarding 0: Frames filtered because no target during Layer 3 forwarding is detected are discarded. 1: Frames filtered because no target during Layer 3 forwarding is detected are forwarded to exceptional path.	R/W
10	—	This bit is read as 0. The write value should be 0.	R/W
11	FLTWNTFEF	Layer 2 No Target Filtering Exceptional Forwarding 0: Frames filtered because the combination on VLAN destination vector and destination MAC destination vector is null are discarded. 1: Frames filtered because the combination on VLAN destination vector and destination MAC destination vector is null are forwarded to exceptional path.	R/W
12	FPBNTFEF	Port Based No Target Filtering Exceptional Forwarding 0: Frames filtered because no target during Port Based forwarding is detected are discarded. 1: Frames filtered because no target during Port Based forwarding is detected are forwarded to exceptional path.	R/W
15:13	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
16	FLTHSLFEF	Layer 3 Source Lock Filtering Exceptional Forwarding 0: Frames filtered because of Layer 3 source lock are discarded. 1: Frames filtered because of Layer 3 source lock are forwarded to exceptional path.	R/W
18:17	—	These bits are read as 0. The write value should be 0.	R/W
19	FDMACSLFEF	Destination MAC Source Lock Filtering Exceptional Forwarding 0: Frames filtered because of Destination MAC source lock are discarded. 1: Frames filtered because of Destination MAC source lock are forwarded to exceptional path.	R/W
20	FSMACSLFEF	Source MAC Source Lock Filtering Exceptional Forwarding 0: Frames filtered because of Source MAC source lock are discarded. 1: Frames filtered because of Source MAC source lock are forwarded to exceptional path.	R/W
21	FVLANSLEF	VLAN Source Lock Filtering Exceptional Forwarding 0: Frames filtered because of VLAN source lock are discarded. 1: Frames filtered because of VLAN source lock are forwarded to exceptional path.	R/W
25:22	—	These bits are read as 0. The write value should be 0.	R/W
26	FWMFEF	Watermark Filtering Exceptional Forwarding 0: Frames filtered because of watermark are discarded. 1: Frames filtered because of watermark are forwarded to exceptional path.	R/W
31:27	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.1.8 FWCLPTC : CPU Learning Path Target Configuration Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x0030

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	LPSL	—	—	—	—	—	—	—	LPCS
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	LPIPV[2:0]			—	—	—	—	—	—	LPCSD[5:0]					
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
5:0	LPCSD[5:0]	Learning Path CPU Sub Destination Configures the CPU sub destination to which Learning frames will be sent.	R/W
11:6	—	These bits are read as 0. The write value should be 0.	R/W
14:12	LPIPV[2:0]	Learning Path Internal Priority Value Configures the priority to which Learning frames will be sent.	R/W
15	—	This bit is read as 0. The write value should be 0.	R/W
16	LPCS	Learning Path CPU Select ^{*1} Configures CPU to which Learning frames will be sent. See Fabric specification for port mapping [MFAB].	R/W
23:17	—	These bits are read as 0. The write value should be 0.	R/W
24	LPSL	Learning Path Security Level Configures descriptor security level for Learning path.	R/W
31:25	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. SW: These bits should not be set to a value greater than 1.

30.3.1.1.9 FWCLPRC : CPU Learning Path Reason Configuration Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x0034

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	UVLA NLF	UPSM ACLF	USMA CLF	UDMA CLF	—	—	—	USIDL F
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	USIDLF	Unknown Stream ID Learning Forwarding 0: Frames received with an unknown stream ID are not forwarded to Learning path. 1: Frames received with an unknown stream ID are forwarded to Learning path.	R/W
3:1	—	These bits are read as 0. The write value should be 0.	R/W
4	UDMACLF	Unknown Destination MAC Learning Forwarding 0: Frames received with an unknown destination MAC address are not forwarded to Learning path. 1: Frames received with an unknown destination MAC address are forwarded to Learning path.	R/W
5	USMACLF	Unknown Source MAC Learning Forwarding 0: Frames received with an unknown Source MAC are not forwarded to Learning path. 1: Frames received with an unknown Source MAC are forwarded to Learning path.	R/W
6	UPSMACLF	Unknown Port for Source MAC Learning Forwarding 0: Frames received with an unknown port for Source MAC are not forwarded to Learning path. 1: Frames received with an unknown port for Source MAC are forwarded to Learning path.	R/W
7	UVLANLF	Unknown VLAN Learning Forwarding 0: Frames received with an unknown VLAN are not forwarded to Learning path. 1: Frames received with an unknown VLAN are forwarded to Learning path.	R/W
31:8	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.1.10 FWCMP TC : CPU Mirroring Path Target Configuration Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x0040

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	CMPS L	—	—	—	—	—	—	—	CMPC S
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	CMPIP U	CMPIPV[2:0]			—	—	—	—	—	—	CMPCSD[5:0]					
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
5:0	CMPCSD[5:0]	CPU Mirroring Path CPU Sub Destination Configures the CPU sub destination to which CPU mirroring frames will be sent.	R/W
11:6	—	These bits are read as 0. The write value should be 0.	R/W
14:12	CMPIPV[2:0]	CPU Mirroring Path Internal Priority Value	R/W
15	CMPIPU	CPU Mirroring Path Internal Priority Update 0: Frames mirrored to CPU path are sent with the same priority as for forwarding. 1: Frames mirrored to CPU path are sent with FWCMPTC.CMPIPV priority.	R/W
16	CMPCS	CPU Mirroring Path CPU Select* ¹ Configures CPU to which CPU mirroring frames will be sent. See Fabric specification for port mapping [MFAB].	R/W
23:17	—	These bits are read as 0. The write value should be 0.	R/W
24	CMPSL	CPU Mirroring Path Security Level Configures descriptor security level for CPU mirroring path.	R/W
31:25	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. SW: These bits should not be set to a value greater than 1.

30.3.1.1.11 FWEMPTC : Ethernet Mirroring Path Target Configuration Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x0044

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	EMPS L	—	—	—	—	—	—	—	EMPP S
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	EMPIP U	EMPIPV[2:0]			—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
11:0	—	These bits are read as 0. The write value should be 0.	R/W
14:12	EMPIPV[2:0]	Ethernet Mirroring Path Internal Priority Value	R/W
15	EMPIPU	Ethernet Mirroring Path Internal Priority Update 0: Frames mirrored to Ethernet path are sent with the same priority as for forwarding. 1: Frames mirrored to Ethernet path are sent with FWEMPTC.EMPIPV priority.	R/W
16	EMPPS	Ethernet Mirroring Path Port Select* ¹ Configures port to which Ethernet mirroring frames will be sent. See Fabric specification for port mapping [MFAB].	R/W
23:17	—	These bits are read as 0. The write value should be 0.	R/W
24	EMPSL	Ethernet Mirroring Path Security Level Configures descriptor security level for Ethernet mirroring path.	R/W
31:25	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. SW: These bits should not be set to a value greater than 1.

30.3.1.1.12 FWSDMPTC : Source-Destination Mirroring Path Target Configuration Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x0050

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	SDMP SL	—	—	—	—	—	—	SDMPPS[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	SDMPI PU	SDMPIPV[2:0]			—	—	—	—	—	—	SDMPCSD[5:0]					
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
5:0	SDMPCSD[5:0]	Source-Destination Mirroring Path CPU Sub Destination Configures the CPU sub destination to which Source-Destination mirroring frames will be sent.	R/W
11:6	—	These bits are read as 0. The write value should be 0.	R/W
14:12	SDMPIPV[2:0]	Source-Destination Mirroring Path Internal Priority Value	R/W
15	SDMPIPU	Source-Destination Mirroring Path Internal Priority Update 0: Frames mirrored to Source-Destination path are sent with the same priority as for forwarding. 1: Frames mirrored to Source-Destination path are sent with FWSDMPTC.SDMPIPV priority.	R/W
17:16	SDMPPS[1:0]	Source-Destination Mirroring Path Port Select ^{*1} Configures Port to which Source-Destination mirroring frames will be sent. See Fabric specification for port mapping [MFAB].	R/W
23:18	—	These bits are read as 0. The write value should be 0.	R/W
24	SDMPSL	Source-Destination Mirroring Path Security Level Configures descriptor security level for Source-Destination mirroring path.	R/W
31:25	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. SW: These bits should not be set to a value greater than 3.

30.3.1.1.13 FWSDMPVC : Source-Destination Mirroring Path Vector Configuration Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x0054

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	SDMSV[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	SDMDV[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	SDMDV[2:0]	Source-Destination Mirroring Destination Vector	R/W
15:3	—	These bits are read as 0. The write value should be 0.	R/W
18:16	SDMSV[2:0]	Source-Destination Mirroring Source Vector	R/W
31:19	—	These bits are read as 0. The write value should be 0.	R/W

SDMDV[2:0] bits (Source-Destination Mirroring Destination Vector)

Frames coming from a port set in FWSDMPVC.SDMSV register and sent to a port set in FWSDMPVC.SDMDV register is also duplicated to Source-Destination mirroring path.

Setting this register to 0 disable Source-Destination mirroring path.

SDMSV[2:0] bits (Source-Destination Mirroring Source Vector)

Frames coming from a port set in FWSDMPVC.SDMSV register and sent to a port set in FWSDMPVC.SDMDV register is also duplicated to Source-Destination mirroring path.

Setting this register to 0 disable Source-Destination mirroring path.

30.3.1.1.14 FWLBWMCi : Port i Level Based Watermark Configuration Register (i = 0 to 2)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x0080 + 0x4 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	WMFLPR15 to WMFLPR0															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	WMCLPR15 to WMCLPR0															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	WMCLPR0 to WMCLPR15	Watermark Critical Level Priority n Reject (n = 0 to 15)* ¹ 0: Frames coming for port i and that should be forwarded with {DEI,IPV} equal to j are forwarded. 1: Frames coming for port i and that should be forwarded with {DEI,IPV} equal to j are discarded when watermark critical level is set for port i (WM.CRITICAL[i] is set [COMA]).	R/W
31:16	WMFLPR0 to WMFLPR15	Watermark Flush Level Priority n Reject (n = 0 to 15)* ¹ 0: Frames coming for port i and that should be forwarded with {DEI,IPV} equal to j are forwarded. 1: Frames coming for port i and that should be forwarded with {DEI,IPV} equal to j are discarded when watermark flush level is set for port i (WM.FLUSH[i] is set [COMA]).	R/W

Note 1. {DEI,IPV} refers to the bit association {DEI,IPV[2:0]}, so in decimal it can be written 8*DEI+IPV.

30.3.1.1.15 FWPCi0 : Port i Configuration Register 0 (i = 0 to 2)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x0100 + 0x10 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	VLAN RUS	VLAN RU	VLAN SA	MACH MA	MACH LA	MACR USSA	MACR USA	MACS SA	MACR UDSA	MACR UDA	MACD SA	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	L2SE	IP6OE	IP6TE	IP6UE	IP4OE	IP4TE	IP4UE	LTHR USS	LTHR US	LTHTA
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	LTHTA	L3 Table Active 0: L3 table is disabled for source port i. 1: L3 table is enabled for source port i.	R/W
1	LTHRUS	L3 Reject Unknown Streams 0: Unknown streams coming from source port i are not rejected. 1: Unknown streams coming from source port i are rejected.	R/W
2	LTHRUS	L3 Reject Unknown Secure Streams 0: Unknown secure streams coming from source port i are not rejected. 1: Unknown secure streams coming from source port i are rejected.	R/W
3	IP4UE	IPv4 UDP Enable 0: IPv4/UDP detection disabled for source port i. 1: IPv4/UDP detection enabled for source port i.	R/W
4	IP4TE	IPv4 TCP Enable 0: IPv4/TCP detection disabled for source port i. 1: IPv4/TCP detection enabled for source port i.	R/W
5	IP4OE	IPv4 Other Enable 0: Non-TCP non-UDP IPv4 detection disabled for source port i. 1: Non-TCP non-UDP IPv4 detection enabled for source port i.	R/W
6	IP6UE	IPv6 UDP Enable 0: IPv6/UDP detection disabled for source port i. 1: IPv6/UDP detection enabled for source port i.	R/W
7	IP6TE	IPv6 TCP Enable 0: IPv6/TCP detection disabled for source port i. 1: IPv6/TCP detection enabled for source port i.	R/W
8	IP6OE	IPv6 Other Enable 0: Non-TCP non-UDP IPv6 detection disabled for source port i. 1: Non-TCP non-UDP IPv6 detection enabled for source port i.	R/W
9	L2SE	L2 Stream Enable 0: L2 stream disabled for source port i. 1: L2 stream enabled for source port i.	R/W
19:10	—	These bits are read as 0. The write value should be 0.	R/W
20	MACDSA	MAC Destination Search Active 0: MAC Destination search is disabled for source port i. 1: MAC Destination search is enabled for source port i.	R/W
21	MACRU	MAC Reject Unknown Destination Addresses 0: Unknown MAC Destination addresses coming from source port i are not rejected. 1: Unknown MAC Destination addresses coming from source port i are rejected.	R/W

Bit	Symbol	Function	R/W
22	MACRUDSA	MAC Reject Unknown Destination Secure Addresses 0: Unknown secure MAC Destination addresses coming from source port i are not rejected. 1: Unknown secure MAC Destination addresses coming from source port i are rejected.	R/W
23	MACSSA	MAC Source Search Active 0: MAC Source search is disabled for source port i. 1: MAC Source search is enabled for source port i.	R/W
24	MACRUSA	MAC Reject Unknown Source Addresses 0: Unknown MAC Source addresses coming from source port i are not rejected. 1: Unknown MAC Source addresses coming from source port i are rejected.	R/W
25	MACRUSSA	MAC Reject Unknown Source Secure Addresses 0: Unknown secure MAC Source addresses coming from source port i are not rejected. 1: Unknown secure MAC Source addresses coming from source port i are rejected.	R/W
26	MACHLA	MAC Hardware Learning Active 0: Unknown unicast MAC Source addresses coming from source port i are not added to the MAC table. 1: Unknown unicast MAC Source addresses coming from source port i are added to the MAC table.	R/W
27	MACHMA	MAC Hardware Migration Active 0: Known unicast MAC Source addresses coming from source port i with a wrong source port in MAC.DV are not overwritten. 1: Known unicast MAC Source addresses coming from source port i with a wrong source port in MAC.DV are overwritten.	R/W
28	VLANSa	VLAN Search Active*1 HW: If VLAN Destination search is disabled for port i, all frames coming from port i are considered as unknown for VLAN search. 0: VLAN search is disabled for source port i. 1: VLAN search is enabled for source port i.	R/W
29	VLANRU	VLAN Reject Unknown*1 0: Unknown VLAN coming from source port i are not rejected. 1: Unknown VLAN coming from source port i are rejected.	R/W
30	VLANRUS	VLAN Reject Unknown Secure*1 0: Unknown secure VLAN coming from source port i are not rejected. 1: Unknown secure VLAN coming from source port i are rejected.	R/W
31	—	This bit is read as 0. The write value should be 0.	R/W

Note 1. SW: This bit should be set to 1 if FWGC.SVM is set to 00b.

30.3.1.1.16 FWPCi1 : Port i Configuration Register 1 (i = 0 to 2)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x0104 + 0x10 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	LTHFM2 to LTHFM0		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	DDSL	DDE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DDE	Direct Descriptor Enable* ¹ 0: Direct Descriptor disabled for port i 1: Direct Descriptor enabled for port i	R/W
1	DDSL	Direct Descriptor Security Level* ¹ 0: Secured Direct Descriptor discarded for port i. 1: Secured Direct Descriptor forwarded for port i.	R/W
15:2	—	These bits are read as 0. The write value should be 0.	R/W
18:16	LTHFM2 to LTHFM0	Layer 3 Forwarding Mask n (n = 0 to 2) 0: Frames forwarded by Layer 3 forwarding can be forwarded to port j. 1: Frames forwarded by Layer 3 forwarding cannot be forwarded to port j.	R/W
31:19	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. HW: This bit is only valid for i = 2, 3.

30.3.1.1.17 FWPCi2 : Port i Configuration Register 2 (i = 0 to 2)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x0108 + 0x10 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	LTWFM2 to LTWFM0		
Value after reset:	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	—	These bits are read as 0. The write value should be 0.	R/W
18:16	LTWFM2 to LTWFM0	Layer 2 Forwarding Mask n (n = 0 to 2) 0: Frames forwarded by Layer 2 forwarding can be forwarded to port j. 1: Frames forwarded by Layer 2 forwarding cannot be forwarded to port j.	R/W
31:19	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.2 Cut-through Forwarding Function Registers

30.3.1.2.1 FWCTGCI0 : Cut-Through General Configuration Register i0 (i = 0 to 7)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x0400 + 0x40 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	CTRT GI	CTVCTRL[1:0]		CTFI	—	—	CTET E	CTSD E	CTSP E	CTSV E	CTCD E	CTCP E	CTCV E	CTMS E	CTMD E
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CTMDE	Cut-Through MAC Destination Enable 0: MAC destination not included in Cut-Through separation for separation rule i. 1: MAC destination included in Cut-Through separation for separation rule i.	R/W
1	CTMSE	Cut-Through MAC Source Enable 0: MAC Source not included in Cut-Through separation for separation rule i. 1: MAC Source included in Cut-Through separation for separation rule i.	R/W
2	CTCVE	Cut-Through C-TAG VLAN Enable* ¹ 0: C-TAG VLAN not included in Cut-Through separation for separation rule i. 1: C-TAG VLAN included in Cut-Through separation for separation rule i.	R/W
3	CTCPE	Cut-Through C-TAG PCP Enable* ¹ 0: C-TAG PCP not included in Cut-Through separation for separation rule i. 1: C-TAG PCP included in Cut-Through separation for separation rule i.	R/W
4	CTCDE	Cut-Through C-TAG DEI Enable* ¹ 0: C-TAG DEI not included in Cut-Through separation for separation rule i. 1: C-TAG DEI included in Cut-Through separation for separation rule i.	R/W
5	CTSVE	Cut-Through S-TAG VLAN Enable* ² 0: S-TAG VLAN not included in Cut-Through separation for separation rule i. 1: S-TAG VLAN included in Cut-Through separation for separation rule i.	R/W
6	CTSPE	Cut-Through S-TAG PCP Enable* ² 0: S-TAG PCP not included in Cut-Through separation for separation rule i. 1: S-TAG PCP included in Cut-Through separation for separation rule i.	R/W
7	CTSDE	Cut-Through S-TAG DEI Enable* ² 0: S-TAG DEI not included in Cut-Through separation for separation rule i. 1: S-TAG DEI included in Cut-Through separation for separation rule i.	R/W
8	CTETE	Cut-Through Ethernet Type Enable 0: Ethernet Type not included in Cut-Through separation for separation rule i. 1: Ethernet Type included in Cut-Through separation for separation rule i.	R/W
10:9	—	These bits are read as 0. The write value should be 0.	R/W
11	CTFI	Cut-Through FCS In* ³ This bit is used as FDESCR.FI for cut-through. 0: The frame matching separation rule i as no FCS included. 1: The frame matching separation rule i as an FCS included.	R/W
13:12	CTVCTRL[1:0]	Cut-Through VLAN Control [GWCA] [ETHA] These bits are used as FDESCR.VCTRL[1:0] for cut-through. 0 0: The ingress matching separation rule i frame is a No TAG frame. 0 1: The ingress matching separation rule i frame is a C-TAG frame. 1 0: The ingress matching separation rule i frame is an SC-TAG frame. 1 1: The ingress matching separation rule i frame is a CoS TAG frame.	R/W
14	CRTGI	Cut-Through R-TAG In [GWCA] [ETHA] This bit is used as FDESCR.RTGI for cut-through. 0: R-TAG is not included in cut-through frame. 1: R-TAG is included in cut-through frame.	R/W
31:15	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. SW: This bit can be set to 1 only if FWGC.SVM is set to 10b or 01b.

Note 2. SW: This bit can be set to 1 only if FWGC.SVM is set to 10b.

Note 3. SW: This bit should be set to the frame source port MRGC.RCPT (same) value [RMAC].

- Case 1: CTFI = 1, RCPT = 1 and correct FCS frame incoming
→ correct FCS frame forwarded
(The FCS attached to the input frame is output as it is. If the tag is converted, the FCS value will be invalid.)
- Case 2: CTFI = 1, RCPT = 1 and incorrect FCS frame incoming
→ incorrect FCS frame forwarded
- Case 3: CTFI = 0, RCPT = 0 and correct FCS frame incoming
→ correct FCS frame forwarded
- Case 4: CTFI = 0, RCPT = 0 and incorrect FCS frame incoming
→ correct FCS frame forwarded (The correct FCS will be added after the incorrect FCS.)

30.3.1.2.2 FWCTGCI1 : Cut-Through General Configuration Register i1 (i = 0 to 7)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x0404 + 0x40 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	CTMT[25:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	CTMT[25:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
25:0	CTMT[25:0]	Cut-Through Maximum time	R/W
31:26	—	These bits are read as 0. The write value should be 0.	R/W

CTMT[25:0] bits (Cut-Through Maximum time)

Describes maximum time on the PHY in ns of frames matching separation rule I.

This time should take in account the whole frame (including preamble, SFD, TAGs and FCS) and IFG.

This size will be used for TAS calculations [ETHA].

- Note:
- If this register does not have the correct value, some frames could cross a closed gate in TAS [ETHA].
 - if TAS is enabled, sending frames smaller than this register value will decrease the PHY throughput because the frame time will be reserved at this register value [ETHA]. Therefore, TAS gate open time must be bigger than CTMT value.

30.3.1.2.3 FWCTTCi0 : Cut-Through Target Configuration Register i0 (i = 0 to 7)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x0408 + 0x40 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CTDFM1 to CTDFM0	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	CTDV[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	CTDV[2:0]	Cut-through Destination Vector* ¹ Select the ports to which the frames matching separation rule i will be sent. Enables cut-through for separation rule i.	R/W
15:3	—	These bits are read as 0. The write value should be 0.	R/W
17:16	CTDFM1 to CTDFM0	Cut-Through Destination n Forwarding Mode (n = 0 to 1)* ² 0: Frame is forwarded in Cut-through mode. 1: Frame is forwarded in Store and forward mode.	R/W
31:18	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. When cut-through is enabled for rule i, at least one bit set to 1 in FWCTTCi0.CTDV[2:0] and should be set FWCTTCi0.CTDFM to 0 for a cut-through able destination agent (when cut-through is enabled, at least one Ethernet destination port should use with cut-through mode).

Note 2. HW: In Cut-through mode, a frame with an error cannot be rejected and is forwarded to its destination ports.

30.3.1.2.4 FWCTTCi1 : Cut-Through Target Configuration Register i1 (i = 0 to 7)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x040C + 0x40 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CTEM E	CTCM E
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	CTIPU	CTIPV[2:0]			—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
11:0	—	These bits are read as 0. The write value should be 0.	R/W
14:12	CTIPV[2:0]	Cut-through Internal Priority Value	R/W
15	CTIPU	Cut-through Internal Priority Update 0: Use the output descriptor priority to forwarding frames matching separation rule i. 1: Use FWCTTCi1.CTIPV priority to forwarding frames matching separation rule i.	R/W
16	CTCME	Cut-through CPU Mirroring Enable 0: CPU mirroring is disabled for frames matching separation rule i. 1: CPU mirroring is enabled for frames matching separation rule i.	R/W
17	CTEME	Cut-through Ethernet Mirroring Enable 0: Ethernet mirroring is disabled for frames matching separation rule i. 1: Ethernet mirroring is enabled for frames matching separation rule i.	R/W
31:18	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.2.5 FWCTTCi20 : Cut-Through Target Configuration Register i20 (i = 0 to 7)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x0410 + 0x40 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	CTCSD[5:0]					
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
5:0	CTCSD[5:0]	Cut-Through CPU Sub Destination Configures the CPU sub destination to which frames matching separation rule i will go for slow port number 0.	R/W
31:6	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.2.6 FWCTSCi0 : Cut-Through Separation Configuration Register i0 (i = 0 to 7)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x0420 + 0x40 × i

Bit position: 31 0

Bit field: CTDMAU[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	CTDMAU[31:0]	Cut-Through Destination MAC Address Upper Part	R/W

30.3.1.2.7 FWCTSCi1 : Cut-Through Separation Configuration Register i1 (i = 0 to 7)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x0424 + 0x40 × i

Bit position: 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16

Bit field: CTDMAL[15:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field: CTSMAL[15:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
15:0	CTSMAL[15:0]	Cut-Through Source MAC Address Upper Part	R/W
31:16	CTDMAL[15:0]	Cut-Through Destination MAC Address Lower Part	R/W

30.3.1.2.8 FWCTSCi2 : Cut-Through Separation Configuration Register i2 (i = 0 to 7)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x0428 + 0x40 × i

Bit position: 31 0

Bit field: CTSMAL[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	CTSMAL[31:0]	Cut-Through Source MAC Address Lower Part	R/W

30.3.1.2.9 FWCTSCi3 : Cut-Through Separation Configuration Register i3 (i = 0 to 7)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x042C + 0x40 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	CTSD	CTSP[2:0]			CTSV[11:0]											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	CTCD	CTCP[2:0]			CTCV[11:0]											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
11:0	CTCV[11:0]	Cut-Through C-TAG VLAN	R/W
14:12	CTCP[2:0]	Cut-Through C-TAG PCP	R/W
15	CTCD	Cut-Through C-TAG DEI	R/W
27:16	CTSV[11:0]	Cut-Through S-TAG VLAN	R/W
30:28	CTSP[2:0]	Cut-Through S-TAG PCP	R/W
31	CTSD	Cut-Through S-TAG DEI	R/W

30.3.1.2.10 FWCTSCi4 : Cut-Through Separation Configuration Register i4 (i = 0 to 7)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x0430 + 0x40 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CTSP N
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	CTET[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	CTET[15:0]	Cut-Through Ethernet Type	R/W
16	CTSPN	Cut-Through Source Port Number*1	R/W
31:17	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. HW: In Cut-through mode (not Store and Forward modes), this bit should be set to a value smaller than 2.

TWBFV0[15:0] bits (Two-Byte Filter Value 0)

- If FWTWBF_{CI}.TWBFUM is 00b: Used as a 2-byte filter value masked by FWTWBFV_{CI}.TWBFV1 with a filter ID of 2*i.
- If FWTWBF_{CI}.TWBFUM is 01b: Used as the first 2-byte of a 4-byte filter extended by FWTWBFV_{CI}.TWBFV1 with a filter ID of 2*i (Compare value is {FWTWBFV_{CI}.TWBFV0, FWTWBFV_{CI}.TWBFV1}).
- If FWTWBF_{CI}.TWBFUM is 10b: Used as 2-byte filter value with a filter ID of 2*i.

TWBFV1[15:0] bits (Two-Byte Filter Value 1)

- If FWTWBF_{CI}.TWBFUM is 00b: Used as a mask for FWTWBFV_{CI}.TWBFV0.
- If FWTWBF_{CI}.TWBFUM is 01b: Used as an extension for FWTWBFV_{CI}.TWBFV0 (Compare value is {FWTWBFV_{CI}.TWBFV0, FWTWBFV_{CI}.TWBFV1}).
- If FWTWBF_{CI}.TWBFUM is 10b: Used as 2-byte filter value with a filter ID of 2*i+1.

30.3.1.3.3 FWTHBFCi : Three-Byte Filter Configuration Register i (i = 0 to 15)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x1400 + 0x10 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	THBFOV[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	THBFUM[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	THBFUM[1:0]	Three-Byte Filter Unit Mode 0 0: Mask mode 0 1: Expand mode 1 0: Precise mode 1 1: Reserved	R/W
15:2	—	These bits are read as 0. The write value should be 0.	R/W
23:16	THBFOV[7:0]	Three-Byte Filter Offset Value Offset in byte where the filter i will start filtering the frame. (The offset ignores all TAGs)	R/W
31:24	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.3.4 FWTHBFV0Ci : Three-Byte Filter Value 0 Configuration Register i (i = 0 to 15)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x1404 + 0x10 × i

Bit position:	31	30	29	28	27	26	25	24	23									0
Bit field:	—	—	—	—	—	—	—	—	—	THBFV0[23:0]								
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
23:0	THBFV0[23:0]	Three-Byte Filter Value 0	R/W
31:24	—	These bits are read as 0. The write value should be 0.	R/W

THBFV0[23:0] bits (Three-Byte Filter Value 0)

- If FWTHBFCi.THBFUM[1:0] is 00b: Used as a 3-byte filter value masked by FWTHBFV1Ci.THBV1[23:0] with a filter ID of $2 \times (i + 16)$.
- If FWTHBFCi.THBFUM[1:0] is 01b: Used as the first 3-byte of a 6-byte filter extended by FWTHBFV1Ci.THBV1[23:0] with a filter ID of $2 \times (i + 16)$ (Compare value is {FWTHBFV0Ci.THBV0[23:0], FWTHBFV1Ci.THBV1[23:0]}).
- If FWTHBFCi.THBFUM[1:0] is 10b: Used as 3-byte filter value with a filter ID of $2 \times (i + 16)$.

30.3.1.3.5 FWTHBFV1Ci : Three-Byte Filter Value 1 Configuration Register i (i = 0 to 15)

Base address: MFWD = 0x403C_0000
MFWD NS = 0x503C 0000

Offset address: $0x1408 + 0x10 \times i$

Bit position: 31 23 0

Bit field:	—	—	—	—	—	—	—	—	—			THBFV1[23:0]
------------	---	---	---	---	---	---	---	---	---	--	--	--------------

Value after reset: 0

Bit	Symbol	Function	R/W
23:0	THBFV1[23:0]	Three-Byte Filter Value 1	R/W
31:24	—	These bits are read as 0. The write value should be 0.	R/W

THBFV1[23:0] bits (Three-Byte Filter Value 1)

- If FWTHBFCi.THBFUM[1:0] is 00b: Used as a mask for FWTHBFV0Ci.THBFV0[23:0].
- If FWTHBFCi.THBFUM[1:0] is 01b: Used as an extension for FWTHBFV0Ci.THBFV0[23:0] (Compare value is { FWTHBFV0Ci.THBFV0[23:0], FWTHBFV1Ci.THBFV1[23:0]}).
- If FWTHBFCi.THBFUM[1:0] is 10b: Used as 3-byte filter value with a filter ID of $2 \times (i + 16) + 1$.

30.3.1.3.6 FWFOBFCi : Four-Byte Filter Configuration Register i (i = 0 to 15)

Base address: MFWD = 0x403C_0000
MFWD NS = 0x503C_0000

Offset address: $0x1800 + 0x10 \times i$

Bit position: 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16

[illegible]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

[illegible]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
1:0	FOBFUM[1:0]	Four-Byte Filter Unit Mode 0 0: Mask mode 0 1: Expand mode 1 0: Precise mode 1 1: Reserved	R/W
15:2	—	These bits are read as 0. The write value should be 0.	R/W
23:16	FOBFOV[7:0]	Four-Byte Filter Offset Value Offset in byte where the filter i will start filtering the frame. (The offset ignores all TAGs)	R/W

Bit	Symbol	Function	R/W
31:24	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.3.7 FWFOBFBV0Ci : Four-Byte Filter Value 0 Configuration Register i (i = 0 to 15)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x1804 + 0x10 × i

Bit position: 31

0

Bit field:

FOBFV0[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	FOBFV0[31:0]	Four-Byte Filter Value 0	R/W

FOBFV0[31:0] bits (Four-Byte Filter Value 0)

- If FWFOBFCi.FOBFUM[1:0] is 00b: Used as a 4-byte filter value masked by FWFOBFBV1Ci.FOBFV1[31:0] with a filter ID of $2 \times (i + 36)$.
- If FWFOBFCi.FOBFUM[1:0] is 01b: Used as the first 4-byte of an 8-byte filter extended by FWFOBFBV1Ci.FOBFV1[31:0] with a filter ID of $2 \times (i + 36)$ (Compare value is {FWFOBFBV0Ci.FOBFV0[31:0], FWFOBFBV1Ci.FOBFV1[31:0]}).
- If FWFOBFCi.FOBFUM[1:0] is 10b: Used as 4-byte filter value with a filter ID of $2 \times (i + 36)$.

30.3.1.3.8 FWFOBFBV1Ci : Four-Byte Filter Value 1 Configuration Register i (i = 0 to 15)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x1808 + 0x10 × i

Bit position: 31

0

Bit field:

FOBFV1[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	FOBFV1[31:0]	Four-Byte Filter Value 1	R/W

FOBFV1[31:0] bits (Four-Byte Filter Value 1)

- If FWFOBFCi.FOBFUM[1:0] is 00b: Used as a mask for FWFOBFBV0Ci.FOBFV0[31:0].
- If FWFOBFCi.FOBFUM[1:0] is 01b: Used as an extension for FWFOBFBV0Ci.FOBFV0[31:0] (Compare value is {FWFOBFBV0Ci.FOBFV0[31:0], FWFOBFBV1Ci.FOBFV1[31:0]}).
- If FWFOBFCi.FOBFUM[1:0] is 10b: Used as 4-byte filter value with a filter ID of $2 \times (i + 36) + 1$.

30.3.1.3.9 FWRFCi : Range Filter Configuration Register i (i = 0 to 15)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x1C00 + 0x10 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	RFOV[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	RFM	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	—	These bits are read as 0. The write value should be 0.	R/W
8	RFM	Range Filtering Mode* ¹ 0: Offset filtering 1: VLAN TAG filtering	R/W
15:9	—	These bits are read as 0. The write value should be 0.	R/W
23:16	RFOV[7:0]	Range Filter Offset Value* ²	R/W
31:24	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. SW: This bit should not be set to 1 if FWGC.SVM is set to 00b.

Note 2. • SW: If FWRFCi.RFM is set to 1, these bits should be set to 0 or 1 to filter S-TAG and set to 2 or 3 to filter C-TAG.
• SW: If FWRFCi.RFM is set to 1 and FWGC.SVM is set to 01b, these bits should not be set to 0 or 1.

RFOV[7:0] bits (Range Filter Offset Value)

- If FWRFCi.RFM is 0: Offset in byte where the filter i will start filtering the frame data.
- If FWRFCi.RFM is 1: Offset in byte where the filter i will start filtering the frame TAGs.

30.3.1.3.10 FWRFVCi : Range Filter Value Configuration Register i (i = 0 to 15)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x1C04 + 0x10 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	RFRV[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	RFSV1[7:0]								RFSV0[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	RFSV0[7:0]	Range Filter Start Value 0	R/W
15:8	RFSV1[7:0]	Range Filter Start Value 1	R/W
19:16	RFRV[3:0]	Range Filter Range Value* ¹ Used as range value for filters values FWRFVCi.RFSV1 and FWRFVCi.RFSV0.	R/W
31:20	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. SW: FWRFVCi.RFSV0+FWRFVCi.RFRV and FWRFVCi.RFSV1+FWRFVCi.RFRV should always be smaller or equal to 255.

RFSV0[7:0] bits (Range Filter Start Value 0)

Used has range filter start value with a filter ID of $2 \times (i + 49)$ (matching values are contained in [FWRFVCi.RFSV0: FWRFVCi.RFSV0+FWRFVCi.RFRV]).

RFSV1[7:0] bits (Range Filter Start Value 1)

Used has range filter start value with a filter ID of $2 \times (i + 48) + 1$ (matching values are contained in [FWRFVCi.RFSV1: FWRFVCi.RFSV1+FWRFVCi.RFRV]).

30.3.1.3.11 FWCFCi : Cascade Filter Configuration Register i (i = 0 to 15)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x2000 + 0x40 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CFPFFV1 to CFPFFV0	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	CFEFFV2 to CFEFFV0		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	CFEFFV2 to CFEFFV0	Cascade Filter E-Frame Filter Valid n (n = 0 to 2) Bit j is 0: Cascade filter i is disabled for port j e-frames (All slow ports frames are e-frames) [GWCA]. Bit j is 1: Cascade filter i is enabled for port j e-frames (All slow ports frames are e-frames) [GWCA].	R/W
15:3	—	These bits are read as 0. The write value should be 0.	R/W
17:16	CFPFFV1 to CFPFFV0	Cascade Filter E-Frame Filter Valid n (n = 0 to 1) Bit j is 0: Cascade filter i is disabled for port j p-frames. Bit j is 1: Cascade filter i is enabled for port j p-frames.	R/W
31:18	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.3.12 FWCFMCij : Cascade Filter Mapping Configuration Register ij (i = 0 to 15) (j = 0 to 6)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x2004 + 0x40 × i + 0x4 × j

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	CFFV	—	—	—	—	—	—	—	—	CFFN[6:0]						
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
6:0	CFFN[6:0]	Cascade Filter Filter Number Map filter number FWCFMCij.CFFN to cascade filter number i.	R/W

Bit	Symbol	Function	R/W
14:7	—	These bits are read as 0. The write value should be 0.	R/W
15	CFFV	Cascade Filter Valid Enables FWCFMCij.CFFN for cascade filter i. If no one of the FWCFMCij.CFFV is equal to 1, cascade filter i is disabled.	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.4 Layer 3 Forwarding/Routing L2 Stream/Hash Filter Function Registers

30.3.1.4.1 FWIP4SC : IPv4 Stream Configuration Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4008

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	IP4IDP TS	IP4IID S	IP4IIS S	IP4IC DS	IP4ICP S	IP4ICV S	IP4ISD S	IP4ISP S	IP4ISV S
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	IP4IDP TH	IP4ISP TH	IP4IPH	IP4IID H	IP4IIS H	IP4IC DH	IP4ICP H	IP4ICV H	IP4ISD H	IP4ISP H	IP4ISV H	IP4IM SH	IP4IM DH
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	IP4IMDH	IPv4 Include MAC Destination in Hash 0: MAC Destination address is not included in hash calculation 1: MAC Destination address is included in hash calculation	R/W
1	IP4IMSH	IPv4 Include MAC Source in Hash 0: MAC Source address is not included in hash calculation 1: MAC Source address is included in hash calculation	R/W
2	IP4ISVH	IPv4 Include S-TAG VLAN ID in Hash*1 0: S-TAG VLAN ID is not included in hash calculation 1: S-TAG VLAN ID is included in hash calculation	R/W
3	IP4ISPH	IPv4 Include S-TAG PCP in Hash*1 0: S-TAG PCP is not included in hash calculation 1: S-TAG PCP is included in hash calculation	R/W
4	IP4ISDH	IPv4 Include S-TAG DEI in Hash*1 0: S-TAG DEI is not included in hash calculation 1: S-TAG DEI is included in hash calculation	R/W
5	IP4ICVH	IPv4 Include C-TAG VLAN ID in Hash*2 0: C-TAG VLAN ID is not included in hash calculation 1: C-TAG VLAN ID is included in hash calculation	R/W
6	IP4ICPH	IPv4 Include C-TAG PCP in Hash*2 0: C-TAG PCP is not included in hash calculation 1: C-TAG PCP is included in hash calculation	R/W
7	IP4ICDH	IPv4 Include C-TAG DEI in Hash*2 0: C-TAG DEI is not included in hash calculation 1: C-TAG DEI is included in hash calculation	R/W
8	IP4IISH	IPv4 Include IP Source in Hash 0: IPv4 Source IP address is not included in hash calculation 1: IPv4 Source IP address is included in hash calculation	R/W
9	IP4IIDH	IPv4 Include IP Destination in Hash 0: IPv4 Destination IP address is not included in hash calculation 1: IP v4 Destination IP address is included in hash calculation	R/W

Bit	Symbol	Function	R/W
10	IP4IPH	IPv4 Include Protocol in Hash 0: IPv4 Protocol is not included in hash calculation 1: IPv4 Protocol is included in hash calculation	R/W
11	IP4ISPTH	IPv4 Include Source Port in Hash 0: TCP/UDP Source port is not included in hash calculation 1: TCP/UDP Source port is included in hash calculation	R/W
12	IP4IDPTH	IPv4 Include Destination Port in Hash 0: TCP/UDP Destination port is not included in hash calculation 1: TCP/UDP Destination port is included in hash calculation	R/W
15:13	—	These bits are read as 0. The write value should be 0.	R/W
16	IP4ISVS	IPv4 Include S-TAG VLAN ID in Stream* ¹ 0: S-TAG VLAN ID is not included in L3 IPv4 stream ID 1: S-TAG VLAN ID is included in L3 IPv4 stream ID	R/W
17	IP4ISPS	IPv4 Include S-TAG PCP in Stream* ¹ 0: S-TAG PCP is not included in L3 IPv4 stream ID 1: S-TAG PCP is included in L3 IPv4 stream ID	R/W
18	IP4ISDS	IPv4 Include S-TAG DEI in Stream* ¹ 0: S-TAG DEI is not included in L3 IPv4 stream ID 1: S-TAG DEI is included in L3 IPv4 stream ID	R/W
19	IP4ICVS	IPv4 Include C-TAG VLAN ID in Stream* ² 0: C-TAG VLAN ID is not included in L3 IPv4 stream ID 1: C-TAG VLAN ID is included in L3 IPv4 stream ID	R/W
20	IP4ICPS	IPv4 Include C-TAG PCP in Stream* ² 0: C-TAG PCP is not included in L3 IPv4 stream ID 1: C-TAG PCP is included in L3 IPv4 stream ID	R/W
21	IP4ICDS	IPv4 Include C-TAG DEI in Stream* ² 0: C-TAG DEI is not included in L3 IPv4 stream ID 1: C-TAG DEI is included in L3 IPv4 stream ID	R/W
22	IP4IIS	IPv4 Include IP Source in Stream 0: IPv4 Source IP address is not included in L3 IPv4 stream ID 1: IPv4 Source IP address is included in L3 IPv4 stream ID	R/W
23	IP4IIDS	IPv4 Include IP Destination in Stream 0: IPv4 Destination IP address is not included in L3 IPv4 stream ID 1: IPv4 Destination IP address is included in L3 IPv4 stream ID	R/W
24	IP4IDPTS	IPv4 Include Destination Port in Stream 0: TCP/UDP Destination port is not included in L3 IPv4 stream ID 1: TCP/UDP Destination port is included in L3 IPv4 stream ID	R/W
31:25	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. SW: If FWGC.SVM is set to 00b or 01b, this bit should be set to 0.

Note 2. SW: If FWGC.SVM is set to 00b, this bit should be set to 0.

30.3.1.4.2 FWIP6SC : IPv6 Stream Configuration Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4018

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	IP6IDP TS	IP6IIS S	IP6IIO S	IP6IC DS	IP6ICP S	IP6ICV S	IP6ISD S	IP6ISP S	IP6ISV S
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	IP6IDP TH	IP6ISP TH	IP6IPH	IP6IID H	IP6IIS H	IP6IC DH	IP6ICP H	IP6ICV H	IP6ISD H	IP6ISP H	IP6ISV H	IP6IM SH	IP6IM DH
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	IP6IMDH	IPv6 Include MAC Destination in Hash 0: MAC Destination address is not included in hash calculation 1: MAC Destination address is included in hash calculation	R/W
1	IP6IMSH	IPv6 Include MAC Source in Hash 0: MAC Source address is not included in hash calculation 1: MAC Source address is included in hash calculation	R/W
2	IP6ISVH	IPv6 Include S-TAG VLAN ID in Hash*1 0: S-TAG VLAN ID is not included in hash calculation 1: S-TAG VLAN ID is included in hash calculation	R/W
3	IP6ISPH	IPv6 Include S-TAG PCP in Hash*1 0: S-TAG PCP is not included in hash calculation 1: S-TAG PCP is included in hash calculation	R/W
4	IP6ISDH	IPv6 Include S-TAG DEI in Hash*1 0: S-TAG DEI is not included in hash calculation 1: S-TAG DEI is included in hash calculation	R/W
5	IP6ICVH	IPv6 Include C-TAG VLAN ID in Hash*2 0: C-TAG VLAN ID is not included in hash calculation 1: C-TAG VLAN ID is included in hash calculation	R/W
6	IP6ICPH	IPv6 Include C-TAG PCP in Hash*2 0: C-TAG PCP is not included in hash calculation 1: C-TAG PCP is included in hash calculation	R/W
7	IP6ICDH	IPv6 Include C-TAG DEI in Hash*2 0: C-TAG DEI is not included in hash calculation 1: C-TAG DEI is included in hash calculation	R/W
8	IP6IISH	IPv6 Include IP Source in Hash 0: IPv6 Source IP address is not included in hash calculation 1: IPv6 Source IP address is included in hash calculation	R/W
9	IP6IIDH	IPv6 Include IP Destination in Hash 0: IPv6 Destination IP address is not included in hash calculation 1: IPv6 Destination IP address is included in hash calculation	R/W
10	IP6IPH	IPv6 Include Protocol in Hash (Next Header) 0: IPv6 Protocol is not included in hash calculation 1: IPv6 Protocol is included in hash calculation	R/W
11	IP6ISPTH	IPv6 Include Source Port in Hash 0: TCP/UDP Source port is not included in hash calculation 1: TCP/UDP Source port is included in hash calculation	R/W
12	IP6IDPTH	IPv6 Include Destination Port in Hash 0: TCP/UDP Destination port is not included in hash calculation 1: TCP/UDP Destination port is included in hash calculation	R/W

Bit	Symbol	Function	R/W
15:13	—	These bits are read as 0. The write value should be 0.	R/W
16	IP6ISVS	IPv6 Include S-TAG VLAN ID in Stream* ¹ 0: S-TAG VLAN ID is not included in L3 IPv6 stream ID 1: S-TAG VLAN ID is included in L3 IPv6 stream ID	R/W
17	IP6ISPS	IPv6 Include S-TAG PCP in Stream* ¹ 0: S-TAG PCP is not included in L3 IPv6 stream ID 1: S-TAG PCP is included in L3 IPv6 stream ID	R/W
18	IP6ISDS	IPv6 Include S-TAG DEI in Stream* ¹ 0: S-TAG DEI is not included in L3 IPv6 stream ID 1: S-TAG DEI is included in L3 IPv6 stream ID	R/W
19	IP6ICVS	IPv6 Include C-TAG VLAN ID in Stream* ² 0: C-TAG VLAN ID is not included in L3 IPv6 stream ID 1: C-TAG VLAN ID is included in L3 IPv6 stream ID	R/W
20	IP6ICPS	IPv6 Include C-TAG PCP in Stream* ² 0: C-TAG PCP is not included in L3 IPv6 stream ID 1: C-TAG PCP is included in L3 IPv6 stream ID	R/W
21	IP6ICDS	IPv6 Include C-TAG DEI in Stream* ² 0: C-TAG DEI is not included in L3 IPv6 stream ID 1: C-TAG DEI is included in L3 IPv6 stream ID	R/W
22	IP6II0S	IPv6 Include IP 0 in Stream IPv6 IP address part 0 is a 4-byte byte length field extracted from the IP source or destination address depending on FWIP6OC configuration. 0: IPv6 IP address part 0 is not included in L3 IPv6 stream ID 1: IPv6 IP address part 0 is included in L3 IPv6 stream ID	R/W
23	IP6II1S	IPv6 Include IP 1 in Stream IPv6 IP address part 1 is a 4-byte byte length field extracted from the IP source or destination address depending on FWIP6OC configuration. 0: IPv6 IP address part 1 is not included in L3 IPv6 stream ID 1: IPv6 IP address part 1 is included in L3 IPv6 stream ID	R/W
24	IP6IDPTS	IPv6 Include Destination Port in Stream 0: TCP/UDP Destination port is not included in L3 IPv6 stream ID 1: TCP/UDP Destination port is included in L3 IPv6 stream ID	R/W
31:25	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. SW: If FWGC.SVM is set to 00b or 01b, this bit should be set to 0.

Note 2. SW: If FWGC.SVM is set to 00b, this bit should be set to 0.

30.3.1.4.3 FWIP6OC : IPv6 Offset Configuration Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x401C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	IP6IPO1[3:0]				—	—	—	IP6IPOM1
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	IP6IPO0[3:0]				—	—	—	IP6IPOM0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	IP6IPOM0	IPv6 IP Offset mode 0 0: IPv6 IP address part 0 is extracted from IP source address 1: IPv6 IP address part 0 is extracted from IP destination address	R/W

Bit	Symbol	Function	R/W
3:1	—	These bits are read as 0. The write value should be 0.	R/W
7:4	IP6IPO0[3:0]	IPv6 IP Offset 0*1 i: Byte [i:i+3] of selected IP address are used for IPv6 IP address part 0.	R/W
15:8	—	These bits are read as 0. The write value should be 0.	R/W
16	IP6IPOM1	IPv6 IP Offset mode 1 0: IPv6 IP address part 1 is extracted from IP source address 1: IPv6 IP address part 1 is extracted from IP destination address	R/W
19:17	—	These bits are read as 0. The write value should be 0.	R/W
23:20	IP6IPO1[3:0]	IPv6 IP Offset 1*1 i: Byte [i:i+3] of selected IP address are used for IPv6 IP address part 1.	R/W
31:24	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. SW: These bits should not be set to a value greater than 12.

30.3.1.4.4 FWL2SC : Layer 2 Stream Configuration Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4020

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	L2ICD S	L2ICP S	L2ICV S	L2ISD S	L2ISP S	L2ISV S	L2IMS S	L2IMD S
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	L2IMDS	Layer 2 Include MAC Destination in Stream 0: MAC Destination address is not included in L2 stream ID 1: MAC Destination address is included in L2 stream ID	R/W
1	L2IMSS	Layer 2 Include MAC Source in Stream 0: MAC Source address is not included in L2 stream ID 1: MAC Source address is included in L2 stream ID	R/W
2	L2ISVS	Layer 2 Include S-TAG VLAN ID in Stream*1 0: MAC S-TAG VLAN ID is not included in L2 stream ID 1: MAC S-TAG VLAN ID is included in L2 stream ID	R/W
3	L2ISPS	Layer 2 Include S-TAG PCP ID in Stream*1 0: MAC S-TAG PCP is not included in L2 stream ID 1: MAC S-TAG PCP is included in L2 stream ID	R/W
4	L2ISDS	Layer 2 Include S-TAG DEI in Stream*1 0: MAC S-TAG DEI is not included in L2 stream ID 1: MAC S-TAG DEI is included in L2 stream ID	R/W
5	L2ICVS	Layer 2 Include C-TAG VLAN ID in Stream*2 0: MAC C-TAG VLAN ID is not included in L2 stream ID 1: MAC C-TAG VLAN ID is included in L2 stream ID	R/W
6	L2ICPS	Layer 2 Include C-TAG PCP ID in Stream*2 0: MAC C-TAG PCP is not included in L2 stream ID 1: MAC C-TAG PCP is included in L2 stream ID	R/W
7	L2ICDS	Layer 2 Include C-TAG DEI in Stream*2 0: MAC C-TAG DEI is not included in L2 stream ID 1: MAC C-TAG DEI is included in L2 stream ID	R/W

Bit	Symbol	Function	R/W
31:8	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. SW: If FWGC.SVM is set to 00b or 01b, this bit should be set to 0.

Note 2. SW: If FWGC.SVM is set to 00b, this bit should be set to 0.

30.3.1.4.5 FWSFHEC : Stream Filter Hash Equation Configuration Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4030

Bit position: 31 16 15 0

Bit field:	IP6HE15 to IP6HE0	IP4HE15 to IP4HE0
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Value after reset: 0

Bit	Symbol	Function	R/W
15:0	IP4HE15 to IP4HE0	IPv4 Stream Filter Hash Equation n (n = 0 to 15) If several Hash equations are activated, the result of all activated equation are XORed to give the final Hash ID. Bit 0 is set: Hash equation $1 + x^{17}$ is activated in Hash function for IPv4 Hash calculation. Bit i (i ≠ 0) is set: Hash equation $1 + x^{i+1} + x^{17}$ is activated in Hash function for IPv4 Hash calculation.	R/W
31:16	IP6HE15 to IP6HE0	IPv6 Stream Filter Hash Equation n (n = 0 to 15) If several Hash equations are activated, the result of all activated equation are XORed to give the final Hash ID. Bit 0 is set: Hash equation $1 + x^{17}$ is activated in Hash function for IPv6 Hash calculation. Bit i (i ≠ 0) is set: Hash equation $1 + x^{i+1} + x^{17}$ is activated in Hash function for IPv6 Hash calculation.	R/W

30.3.1.4.6 FWSHCR0 : Software Hash Calculation Request Register 0

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4040

Bit position: 31 0

Bit field:	SHCMDP0[31:0]
------------	---------------

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	SHCMDP0[31:0]	Software Hash Calculation MAC Destination Part 0 SW: This register is used to accelerate hash calculation.	R/W

30.3.1.4.7 FWSHCR1 : Software Hash Calculation Request Register 1

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4044

Bit position: 31 16 15 0

Bit field:	SHCMDP1[15:0]	SHCMSP0[15:0]
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Value after reset: 0

Bit	Symbol	Function	R/W
15:0	SHCMSP0[15:0]	Software Hash Calculation MAC Source Part 0 SW: These bits are used to accelerate hash calculation.	R/W
31:16	SHCMSP1[15:0]	Software Hash Calculation MAC Destination Part 1 SW: These bits are used to accelerate hash calculation.	R/W

30.3.1.4.8 FWSHCR2 : Software Hash Calculation Request Register 2

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4048

Bit position: 31

0

Bit field:

SHCMSP1[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	SHCMSP1[31:0]	Software Hash Calculation MAC Source Part 1 SW: This register is used to accelerate hash calculation.	R/W

30.3.1.4.9 FWSHCR3 : Software Hash Calculation Request Register 3

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x404C

Bit position: 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16

Bit field:

SHCSP[2:0]

SHCS
D

SHCSV[11:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field:

SHCCP[2:0]

SHCC
D

SHCCV[11:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
11:0	SHCCV[11:0]	Software Hash Calculation C-TAG VLAN SW: These bits are used to accelerate hash calculation.	R/W
12	SHCCD	Software Hash Calculation C-TAG DEI SW: This bit is used to accelerate hash calculation.	R/W
15:13	SHCCP[2:0]	Software Hash Calculation C-TAG PCP SW: These bits are used to accelerate hash calculation.	R/W
27:16	SHCSV[11:0]	Software Hash Calculation S-TAG VLANs SW: These bits are used to accelerate hash calculation.	R/W
28	SHCSD	Software Hash Calculation S-TAG DEI SW: This bit is used to accelerate hash calculation.	R/W
31:29	SHCSP[2:0]	Software Hash Calculation S-TAG PCP SW: These bits are used to accelerate hash calculation.	R/W

30.3.1.4.10 FWSHCR4 : Software Hash Calculation Request Register 4

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4050

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	SHCF F
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	SHCP[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	SHCP[7:0]	Software Hash Calculation Protocol (NextHeader for IPv6) SW: These bits are used to accelerate hash calculation.	R/W
15:8	—	These bits are read as 0. The write value should be 0.	R/W
16	SHCFF	Software Hash Calculation Frame Format SW: This bit is used to accelerate hash calculation. 0: IPv4 Hash Calculation 1: IPv6 Hash Calculation	R/W
31:17	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.4.11 FWSHCR5 : Software Hash Calculation Request Register 5

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4054

Bit position:	31	0																															
Bit field:	SHCISP0[31:0]																																
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
31:0	SHCISP0[31:0]	Software Hash Calculation IP Source Part 0*1 SW: This register is used to accelerate hash calculation.	R/W

Note 1. SW: Should be set to 0 for IPv4 Hash calculation.

30.3.1.4.12 FWSHCR6 : Software Hash Calculation Request Register 6

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4058

Bit position:	31	0																															
Bit field:	SHCISP1[31:0]																																
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
31:0	SHCISP1[31:0]	Software Hash Calculation IP Source Part 1*1 SW: This register is used to accelerate hash calculation.	R/W

Note 1. SW: Should be set to 0 for IPv4 Hash calculation.

30.3.1.4.13 FWSHCR7 : Software Hash Calculation Request Register 7

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x405C

Bit position: 31

0

Bit field: SHCISP2[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	SHCISP2[31:0]	Software Hash Calculation IP Source Part 2*1 SW: This register is used to accelerate hash calculation.	R/W

Note 1. SW: Should be set to 0 for IPv4 Hash calculation.

30.3.1.4.14 FWSHCR8 : Software Hash Calculation Request Register 8

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4060

Bit position: 31

0

Bit field: SHCISP3[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	SHCISP3[31:0]	Software Hash Calculation IP Source Part 3*1 SW: This register is used to accelerate hash calculation.	R/W

Note 1. SW: Should be set to IP source address for IPv4 Hash calculation.

30.3.1.4.15 FWSHCR9 : Software Hash Calculation Request Register 9

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4064

Bit position: 31

0

Bit field: SHCIDP0[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	SHCIDP0[31:0]	Software Hash Calculation IP Destination Part 0*1 SW: This register is used to accelerate hash calculation.	R/W

Note 1. SW: Should be set to 0 for IPv4 Hash calculation.

30.3.1.4.16 FWSHCR10 : Software Hash Calculation Request Register 10

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4068

Bit position: 31 0

Bit field: SHCIDP1[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	SHCIDP1[31:0]	Software Hash Calculation IP Source Destination Part 1 ^{*1} SW: This register is used to accelerate hash calculation.	R/W

Note 1. SW: Should be set to 0 for IPv4 Hash calculation.

30.3.1.4.17 FWSHCR11 : Software Hash Calculation Request Register 11

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x406C

Bit position: 31 0

Bit field: SHCIDP2[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	SHCIDP2[31:0]	Software Hash Calculation IP Destination Part 2 ^{*1} SW: This register is used to accelerate hash calculation.	R/W

Note 1. - SW: Should be set to 0 for IPv4 Hash calculation.

30.3.1.4.18 FWSHCR12 : Software Hash Calculation Request Register 12

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4070

Bit position: 31 0

Bit field: SHCIDP3[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	SHCIDP3[31:0]	Software Hash Calculation IP Destination Part 3 ^{*1} SW: This register is used to accelerate hash calculation.	R/W

Note 1. SW: Should be set to IP destination address for IPv4 Hash calculation.

30.3.1.4.19 FWSHCR13 : Software Hash Calculation Request Register 13

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4074

Bit position:	31	16	15	0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																					
Bit field:	SHCSP[15:0]															SHCDP[15:0]																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	SHCDP[15:0]	Software Hash Calculation Destination Port SW: These bits are used to accelerate hash calculation.	R/W
31:16	SHCSP[15:0]	Software Hash Calculation Source Port SW: These bits are used to accelerate hash calculation.	R/W

30.3.1.4.20 FWSHCRR : Software Hash Calculation Request Result Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4078

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	SHC	—														
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	SHCR[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	SHCR[15:0]	Software Hash Calculation Result SW: These bits are used to accelerate hash calculation.	R
30:16	—	These bits are read as 0.	R
31	SHC	Software Hash Calculation	R

SHC bit (Software Hash Calculation)

SW: This bit is used to accelerate hash calculation.

[Setting condition]

HW: Writing FWSHCR13 register will set this bit.

[Clearing condition]

HW: This bit is deasserted when hash calculation is completed.

30.3.1.5 Layer 3/4 Forwarding/Routing Hash Function Registers

30.3.1.5.1 FWLTHHEC : L3 Hash Entry Configuration Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4090

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	LTHHMUE[8:0]								
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	LTHHMC[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	LTHHMC[7:0]	L3 Hash Maximum Collision Maximum collision number from which hash search by HW will be given up.	R/W
15:8	—	These bits are read as 0. The write value should be 0.	R/W
24:16	LTHHMUE[8:0]	L3 Hash Maximum Unsecure Entry SW: Maximum unsecure entry number that can be set in table.	R/W
31:25	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.5.2 FWLTHHC : L3 Hash Configuration Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4094

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	LTHHE7 to LTHHE0							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	LTHHE7 to LTHHE0	L3 Hash Equation n (n = 0 to 7) - Bit 0 is set: Hash equation $1 + x^9$ is activated in Hash function for L3 Hash calculation. - Bit i (i ≠ 0) is set: Hash equation $1 + x^{i+1} + x^9$ is activated in Hash function for L3 Hash calculation.	R/W
31:8	—	These bits are read as 0. The write value should be 0.	R/W

LTHHEn bits (L3 Hash Equation n (n = 0 to 7))

HW: If several Hash equations are activated, the result of all activated equation are XORed to give the final Hash ID.

30.3.1.5.3 FWLTHTL0 : L3 Table Learn Register 0

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x40A0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LTHED
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	LTHSL L	—	—	—	—	—	—	LTHSLP0[2:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	LTHSLP0[2:0]	L3 Stream Learn Part 0	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W
8	LTHSLL	L3 Security Level Learn Used for learning/overwriting an entry in L3 table.	R/W
15:9	—	These bits are read as 0. The write value should be 0.	R/W
16	LTHED	L3 Entry Delete 0: Learn/overwrite the set stream ID in L3 table. 1: Delete the set stream ID in L3 table.	R/W
31:17	—	These bits are read as 0. The write value should be 0.	R/W

LTHSLP0[2:0] bits (L3 Stream Learn Part 0)

Used for learning/overwriting an entry in L3 table.

Stream ID correspond to {FWLTHTL0. LTHSLP0, FWLTHTL1. LTHSLP1, FWLTHTL2. LTHSLP2, FWLTHTL3. LTHSLP3, FWLTHTL4. LTHSLP4}.

30.3.1.5.4 FWLTHTL1 : L3 Table Learn Register 1

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x40A4

Bit position:	31	0																																	
Bit field:	LTHSLP1[31:0]																																		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
31:0	LTHSLP1[31:0]	L3 Stream Learn Part 1	R/W

LTHSLP1[31:0] bits (L3 Stream Learn Part 1)

Used for learning/overwriting an entry in L3 table.

Stream ID correspond to {FWLTHTL0. LTHSLP0, FWLTHTL1. LTHSLP1, FWLTHTL2. LTHSLP2, FWLTHTL3. LTHSLP3, FWLTHTL4. LTHSLP4}.

30.3.1.5.5 FWLTHTL2 : L3 Table Learn Register 2

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x40A8

Bit position: 31 0

Bit field: LTHSLP2[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	LTHSLP2[31:0]	L3 Stream Learn Part 2	R/W

LTHSLP2[31:0] bits (L3 Stream Learn Part 2)

Used for learning/overwriting an entry in L3 table.

Stream ID correspond to {FWLTHTL0. LTHSLP0, FWLTHTL1. LTHSLP1, FWLTHTL2. LTHSLP2, FWLTHTL3. LTHSLP3, FWLTHTL4. LTHSLP4}.

30.3.1.5.6 FWLTHTL3 : L3 Table Learn Register 3

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x40AC

Bit position: 31 0

Bit field: LTHSLP3[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	LTHSLP3[31:0]	L3 Stream Learn Part 3	R/W

LTHSLP3[31:0] bits (L3 Stream Learn Part 3)

Used for learning/overwriting an entry in L3 table.

Stream ID correspond to {FWLTHTL0. LTHSLP0, FWLTHTL1. LTHSLP1, FWLTHTL2. LTHSLP2, FWLTHTL3. LTHSLP3, FWLTHTL4. LTHSLP4}.

30.3.1.5.7 FWLTHTL4 : L3 Table Learn Register 4

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x40B0

Bit position: 31 0

Bit field: LTHSLP4[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	LTHSLP4[31:0]	L3 Stream Learn Part 4	R/W

LTHSLP4[31:0] bits (L3 Stream Learn Part 4)

Used for learning/overwriting an entry in L3 table.

Stream ID correspond to {FWLTHTL0. LTHSLP0, FWLTHTL1. LTHSLP1, FWLTHTL2. LTHSLP2, FWLTHTL3. LTHSLP3, FWLTHTL4. LTHSLP4}.

30.3.1.5.8 FWLTHTL5 : L3 Table Learn Register 5

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x40B4

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	LTHMSDUVL	—	—	—	—	—	—	—	—	—	—	—	LTHMSDUNL[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	—	These bits are read as 0. The write value should be 0.	R/W
19:16	LTHMSDUNL[3:0]	L3 MSDU Number Learn SW: Used for learning/overwriting an entry in L3 table.	R/W
30:20	—	These bits are read as 0. The write value should be 0.	R/W
31	LTHMSDUVL	L3 MSDU Valid Learn Used for learning/overwriting an entry in L3 table.	R/W

30.3.1.5.9 FWLTHTL6 : L3 Table Learn Register 6

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x40B8

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	LTHMTRVL	—	—	—	—	—	—	—	—	—	—	LTHMTRNL[4:0]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	LTHFRERVL	—	—	—	—	—	—	—	—	LTHFRERNL[6:0]						
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
6:0	LTHFRERNL[6:0]	L3 FRER Number Learn Used for learning/overwriting an entry in L3 table.	R/W
14:7	—	These bits are read as 0. The write value should be 0.	R/W
15	LTHFRERVL	L3 FRER Valid Learn Used for learning/overwriting an entry in L3 table.	R/W
20:16	LTHMTRNL[4:0]	L3 Meter Number Learn Used for learning/overwriting an entry in L3 table.	R/W
30:21	—	These bits are read as 0. The write value should be 0.	R/W
31	LTHMTRVL	L3 Meter Valid Learn Used for learning/overwriting an entry in L3 table.	R/W

30.3.1.5.10 FWLTHTL7 : L3 Table Learn Register 7

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x40BC

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	LTHSLVL[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	LTHRV L	—	—	—	—	—	—	—	LTHRNL[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	LTHRNL[7:0]	L3 Routing Number Learn Used for learning/overwriting an entry in L3 table.	R/W
14:8	—	These bits are read as 0. The write value should be 0.	R/W
15	LTHRV L	L3 Routing Valid Learn Used for learning/overwriting an entry in L3 table.	R/W
18:16	LTHSLVL[2:0]	L3 Source Lock Vector Learn Used for learning/overwriting an entry in L3 table.	R/W
31:19	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.5.11 FWLTHTL80 : L3 Table Learn Register 80

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x40C0 + 0x4 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	LTHCSDL[5:0]					
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
5:0	LTHCSDL[5:0]	L3 CPU Sub-Destination Learn Used for learning/overwriting an entry in L3 table.	R/W
31:6	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.5.12 FWLTHTL9 : L3 Table Learn Register 9

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x40D0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	LTHC MEL	LTHE MEL	LTHIP UL	LTHIPVL[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	LTHDVL[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	LTHDVL[2:0]	L3 Destination Vector Learn Used for learning/overwriting an entry in L3 table.	R/W
15:3	—	These bits are read as 0. The write value should be 0.	R/W
18:16	LTHIPVL[2:0]	L3 Internal Priority Value Learn Used for learning/overwriting an entry in L3 table.	R/W
19	LTHIPUL	L3 Internal Priority Update Learn Used for learning/overwriting an entry in L3 table.	R/W
20	LTHEMEL	L3 Ethernet Mirroring Enable Learn SW: Used for learning/overwriting an entry in L3 table.	R/W
21	LTHCMEL	L3 CPU Mirroring Enable Learn Used for learning/overwriting an entry in L3 table.	R/W
31:22	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.5.13 FWLTHTLR : L3 Table Learn Result Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x40D4

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	LHTL	—	—	—	—	—	—	—	LTHLCN[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	LTHLO	—	LTHLS F	LTHLF
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	LTHLF	L3 Learn Fail Used for learning/overwriting an entry in L3 table. 0: Entry learning did not fail. (See (2) SW Learning.) 1: Entry learning failed. (See (2) SW Learning.)	R
1	LTHLSF	L3 Learn Security Fail Used for learning/overwriting an entry in L3 table. 0: Entry learning did not fail. (See (2) SW Learning.) 1: Entry learning failed. (See (2) SW Learning.)	R
2	—	This bit is read as 0.	R

Bit	Symbol	Function	R/W
3	LTHLO	L3 Learn Overwrite Used for learning/overwriting an entry in L3 table. 0: The entry learning did not overwrite an existing entry. 1: The entry learning overwrote an existing entry.	R
15:4	—	These bits are read as 0.	R
23:16	LTHLCN[7:0]	L3 Learn Collision Number Used for learning/overwriting an entry in L3 table.	R
30:24	—	These bits are read as 0.	R
31	LHTHL	L3 Table Learn	R

LHTHL bit (L3 Table Learn)

Used for learning/overwriting an entry in L3 table.

[Setting condition]

HW: Writing FWLHTHL9 register will set this bit.

[Clearing condition]

HW: This bit will be de-asserted when learning is completed.

30.3.1.5.14 FWLHTIM : L3 Table Initialization Monitoring Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x40E0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LHTR	LHTIOG
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	LHTIOG	L3 Table Initialization Ongoing	R/W
1	LHTR	L3 Table Ready	R
31:2	—	These bits are read as 0. The write value should be 0.	R/W

LHTIOG bit (L3 Table Initialization Ongoing)

[Setting condition]

SW: By writing 1 to this bit. It starts L3 Table initialization.

[Clearing condition]

HW: This bit is cleared when L3 Table initialization is finished.

LHTR bit (L3 Table Ready)

[Setting condition]

- HW: When FWLHTIM.LHTIOG is getting cleared.
- HW: This bit is set at $\text{clk_period [ns]} \times 256$ time from L3 table initialization starting.

[Clearing condition]

SW: By writing 1 to FWLHTIM.LHTIOG.

30.3.1.5.15 FWLTHTEM : L3 Table Entry Monitoring Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x40E4

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	LTHTUEN[8:0]								
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	LTHTEN[8:0]								
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
8:0	LTHTEN[8:0]	L3 Table Entry Number	R
15:9	—	These bits are read as 0.	R
24:16	LTHTUEN[8:0]	L3 Table Unsecure Entry Number	R
31:25	—	These bits are read as 0.	R

LTHTEN[8:0] bits (L3 Table Entry Number)

[Increment condition]

HW: An entry is learned successfully.

[Decrement condition]

HW: An entry is deleted successfully.

LTHTUEN[8:0] bits (L3 Table Unsecure Entry Number)

[Increment condition]

HW: An unsecure entry is learned successfully.

[Decrement condition]

HW: An unsecure entry is deleted successfully.

30.3.1.5.16 FWLTHTS0 : L3 Table Search Register 0

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4100

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	LTHSSP0[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	LTHSSP0[2:0]	L3 Stream Search Part 0 Used for searching an entry in L3 table.	R/W

Bit	Symbol	Function	R/W
31:3	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.5.17 FWLTHTS1 : L3 Table Search Register 1

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4104

Bit position: 31 0

Bit field: LTHSSP1[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	LTHSSP1[31:0]	L3 Stream Search Part 1 Used for searching an entry in L3 table.	R/W

30.3.1.5.18 FWLTHTS2 : L3 Table Search Register 2

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4108

Bit position: 31 0

Bit field: LTHSSP2[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	LTHSSP2[31:0]	L3 Stream Search Part 2 Used for searching an entry in L3 table.	R/W

30.3.1.5.19 FWLTHTS3 : L3 Table Search Register 3

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x410C

Bit position: 31 0

Bit field: LTHSSP3[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	LTHSSP3[31:0]	L3 Stream Search Part 3 Used for searching an entry in L3 table.	R/W

30.3.1.5.20 FWLTHTS4 : L3 Table Search Register 4

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4110

Bit position: 31

0

Bit field:

LTHSSP4[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	LTHSSP4[31:0]	L3 Stream Search Part 4 Used for searching an entry in L3 table.	R/W

30.3.1.5.21 FWLTHTSR0 : L3 Table Search Result Register 0

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4120

Bit position: 31

30

29

28

27

26

25

24

23

22

21

20

19

18

17

16

Bit field:

LTHTS

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

LTHSCN[7:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit position: 15

14

13

12

11

10

9

8

7

6

5

4

3

2

1

0

Bit field:

—

—

—

—

—

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—

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—

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—

—

—

—

—

LTHSL

S

LTHS

NF

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	—	This bit is read as 0. The write value should be 0.	R/W
1	LTHSNF	L3 Search Not found Used for searching an entry in L3 table. 0: Stream ID found in L3 Table. 1: Stream ID not found in L3 Table.	R
7:2	—	These bits are read as 0. The write value should be 0.	R/W
8	LTHSLS	L3 Security Level Search Used for searching an entry in L3 table.	R
15:9	—	These bits are read as 0. The write value should be 0.	R/W
23:16	LTHSCN[7:0]	L3 Search Collision Number Used for searching an entry in L3 table.	R
30:24	—	These bits are read as 0. The write value should be 0.	R/W
31	LTHTS	L3 Table Search	R

LTHTS bit (L3 Table Search)

Used for searching an entry in L3 table.

[Setting condition]

HW: Writing FWLTHTS4 register will set this bit.

[Clearing condition]

HW: This bit will be de-asserted when searching is completed.

30.3.1.5.22 FWLHTSR1 : L3 Table Search Result Register 1

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4124

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	LTHM SDUV S	—	—	—	—	—	—	—	—	—	—	—	LTHMSDUNS[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	—	These bits are read as 0.	R
19:16	LTHMSDUNS[3:0]	L3 MSDU Number Search Used for searching an entry in L3 table.	R
30:20	—	These bits are read as 0.	R
31	LTHMSDUVS	L3 MSDU Valid Search Used for searching an entry in L3 table.	R

30.3.1.5.23 FWLHTSR2 : L3 Table Search Result Register 2

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4128

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	LTHM TRVS	—	—	—	—	—	—	—	—	—	—	LTHMTRNS[4:0]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	LTHFR ERVS	—	—	—	—	—	—	—	—	LTHFRERNS[6:0]						
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
6:0	LTHFRERNS[6:0]	L3 FRER Number Search Used for searching an entry in L3 table.	R
14:7	—	These bits are read as 0.	R
15	LTHFRERVS	L3 FRER Valid Search Used for searching an entry in L3 table.	R
20:16	LTHMTRNS[4:0]	L3 Meter Number Search Used for searching an entry in L3 table.	R
30:21	—	These bits are read as 0.	R
31	LTHMTRVS	L3 Meter Valid Search Used for searching an entry in L3 table.	R

30.3.1.5.24 FWLTHTSR3 : L3 Table Search Result Register 3

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x412C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	LTHSLVS[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	LTHRVS	—	—	—	—	—	—	—	—	—	—	—	—	LTHRNS[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	LTHRNS[2:0]	L3 Routing Number Search Used for searching an entry in L3 table.	R
14:3	—	These bits are read as 0.	R
15	LTHRVS	L3 Routing Valid Search Used for searching an entry in L3 table.	R
18:16	LTHSLVS[2:0]	L3 Source Lock Vector Search Used for searching an entry in L3 table.	R
31:19	—	These bits are read as 0.	R

30.3.1.5.25 FWLTHTSR40 : L3 Table Search Result Register 40

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4130 + 0x4 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	LTHCSDS[5:0]					
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
5:0	LTHCSDS[5:0]	L3 CPU Sub-Destination Search Used for searching an entry in L3 table.	R
31:6	—	These bits are read as 0.	R

30.3.1.5.26 FWLHTSR5 : L3 Table Search Result Register 5

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4140

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	LTHC MES	LTHE MES	LTHIP US	LTHIPVS[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	LTHDVS[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	LTHDVS[2:0]	L3 Destination Vector Search Used for searching an entry in L3 table.	R
15:3	—	These bits are read as 0.	R
18:16	LTHIPVS[2:0]	L3 Internal Priority Value Search Used for searching an entry in L3 table.	R
19	LTHIPUS	L3 Internal Priority Update Search Used for searching an entry in L3 table.	R
20	LTHEMES	L3 Ethernet Mirroring Enable Search Used for searching an entry in L3 table.	R
21	LTHCMES	L3 CPU Mirroring Enable Search Used for searching an entry in L3 table.	R
31:22	—	These bits are read as 0.	R

30.3.1.5.27 FWLHTTR : L3 Table Read Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4150

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	LTHAR[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	LTHAR[7:0]	L3 Address Read Used for reading an entry in L3 table.	R/W
31:8	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.5.28 FWLTHTRR0 : L3 Table Read Result Register 0

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4154

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	LTHTR	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LTHEVR	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	—	This bit is read as 0.	R
1	LTHEVR	L3 Entry Valid Read Used for reading an entry in L3 table.	R
30:2	—	These bits are read as 0.	R
31	LTHTR	L3 Table Read	R

LTHTR bit (L3 Table Read)

Used for reading an entry in L3 table.

[Setting condition]

HW: Writing FWLTHTR register will set this bit.

[Clearing condition]

HW: This bit will be de-asserted when reading is completed.

30.3.1.5.29 FWLTHTRR1 : L3 Table Read Result Register 1

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4158

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	LTHSLR	—	—	—	—	—	—	—	LTHSRP0[2:0]
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	LTHSRP0[2:0]	L3 Stream Read Part 0 Used for reading an entry in L3 table.	R
7:3	—	These bits are read as 0.	R
8	LTHSLR	L3 Security Level Read Used for reading an entry in L3 table.	R
31:9	—	These bits are read as 0.	R

30.3.1.5.30 FWLTHTRR2 : L3 Table Read Result Register 2

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x415C

Bit position: 31 0

Bit field: LTHSRP1[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	LTHSRP1[31:0]	L3 Stream Read Part 1 Used for reading an entry in L3 table.	R

30.3.1.5.31 FWLTHTRR3 : L3 Table Read Result Register 3

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4160

Bit position: 31 0

Bit field: LTHSRP2[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	LTHSRP2[31:0]	L3 Stream Read Part 2 Used for reading an entry in L3 table.	R

30.3.1.5.32 FWLTHTRR4 : L3 Table Read Result Register 4

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4164

Bit position: 31 0

Bit field: LTHSRP3[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	LTHSRP3[31:0]	L3 Stream Read Part 3 Used for reading an entry in L3 table.	R

30.3.1.5.33 FWLTHTRR5 : L3 Table Read Result Register 5

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4168

Bit position: 31 0

Bit field: LTHSRP4[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	LTHSRP4[31:0]	L3 Stream Read Part 4 Used for reading an entry in L3 table.	R

30.3.1.5.34 FWLTHTRR6 : L3 Table Read Result Register 6

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x416C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	LTHM SDUV R	—	—	—	—	—	—	—	—	—	—	—	LTHMSDUNR[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	—	These bits are read as 0.	R
19:16	LTHMSDUNR[3:0]	L3 MSDU Number Read Used for reading an entry in L3 table.	R
30:20	—	These bits are read as 0.	R
31	LTHMSDUVR	L3 MSDU Valid Read Used for reading an entry in L3 table.	R

30.3.1.5.35 FWLTHTRR7 : L3 Table Read Result Register 7

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4170

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	LTHM TRVR	—	—	—	—	—	—	—	—	—	—	LTHMTRNR[4:0]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	LTHFR ERVR	—	—	—	—	—	—	—	—	LTHFRERNR[6:0]						
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
6:0	LTHFRERNR[6:0]	L3 FRER Number Read Used for reading an entry in L3 table.	R
14:7	—	These bits are read as 0.	R
15	LTHFRERVR	L3 FRER Valid Read Used for reading an entry in L3 table.	R
20:16	LTHMTRNR[4:0]	L3 Meter Number Read Used for reading an entry in L3 table.	R
30:21	—	These bits are read as 0.	R

Bit	Symbol	Function	R/W
31	LTHMTRVR	L3 Meter Valid Read Used for reading an entry in L3 table.	R

30.3.1.5.36 FWLTHTRR8 : L3 Table Read Result Register 8

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4174

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	LTHSLVR[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	LTHRV R	—	—	—	—	—	—	—	LTHRNR[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	LTHRNR[7:0]	L3 Routing Number Read Used for reading an entry in L3 table.	R
14:8	—	These bits are read as 0.	R
15	LTHRVR	L3 Routing Valid Read Used for reading an entry in L3 table.	R
18:16	LTHSLVR[2:0]	L3 Source Lock Vector Read Used for reading an entry in L3 table.	R
31:19	—	These bits are read as 0.	R

30.3.1.5.37 FWLTHTRR90 : L3 Table Read Result Register 90

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4180

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	LTHCSDR[5:0]					
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
5:0	LTHCSDR[5:0]	L3 CPU Sub-Destination Read Used for reading an entry in L3 table.	R
31:6	—	These bits are read as 0.	R

30.3.1.5.38 FWLTHTRR10 : L3 Table Read Result Register 10

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4190

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	LTHC MER	LTHE MER	LTHIP UR	LTHIPVR[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	LTHDVR[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	LTHDVR[2:0]	L3 Destination Vector Read Used for reading an entry in L3 table.	R
15:3	—	These bits are read as 0.	R
18:16	LTHIPVR[2:0]	L3 Internal Priority Value Read Used for reading an entry in L3 table.	R
19	LTHIPUR	L3 Internal Priority Update Read Used for reading an entry in L3 table.	R
20	LTHEMER	L3 Ethernet Mirroring Enable Read Used for reading an entry in L3 table.	R
21	LTHCMER	L3 CPU Mirroring Enable Read Used for reading an entry in L3 table.	R
31:22	—	These bits are read as 0.	R

30.3.1.6 Layer 2 Forwarding Function Registers

30.3.1.6.1 FWMACHEC : MAC Hash Entry Configuration Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4620

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	MACHMUE[11:0]											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	MACHMC[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
10:0	MACHMC[10:0]	MAC Hash Maximum Collision Maximum collision number from which hash search by HW will be given up.	R/W
15:11	—	These bits are read as 0. The write value should be 0.	R/W
27:16	MACHMUE[11:0]	MAC Hash Maximum Unsecure Entry SW: Maximum unsecure entry number that can be set in table.	R/W
31:28	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.6.2 FWMACHC : MAC Hash Configuration Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4624

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	MACHE10 to MACHE0										
Value after reset:	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
10:0	MACHE10 to MACHE0	MAC Hash Equation n (n = 0 to 10) - Bit 0 is set: Hash equation $1 + x^{12}$ is activated in Hash function for MAC Hash calculation. - Bit i (i ≠ 0) is set: Hash equation $1 + x^{i+1} + x^{12}$ is activated in Hash function for MAC Hash calculation.	R/W
31:11	—	These bits are read as 0. The write value should be 0.	R/W

MACHEn bits (MAC Hash Equation n (n = 0 to 10))

HW: If several Hash equations are activated, the result of all activated equation are XORed to give the final Hash ID.

30.3.1.6.3 FWMACTL0 : MAC Table Learn Register 0

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4630

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	MACE D
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	MACH LDL	MACD EL	MACS LL	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	—	These bits are read as 0. The write value should be 0.	R/W
8	MACSLL	MAC Security Level Learn Used for learning/overwriting an entry in MAC table.	R/W
9	MACDEL	MAC Dynamic Entry Learn Used for learning/overwriting an entry in MAC table.	R/W
10	MACHLDL	MAC Hardware Learning Disable Learn Used for learning/overwriting an entry in MAC table.	R/W
15:11	—	These bits are read as 0. The write value should be 0.	R/W
16	MACED	MAC Entry Delete 0: Learn/overwrite the set MAC address in MAC table. 1: Delete the set MAC address in MAC table.	R/W
31:17	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.6.4 FWMACTL1 : MAC Table Learn Register 1

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4634

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	MACMALP0[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	MACMALP0[15:0]	MAC MAC address Learn Part 0 Used for learning/overwriting an entry in MAC table.	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.6.5 FWMACTL2 : MAC Table Learn Register 2

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4638

Bit position:	31																																	0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																										
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Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
31:0	MACMALP1[31:0]	MAC MAC address Learn Part 1 Used for learning/overwriting an entry in MAC table.	R/W

30.3.1.6.6 FWMACTL3 : MAC Table Learn Register 3

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x463C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	MACDSLVL[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	MACSSLVL[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	MACSSLVL[2:0]	MAC Source Source Lock Vector Learn Used for learning/overwriting an entry in MAC table.	R/W
15:3	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
18:16	MACDSLVL[2:0]	MAC Destination Source Lock Vector Learn Used for learning/overwriting an entry in MAC table.	R/W
31:19	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.6.7 FWMACTL40 : MAC Table Learn Register 40

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4640 + 0x4 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	MACCSDL[5:0]					
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
5:0	MACCSDL[5:0]	MAC CPU Sub-Destination Learn Used for learning/overwriting an entry in MAC table.	R/W
31:6	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.6.8 FWMACTL5 : MAC Table Learn Register 5

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4650

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	MACC MEL	MACE MEL	MACIP UL	MACIPVL[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	MACDVL[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	MACDVL[2:0]	MAC Destination Vector Learn Used for learning/overwriting an entry in MAC table.	R/W
15:3	—	These bits are read as 0. The write value should be 0.	R/W
18:16	MACIPVL[2:0]	MAC Internal Priority Value Learn Used for learning/overwriting an entry in MAC table.	R/W
19	MACIPUL	MAC Internal Priority Update Learn Used for learning/overwriting an entry in MAC table.	R/W
20	MACEMEL	MAC Ethernet Mirroring Enable Learn Used for learning/overwriting an entry in MAC table.	R/W
21	MACCMEL	MAC CPU Mirroring Enable Learn Used for learning/overwriting an entry in MAC table.	R/W
31:22	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.6.9 FWMACTLR : MAC Table Learn Result Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4654

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	MACTL	—	—	—	—	MACLCN[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	MACLO	—	MACLSF	MACLF
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	MACLF	MAC Learn Fail Used for learning/overwriting an entry in MAC table. 0: Entry learning did not fail. (See (2) SW Learning). 1: Entry learning failed. (See (2) SW Learning).	R
1	MACLSF	MAC Learn Security Fail 0: Entry learning did not fail because of a security error. (See (2) SW Learning). 1: Entry learning failed because of a security error. (See (2) SW Learning).	R
2	—	This bit is read as 0.	R
3	MACLO	MAC Learn Overwrite Used for learning/overwriting an entry in MAC table. 0: The entry learning did not overwrite an existing entry. 1: The entry learning overwrote an existing entry.	R
15:4	—	These bits are read as 0.	R
26:16	MACLCN[10:0]	MAC Learn Collision Number Used for learning/overwriting an entry in MAC table.	R
30:27	—	These bits are read as 0.	R
31	MACTL	MAC Table Learn	R

MACTL bit (MAC Table Learn)

Used for learning/overwriting an entry in MAC table.

[Setting condition]

HW: Writing FWMACTL5 register will set this bit.

[Clearing condition]

HW: This bit will be de-asserted when learning is completed.

30.3.1.6.10 FWMACTIM : MAC Table Initialization Monitoring Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4660

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	MACT R	MACTI OG
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	MACTIOG	MAC Table Initialization Ongoing	R/W
1	MACTR	MAC Table Ready	R
31:2	—	These bits are read as 0. The write value should be 0.	R/W

MACTIOG bit (MAC Table Initialization Ongoing)

[Setting condition]

SW: By writing 1 to this bit. It starts MAC Table initialization.

[Clearing condition]

HW: This bit is cleared when MAC Table initialization is finished.

MACTR bit (MAC Table Ready)

[Setting condition]

- HW: When FWMACTIM.MACTIOG is getting cleared.
- HW: This bit is set at $\text{clk_period [ns]} \times 2048$ time from MAC table initialization starting.

[Clearing condition]

SW: By writing 1 to FWMACTIM.MACTIOG.

30.3.1.6.11 FWMACTEM : MAC Table Entry Monitoring Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4664

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	MACTUEN[11:0]											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	MACTEN[11:0]											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
11:0	MACTEN[11:0]	MAC Table Entry Number	R

Bit	Symbol	Function	R/W
15:12	—	These bits are read as 0.	R
27:16	MACTUEN[11:0]	MAC Table Unsecure Entry Number	R
31:28	—	These bits are read as 0.	R

MACTEN[11:0] bits (MAC Table Entry Number)

[Increment condition]

HW: An entry is learned successfully.

[Decrement condition]

HW: An entry is deleted successfully.

MACTUEN[11:0] bits (MAC Table Unsecure Entry Number)

[Increment condition]

HW: An unsecure entry is learned successfully.

[Decrement condition]

HW: An unsecure entry is deleted successfully.

30.3.1.6.12 FWMACTSO : MAC Table Search Register 0

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4670

[illegible]

Bit	Symbol	Function	R/W
15:0	MACMASP0[15:0]	MAC MAC Address Search Part 0 Used for searching an entry in MAC table.	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.6.13 FWMACTS1 : MAC Table Search Register 1

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4674

[illegible]

Bit	Symbol	Function	R/W
31:0	MACMASP1[31:0]	MAC MAC Address Search Part 1 Used for searching an entry in MAC table.	R/W

30.3.1.6.14 FWMACTSR0 : MAC Table Search Result Register 0

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4678

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	MACT S	—	—	—	—	MACSCN[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	MACH LDS	MACD ES	MACS LS	—	—	—	—	—	—	MACS NF	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	—	This bit is read as 0. The write value should be 0.	R/W
1	MACSNF	MAC Search Not found Used for searching an entry in MAC table. 0: MAC address found in MAC Table. 1: MAC address not found in MAC Table.	R
7:2	—	These bits are read as 0. The write value should be 0.	R/W
8	MACSLS	MAC Security Level Search Used for searching an entry in MAC table.	R
9	MACDES	MAC Dynamic Entry Search Used for searching an entry in MAC table.	R
10	MACHLDS	MAC Hardware Learning Disable Search Used for searching an entry in MAC table.	R
15:11	—	These bits are read as 0. The write value should be 0.	R/W
26:16	MACSCN[10:0]	MAC Search Collision Number Used for searching an entry in MAC table.	R
30:27	—	These bits are read as 0. The write value should be 0.	R/W
31	MACTS	MAC Table Search	R

MACTS bit (MAC Table Search)

Used for searching an entry in MAC table.

[Setting condition]

HW: Writing FWMACTS1 register will set this bit.

[Clearing condition]

HW: This bit will be de-asserted when searching is completed.

30.3.1.6.15 FWMACTSR1 : MAC Table Search Result Register 1

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x467C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	MACDSLVS[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	MACSSLVS[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	MACSSLVS[2:0]	MAC Source Source Lock Vector Search Used for searching an entry in MAC table.	R
15:3	—	These bits are read as 0.	R
18:16	MACDSLVS[2:0]	MAC Destination Source Lock Vector Search Used for searching an entry in MAC table.	R
31:19	—	These bits are read as 0.	R

30.3.1.6.16 FWMACTSR20 : MAC Table Search Result Register 20

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4680 + 0x4 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	MACCSDS[5:0]					
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
5:0	MACCSDS[5:0]	MAC CPU Sub-Destination Search Used for searching an entry in MAC table.	R
31:6	—	These bits are read as 0.	R

30.3.1.6.17 FWMACTSR3 : MAC Table Search Result Register 3

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4690

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	MACC MES	MACE MES	MACIP US	MACIPVS[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	MACDVS[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	MACDVS[2:0]	MAC Destination Vector Search Used for searching an entry in MAC table.	R
15:3	—	These bits are read as 0.	R
18:16	MACIPVS[2:0]	MAC Internal Priority Value Search Used for searching an entry in MAC table.	R
19	MACIPUS	MAC Internal Priority Update Search Used for searching an entry in MAC table.	R
20	MACEMES	MAC Ethernet Mirroring Enable Search Used for searching an entry in MAC table.	R
21	MACCMES	MAC CPU Mirroring Enable Search Used for searching an entry in MAC table.	R
31:22	—	These bits are read as 0.	R

30.3.1.6.18 FWMACTR : MAC Table Read Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x46A0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	MACAR[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
10:0	MACAR[10:0]	MAC Address Read Used for reading an entry in MAC table	R/W
31:11	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.6.19 FWMACTRR0 : MAC Table Read Result Register 0

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x46A4

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	MACT R	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	MACE VR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	MACEVR	MAC Entry Valid Read - Used for reading an entry in MAC table.	R
30:1	—	These bits are read as 0.	R
31	MACTR	MAC Table Read Used for reading an entry in MAC table.	R

MACTR bit (MAC Table Read)

[Setting condition]

HW: Writing FWMACTR register will set this bit.

[Clearing condition]

HW: This bit will be de-asserted when reading is completed.

30.3.1.6.20 FWMACTRR1 : MAC Table Read Result Register 1

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x46A8

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	MACA BR	MACH LDR	MACD ER	MACS LR	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	—	These bits are read as 0.	R
8	MACSLR	MAC Security Level Read Used for reading an entry in MAC table.	R
9	MACDER	MAC Dynamic Entry Read Used for reading an entry in MAC table.	R
10	MACHLDR	MAC Hardware Learn Disable Read Used for reading an entry in MAC table.	R
11	MACABR	MAC Aging Bit Read Used for reading an entry in MAC table.	R

Bit	Symbol	Function	R/W
31:12	—	These bits are read as 0.	R

30.3.1.6.21 FWMACTRR2 : MAC Table Read Result Register 2

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x46AC

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	MACMARP0[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	MACMARP0[15:0]	MAC MAC address Read Part 0 Used for reading an entry in MAC table.	R
31:16	—	These bits are read as 0.	R

30.3.1.6.22 FWMACTRR3 : MAC Table Read Result Register 3

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x46B0

Bit position:	31																																	0		
Bit field:	MACMARP1[31:0]																																			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
31:0	MACMARP1[31:0]	MAC MAC Address Read Part 1 Used for reading an entry in MAC table.	R

30.3.1.6.23 FWMACTRR4 : MAC Table Read Result Register 4

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x46B4

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	MACDSLVR[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	MACSSLVR[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	MACSSLVR[2:0]	MAC Source Source Lock Vector Read Used for reading an entry in MAC table.	R
15:3	—	These bits are read as 0.	R
18:16	MACDSLVR[2:0]	MAC Destination Source Lock Vector Read Used for reading an entry in MAC table.	R
31:19	—	These bits are read as 0.	R

30.3.1.6.24 FWMACTRR50 : MAC Table Read Result Register 50

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x46C0 + 0x4 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	MACCSDR[5:0]					
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
5:0	MACCSDR[5:0]	MAC CPU Sub-Destination Read Used for reading an entry in MAC table.	R
31:6	—	These bits are read as 0.	R

30.3.1.6.25 FWMACTRR6 : MAC Table Read Result Register 6

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x46D0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	MACC MER	MACE MER	MACIP UR	MACIPVR[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	MACDVR[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	MACDVR[2:0]	MAC Destination Vector Read Used for reading an entry in MAC table.	R
15:3	—	These bits are read as 0.	R
18:16	MACIPVR[2:0]	MAC Internal Priority Value Read Used for reading an entry in MAC table.	R
19	MACIPUR	MAC Internal Priority Update Read Used for reading an entry in MAC table.	R
20	MACEMER	MAC Ethernet Mirroring Enable Read Used for reading an entry in MAC table.	R

Bit	Symbol	Function	R/W
21	MACCMER	MAC CPU Mirroring Enable Read Used for reading an entry in MAC table.	R
31:22	—	These bits are read as 0.	R

30.3.1.6.26 FWMACAGUSPC : MAC Aging US Prescaler Configuration Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4880

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	MACAGUSP[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	MACAGUSP[9:0]*1	MAC Aging US prescaler Used to creates an internal clock for aging at 1 MHz to derive the aging 1 Hz clock.	R/W
31:10	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. SW: These bits should be set to the number of clocks clk contained in 1 μ s.

30.3.1.6.27 FWMACAGC : MAC Aging Configuration Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4884

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	MACD ESOG	MACA GOG	—	—	—	MACD ES	—	—	—	—	—	MACA GPM	MACA GSL	MACA GE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	MACAGT[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	MACAGT[15:0]	MAC Aging Time Defines the time in seconds the MFWD will wait between two MAC address aging.	R/W
16	MACAGE	MAC Aging Enable 0: MAC aging is disabled 1: MAC aging is enabled and will happen every FWMACAGC.MACAGT seconds.	R/W
17	MACAGSL	MAC Aging Security Level 0: Only unsecure entries will be aged (MAC.SL == 0) 1: All entries will be aged.	R/W
18	MACAGPM	MAC Aging Polling Mode When set, allows software to check all MAC addresses suppresses by aging. 0: MAC aging polling mode disabled 1: MAC aging polling mode enabled	R/W

Bit	Symbol	Function	R/W
23:19	—	These bits are read as 0. The write value should be 0.	R/W
24	MACDES*1	MAC Dynamic Entry Suppression Writing 1 to this bit will start dynamic entry suppression.	R
27:25	—	These bits are read as 0. The write value should be 0.	R/W
28	MACAGOG	MAC Aging Ongoing 0: MAC aging is not ongoing. 1: MAC aging is ongoing.	R
29	MACDESOG	MAC Dynamic Entry Suppression Ongoing 0: MAC Dynamic Entry Suppression is not ongoing. 1: MAC Dynamic Entry Suppression is ongoing.	R
31:30	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Dynamic Entry Suppression can only happen when aging is disabled (FWMACAGC.MACAGE is not set)

30.3.1.6.28 FWMACAGM0 : MAC Aging Monitoring Register 0

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4888

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	AGMACAP0[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	AGMACAP0[15:0]	Aged MAC Address Part 0 Shows the MAC address part 0 of the latest aged entry.	R
31:16	—	These bits are read as 0.	R

30.3.1.6.29 FWMACAGM1 : MAC Aging Monitoring Register 1

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x488C

Bit position:	31																																	0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
Bit field:	AGMACAP1[31:0]																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
31:0	AGMACAP1[31:0]	Aged MAC Address Part 1 Shows the MAC address part 1 of the latest aged entry.	R

30.3.1.6.30 FWVLANTEC : VLAN Table Entry Configuration Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4900

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	VLANTMUE[12:0]												
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	—	These bits are read as 0. The write value should be 0.	R/W
28:16	VLANTMUE[12:0]	VLAN Table Maximum Unsecure Entry SW: Maximum unsecure entry number that can be set in table	R/W
31:29	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.6.31 FWVLANTL0 : VLAN Table Learn Register 0

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4910

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	VLAN ED
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	VLAN HLDL	—	VLAN SLL	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	—	These bits are read as 0. The write value should be 0.	R/W
8	VLANSLL	VLAN Security Level Learn Used for learning/overwriting an entry in VLAN table.	R/W
9	—	This bit is read as 0. The write value should be 0.	R/W
10	VLANHLDL	VLAN Hardware Learning Disable Learn Used for learning/overwriting an entry in VLAN table.	R/W
15:11	—	These bits are read as 0. The write value should be 0.	R/W
16	VLANED	VLAN Entry Delete Used for learning/overwriting an entry in VLAN table. 0: Learn/overwrite the set VLAN address in VLAN table. 1: Delete the set VLAN address in VLAN table.	R/W
31:17	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.6.32 FWVLANTL1 : VLAN Table Learn Register 1

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4914

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	VLANVIDL[11:0]											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
11:0	VLANVIDL[11:0]	VLAN VID Learn Used for learning/overwriting an entry in VLAN table.	R/W
31:12	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.6.33 FWVLANTL2 : VLAN Table Learn Register 2

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4918

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	VLANSVL[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	VLANSVL[2:0]	VLAN Source Lock Vector Learn Used for learning/overwriting an entry in VLAN table.	R/W
31:3	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.6.34 FWVLANTL30 : VLAN Table Learn Register 30

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4920 + 0x4 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	VLANCSDL[5:0]					
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
5:0	VLANCSDL[5:0]	VLAN CPU Sub-Destination Learn Used for learning/overwriting an entry in VLAN table.	R/W
31:6	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.6.35 FWVLANTL4 : VLAN Table Learn Register 4

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4930

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	VLAN CMEL	VLAN EMEL	VLANI PUL	VLANIPVL[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	VLANDVL[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	VLANDVL[2:0]	VLAN Destination Vector Learn Used for learning/overwriting an entry in VLAN table.	R/W
15:3	—	These bits are read as 0. The write value should be 0.	R/W
18:16	VLANIPVL[2:0]	VLAN Internal Priority Value Learn Used for learning/overwriting an entry in VLAN table.	R/W
19	VLANIPUL	VLAN Internal Priority Update Learn Used for learning/overwriting an entry in VLAN table.	R/W
20	VLANEMEL	VLAN Ethernet Mirroring Enable Learn Used for learning/overwriting an entry in VLAN table.	R/W
21	VLANCMEL	VLAN CPU Mirroring Enable Learn Used for learning/overwriting an entry in VLAN table.	R/W
31:22	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.6.36 FWVLANTLR : VLAN Table Learn Result Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4934

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	VLAN TL	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	VLANL O	—	VLANL SF	VLANL F
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	VLANLF	VLAN Learn Fail Used for learning/overwriting an entry in VLAN table. 0: Entry learning did not fail. (See (2) SW Learning.) 1: Entry learning failed. (See (2) SW Learning.)	R
1	VLANLSF	VLAN Learn Security Fail Used for learning/overwriting an entry in VLAN table. 0: Entry learning did not fail because of a security error. (See (2) SW Learning.) 1: Entry learning failed because of a security error. (See (2) SW Learning.)	R
2	—	This bit is read as 0.	R
3	VLANLO	VLAN Learn Overwrite Used for learning/overwriting an entry in VLAN table. 0: The entry learning did not overwrite an existing entry. 1: The entry learning overwrote an existing entry.	R
30:4	—	These bits are read as 0.	R
31	VLANTL	VLAN Table Learn	R

VLANTL bit (VLAN Table Learn)

Used for learning/overwriting an entry in VLAN table.

[Setting condition]

HW: Writing FWVLANTL4 register will set this bit.

[Clearing condition]

HW: This bit will be de-asserted when learning is completed.

30.3.1.6.37 FWVLANTIM : VLAN Table Initialization Monitoring Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4940

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	VLAN TR	VLAN TIOG
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	VLANTIIOG	VLAN Table Initialization Ongoing	R/W ^{*1}
1	VLANTR	VLAN Table Ready	R
31:2	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Read value differs from written value.

VLANTIIOG bit (VLAN Table Initialization Ongoing)

[Setting condition]

SW: By writing 1 to this bit. It starts VLAN Table initialization.

[Clearing condition]

HW: This bit is cleared when VLAN Table initialization is finished.

VLANTR bit (VLAN Table Ready)

[Setting condition]

HW: When FWVLANTIM.VLANTIOG is getting cleared.

HW: This bit is set at “clk_period[ns]*4096” time from VLAN table initialization starting.

[Clearing condition]

SW: By writing 1 to FWVLANTIM.VLANTIOG.

30.3.1.6.38 FWVLANTEM : VLAN Table Entry Monitoring Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4944

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	VLANTUEN[12:0]												
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	VLANTEN[12:0]												
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
12:0	VLANTEN[12:0]	VLAN Table Entry Number	R
15:13	—	These bits are read as 0.	R
28:16	VLANTUEN[12:0]	VLAN Table Unsecure Entry Number	R
31:29	—	These bits are read as 0.	R

VLANTEN[12:0] bits (VLAN Table Entry Number)

[Increment condition]

HW: An unsecure entry is learned successfully.

[Decrement condition]

HW: An unsecure entry is deleted successfully.

VLANTUEN[12:0] bits (VLAN Table Unsecure Entry Number)

[Increment condition]

HW: An entry is learned successfully.

[Decrement condition]

HW: An entry is deleted successfully.

30.3.1.6.39 FWVLANTS : VLAN Table Search Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4950

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	VLANVIDS[11:0]											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
11:0	VLANVIDS[11:0]	VLAN VID Search Used for searching an entry in VLAN table.	R/W
31:12	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.6.40 FWVLANTSR0 : VLAN Table Search Result Register 0

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4954

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	VLAN TS	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	VLAN HLDS	—	VLAN SLS	—	—	—	—	—	—	VLAN SNF	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	—	This bit is read as 0.	R
1	VLANSNF	VLAN Search Not found Used for searching an entry in VLAN table. 0: VLAN found in VLAN Table. 1: VLAN not found in VLAN Table.	R
7:2	—	These bits are read as 0.	R
8	VLANSLS	VLAN Security Level Search Used for searching an entry in VLAN table.	R
9	—	These bits are read as 0.	R
10	VLANHLDS	VLAN Hardware Learning Disable Search Used for searching an entry in VLAN table.	R
30:11	—	These bits are read as 0.	R
31	VLANTS	VLAN Table Search	R

VLANTS bit (VLAN Table Search)

Used for searching an entry in VLAN table.

[Setting condition]

HW: Writing FWVLANTS register will set this bit.

[Clearing condition]

HW: This bit will be de-asserted when searching is completed.

30.3.1.6.41 FWVLANTSR1 : VLAN Table Search Result Register 1

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4958

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	VLANSLSVS[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	VLANSLSVS[2:0]	VLAN Source Lock Vector Search Used for searching an entry in VLAN table.	R
31:3	—	These bits are read as 0.	R

30.3.1.6.42 FWVLANTSR20 : VLAN Table Search Result Register 20

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4960 + 0x4 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	VLANCSDS[5:0]					
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
5:0	VLANCSDS[5:0]	VLAN CPU Sub-Destination Search Used for searching an entry in VLAN table.	R
31:6	—	These bits are read as 0.	R

30.3.1.6.43 FWVLANTSR3 : VLAN Table Search Result Register 3

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4970

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	VLAN CMES	VLAN EMES	VLANI PUS	VLANIPVS[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	VLANDVS[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	VLANDVS[2:0]	VLAN Destination Vector Search Used for searching an entry in VLAN table.	R
15:3	—	These bits are read as 0.	R
18:16	VLANIPVS[2:0]	VLAN Internal Priority Value Search Used for searching an entry in VLAN table.	R
19	VLANI PUS	VLAN Internal Priority Update Search Used for searching an entry in VLAN table.	R
20	VLANEMES	VLAN Ethernet Mirroring Enable Search Used for searching an entry in VLAN table.	R
21	VLANCMES	VLAN CPU Mirroring Enable Search Used for searching an entry in VLAN table.	R
31:22	—	These bits are read as 0.	R

30.3.1.7 Port Based Forwarding Function Registers

30.3.1.7.1 FWPBFCi : Port i Port Based Forwarding Configuration Register (i = 0 to 2)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4A00 + 0x10 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	FAIFP	IP6PD E	IP4PD M	IP4PD E	PBSL	PBCM E	PBEM E	PBIPU	PBIPV[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	PBDV[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	PBDV[2:0]	Port Based Destination Vector Destination vector for port i port-based forwarding.	R/W
15:3	—	These bits are read as 0. The write value should be 0.	R/W
18:16	PBIPV[2:0]	Port Based Internal Priority Value Internal priority value for port i port-based forwarding.	R/W

Bit	Symbol	Function	R/W
19	PBIPU	Port Based Internal Priority Update Enables frame priority update to internal priority value (FWPBFCi.PBIPV) for port i.	R/W
20	PBEME	Port Based Ethernet Mirroring Enabled Enables ethernet mirroring for port i.	R/W
21	PBCME	Port Based CPU Mirroring Enabled Enables CPU mirroring for port i.	R/W
22	PBSL	Port Based Security Level Set descriptor security level for port i.	R/W
23	IP4PDE	IPv4 Priority Decode Enable 0: For port i, Input frame priority is taken from VLAN TAGs for port i. 1: For port i, If an IPv4 frame is detected, input frame priority is taken from precedence ToS. Else Input frame priority is taken from VLAN TAGs.	R/W
24	IP4PDM	IPv4 Priority Decode Mode 0: For port i, Input frame priority is taken from precedent field in ToS. 1: For port i, Input frame priority is decoded from DSCP field in ToS.	R/W
25	IP6PDE	IPv6 Priority Decode Enable 0: For port i, Input frame priority is taken from VLAN TAGs. 1: For port i, If an IPv6 frame is detected, input frame priority is decoded from DSCP field in ToS. Else Input frame priority is taken from VLAN TAGs.	R/W
26	FAIFP	Force All Input Frame Priority Enable 0: For port i, Input frame priority is given by frame decoding. 1: For port i, Input frame priority is forced to FWPBFCi.PBIPV.	R/W
31:27	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.7.2 FWPBFCSDCi0 : Port i Port Based Forwarding CSD Configuration Register 0 (i = 0 to 2)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4A04 + 0x10 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	PBCSD[5:0]					
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
5:0	PBCSD[5:0]	Port Based CPU Sub Destination CPU Sub Destination from GWCAj [GWCA] for port i port-based forwarding.	R/W
31:6	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.8 Layer 2/Layer 3 Update Function Registers

30.3.1.8.1 FWL23URL0 : Layer 2/Layer 3 Update Rule Learn Register 0

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4E00

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	L23URPVL[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	L23URNL[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	L23URNL[7:0]	Layer 2/Layer 3 Update Routing Number Learn Used to learn a Layer 2/Layer 3 update rule	R/W
15:8	—	These bits are read as 0. The write value should be 0.	R/W
18:16	L23URPVL[2:0]	Layer 2/Layer 3 Update Routing Port Valid Learn Used to learn a Layer 2/Layer 3 update rule	R/W
31:19	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.8.2 FWL23URL1 : Layer 2/Layer 3 Update Rule Learn Register 1

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4E04

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	L23URTUL[1:0]	L23US DEIUL	L23US PCPU L	L23US VIDUL	L23UC DEIUL	L23UC PCPU L	L23UC VIDUL	L23U MSAUL	L23U MDAUL	L23UT TLUL	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	L23UMDALP0[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	L23UMDALP0[15:0]	Layer 2/Layer 3 Update MAC Destination Address Learn Part 0 Used to learn a Layer 2/Layer 3 update rule	R/W
16	L23UTTLUL	Layer 2/Layer 3 Update Time To Live Update Learn Used to learn a Layer 2/Layer 3 update rule	R/W
17	L23UMDAUL	Layer 2/Layer 3 Update MAC Destination Address Update Learn Used to learn a Layer 2/Layer 3 update rule	R/W
18	L23UMSAUL	Layer 2/Layer 3 Update MAC Source Address Update Learn Used to learn a Layer 2/Layer 3 update rule	R/W
19	L23UCVIDUL	Layer 2/Layer 3 Update C-TAG VID Update Learn Used to learn a Layer 2/Layer 3 update rule	R/W
20	L23UCPCPUL	Layer 2/Layer 3 Update C-TAG PCP Update Learn Used to learn a Layer 2/Layer 3 update rule	R/W

Bit	Symbol	Function	R/W
21	L23UCDEIUL	Layer 2/Layer 3 Update C-TAG DEI Update Learn Used to learn a Layer 2/Layer 3 update rule	R/W
22	L23USVIDUL	Layer 2/Layer 3 Update S-TAG VID Update Learn Used to learn a Layer 2/Layer 3 update rule	R/W
23	L23USPCPUL	Layer 2/Layer 3 Update S-TAG PCP Update Learn Used to learn a Layer 2/Layer 3 update rule	R/W
24	L23USDEIUL	Layer 2/Layer 3 Update S-TAG DEI Update Learn Used to learn a Layer 2/Layer 3 update rule	R/W
26:25	L23URTUL[1:0]	Layer 2/Layer 3 Update R-TAG Update Learn Used to learn a Layer 2/Layer 3 update rule	R/W
31:27	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.8.3 FWL23URL2 : Layer 2/Layer 3 Update Rule Learn Register 2

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4E08

Bit position: 31 0

Bit field: L23UMDALP1[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	L23UMDALP1[31:0]	Layer 2/Layer 3 Update MAC Destination Address Learn Part 1 Used to learn a Layer 2/Layer 3 update rule	R/W

30.3.1.8.4 FWL23URL3 : Layer 2/Layer 3 Update Rule Learn Register 3

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4E0C

Bit position: 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16

Bit field: L23USDEIL L23USPCPL[2:0] L23USVIDL[11:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field: L23UCDEIL L23UCPCPL[2:0] L23UCVIDL[11:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
11:0	L23UCVIDL[11:0]	Layer 2/Layer 3 Update C-TAG VID Learn Used to learn a Layer 2/Layer 3 update rule	R/W
14:12	L23UCPCPL[2:0]	Layer 2/Layer 3 Update C-TAG PCP Learn Used to learn a Layer 2/Layer 3 update rule	R/W
15	L23UCDEIL	Layer 2/Layer 3 Update C-TAG DEI Learn Used to learn a Layer 2/Layer 3 update rule	R/W
27:16	L23USVIDL[11:0]	Layer 2/Layer 3 Update S-TAG VID Learn Used to learn a Layer 2/Layer 3 update rule	R/W
30:28	L23USPCPL[2:0]	Layer 2/Layer 3 Update S-TAG PCP Learn Used to learn a Layer 2/Layer 3 update rule	R/W

Bit	Symbol	Function	R/W
31	L23USDEIL	Layer 2/Layer 3 Update S-TAG DEI Learn Used to learn a Layer 2/Layer 3 update rule	R/W

30.3.1.8.5 FWL23URLR : Layer 2/Layer 3 Update Rule Learn Result Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4E10

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	L23URL	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	L23ULF
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	L23ULF	Layer 2/Layer 3 Update Learn Fail 0: Entry learning did not fail because the Layer 2/Layer 3 table is ready. 1: Entry learning failed because the Layer 2/Layer 3 table is not ready.	R
30:1	—	These bits are read as 0.	R
31	L23URL	Layer 2/Layer 3 Update Rule Learn	R

L23URL bit (Layer 2/Layer 3 Update Rule Learn)

[Setting condition]

HW: Writing FWL23URL3 register will set this bit.

[Clearing condition]

HW: This bit will be de-asserted when learning is completed.

30.3.1.8.6 FWL23UTIM : Layer 2/Layer 3 Update Table Initialization Monitoring Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4E20

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	L23UTR	L23UTIOG
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	L23UTIOG	Layer 2/Layer 3 Update Table Initialization Ongoing	R/W ¹
1	L23UTR	Layer 2/Layer 3 Update Table Ready	R
31:2	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Read value differs from written value.

L23UTIOG bit (Layer 2/Layer 3 Update Table Initialization Ongoing)

[Setting condition]

SW: By writing 1 to this bit. It starts Layer 2/Layer 3 Update Table initialization.

[Clearing condition]

HW: This bit is cleared when Layer 2/Layer 3 Update Table initialization is finished.

L23UTR bit (Layer 2/Layer 3 Update Table Ready)

[Setting condition]

- HW: When FWL23UTIM.L23UTIOG is getting cleared.
- HW: This bit is set at $\text{clk_period [ns]} \times 256$ time from L23 update table initialization starting.

[Clearing condition]

SW: By writing 1 to FWL23UTIM.L23UTIOG.

30.3.1.8.7 FWL23URR : Layer 2/Layer 3 Update Rule Read Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4E30

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	L23RNR[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	L23RNR[7:0]	Layer 2/Layer 3 Routing Number Read Used to read a Layer 2/Layer 3 update rule	R/W
31:8	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.8.8 FWL23URRR0 : Layer 2/Layer 3 Update Rule Read Result Register 0

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4E34

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	L23UR R	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	L23URPVR[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	L23URPVR[2:0]	Layer 2/Layer 3 Update Routing Port Valid Read Used to read a Layer 2/Layer 3 update rule	R

Bit	Symbol	Function	R/W
30:3	—	These bits are read as 0.	R
31	L23URR	Layer 2/Layer 3 Update Rule Read	R

L23URR bit (Layer 2/Layer 3 Update Rule Read)

[Setting condition]

HW: Writing FWL23URR register will set this bit.

[Clearing condition]

HW: This bit will be de-asserted when reading is completed.

30.3.1.8.9 FWL23URRR1 : Layer 2/Layer 3 Update Rule Read Result Register 1

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4E38

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	L23URTUR[1:0]	L23US DEIUR	L23US PCPUR	L23US VIDUR	L23UC DEIUR	L23UC PCPUR	L23UC VIDUR	L23U MSAUR	L23U MDAUR	L23UT TLUR	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	L23UMDARP0[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	L23UMDARP0[15:0]	Layer 2/Layer 3 MAC Destination Address Read Part 0 Used to read a Layer 2/Layer 3 update rule	R
16	L23UTTLUR	Layer 2/Layer 3 Time To Live Update Read Used to read a Layer 2/Layer 3 update rule	R
17	L23UMDAUR	Layer 2/Layer 3 MAC Destination Address Update Read Used to read a Layer 2/Layer 3 update rule	R
18	L23UMSAUR	Layer 2/Layer 3 MAC Source Address Update Read Used to read a Layer 2/Layer 3 update rule	R
19	L23UCVIDUR	Layer 2/Layer 3 C-TAG VID Update Read Used to read a Layer 2/Layer 3 update rule	R
20	L23UCPCPUR	Layer 2/Layer 3 C-TAG PCP Update Read Used to read a Layer 2/Layer 3 update rule	R
21	L23UCDEIUR	Layer 2/Layer 3 C-TAG DEI Update Read Used to read a Layer 2/Layer 3 update rule	R
22	L23USVIDUR	Layer 2/Layer 3 S-TAG VID Update Read Used to read a Layer 2/Layer 3 update rule	R
23	L23USPCPUR	Layer 2/Layer 3 S-TAG PCP Update Read Used to read a Layer 2/Layer 3 update rule	R
24	L23USDEIUR	Layer 2/Layer 3 S-TAG DEI Update Read Used to read a Layer 2/Layer 3 update rule	R
26:25	L23URTUR[1:0]	Layer 2/Layer 3 R-TAG Update Read Used to read a Layer 2/Layer 3 update rule	R
31:27	—	These bits are read as 0.	R

30.3.1.8.10 FWL23URRR2 : Layer 2/Layer 3 Update Rule Read Result Register 2

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4E3C

Bit position: 31

0

Bit field:

L23UMDARP1[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	L23UMDARP1[31:0]	Layer 2/Layer 3 Update MAC Destination Address Read Part 1 Used to read a Layer 2/Layer 3 update rule	R

30.3.1.8.11 FWL23URRR3 : Layer 2/Layer 3 Update Rule Read Result Register 3

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4E40

Bit position:

31

30

29

28

27

26

25

24

23

22

21

20

19

18

17

16

Bit field:

L23US
DEIR

L23USPCPR[2:0]

L23USVIDR[11:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit position:

15

14

13

12

11

10

9

8

7

6

5

4

3

2

1

0

Bit field:

L23UC
DEIR

L23UCPCPR[2:0]

L23UCVIDR[11:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
11:0	L23UCVIDR[11:0]	Layer 2/Layer 3 Update MAC C-TAG VID Read Used to read a Layer 2/Layer 3 update rule	R
14:12	L23UCPCPR[2:0]	Layer 2/Layer 3 Update MAC C-TAG PCP Read Used to read a Layer 2/Layer 3 update rule	R
15	L23UCDEIR	Layer 2/Layer 3 Update MAC C-TAG DEI Read Used to read a Layer 2/Layer 3 update rule	R
27:16	L23USVIDR[11:0]	Layer 2/Layer 3 Update MAC S-TAG VID Read Used to read a Layer 2/Layer 3 update rule	R
30:28	L23USPCPR[2:0]	Layer 2/Layer 3 Update MAC S-TAG PCP Read Used to read a Layer 2/Layer 3 update rule	R
31	L23USDEIR	Layer 2/Layer 3 Update MAC S-TAG DEI Read Used to read a Layer 2/Layer 3 update rule	R

30.3.1.8.12 FWL23URMCi : Layer 2/Layer 3 Update Remapping Configuration Register i (i = 0 to 31)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x4F00 + 0x4 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	RME	—	—	—	—	RMNRN[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	RMDPN[1:0]	—	—	—	—	—	RMRN[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	RMRN[7:0]	Remapping Rule Number Set the rule number that should be remapped for remapping rule i.	R/W
11:8	—	These bits are read as 0. The write value should be 0.	R/W
13:12	RMDPN[1:0]	Remapping Destination Port Number Set the port number for which rule number should be remapped for remapping rule i.	R/W
15:14	—	These bits are read as 0. The write value should be 0.	R/W
23:16	RMNRN[7:0]	Remapping New Rule Number Set the rule number to which routing should be remapped for remapping rule i.	R/W
27:24	—	These bits are read as 0. The write value should be 0.	R/W
28	RME	Remapping Enable 0: Remapping rule i disabled 1: Remapping rule i enabled	R/W ^{*1}
31:29	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Read value differs from written value.

RME bit (Remapping Enable)

[Setting condition]

SW: Writing 1 to this bit will set it.

30.3.1.9 PSFP (Per Stream Filtering and Policing) [802.1Qci]

30.3.1.9.1 FWPMFGCi : PSFP MSDU Filter Global Configuration Register i (i = 0 to 15)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x5000 + 0x4 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	MFM	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	MSDUV[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	MSDUV[15:0]	MSDU Value HW: Maximum frame size for stream linked to MSDU filter i thanks to L3 table.	R/W
30:16	—	These bits are read as 0. The write value should be 0.	R/W
31	MFM	MSDU Filter Mode 0: Normal mode: Any frame linked to MSDU filter i thanks to L3 table received with LDESCR.TPL [GWCA] bigger than FWPMPCi.MSDUV will be filtered. 1: Throttle mode: Any frame linked to MSDU filter i thanks to L3 table received with LDESCR.TPL [GWCA] bigger than FWPMPCi.MSDUV will be filtered. Any frame linked to MSDU filter i thanks to L3 table received when FWEIS2.PMFSi is set will be filtered.	R/W

30.3.1.9.2 FWPMTRFCi : PSFP Meter i Filter Configuration Register (i = 0 to 31)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x5600 + 0x20 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	MTRCM15 to MTRCM0															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	MTRCF	MTRFRFD	MTRFM[1:0]	MTRFE	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	MTRFE	Meter Filter Enable 0: Meter i disabled (Pass through all frames.) 1: Meter i enabled (Reject target stream frames by meter filter.)	R/W
2:1	MTRFM[1:0]	Meter Filter Mode 0 0: Normal mode: Any frame linked to Meter filter i thanks to L3 table received when not enough token is available will be red. 0 1: Throttle mode: Any frame linked to Meter filter i thanks to L3 table received when not enough token is available or when FWEIS5.PMRFSi is set will be red. 1 0: ATS mode: Any frame linked to Meter filter i thanks to L3 table received when not enough token is available will be stored in ATS RAM until enough tokens are available. When enough tokens are available, frame will be forwarded as yellow or green. 1 1: ATS throttle mode: Any frame linked to Meter filter i thanks to L3 table received when not enough token is available will be stored in ATS RAM until enough tokens are available. When enough tokens are available, frame will be forwarded as yellow or green if FWEIS5.PMRFSi is not set and will be red if FWEIS5.PMRFSi is set.	R/W
3	MTRFRFD	Meter Filter Red Frame Drop 0: Red frames are not dropped by meter i. 1: Red frames are dropped by meter i.	R/W
4	MTRCF	Meter Coupling Flag ¹ 0: When green bucket is full, CIR tokens are lost. 1: When green bucket is full, CIR tokens are added to the yellow bucket.	R/W
15:5	—	These bits are read as 0. The write value should be 0.	R/W
31:16	MTRCM15 to MTRCM0	Meter Color Mode n (n = 0 to 15) ²	R/W

Note 1. HW: This bit is only valid if i = 0 to 7.

Note 2. HW: These bits are only valid if i = 0 to 7.

SW: in ATS mode (FWPMTRFCi.MTRFM set to 10b) or in ATS throttle mode (FWPMTRFCi.MTRFM set to 11b) Color mode is not available, so these bits should be set to All0.

MTRCMn bits (Meter Color Mode n (n = 0 to 15))

If bit j is set to 0: Incoming frame with {PCP,DEI} == j is coming as green (Green or yellow bucket can be used by the frame).

If bit j is set to 1: Incoming frame with {PCP,DEI} == j is coming as Yellow (Only yellow bucket can be used by the frame).

30.3.1.9.3 FWPMTRCBSCi : PSFP Meter i CBS Configuration Register (i = 0 to 31)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x5604 + 0x20 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CBS[17:16]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	CBS[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
17:0	CBS[17:0]	CBS Maximum number of tokens (in bytes) contained in meter i green bucket.	R/W
31:18	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.9.4 FWPMTRCIRCi : PSFP Meter i CIR Configuration Register (i = 0 to 31)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x5608 + 0x20 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	CIR[19:16]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	CIR[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
19:0	CIR[19:0]	CIR	R/W
31:20	—	These bits are read as 0. The write value should be 0.	R/W

CIR[19:0] bits (CIR)

Used to increment green token tokens when meter i green bucket is not full.

CIR[19:16] is the credit increment in byte per clock and CIR[15:0] is the credit increment in sub-byte per clock.

Note: SW: In ATS mode, setting these bits to 0 or a value close to 0 would set corresponding queue to a very low throughput or could stick the corresponding queue and lead to a queue overflow.

30.3.1.9.5 FWPMTREBSCi : PSFP Meter i EBS Configuration Register (i = 0 to 7)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x560C + 0x20 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	EBS[17:16]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	EBS[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
17:0	EBS[17:0]	EBS Maximum number of tokens (in bytes) contained in meter i yellow bucket.	R/W
31:18	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.9.6 FWPMTREIRCi : PSFP Meter i EIR Configuration Register (i = 0 to 7)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x5610 + 0x20 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	EIR[19:16]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	EIR[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
19:0	EIR[19:0]	EIR	R/W
31:20	—	These bits are read as 0. The write value should be 0.	R/W

EIR[19:0] bits (EIR)

Used to increment green token tokens when meter i yellow bucket is not full.

EIR[19:16] is the credit increment in byte per clock and EIR[15:0] is the credit increment in sub-byte per clock.

Note: SW: In ATS mode, setting these bits to 0 or a value close to 0 would set the corresponding queue to a very low throughput or could stick the corresponding queue and lead to a queue overflow.

30.3.1.9.7 FWPMTRFMi : PSFP Meter i Filter Monitoring Register (i = 0 to 31)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x5614 + 0x20 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	MTRARDNMN[4:0]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	MTRARDN[4:0]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
4:0	MTRARDN[4:0]	Meter ATS RAM Descriptor Number	R
15:5	—	These bits are read as 0.	R
20:16	MTRARDNMN[4:0]	Meter ATS RAM Descriptor Number Maximum Number	R
31:21	—	These bits are read as 0.	R

MTRARDN[4:0] bits (Meter ATS RAM Descriptor Number)

Shown the number of descriptors in the ATS RAM for meter number i.

[Increment condition]

A descriptor is received for meter i but there is already a descriptor in the ATS RAM for meter i or there is not even token available for the frame.

30.3.1.10 FRER (Frame Replication and Elimination for Reliability) [802.1CB]

30.3.1.10.1 FWFTL0 : FRER Table Learn Register 0

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x6000

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	FSRPL[6:0]						
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	FEAL[6:0]						
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
6:0	FEAL[6:0]	FRER Entry Address Learn Used to set an FRER entry in FRER RAM.	R/W
15:7	—	These bits are read as 0. The write value should be 0.	R/W
22:16	FSRPL[6:0]	FRER Sequence Recovery Pointer Learn Used to set an FRER entry in FRER RAM.	R/W
31:23	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.10.2 FWFTL1 : FRER Table Learn Register 1

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x6004

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	FSRRTL[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	FSRP VL	FTNSL	—	—	—	—	FSHLL[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	FSHLL[3:0]	FRER Sequence History Length Learn*1 Used to set an FRER entry in FRER RAM.	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W
8	FTNSL	FRER Take No Sequence Learn Used to set an FRER entry in FRER RAM.	R/W
9	FSRPVL	FRER Sequence Recovery Pointer Valid Learn Used to set an FRER entry in FRER RAM.	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W
25:16	FSRRTL[9:0]	FRER Sequence Recovery Remaining Ticks Learn Used to set an FRER entry in FRER RAM.	R/W
31:26	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. SW: These bits should not be set to a value greater than 14.

30.3.1.10.3 FWFTLR : FRER Table Learn Result Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x6008

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	FTL	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	FLF
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	FLF	FRER Learn Fail 0: Entry learning did not fail because the FRER RAM is ready. 1: Entry learning failed because the FRER RAM is not ready.	R
30:1	—	These bits are read as 0.	R
31	FTL	FRER Table Learn	R

FTL bit (FRER Table Learn)

[Setting condition]

HW: Writing FWFTL1 register will set this bit.

[Clearing condition]

HW: This bit will be deasserted when learning is completed.

30.3.1.10.4 FWFTOC : FRER Timeout Configuration Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x6010

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	TOOG	TOCE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	TOT[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	TOT[15:0]	Timeout Time (ms) When Timeout check is enabled (FWFTOC.TOCE set to 1), timeout check will happen every FWFTOC.TOT milliseconds.	R/W
16	TOCE	Timeout Check Enable 0: Timeout check is disabled. 1: Timeout check is enabled.	R/W
17	TOOG	Timeout Ongoing 0: Timeout check is not ongoing. 1: Timeout check is ongoing.	R/W
31:18	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.10.5 FWFTOPC : FRER Timeout Prescaler Configuration Register 0

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x6014

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	USP[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	USP[9:0]	Microsecond Prescaler* ¹ Used to creates an internal clock for aging at 1 MHz to derive the Timeout 1 kHz clock.	R/W
31:10	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. SW: These bits should be set to the number of clk clocks contained in 1 μ s.

30.3.1.10.6 FWFTIM : FRER Table Initialization Monitoring Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x6020

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	FTR	FTIOG
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	FTIOG	FRER Table Initialization Ongoing	R/W ^{*1}
1	FTR	FRER Table Ready	R
31:2	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Read value differs from written value.

FTIOG bit (FRER Table Initialization Ongoing)

[Setting condition]

SW: By writing 1 to this bit. It starts FRER Table initialization.

[Clearing condition]

HW: This bit is cleared when FRER Table initialization is finished.

FTR bit (FRER Table Ready)

[Setting condition]

- HW: When FWFTIM.FTIOG is getting cleared.
- HW: This bit is set at $\text{clk_period [ns]} \times 128$ time from buffer pool initialization starting.

[Clearing condition]

SW: By writing 1 to FWFTIM.FTIOG.

30.3.1.10.7 FWFTR : FRER Table Read Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x6030

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	FEAR[6:0]						
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
6:0	FEAR[6:0]	FRER Entry Address Read Used to read an FRER entry in FRER RAM.	R/W
31:7	—	These bits are read as 0. The write value should be 0.	R/W

30.3.1.10.8 FWFTRR0 : FRER Table Read Result Register 0

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x6034

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	FTR	—	—	—	—	—	FSRRTR[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	FSRP VR	FTNS R	—	—	—	—	FSHLR[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	FSHLR[3:0]	FRER Sequence History Length Read	R
7:4	—	These bits are read as 0.	R
8	FTNSR	FRER Take No Sequence Read	R
9	FSRPVR	FRER Sequence Recovery Pointer Valid Read	R
15:10	—	These bits are read as 0.	R
25:16	FSRRTR[9:0]	FRER Set Recovery Remaining Ticks Read	R
30:26	—	These bits are read as 0.	R
31	FTR	FRER Table Read	R

FTR bit (FRER Table Read)

[Setting condition]

HW: Writing FWFTR register will set this bit.

[Clearing condition]

HW: This bit will be de-asserted when reading is completed.

30.3.1.10.9 FWFTRR1 : FRER Table Read Result Register 1

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x6038

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	FSRPR[6:0]						
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	FSHR[14:0]														
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
14:0	FSHR[14:0]	FRER Sequence History Read	R
15	—	This bit is read as 0.	R
22:16	FSRPR[6:0]	FRER Sequence Recovery Pointer Read	R
31:23	—	These bits are read as 0.	R

30.3.1.10.10 FWFTRR2 : FRER Table Read Result Register 2

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x603C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	FRTRR[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	FRSNR[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	FRSNR[15:0]	FRER Recovery Sequence Number Read	R
25:16	FRTRR[9:0]	FRER Recovery Remaining Ticks Read	R
31:26	—	These bits are read as 0.	R

30.3.1.10.11 FWSEQNGCi : Sequence Number Generation i Configuration Register (i = 0 to 31)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x6100 + 0x8 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	SEQNGE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	SEQNGRN[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	SEQNGRN[7:0]	SEQUence Number Generation Routing Number Specifies the routing number for which sequence number will be generated.	R/W
15:8	—	These bits are read as 0. The write value should be 0.	R/W
16	SEQNGE	SEQUence Number Generation Enable 0: Sequence number generation disabled for routing number FWSEQNGCi.SEQNGRN 1: Sequence number generation enabled for routing number FWSEQNGCi.SEQNGRN	R/W
31:17	—	These bits are read as 0. The write value should be 0.	R/W

30.3.2.1 FWCTFDCNi : Port i Cut-Through Forwarded Descriptor Counter Register (i = 0, 1)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x6300 + 0x20 × i

Bit position: 31

0

Bit field:

CTFDN[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	CTFDN[31:0]	Cut-Through Forwarded Descriptor Number	R

CTFDN[31:0] bits (Cut-Through Forwarded Descriptor Number)

These bits count the number of descriptors forwarded by Cut-Through forwarding for port i.

[Clearing condition]

SW: Reading these bits clears it.

[Increment condition]

HW: Incremented by 1 when a frame coming from port i is forwarded by Cut-Through Forwarding in store and forward mode and if these bits have a value other than 0xFFFFFFFF.

Note: HW: These bits do not take in account the cut-through path of cut-through forwarding but only the store and forward path so, any frame forwarded in store and forward with cut-through (for at least one value of j, if FWCTTCi0.CTDV[j] is set, FWCTTCi0.CTDFMj is also set) with be counted here.

30.3.2.2 FWDDFDCN2 : Port 2 Direct Descriptor Forwarded Descriptor Counter Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x6340

Bit position: 31

0

Bit field:

DDFDN[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	DDFDN[31:0]	Direct Descriptor Forwarded Descriptor Number	R

DDFDN[31:0] bits (Direct Descriptor Forwarded Descriptor Number)

These bits count the number of descriptors forwarded by Direct Descriptor forwarding for port 2.

[Clearing condition]

SW: Reading these bits clears it.

[Increment condition]

HW: Incremented by 1 when a frame coming from port 2 is forwarded by Direct Descriptor Forwarding and if these bits have a value other than 0xFFFFFFFF.

30.3.2.3 FWLTHFDCNi : Port i Layer 3 Forwarded Descriptor Counter Register (i = 0 to 2)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x6304 + 0x20 × i

Bit position: 31

0

Bit field:

LTHFDN[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	LTHFDN[31:0]	Layer 3 Forwarded Descriptor Number	R

LTHFDN[31:0] bits (Layer 3 Forwarded Descriptor Number)

These bits count the number of descriptors forwarded by Layer 3 forwarding for port i.

[Clearing condition]

SW: Reading these bits clears it.

[Increment condition]

HW: Incremented by 1 when a frame coming from port i is forwarded by Layer 3 Forwarding and if these bits have a value different from 0xFFFFFFFF.

Note: HW: These bits count the frame selected by Layer 3 forwarding mechanism before PSFP/ATS and FRER filtering functions. Frame rejected by these filtering functions are still taken in account here.

30.3.2.4 FWLTWFDNCNi : Port i Layer 2 Forwarded Descriptor Counter Register (i = 0 to 2)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x630C + 0x20 × i

Bit position: 31

0

Bit field:

LTWFDN[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	LTWFDN[31:0]	Layer 2 Forwarded Descriptor Number	R

LTWFDN[31:0] bits (Layer 2 Forwarded Descriptor Number)

These bits count the number of descriptors forwarded by Layer 2 forwarding for port i.

[Clearing condition]

SW: Reading these bits clears it.

[Increment condition]

HW: Incremented by 1 when a frame coming from port i is forwarded by Layer 2 Forwarding and if these bits have a value different from 0xFFFFFFFF.

30.3.2.5 FWPBFDCNi : Port i Port Based Forwarded Descriptor Counter Register (i = 0 to 2)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x6310 + 0x20 × i

Bit position: 31

0

Bit field:

PBFDN[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	PBFDN[31:0]	Port Based Forwarded Descriptor Number	R

PBFDN[31:0] bits (Port Based Forwarded Descriptor Number)

These bits count the number of descriptors forwarded by Port Based forwarding for port i.

[Clearing condition]

SW: Reading these bits clears it.

[Increment condition]

HW: Incremented by 1 when a frame coming from port i is forwarded by Port Based Forwarding and if these bits have a value different from 0xFFFFFFFF.

30.3.2.6 FWMHLCNi : Port i MAC Hardware Learn Counter Register (i = 0 to 2)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x6314 + 0x20 × i

Bit position: 31

0

Bit field:

MHLN[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	MHLN[31:0]	MAC Hardware Learn Number	R

MHLN[31:0] bits (MAC Hardware Learn Number)

These bits count the number of MAC source addresses that has been learnt or migrated for port i.

[Clearing condition]

SW: Reading these bits clears it.

[Increment condition]

HW: Incremented by 1 when a frame is selected for source MAC address HW learning or HW migrated ([section 30.5.1.5.2. CPU Learning Path](#)) and learning rate is not too high and if these bits have a value different from {32 {1}}.

This counter is incremented regardless of the MAC learn result [Table 30.45](#). (It counts even if it fails)

This counter is incremented only learning or migrated HW request. So, the counter increments twice even for consecutive requests with the same content. Since the processing is the same, it looks like one time.

30.3.2.7 FWWMRDCNi : Port i Watermark Rejected Descriptor Counter Register (i = 0 to 2)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x6504 + 0x20 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	WMRDN[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	WMRDN[15:0]	Watermark rejected Descriptor Number	R
31:16	—	These bits are read as 0.	R

WMRDN[15:0] bits (Watermark rejected Descriptor Number)

These bits count the number of descriptors rejected by Watermark for port i.

[Clearing condition]

SW: Reading these bits clears it.

[Increment condition]

HW: Incremented by 1 when a frame coming from port i is rejected by Watermark and if these bits have a value different from 0xFFFFFFFF.

30.3.2.8 FWCTRDCNi : Port i Cut-Through Rejected Descriptor Counter Register (i = 0, 1)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x6508 + 0x20 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	CTRDN[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	CTRDN[15:0]	Cut-through rejected Descriptor Number	R
31:16	—	These bits are read as 0.	R

CTRDN[15:0] bits (Cut-through rejected Descriptor Number)

These bits count the number of descriptors rejected by Cut-through forwarding for port i.

[Clearing condition]

SW: Reading these bits clears it.

[Increment condition]

HW: Incremented by 1 when a frame coming from port *i* is forwarded by Cut-Through Forwarding but not in store and forward mode and if these bits have a value different from 0xFFFFFFFF.

Note: HW: These bits do not take in account the cut-through path of cut-through forwarding but only the store and forward path so, any frame not forwarded in store and forward with cut-through (for any *j*, if FWCTTCi0.CTDV[j] is set, FWCTTCi0.CTDFMj is not set) will be counted here.

30.3.2.9 FWDDRDCN2 : Port 2 Direct Descriptor Rejected Descriptor Counter Register

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x6548

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DDRDN[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	DDRDN[15:0]	Direct Descriptor rejected Descriptor Number	R
31:16	—	These bits are read as 0.	R

DDRDN[15:0] bits (Direct Descriptor rejected Descriptor Number)

These bits count the number of descriptors rejected by Direct Descriptor forwarding for port 2.

[Clearing condition]

SW: Reading these bits clears it.

[Increment condition]

HW: Incremented by 1 when a frame coming from port 2 is rejected by Direct Descriptor forwarding and if these bits have a value different from 0xFFFFFFFF.

30.3.2.10 FWLTHRDCNi : Port *i* Layer 3 Rejected Descriptor Counter Register (*i* = 0 to 2)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x650C + 0x20 × *i*

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	LTHRDN[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	LTHRDN[15:0]	Layer 3 rejected Descriptor Number	R
31:16	—	These bits are read as 0.	R

LTHRDN[15:0] bits (Layer 3 rejected Descriptor Number)

These bits count the number of descriptors rejected by Layer 3 forwarding for port i.

[Clearing condition]

SW: Reading these bits clears it.

[Increment condition]

HW: Incremented by 1 when a frame coming from port i is rejected by Layer 3 forwarding and if these bits have a value other than 0xFFFF.

30.3.2.11 FWLTWRDCNi : Port i Layer 2 Rejected Descriptor Counter Register (i = 0 to 2)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x6514 + 0x20 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	LTWRDN[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	LTWRDN[15:0]	Layer 2 rejected Descriptor Number	R
31:16	—	These bits are read as 0.	R

LTWRDN[15:0] bits (Layer 2 rejected Descriptor Number)

These bits count the number of descriptors rejected by Layer 2 forwarding for port i.

[Clearing condition]

SW: Reading these bits clears it.

[Increment condition]

HW: Incremented by 1 when a frame coming from port i is rejected by Layer 2 forwarding and if these bits have a value other than 0xFFFF.

30.3.2.12 FWPBRDCNi : Port i Port Based Rejected Descriptor Counter Register (i = 0 to 2)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x6518 + 0x20 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PBRDN[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	PBRDN[15:0]	Port Based rejected Descriptor Number	R
31:16	—	These bits are read as 0.	R

PBRDN[15:0] bits (Port Based rejected Descriptor Number)

These bits count the number of descriptors rejected by Port Based forwarding for port i.

[Clearing condition]

SW: Reading these bits clears it.

[Increment condition]

HW: Incremented by 1 when a frame coming from port i is rejected by Port Based forwarding and if these bits have a value other than 0xFFFF.

30.3.2.13 FWPMFDCNi : PSFP MSDU i Filtered Descriptor Counter Register (i = 0 to 15)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x6700 + 0x4 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PMFDN[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	PMFDN[15:0]	PSFP MSDU Filtered Descriptor Number	R
31:16	—	These bits are read as 0.	R

PMFDN[15:0] bits (PSFP MSDU Filtered Descriptor Number)

These bits count the number of descriptors rejected by PSFP MSDU Filter i.

[Clearing condition]

SW: Reading these bits clears it.

[Increment condition]

HW: Incremented by 1 when a frame is rejected by PSFP MSDU Filter i and if these bits have a value different from 0xFFFF.

30.3.2.14 FWPMGDCNi : PSFP Meter i Green Descriptor Counter Register (i = 0 to 31)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x6800 + 0x10 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PMGDN[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	PMGDN[15:0]	PSFP Meter Green Descriptor Number	R
31:16	—	These bits are read as 0.	R

PMGDN[15:0] bits (PSFP Meter Green Descriptor Number)

These bits count the number of descriptors with the color green given by PSFP meter Filter i.

[Clearing condition]

SW: Reading these bits clears it.

[Increment condition]

HW: Incremented by 1 when a frame has green color associated by PSFP meter Filter i and if these bits have a value different from 0xFFFF.

30.3.2.15 FWPMYDCNi : PSFP Meter i Yellow Descriptor Counter Register (i = 0 to 7)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x6804 + 0x10 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PMYDN[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	PMYDN[15:0]	PSFP Meter Yellow Descriptor Number	R
31:16	—	These bits are read as 0.	R

PMYDN[15:0] bits (PSFP Meter Yellow Descriptor Number)

These bits count the number of descriptors with the color Yellow given by PSFP meter Filter i.

[Clearing condition]

SW: Reading these bits clears it.

[Increment condition]

HW: Incremented by 1 when a frame has yellow color associated by PSFP meter Filter i and if these bits have a value different from 0xFFFF.

30.3.2.16 FWPMRDCNi : PSFP Meter i Red Descriptor Counter Register (i = 0 to 31)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x6808 + 0x10 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PMRDN[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	PMRDN[15:0]	PSFP Meter Red Descriptor Number	R
31:16	—	These bits are read as 0.	R

PMRDN[15:0] bits (PSFP Meter Red Descriptor Number)

These bits count the number of descriptors with the color Red given by PSFP meter Filter i. (All and only frames rejected by meter filter are accounted)

[Clearing condition]

SW: Reading these bits clears it.

[Increment condition]

- HW: Incremented by 1 when a frame has Red color associated and is rejected by PSFP meter Filter i and if these bits have a value different from 0xFFFF.
- HW: Incremented by 1 when a frame is rejected by PSFP meter Filter i because ATS RAM corresponding queue is full and if these bits have a value different from 0xFFFF.
- HW: Incremented by 1 when a frame is rejected by PSFP meter Filter i because corresponding meter is disabled and if these bits have a value different from 0xFFFF.

30.3.2.17 FWFRPPCNI : FRER i Passed Packet Counter Register (i = 0 to 127)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x6A00 + 0x8 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PPC[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	PPC[15:0]	Passed Packet Count	R

Bit	Symbol	Function	R/W
31:16	—	These bits are read as 0.	R

PPC[15:0] bits (Passed Packet Count)

These bits count the number of packets that passed FRER recovery rule number i.

[Clearing condition]

SW: Reading these bits clears it.

[Increment condition]

HW: Incremented by 1 when a frame passed FRER recovery rule number i and if these bits have a value different from 0xFFFF.

30.3.2.18 FWFRDPCNi : FRER i Discarded Packet Counter Register (i = 0 to 127)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x6A04 + 0x8 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DPC[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	DPC[15:0]	Discarded Packet Count	R
31:16	—	These bits are read as 0.	R

DPC[15:0] bits (Discarded Packet Count)

These bits count the number of packets that has been discarded by FRER recovery rule number i.

[Clearing condition]

SW: Reading these bits clears it.

[Increment condition]

HW: Incremented by 1 when a frame has been discarded by FRER recovery rule number i and if these bits have a value different from 0xFFFF.

30.3.3 MFWD Interrupt Registers

This section describes MFWD interrupt registers.

30.3.3.1 FWEIS0i : Port i Error Interrupt Status Register 0 (i = 0 to 2)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7900 + 0x10 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	DDNT FS	DDSE S	—	DDES	WMIU FS	WMIS FS	WMFF S	WMCFS	—	—	SMHM FS	SMHL FS	PBNT FS	LTWV UFS
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	LTWD UFS	LTWS UFS	LTWN TFS	LTWV SPFS	LTWS SPFS	LTWD SPFS	—	—	—	—	—	—	LTHUF S	LTHNT FS	—	LTHSP FS
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	LTHSPFS	Layer 3 Source Port Filtering Status Flag	R/W ^{*1}
1	—	These bits are read as 0. The write value should be 0.	R/W
2	LTHNTFS	Layer 3 No Target Filtering Status Flag	R/W ^{*1}
3	LTHUFS	Layer 3 Unknown Filtering Status Flag	R/W ^{*1}
9:4	—	These bits are read as 0. The write value should be 0.	R/W
10	LTWDSPFS	Layer 2 Destination Source Port Filtering Status Flag	R/W ^{*1}
11	LTWSSPFS	Layer 2 Source Source Port Filtering Status Flag	R/W ^{*1}
12	LTWVSPFS	Layer 2 VLAN Source Port Filtering Status Flag	R/W ^{*1}
13	LTWNTFS	Layer 2 No Target Filtering Status Flag	R/W ^{*1}
14	LTWSUFS	Layer 2 Source Unknown Filtering Status Flag	R/W ^{*1}
15	LTWDUFS	Layer 2 Destination Unknown Filtering Status Flag	R/W ^{*1}
16	LTWVUFS	Layer 2 VLAN Unknown Filtering Status Flag	R/W ^{*1}
17	PBNTFS	Port Based No Target Filtering Status Flag	R/W ^{*1}
18	SMHLFS	Source MAC Hardware Learning Fail Status Flag	R/W ^{*1}
19	SMHMF	Source MAC Hardware Migration Fail Status Flag	R/W ^{*1}
21:20	—	These bits are read as 0. The write value should be 0.	R/W
22	WMCFS	Watermark Critical Filtering Status Flag	R/W ^{*1}
23	WMFFS	Watermark Flush Filtering Status Flag	R/W ^{*1}
24	WMISFS	Watermark IPV Secure Filtering Status Flag	R/W ^{*1}
25	WMIUFS	Watermark IPV Unsecure Filtering Status Flag	R/W ^{*1}
26	DDES	Direct Descriptor Error Status Flag ^{*2}	R/W ^{*1}
27	—	This bit is read as 0. The write value should be 0.	R/W
28	DDSES	Direct Descriptor Security Error Status Flag ^{*2}	R/W ^{*1}
29	DDNTFS	Direct Descriptor No Target Filtering Status Flag ^{*2}	R/W ^{*1}
31:30	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Read value differs from written value.

Note 2. HW: This bit is only valid for i = 2, 3.

LTHSPFS bit (Layer 3 Source Port Filtering Status Flag)

[Setting condition]

- HW: When forwarding flow for a frame coming from port i goes until “Layer 3 routing secure error” block in [Figure 30.85](#) and L3.SLV[i] is not set.
- HW: When forwarding flow for a frame coming from port i goes until “Layer 3 routing error” block in [Figure 30.86](#) and L3.SLV[i] is not set.

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

HW: Descriptor corresponding frame is rejected.

LTHNTFS bit (Layer 3 No Target Filtering Status Flag)

[Setting condition]

- HW: When forwarding flow for a frame coming from port i goes until “Layer 3 routing secure error” block in [Figure 30.85](#) and L3.DV & !FWPCi1.LTHFM is null.
- HW: When forwarding flow for a frame coming from port i goes until “Layer 3 routing error” block in [Figure 30.86](#) and L3.DV & !FWPCi1.LTHFM is null.

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

HW: Descriptor corresponding frame is rejected.

LTHUFS bit (Layer 3 Unknown Filtering Status Flag)

[Setting condition]

- HW: When forwarding flow for a frame coming from port i goes until “Layer 3 routing unknown secure stream error” block in [Figure 30.85](#) and FWPCi0.LTHRUSS is set.
- HW: When forwarding flow for a frame coming from port i goes until “Layer 3 routing error” block in [Figure 30.86](#) and FWPCi0.LTHRUS is set.

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

HW: Descriptor corresponding frame is rejected.

LTWDSPFS bit (Layer 2 Destination Source Port Filtering Status Flag)

[Setting condition]

- HW: When forwarding flow for a frame coming from port i goes until “L2 forwarding secure error” block in [Figure 30.95](#) and, FWPCi0.MACDSA is set, Destination MAC is set in the MAC table, MAC.SL is set and MAC.DSLV is not set for destination MAC.
- HW: When forwarding flow for a frame coming from port i goes until “L2 forwarding error” block in [Figure 30.96](#) and, FWPCi0.MACDSA is set, Destination MAC is set in the MAC table and MAC.DSLV is not set for destination MAC.

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

HW: Descriptor corresponding frame is rejected.

LTWSSPFS bit (Layer 2 Source Source Port Filtering Status Flag)

[Setting condition]

- HW: When forwarding flow for a frame coming from port i goes until “L2 forwarding secure error” block in [Figure 30.95](#) and, FWPCi0.MACSSA is set, Source MAC is set in the MAC table, MAC.SL is set and MAC.SSLV is not set for source MAC.
- HW: When forwarding flow for a frame coming from port i goes until “L2 forwarding error” block in [Figure 30.96](#) and, FWPCi0.MACSSA is set, Source MAC is set in the MAC table and MAC.SSLV is not set for source MAC.

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

HW: Descriptor corresponding frame is rejected.

LTWVSPFS bit (Layer 2 VLAN Source Port Filtering Status Flag)

[Setting condition]

- HW: When forwarding flow for a frame coming from port i goes until “L2 forwarding secure error” block in [Figure 30.95](#) and, FWPCi0.VLANSA is set, VLAN ID is valid, VLAN ID is set in the VLAN table, VLAN.SL is set and VLAN.SLV is not set for VLAN ID.
- HW: When forwarding flow for a frame coming from port i goes until “L2 forwarding error” block in [Figure 30.96](#) and, FWPCi0.VLANSA is set, VLAN ID is valid, VLAN ID is set in the VLAN table and VLAN.SLV is not set for VLAN ID.

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

HW: Descriptor corresponding frame is rejected.

LTWNTFS bit (Layer 2 No Target Filtering Status Flag)

[Setting condition]

- HW: When forwarding flow for a frame coming from port i goes until “L2 forwarding secure error” block in [Figure 30.95](#) and, FWPCi0.MACDSA is set, Destination MAC is set in the MAC table, MAC.SL is set, FWPCi0.VLANSA is set, VLAN ID is valid, VLAN ID is set in the VLAN table, VLAN.SL is set and MAC.DV & VLAN.DV & !FWPCi2.LTWFM is null.
- HW: When forwarding flow for a frame coming from port i goes until “L2 forwarding secure error” block in [Figure 30.95](#) and, FWPCi0.MACDSA is set, Destination MAC is set in the MAC table, MAC.SL is set, { FWPCi0.VLANSA is not set or VLAN ID is not valid or VLAN ID is not set in the VLAN table or VLAN.SL is not set } and MAC.DV & !FWPCi2.LTWFM is null.
- HW: When forwarding flow for a frame coming from port i goes until “L2 forwarding secure error” block in [Figure 30.95](#) and, {FWPCi0.MACDSA is not set or Destination MAC is not set in the MAC table or MAC.SL is not set}, VLAN ID is valid, VLAN ID is set in the VLAN table, VLAN.SL is set and VLAN.DV & !FWPCi2.LTWFM is null.
- HW: When forwarding flow for a frame coming from port i goes until “L2 forwarding error” block in [Figure 30.96](#) and, FWPCi0.MACDSA is set, Destination MAC is set in the MAC table, FWPCi0.VLANSA is set, VLAN ID is valid, VLAN ID is set in the VLAN table and MAC.DV & VLAN.DV & !FWPCi2.LTWFM is null.
- HW: When forwarding flow for a frame coming from port i goes until “L2 forwarding error” block in [Figure 30.96](#) and, FWPCi0.MACDSA is set, Destination MAC is set in the MAC table, { FWPCi0.VLANSA is not set or VLAN ID is not valid or VLAN ID is not set in the VLAN table } and MAC.DV & !FWPCi2.LTWFM is null.
- HW: When forwarding flow for a frame coming from port i goes until “L2 forwarding error” block in [Figure 30.96](#) and, {FWPCi0.MACDSA is not set or Destination MAC is not set in the MAC table}, VLAN ID is valid, VLAN ID is set in the VLAN table and VLAN.DV & !FWPCi2.LTWFM is null.

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

HW: Descriptor corresponding frame is rejected.

LTWSUFS bit (Layer 2 Source Unknown Filtering Status Flag)

[Setting condition]

- HW: When forwarding flow for a frame coming from port i goes until “L2 forwarding secure error” block in [Figure 30.95](#) and, {FWPCi0.MACSSA is not set or Source MAC is not set in the MAC table or MAC.SL is not set for Source MAC} and FWPCi0.MACRUSSA is set.
- HW: When forwarding flow for a frame coming from port i goes until “L2 forwarding error” block in [Figure 30.96](#) and, {FWPCi0.MACSSA is not set or Source MAC is not set in the MAC table} and FWPCi0.MACRUSA is set.

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

HW: Descriptor corresponding frame is rejected.

LTWDUFS bit (Layer 2 Destination Unknown Filtering Status Flag)

[Setting condition]

- HW: When forwarding flow for a frame coming from port i goes until “L2 forwarding secure error” block in [Figure 30.95](#) and, {FWPCi0.MACDSA is not set or Destination MAC is not set in the MAC table or MAC.SL is not set for destination MAC} and FWPCi0.MACRUSSA is set.
- HW: When forwarding flow for a frame coming from port i goes until “L2 forwarding error” block in [Figure 30.96](#) and, {FWPCi0.MACDSA is not set or Destination MAC is not set in the MAC table} and FWPCi0.MACRUDA is set.

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

HW: Descriptor corresponding frame is rejected.

LTWVUFS bit (Layer 2 VLAN Unknown Filtering Status Flag)

[Setting condition]

- HW: When forwarding flow for a frame coming from port i goes until “L2 forwarding secure error” block in [Figure 30.95](#) and, {FWPCi0.VLANSA is not set or VLAN is not valid or VLAN ID is not set in the VLAN table or VLAN.SL is not set for VLAN ID} and FWPCi0.VLANRUS is set.
- HW: When forwarding flow for a frame coming from port i goes until “L2 forwarding error” block in [Figure 30.96](#) and, {FWPCi0.VLANSA is not set or VLAN is not valid or VLAN ID is not set in the VLAN table} and FWPCi0.VLANRU is set.

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

HW: Descriptor corresponding frame is rejected.

PBNTFS bit (Port Based No Target Filtering Status Flag)

[Setting condition]

- HW: When forwarding flow for a frame coming from port i goes until “Port based forwarding secure error” block in [Figure 30.99](#) and FWPBFCi.PBDV is null.
- HW: When forwarding flow for a frame coming from port i goes until “Port based forwarding unsecure error” block in [Figure 30.100](#) and FWPBFCi.PBDV is null.

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

HW: Descriptor corresponding frame is rejected.

SMHLFS bit (Source MAC Hardware Learning Fail Status Flag)

[Setting condition]

HW: When a frame is selected for source MAC address HW learning ([section 30.5.1.5.2. CPU Learning Path](#)) and it could not be learnt because the learning rate is too high.

This error has nothing to do with fail of [Table 30.45](#).

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

HW: Descriptor corresponding frame is rejected.

SMHMFS bit (Source MAC Hardware Migration Fail Status Flag)

[Setting condition]

- HW: When a frame is selected for source MAC address HW migration ([section 30.5.1.5.2. CPU Learning Path](#)) and it could not be learnt because the learning rate is too high.
- HW: When a frame is selected for source MAC address HW migration and its source MAC address contained in the table in static (MAC.DE is not set).
- HW: When a frame is selected for source MAC address HW migration and its source MAC address contained in the table is secure (MAC.SL is set).
- HW: When a frame is selected for source MAC address HW migration and its source MAC address contained in the table has one of its destination port which does not allow migration (for at least one i, MAC.DV[i] is set and FWPCi0.MACHMA is not set).

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

HW: Descriptor corresponding frame is rejected.

WMCFS bit (Watermark Critical Filtering Status Flag)

[Setting condition]

HW: A frame coming from port i is forwarded with a Layer 3/IP/Layer 2/Port based normal descriptor and FWLBWMCi.WMCLPR bit {DEI,IPV} is set and coa_watermark_critical[i] pin is set (where DEI is set to input frame C-TAG DEI if FWGC.SVM is set to 01b, input frame S-TAG DEI if FWGC.SVM is set to 10b and set to 0 if FWGC.SVM is set to 00b and IPV is the Frame corresponding normal descriptor, selected by the forwarding flow, FDESCR.IPV value).

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

HW: Descriptor corresponding frame is rejected.

WMFFS bit (Watermark Flush Filtering Status Flag)

[Setting condition]

HW: A frame coming from port i is forwarded with a Layer 3/IP/Layer 2/Port based normal descriptor and FWLBWMCi.WMFLPR bit {DEI,IPV} is set and coa_watermark_flush[i] pin is set (where DEI is set to input frame C-TAG DEI if FWGC.SVM is set to 01b, input frame S-TAG DEI if FWGC.SVM is set to 10b and set to 0 if FWGC.SVM is set to 00b and IPV is the Frame corresponding normal descriptor, selected by the forwarding flow, FDESCR.IPV value).

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

HW: Descriptor corresponding frame is rejected.

WMISFS bit (Watermark IPV Secure Filtering Status Flag)

[Setting condition]

HW: A frame coming from port i is forwarded with a Layer 3/IP/Layer 2/Port based normal descriptor and corresponding IPV is rejected by IPV based unsecure watermark [COMA] (where IPV is Frame corresponding normal descriptor, selected by the forwarding flow, FDESCR.IPV value before filtering).

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

HW: Descriptor corresponding frame is rejected.

WMIUFS bit (Watermark IPV Unsecure Filtering Status Flag)

[Setting condition]

HW: A frame coming from port i is forwarded with a Layer 3/IP/Layer 2/Port based normal descriptor and SEC is not set and corresponding IPV is rejected by IPV based secure watermark [COMA] (where SEC and IPV are respectively Frame corresponding normal descriptor, selected by the forwarding flow, FDESCR.SEC and FDESCR.IPV values before filtering).

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

HW: Descriptor corresponding frame is rejected.

DDES bit (Direct Descriptor Error Status Flag)

[Setting condition]

HW: When a direct descriptor (LDESCR.FMT [GWCA] set to 1) is received from port $2 + i$ and FWPCi1.DDE is not set.

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

HW: Descriptor corresponding frame is rejected.

DDSES bit (Direct Descriptor Security Error Status Flag)

[Setting condition]

HW: When a secure direct descriptor (LDESCR.FMT [GWCA] and LDESCR.SEC [GWCA] are both set to 1) is received from port $2 + i$, FWPCi1.DDE is set and FWPCi1.DDSL is not set.

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

HW: Descriptor corresponding frame is rejected.

DDNTFS bit (Direct Descriptor No Target Filtering Status Flag)

[Setting condition]

HW: When a directed descriptor (LDESCR.FMT [GWCA] set to 1) is received from port $2 + i$ and LDESCR.DV [GWCA] is set to All0.

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

HW: Descriptor corresponding frame is rejected.

30.3.3.2 FWEIE0i : Port i Error Interrupt Enable Register 0 (i = 0 to 2)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7904 + 0x10 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	DDNT FE	DDSE E	DDFE E	DDEE	WMIU FE	WMIS FE	WMFF E	WMCF E	—	—	SMHM FE	SMHL FE	PBNT FE	LTWV UFE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	LTWD UFE	LTWS UFE	LTWN TFE	LTWV SPFE	LTWS SPFE	LTWD SPFE	—	—	—	—	—	—	LTHUF E	LTHNT FE	—	LTHSP FE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	LTHSPFE	Layer 3 Source Port Filtering Enable 0: Interrupt i disabled. 1: Interrupt i Enabled.	R/W ¹
1	—	This bit is read as 0. The write value should be 0.	R/W
2	LTHNTFE	Layer 3 No Target Filtering Enable 0: Interrupt i disabled. 1: Interrupt i Enabled.	R/W ¹
3	LTHUFE	Layer 3 Unknown Filtering Enable 0: Interrupt i disabled. 1: Interrupt i Enabled.	R/W ¹
9:4	—	These bits are read as 0. The write value should be 0.	R/W
10	LTWDSPFE	Layer 2 Destination Source Port Filtering Enable 0: Interrupt i disabled. 1: Interrupt i Enabled.	R/W ¹
11	LTWSSPFE	Layer 2 Source Source Port Filtering Enable 0: Interrupt i disabled. 1: Interrupt i Enabled.	R/W ¹
12	LTWVSPFE	Layer 2 VLAN Source Port Filtering Enable 0: Interrupt i disabled. 1: Interrupt i Enabled.	R/W ¹
13	LTWNTFE	Layer 2 No Target Filtering Enable 0: Interrupt i disabled. 1: Interrupt i Enabled.	R/W ¹
14	LTWSUFE	Layer 2 Source Unknown Filtering Enable 0: Interrupt i disabled. 1: Interrupt i Enabled.	R/W ¹
15	LTWDUFE	Layer 2 Destination Unknown Filtering Enable 0: Interrupt i disabled. 1: Interrupt i Enabled.	R/W ¹
16	LTWVUFE	Layer 2 VLAN Unknown Filtering Enable 0: Interrupt i disabled. 1: Interrupt i Enabled.	R/W ¹
17	PBNTFE	Port Based No Target Filtering Enable 0: Interrupt i disabled. 1: Interrupt i Enabled.	R/W ¹
18	SMHLFE	Source MAC Hardware Learning Fail Enable 0: Interrupt i disabled. 1: Interrupt i Enabled.	R/W ¹

Bit	Symbol	Function	R/W
19	SMHMF	Source MAC Hardware Migration Fail Enable 0: Interrupt i disabled. 1: Interrupt i Enabled.	R/W ^{*1}
21:20	—	These bits are read as 0. The write value should be 0.	R/W
22	WMCFF	Watermark Critical Filtering Enable 0: Interrupt i disabled. 1: Interrupt i Enabled.	R/W ^{*1}
23	WMFFF	Watermark Flush Filtering Enable 0: Interrupt i disabled. 1: Interrupt i Enabled.	R/W ^{*1}
24	WMISFF	Watermark IPV Secure Filtering Enable 0: Interrupt i disabled. 1: Interrupt i Enabled.	R/W ^{*1}
25	WMIUFF	Watermark IPV Unsecure Filtering Enable 0: Interrupt i disabled. 1: Interrupt i Enabled.	R/W ^{*1}
26	DDEE	Direct Descriptor Error Enable 0: Interrupt i disabled. 1: Interrupt i Enabled.	R/W ^{*1}
27	DDFEE	Direct Descriptor Format Error Enable 0: Interrupt i disabled. 1: Interrupt i Enabled.	R/W ^{*1}
28	DDSEE	Direct Descriptor Security Error Enable 0: Interrupt i disabled. 1: Interrupt i Enabled.	R/W ^{*1}
29	DDNTFF	Direct Descriptor No Target Filtering Enable 0: Interrupt i disabled. 1: Interrupt i Enabled.	R/W ^{*1}
31:30	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Read value differs from written value.

LTHSPFE bit (Layer 3 Source Port Filtering Enable)

[Setting condition]

Writing 1 to this bit will set it.

[Clearing condition]

Writing 1 to FWEID0i.LTHSPFD register will clear this bit.

LTHNTFE bit (Layer 3 No Target Filtering Enable)

[Setting condition]

Writing 1 to this bit will set it.

[Clearing condition]

Writing 1 to FWEID0i.LTHNTFD register will clear this bit.

LTHUFE bit (Layer 3 Unknown Filtering Enable)

[Setting condition]

Writing 1 to this bit will set it.

[Clearing condition]

Writing 1 to FWEID0i.LTHUFD register will clear this bit.

LTDWSPFE bit (Layer 2 Destination Source Port Filtering Enable)

[Setting condition]

Writing 1 to this bit will set it.

[Clearing condition]

Writing 1 to FWEID0i.LTWDSFPD register will clear this bit.

LTWSSPFE bit (Layer 2 Source Source Port Filtering Enable)

[Setting condition]

Writing 1 to this bit will set it.

[Clearing condition]

Writing 1 to FWEID0i.LTWSSFPD register will clear this bit.

LTWVSPFE bit (Layer 2 VLAN Source Port Filtering Enable)

[Setting condition]

Writing 1 to this bit will set it.

[Clearing condition]

Writing 1 to FWEID0i.LTWVSPFD register will clear this bit.

LTWNTE bit (Layer 2 No Target Filtering Enable)

[Setting condition]

Writing 1 to this bit will set it.

[Clearing condition]

Writing 1 to FWEID0i.LTWNTEFD register will clear this bit.

LTWSUFE bit (Layer 2 Source Unknown Filtering Enable)

[Setting condition]

Writing 1 to this bit will set it.

[Clearing condition]

Writing 1 to FWEID0i.LTWSUFD register will clear this bit.

LTWDUFE bit (Layer 2 Destination Unknown Filtering Enable)

[Setting condition]

Writing 1 to this bit will set it.

[Clearing condition]

Writing 1 to FWEID0i.LTWDUFD register will clear this bit.

LTWVUFE bit (Layer 2 VLAN Unknown Filtering Enable)

[Setting condition]

Writing 1 to this bit will set it.

[Clearing condition]

Writing 1 to FWEID0i.LTWVUFD register will clear this bit.

PBNTFE bit (Port Based No Target Filtering Enable)

[Setting condition]

Writing 1 to this bit will set it.

[Clearing condition]

Writing 1 to FWEID0i.PBNTFD register will clear this bit.

SMHLFE bit (Source MAC Hardware Learning Fail Enable)

[Setting condition]

Writing 1 to this bit will set it.

[Clearing condition]

Writing 1 to FWEID0i.SMHLFD register will clear this bit.

SMHMF bit (Source MAC Hardware Migration Fail Enable)

[Setting condition]

Writing 1 to this bit will set it.

[Clearing condition]

Writing 1 to FWEID0i.SMHMFD register will clear this bit.

WMCFE bit (Watermark Critical Filtering Enable)

[Setting condition]

Writing 1 to this bit will set it.

[Clearing condition]

Writing 1 to FWEID0i.WMFFD register will clear this bit.

WMFFE bit (Watermark Flush Filtering Enable)

[Setting condition]

Writing 1 to this bit will set it.

[Clearing condition]

Writing 1 to FWEID0i.WMFFE register will clear this bit.

WMISFE bit (Watermark IPV Secure Filtering Enable)

[Setting condition]

Writing 1 to this bit will set it.

[Clearing condition]

Writing 1 to FWEID0i.WMISFD register will clear this bit.

WMIUFE bit (Watermark IPV Unsecure Filtering Enable)

[Setting condition]

Writing 1 to this bit will set it.

[Clearing condition]

Writing 1 to FWEID0i.WMIUFD register will clear this bit.

DDEE bit (Direct Descriptor Error Enable)

[Setting condition]

Writing 1 to this bit will set it.

[Clearing condition]

Writing 1 to FWEID0i.DDED register will clear this bit.

DDFEE bit (Direct Descriptor Format Error Enable)

[Setting condition]

Writing 1 to this bit will set it.

[Clearing condition]

Writing 1 to FWEID0i.DDFED register will clear this bit.

DDSEE bit (Direct Descriptor Security Error Enable)

[Setting condition]

Writing 1 to this bit will set it.

[Clearing condition]

Writing 1 to FWEID0i.DDSED register will clear this bit.

DDNTFE bit (Direct Descriptor No Target Filtering Enable)

[Setting condition]

Writing 1 to this bit will set it.

[Clearing condition]

Writing 1 to FWEID0i.DDNTFD register will clear this bit.

30.3.3.3 FWEID0i : Port i Error Interrupt Disable Register 0 (i = 0 to 2)

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7908 + 0x10 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	DDNT FD	DDSE D	DDFE D	DDSD	WMIU FD	WMIS FD	WMFF D	WMCFF D	—	—	SMHM FD	SMHL FD	PBNT FD	LTWV UFD
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	LTWD UFD	LTWS UFD	LTWN TFD	LTWV SPFD	LTWS SPFD	LTWD SPFD	—	—	—	—	—	—	LTHUF D	LTHNT FD	—	LTHSP FD
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	LTHSPFD	Layer 3 Source Port Filtering Disable Writing 1 to this bit clears FWEIE0.LTHSPFE register.	R/W
1	—	This bit is read as 0. The write value should be 0.	R/W
2	LTHNTFD	Layer 3 No Target Filtering Disable Writing 1 to this bit clears FWEIE0.LTHNTFE register.	R/W
3	LTHUFD	Layer 3 Unknown Filtering Disable Writing 1 to this bit clears FWEIE0.LTHNTFE register.	R/W
9:4	—	These bits are read as 0. The write value should be 0.	R/W
10	LTWDSPFD	Layer 2 Destination Source Port Filtering Disable Writing 1 to this bit clears FWEIE0.LTWDSPFE register.	R/W
11	LTWSSPFD	Layer 2 Source Source Port Filtering Disable Writing 1 to this bit clears FWEIE0.LTWSSPFE register.	R/W
12	LTWVSPFD	Layer 2 VLAN Source Port Filtering Disable Writing 1 to this bit clears FWEIE0.LTWVSPFE register.	R/W
13	LTWNTFD	Layer 2 No Target Filtering Disable Writing 1 to this bit clears FWEIE0.LTWNTFE register.	R/W
14	LTWSUFD	Layer 2 Source Unknown Filtering Disable Writing 1 to this bit clears FWEIE0.LTWSUFE register.	R/W
15	LTWDUFD	Layer 2 Destination Unknown Filtering Disable Writing 1 to this bit clears FWEIE0.LTWDUFE register.	R/W
16	LTWVUFD	Layer 2 VLAN Unknown Filtering Disable Writing 1 to this bit clears FWEIE0.LTWVUFE register.	R/W
17	PBNTFD	Port Based No Target Filtering Disable Writing 1 to this bit clears FWEIE0.PBNTFE register.	R/W
18	SMHLFD	Source MAC Hardware Learning Fail Disable Writing 1 to this bit clears FWEIE0.SMHLFE register.	R/W

Bit	Symbol	Function	R/W
19	SMHMF	Source MAC Hardware Migration Fail Disable Writing 1 to this bit clears FWEIE0.SMHMFE register.	R/W
21:20	—	These bits are read as 0. The write value should be 0.	R/W
22	WMC	Watermark Critical Filtering Disable Writing 1 to this bit clears FWEIE0.WMCFE register.	R/W
23	WMF	Watermark Flush Filtering Disable Writing 1 to this bit clears FWEIE0.WMFFE register.	R/W
24	WMIS	Watermark IPV Secure Filtering Disable Writing 1 to this bit clears FWEIE0.WMISFE register.	R/W
25	WMIU	Watermark IPV Unsecure Filtering Disable Writing 1 to this bit clears FWEIE0.WMIUFE register.	R/W
26	DDE	Direct Descriptor Error Disable Writing 1 to this bit clears FWEIE0.DDEE register.	R/W
27	DDF	Direct Descriptor Format Error Disable Writing 1 to this bit clears FWEIE0.DDFEE register.	R/W
28	DDSE	Direct Descriptor Security Error Disable Writing 1 to this bit clears FWEIE0.DDSEE register.	R/W
29	DDNT	Direct Descriptor No Target Filtering Disable Writing 1 to this bit clears FWEIE0.DDNTFE register.	R/W
31:30	—	These bits are read as 0. The write value should be 0.	R/W

30.3.3.4 FWEIS1 : Error Interrupt Status Register 1

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7A00

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	VLAN TSES	—	MACT SES	—	—	—	LTHTS ES	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	—	This bit is read as 0. The write value should be 0.	R/W
1	LTHTSES	L3 Table Security Error Status Flag	R/W ^{*1}
4:2	—	These bits are read as 0. The write value should be 0.	R/W
5	MACTSES	MAC Table Security Error Status Flag	R/W ^{*1}
6	—	This bit is read as 0. The write value should be 0.	R/W
7	VLANTSES	VLAN Table Security Error Status Flag	R/W ^{*1}
31:8	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Read value differs from written value.

LTHTSES bit (L3 Table Security Error Status Flag)

[Setting condition]

HW: A CPU tried to learn an unsecure entry, but the unsecure entry maximum number is already reached.

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

- HW: Unsecure entry won't be learned
- SW: System dependent, cannot be defined here.

MACTSES bit (MAC Table Security Error Status Flag)

[Setting condition]

HW: A CPU tried to learn an unsecure entry, but the unsecure entry maximum number is already reached.

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

- HW: Unsecure entry won't be learned
- SW: System dependent, cannot be defined here.

VLANTSES bit (VLAN Table Security Error Status Flag)

[Setting condition]

HW: A CPU tried to learn an unsecure entry, but the unsecure entry maximum number is already reached.

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

- HW: Unsecure entry won't be learned
- SW: System dependent, cannot be defined here.

30.3.3.5 FWEIE1 : Error Interrupt Enable Register 1

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7A04

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	VLAN TSEE	—	MACT SEE	—	—	—	LTHTS EE	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	—	This bit is read as 0. The write value should be 0.	R/W
1	LTHTSEE	L3 Table Security Error Enable 0: Interrupt disabled. 1: Interrupt Enabled.	R/W ¹
4:2	—	These bits are read as 0. The write value should be 0.	R/W
5	MACTSEE	MAC Table Security Error Enable 0: Interrupt disabled. 1: Interrupt Enabled.	R/W ¹
6	—	This bit is read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
7	VLANTSEE	VLAN Table Security Error Enable 0: Interrupt disabled. 1: Interrupt Enabled.	R/W ^{*1}
31:8	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Read value differs from written value.

LTHTSEE bit (L3 Table Security Error Enable)

[Setting condition]

Writing 1 to this bit will set it.

[Clearing condition]

Writing 1 to FWEID1.LTHTSED register will clear this bit.

MACTSEE bit (MAC Table Security Error Enable)

[Setting condition]

Writing 1 to this bit will set it.

[Clearing condition]

Writing 1 to FWEID1.MACTSED register will clear this bit.

VLANTSEE bit (VLAN Table Security Error Enable)

[Setting condition]

Writing 1 to this bit will set it.

[Clearing condition]

Writing 1 to FWEID1.VLANTSED register will clear this bit.

30.3.3.6 FWEID1 : Error Interrupt Disable Register 1

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7A08

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	L23UE ED	VLAN TSED	—	MACT SED	—	—	—	LTHTS ED	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	—	This bit is read as 0. The write value should be 0.	R/W
1	LTHTSED	L3 Table Security Error Disable Writing 1 to this bit will clear FWEIE1.LTHTSEE.	R/W
4:2	—	These bits are read as 0. The write value should be 0.	R/W
5	MACTSED	MAC Table Security Error Disable Writing 1 to this bit will clear FWEIE1.MACTSEE.	R/W
6	—	This bit is read as 0. The write value should be 0.	R/W
7	VLANTSED	VLAN Table Security Error Disable Writing 1 to this bit will clear FWEIE1.VLANTSEE.	R/W

Bit	Symbol	Function	R/W
8	L23UEED	Layer 2/Layer 3 Update Error Disable Writing 1 to this bit will clear FWEIE1.L23UEEE.	R/W
31:9	—	These bits are read as 0. The write value should be 0.	R/W

30.3.3.7 FWEIS2 : Error Interrupt Status Register 2

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7A10

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PMFS15 to PMFS0															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	PMFS15 to PMFS0	PSFP MSDU n Filtering Status Flag (n = 0 to 15)	R/W ¹
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Read value differs from written value.

PMFSn bits (PSFP MSDU n Filtering Status Flag (n = 0 to 15))

[Setting condition]

HW: Bit i is set when a descriptor is rejected by MSDU filter number i. (See [section 30.5.3.1.1. MSDU Filter](#)).

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

HW: Descriptor corresponding frame is rejected.

30.3.3.8 FWEIE2 : Error Interrupt Enable Register 2

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7A14

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PMFE15 to PMFE0															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	PMFE15 to PMFE0	PSFP MSDU n Filtering Enable (n = 0 to 15) 0: Interrupt disabled. 1: Interrupt Enabled.	R/W ¹

Note 1. Read value differs from written value.

[Setting condition]

Writing 1 to one of these bits will set it.

[Clearing condition]

Writing 1 to FWEID2.PMFD register bit i will clear this register bit i.

Base address: MFWD = 0x403C_0000
MFWD NS = 0x503C_0000

Offset address: 0x7A18

Bit	Symbol	Function	R/W
15:0	PMFD15 to PMFD0	PSFP MSDU n Filtering Disable (n = 0 to 15) Writing 1 to this register bit i will clear FWEIE2.PMFE register bit i.	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Base address: MFWD = 0x403C_0000
MFWD NS = 0x503C_0000

Offset address: 0x7A40

Bit	Symbol	Function	R/W
31:0	PMRFS31 to PMRFS0	PSFP Meter n Filtering Status Flag (n = 0 to 31)	R/W*1

Note 1. Read value differs from written value.

[Setting condition]

HW: Bit i is set when a descriptor is filtered by PSFP Meter filter number i. (See [section 30.5.3.1.2. Meter Filter/ATS](#)).

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

HW: Descriptor corresponding frame is rejected.

30.3.3.11 FWEIE5 : Error Interrupt Enable Register 5

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7A44

Bit position: 31

0

Bit field:

PMRFE31 to PMRFE0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	PMRFE31 to PMRFE0	PSFP Meter n Filtering Enable (n = 0 to 31)	R/W ^{*1}

Note 1. Read value differs from written value.

PMRFE_n bits (PSFP Meter n Filtering Enable (n = 0 to 31))

- Bit i set to 0: Interrupt i disabled.
- Bit i set to 1: Interrupt i Enabled.

[Setting condition]

Writing 1 to one of these bits will set it.

[Clearing condition]

Writing 1 to FWEID5.PMRFD register bit i will clear this register bit i.

30.3.3.12 FWEID5 : Error Interrupt Disable Register 5

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7A48

Bit position: 31

0

Bit field:

PMRFD31 to PMRFD0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	PMRFD31 to PMRFD0	PSFP Meter n Filtering Disable (n = 0 to 31) Writing 1 to this register bit i will clear FWEIE5.PMRFD register bit i.	R/W

30.3.3.13 FWEIS60 : Error Interrupt Status Register 60

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7A50

Bit position: 31

0

Bit field:

FFS31 to FFS0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	FFS31 to FFS0	FRER n Filtering Status Flag (n = 96 to 127)	R/W ^{*1}

Note 1. Read value differs from written value.

FFSn bits (FRER n Filtering Status Flag (n = 96 to 127))

[Setting condition]

- HW: If FRER rule i+96 is used for individual recovery, bit i is set when a descriptor is filtered by FRER rule number i+96 because of no sequence error or struck error. (See (e) [Individual recovery algorithms](#).)
- HW: If FRER rule i+96 is used for sequence recovery, bit i is set when a descriptor is filtered by FRER rule number i+96 because of no sequence error. (See (f) [Sequence recovery algorithms](#).)

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

HW: Descriptor corresponding frame is rejected.

30.3.3.14 FWEIE60 : Error Interrupt Enable Register 60

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7A54

Bit position: 31

0

Bit field:

FFE31 to FFE0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	FFE31 to FFE0	FRER n Filtering Enable (n = 96 to 127)	R/W ¹

Note 1. Read value differs from written value.

FFEn bits (FRER n Filtering Enable (n = 96 to 127))

- Bit i set to 0: Interrupt i disabled.
- Bit i set to 1: Interrupt i Enabled.

[Setting condition]

Writing 1 to one of these bits will set it.

[Clearing condition]

Writing 1 to FWEID60.FFDi register bit i will clear this register bit i.

30.3.3.15 FWEID60 : Error Interrupt Disable Register 60

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7A58

Bit position: 31

0

Bit field:

FFD31 to FFD0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	FFD31 to FFD0	FRER n Filtering Disable (n = 96 to 127) Writing 1 to this register bit i will clear FWEIE60.FFEi register bit i.	R/W

30.3.3.16 FWEIS61 : Error Interrupt Status Register 61

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7A60

Bit position: 31

0

Bit field:

FFS31 to FFS0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	FFS31 to FFS0	FRER n Filtering Status Flag (n = 64 to 95)	R/W ^{*1}

Note 1. Read value differs from written value.

FFSn bits (FRER n Filtering Status Flag (n = 64 to 95))

[Setting condition]

- HW: If FRER rule i+64 is used for individual recovery, bit i is set when a descriptor is filtered by FRER rule number i+64 because of no sequence error or struck error. (See [\(e\) Individual recovery algorithms.](#))
- HW: If FRER rule i+64 is used for sequence recovery, bit i is set when a descriptor is filtered by FRER rule number i+64 because of no sequence error. (See [\(f\) Sequence recovery algorithms.](#))

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

HW: Descriptor corresponding frame is rejected.

30.3.3.17 FWEIE61 : Error Interrupt Enable Register 61

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7A64

Bit position: 31

0

Bit field:

FFE31 to FFE0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	FFE31 to FFE0	FRER n Filtering Enable (n = 64 to 95)	R/W ^{*1}

Note 1. Read value differs from written value.

FFEn bits (FRER n Filtering Enable (n = 64 to 95))

- Bit i set to 0: Interrupt i disabled.
- Bit i set to 1: Interrupt i Enabled.

[Setting condition]

Writing 1 to one of these bits will set it.

[Clearing condition]

Writing 1 to FWEID61.FFDi register bit i will clear this register bit i.

30.3.3.18 FWEID61 : Error Interrupt Disable Register 61

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7A68

Bit position: 31 0

Bit field: FFD31 to FFD0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	FFD31 to FFD0	FRER n Filtering Disable (n = 64 to 95) Writing 1 to this register bit i will clear FWEIE61.FFEi register bit i.	R/W

30.3.3.19 FWEIS62 : Error Interrupt Status Register 62

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7A70

Bit position: 31 0

Bit field: FFS31 to FFS0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	FFS31 to FFS0	FRER n Filtering Status Flag (n = 32 to 63)	R/W ¹

Note 1. Read value differs from written value.

FFSn bits (FRER n Filtering Status Flag (n = 32 to 63))

[Setting condition]

- HW: If FRER rule i+32 is used for individual recovery, bit i is set when a descriptor is filtered by FRER rule number i+32 because of no sequence error or struck error. (See (e) [Individual recovery algorithms.](#))
- HW: If FRER rule i+32 is used for sequence recovery, bit i is set when a descriptor is filtered by FRER rule number i+32 because of no sequence error. (See (f) [Sequence recovery algorithms.](#))

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

HW: Descriptor corresponding frame is rejected.

30.3.3.20 FWEIE62 : Error Interrupt Enable Register 62

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7A74

Bit position: 31 0

Bit field: FFE31 to FFE0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	FFE31 to FFE0	FRER n Filtering Enable (n = 32 to 63)	R/W ^{*1}

Note 1. Read value differs from written value.

FFEn bits (FRER n Filtering Enable (n = 32 to 63))

- Bit i set to 0: Interrupt i disabled.
- Bit i set to 1: Interrupt i Enabled.

[Setting condition]

Writing 1 to one of these bits will set it.

[Clearing condition]

Writing 1 to FWEID62.FFDi register bit i will clear this register bit i.

30.3.3.21 FWEID62 : Error Interrupt Disable Register 62

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7A78

Bit position: 31

0

Bit field:

FFD31 to FFD0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	FFD31 to FFD0	FRER n Filtering Disable (n = 32 to 63) Writing 1 to this register bit i will clear FWEIE62.FFEi register bit i.	R/W

30.3.3.22 FWEIS63 : Error Interrupt Status Register 63

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7A80

Bit position: 31

0

Bit field:

FFS31 to FFS0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	FFS31 to FFS0	FRER n Filtering Status Flag (n = 0 to 31)	R/W ^{*1}

Note 1. Read value differs from written value.

FFSn bits (FRER n Filtering Status Flag (n = 0 to 31))

[Setting condition]

- HW: If FRER rule i is used for individual recovery, bit i is set when a descriptor is filtered by FRER rule number i because of no sequence error or struck error. (See [\(e\) Individual recovery algorithms.](#))
- HW: If FRER rule i is used for sequence recovery, bit i is set when a descriptor is filtered by FRER rule number i because of no sequence error. (See [\(f\) Sequence recovery algorithms.](#))

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

Error recovery:

HW: Descriptor corresponding frame is rejected.

30.3.3.23 FWEIE63 : Error Interrupt Enable Register 63

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7A84

Bit position: 31 0

Bit field: FFE31 to FFE0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	FFE31 to FFE0	FRER n Filtering Enable (n = 0 to 31)	R/W ^{*1}

Note 1. Read value differs from written value.

FFEn bits (FRER n Filtering Enable (n = 0 to 31))

- Bit i set to 0: Interrupt i disabled.
- Bit i set to 1: Interrupt i Enabled.

[Setting condition]

Writing 1 to one of these bits will set it.

[Clearing condition]

Writing 1 to FWEID63.FFDi register bit i will clear this register bit i.

30.3.3.24 FWEID63 : Error Interrupt Disable Register 63

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7A88

Bit position: 31 0

Bit field: FFD31 to FFD0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	FFD31 to FFD0	FRER n Filtering Disable (n = 0 to 31) Writing 1 to this register bit i will clear FWEIE63.FFEi register bit i.	R/W

30.3.3.25 FWEIS70 : Error Interrupt Status Register 70

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7A90

Bit position: 31 0

Bit field: FOORS31 to FOORS0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	FOORS31 to FOORS0	FRER n Out Of Range Status Flag (n = 96 to 127)	R/W ^{*1}

Note 1. Read value differs from written value.

FOORSn bits (FRER n Out Of Range Status Flag (n = 96 to 127))

[Setting condition]

- HW: If FRER rule i+96 is used with match algorithm, bit i is set when a descriptor an out of order error occurs. (See (e) [Individual recovery algorithms](#) and (f) [Sequence recovery algorithms](#)).
- HW: : If FRER rule i+96 is used with vector algorithm, bit i is set when a descriptor is filtered by FRER rule number i+96 because of an out of range error. (See (e) [Individual recovery algorithms](#) and (f) [Sequence recovery algorithms](#)).

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

HW: Descriptor corresponding frame is rejected.

30.3.3.26 FWEIE70 : Error Interrupt Enable Register 70

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7A94

Bit position: 31

0

Bit field:

FOORE31 to FOORE0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	FOORE31 to FOORE0	FRER n Out Of Range Enable (n = 96 to 127)	R/W ¹

Note 1. Read value differs from written value.

FOOREn bits (FRER n Out Of Range Enable (n = 96 to 127))

- Bit i set to 0: Interrupt i disabled.
- Bit i set to 1: Interrupt i Enabled.

[Setting condition]

Writing 1 to one of these bits will set it.

[Clearing condition]

Writing 1 to FWEID70.FOORDi register bit i will clear this register bit i.

30.3.3.27 FWEID70 : Error Interrupt Disable Register 70

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7A98

Bit position: 31

0

Bit field:

FOORD31 to FOORD0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	FOORD31 to FOORD0	FRER n Out Of Range Disable (n = 96 to 127) Writing 1 to this register bit i will clear FWEIE70.FOOREi register bit i.	R/W

30.3.3.28 FWEIS71 : Error Interrupt Status Register 71

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7AA0

Bit position: 31

0

Bit field:

FOORS31 to FOORS0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	FOORS31 to FOORS0	FRER n Out Of Range Status Flag (n = 64 to 95)	R/W ^{*1}

Note 1. Read value differs from written value.

FOORSn bits (FRER n Out Of Range Status Flag (n = 64 to 95))

[Setting condition]

- HW: If FRER rule i+64 is used with match algorithm, bit i is set when a descriptor an out of order error occurs. (See [\(e\) Individual recovery algorithms](#) and [\(f\) Sequence recovery algorithms](#)).
- HW: : If FRER rule i+64 is used with vector algorithm, bit i is set when a descriptor is filtered by FRER rule number i+64 because of an out of range error. (See [\(e\) Individual recovery algorithms](#) and [\(f\) Sequence recovery algorithms](#)).

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

HW: Descriptor corresponding frame is rejected.

30.3.3.29 FWEIE71 : Error Interrupt Enable Register 71

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7AA4

Bit position: 31

0

Bit field:

FOORE31 to FOORE0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	FOORE31 to FOORE0	FRER n Out Of Range Enable (n = 64 to 95)	R/W ^{*1}

Note 1. Read value differs from written value.

FOOREn bits (FRER n Out Of Range Enable (n = 64 to 95))

- Bit i set to 0: Interrupt i disabled.
- Bit i set to 1: Interrupt i Enabled.

[Setting condition]

Writing 1 to one of these bits will set it.

[Clearing condition]

Writing 1 to FWEID71.FOORDi register bit i will clear this register bit i.

30.3.3.30 FWEID71 : Error Interrupt Disable Register 71

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7AA8

Bit position: 31 0

Bit field: FOORD31 to FOORD0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	FOORD31 to FOORD0	FRER n Out Of Range Disable (n = 64 to 95) Writing 1 to this register bit i will clear FWEIE71.FOOREi register bit i.	R/W

30.3.3.31 FWEIS72 : Error Interrupt Status Register 72

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7AB0

Bit position: 31 0

Bit field: FOORS31 to FOORS0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	FOORS31 to FOORS0	FRER n Out Of Range Status Flag (n = 32 to 63)	R/W ¹

Note 1. Read value differs from written value.

FOORSn bits (FRER n Out Of Range Status Flag (n = 32 to 63))

[Setting condition]

- HW: If FRER rule i+32 is used with match algorithm, bit i is set when a descriptor an out of order error occurs. (See [\(e\) Individual recovery algorithms](#) and [\(f\) Sequence recovery algorithms](#)).
- HW: If FRER rule i+32 is used with vector algorithm, bit i is set when a descriptor is filtered by FRER rule number i+32 because of an out of range error. (See [\(e\) Individual recovery algorithms](#) and [\(f\) Sequence recovery algorithms](#)).

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

HW: Descriptor corresponding frame is rejected.

30.3.3.32 FWEIE72 : Error Interrupt Enable Register 72

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7AB4

Bit position: 31 0

Bit field: FOORE31 to FOORE0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	FOORE31 to FOORE0	FRER n Out Of Range Enable (n = 32 to 63)	R/W ^{*1}

Note 1. Read value differs from written value.

FOOREn bits (FRER n Out Of Range Enable (n = 32 to 63))

- Bit i set to 0: Interrupt i disabled.
- Bit i set to 1: Interrupt i Enabled.

[Setting condition]

Writing 1 to one of these bits will set it.

[Clearing condition]

Writing 1 to FWEID72.FOORDi register bit i will clear this register bit i.

30.3.3.33 FWEID72 : Error Interrupt Disable Register 72

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7AB8

Bit position: 31 0

Bit field: FOORD31 to FOORD0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	FOORD31 to FOORD0	FRER n Out Of Range Disable (n = 32 to 63) Writing 1 to this register bit i will clear FWEIE72.FOOREi register bit i.	R/W

30.3.3.34 FWEIS73 : Error Interrupt Status Register 73

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7AC0

Bit position: 31 0

Bit field: FOORS31 to FOORS0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	FOORS31 to FOORS0	FRER n Out Of Range Status Flag (n = 0 to 31)	R/W ^{*1}

Note 1. Read value differs from written value.

FOORSn bits (FRER n Out Of Range Status Flag (n = 0 to 31))

[Setting condition]

- HW: If FRER rule i is used with match algorithm, bit i is set when a descriptor an out of order error occurs. (See [\(e\) Individual recovery algorithms](#) and [\(f\) Sequence recovery algorithms](#)).
- HW: If FRER rule i is used with vector algorithm, bit i is set when a descriptor is filtered by FRER rule number i because of an out of range error. (See [\(e\) Individual recovery algorithms](#) and [\(f\) Sequence recovery algorithms](#)).

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

HW: Descriptor corresponding frame is rejected.

30.3.3.35 FWEIE73 : Error Interrupt Enable Register 73

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7AC4

Bit position: 31

0

Bit field:

FOORE31 to FOORE0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	FOORE31 to FOORE0	FRER n Out Of Range Enable (n = 0 to 31)	R/W ^{*1}

Note 1. Read value differs from written value.

FOOREn bits (FRER n Out Of Range Enable (n = 0 to 31))

- Bit i set to 0: Interrupt i disabled.
- Bit i set to 1: Interrupt i Enabled.

[Setting condition]

Writing 1 to one of these bits will set it.

[Clearing condition]

Writing 1 to FWEID73.FOORDi register bit i will clear this register bit i.

30.3.3.36 FWEID73 : Forwarding Engine Error Interrupt Disable 73

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7AC8

Bit position: 31

0

Bit field:

FOORD31 to FOORD0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	FOORD31 to FOORD0	FRER n Out Of Range Disable (n = 0 to 31) Writing 1 to this register bit i will clear FWEIE73.FOOREi register bit i.	R/W

30.3.3.37 FWEIS80 : Error Interrupt Status Register 80

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7AD0

Bit position: 31

0

Bit field:

TOS31 to TOS0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	TOS31 to TOS0	Timeout Status Flag n (n = 96 to 127)	R/W ^{*1}

Note 1. Read value differs from written value.

TOSn bits (Timeout Status Flag n (n = 96 to 127))

[Setting condition]

HW: When timeout happens for FRER rule i+96. (See (3) Timeout.)

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

HW: Descriptor corresponding frame is rejected.

30.3.3.38 FWEIE80 : Error Interrupt Enable Register 80

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7AD4

Bit position: 31

0

Bit field:

TOE31 to TOE0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	TOE31 to TOE0	Timeout Enable n (n = 96 to 127)	R/W ^{*1}

Note 1. Read value differs from written value.

TOEn bits (Timeout Enable n (n = 96 to 127))

- Bit i set to 0: Interrupt i disabled.
- Bit i set to 1: Interrupt i Enabled.

[Setting condition]

Writing 1 to one of these bits will set it.

[Clearing condition]

Writing 1 to FWEID80.TODi register bit i will clear this register bit i.

30.3.3.39 FWEID80 : Error Interrupt Disable Register 80

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7AD8

Bit position: 31

0

Bit field:

TOD31 to TOD0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	TOD31 to TOD0	Timeout Disable n (n = 96 to 127) Writing 1 to this register bit i will clear FWEIE80.TOEi register bit i.	R/W

30.3.3.40 FWEIS81 : Error Interrupt Status Register 81

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7AE0

Bit position: 31

0

Bit field:

TOS31 to TOS0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	TOS31 to TOS0	Timeout Status Flag n (n = 64 to 95)	R/W ^{*1}

Note 1. Read value differs from written value.

TOSn bits (Timeout Status Flag n (n = 64 to 95))

[Setting condition]

HW: When timeout happens for FRER rule i+64. (See (3) Timeout.)

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

HW: Descriptor corresponding frame is rejected.

30.3.3.41 FWEIE81 : Error Interrupt Enable Register 81

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7AE4

Bit position: 31

0

Bit field:

TOE31 to TOE0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	TOE31 to TOE0	Timeout Enable n (n = 64 to 95)	R/W ^{*1}

Note 1. Read value differs from written value.

TOEn bits (Timeout Enable n (n = 64 to 95))

- Bit i set to 0: Interrupt i disabled.
- Bit i set to 1: Interrupt i Enabled.

[Setting condition]

Writing 1 to one of these bits will set it.

[Clearing condition]

Writing 1 to FWEID81.TODi register bit i will clear this register bit i.

30.3.3.42 FWEID81 : Error Interrupt Disable Register 81

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7AE8

Bit position: 31 0

Bit field: TOD31 to TOD0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	TOD31 to TOD0	Timeout Disable n (n = 64 to 95) Writing 1 to this register bit i will clear FWEIE81.TOEi register bit i.	R/W

30.3.3.43 FWEIS82 : Error Interrupt Status Register 82

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7AF0

Bit position: 31 0

Bit field: TOS31 to TOS0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	TOS31 to TOS0	Timeout Status Flag n (n = 32 to 63)	R/W ¹

Note 1. Read value differs from written value.

TOSn bits (Timeout Status Flag n (n = 32 to 63))

[Setting condition]

HW: When timeout happens for FRER rule i+32. (See (3) Timeout.)

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

HW: Descriptor corresponding frame is rejected.

30.3.3.44 FWEIE82 : Error Interrupt Enable Register 82

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7AF4

Bit position: 31 0

Bit field: TOE31 to TOE0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	TOE31 to TOE0	Timeout Enable n (n = 32 to 63)	R/W ¹

Note 1. Read value differs from written value.

TOEn bits (Timeout Enable n (n = 32 to 63))

- Bit i set to 0: Interrupt i disabled.
- Bit i set to 1: Interrupt i Enabled.

[Setting condition]

Writing 1 to one of these bits will set it.

[Clearing condition]

Writing 1 to FWEID82.TODi register bit i will clear this register bit i.

30.3.3.45 FWEID82 : Error Interrupt Disable Register 82

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7AF8

Bit position: 31

0

Bit field:

TOD31 to TOD0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	TOD31 to TOD0	Timeout Disable n (n = 32 to 63) Writing 1 to this register bit i will clear FWEIE82.TOEi register bit i.	R/W

30.3.3.46 FWEIS83 : Error Interrupt Status Register 83

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7B00

Bit position: 31

0

Bit field:

TOS31 to TOS0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	TOS31 to TOS0	Timeout Status Flag n (n = 0 to 31)	R/W ¹

Note 1. Read value differs from written value.

TOSn bits (Timeout Status Flag n (n = 0 to 31))

[Setting condition]

HW: When timeout happens for FRER rule i. (See [\(3\) Timeout.](#))

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

[Error recovery]

HW: Descriptor corresponding frame is rejected.

30.3.3.47 FWEIE83 : Error Interrupt Enable Register 83

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7B04

Bit position: 31

0

Bit field:

TOE31 to TOE0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	TOE31 to TOE0	Timeout Enable n (n = 0 to 31)	R/W ^{*1}

Note 1. Read value differs from written value.

TOEn bits (Timeout Enable n (n = 0 to 31))

- Bit i set to 0: Interrupt i disabled.
- Bit i set to 1: Interrupt i Enabled.

[Setting condition]

Writing 1 to one of these bits will set it.

[Clearing condition]

Writing 1 to FWEID83.TODi register bit i will clear this register bit i.

30.3.3.48 FWEID83 : Error Interrupt Disable Register 83

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7B08

Bit position: 31

0

Bit field:

TOD31 to TOD0

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	TOD31 to TOD0	Timeout Disable n (n = 0 to 31) Writing 1 to this register bit i will clear FWEIE83.TOEi register bit i.	R/W

30.3.3.49 FWMIS0 : Monitoring Interrupt Status Register 0

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7C00

Bit position:

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16

Bit field:

—	—	—	—	—	—	—	—	—	—	—	—	—	—	MACA DAS	—
---	---	---	---	---	---	---	---	---	---	---	---	---	---	-------------	---

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit position:

15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field:

—	—	—	—	—	—	—	—	—	—	—	—	VLAN TFS	MACT FS	—	LTHTF S
---	---	---	---	---	---	---	---	---	---	---	---	-------------	------------	---	------------

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	LTHTFS	L3 Table Full Status Flag	R/W ^{*1}
1	—	This bit is read as 0. The write value should be 0.	R/W
2	MACTFS	MAC Table Full Status Flag	R/W ^{*1}
3	VLANTFS	VLAN Table Full Status Flag	R/W ^{*1}
16:4	—	These bits are read as 0. The write value should be 0.	R/W
17	MACADAS	MAC Address Deleted Aging Status Flag	R/W ^{*1}
31:18	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Read value differs from written value.

LTHTFS bit (L3 Table Full Status Flag)

[Setting condition]

HW: L3 table becomes full

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

MACTFS bit (MAC Table Full Status Flag)

[Setting condition]

HW: MAC table becomes full

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

VLANTFS bit (VLAN Table Full Status Flag)

[Setting condition]

HW: VLAN table becomes full

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

MACADAS bit (MAC Address Deleted Aging Status Flag)

[Setting condition]

HW: A MAC address has been suppressed by aging or dynamic entry delete

[Clearing condition]

SW: Writing 1 to one of these bits will clear it.

30.3.3.50 FWMIE0 : Monitoring Interrupt Enable Register 0

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7C04

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	MACA DAE	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	VLAN TFE	MACT FE	—	LTHTF E
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	LTHTFE	L3 Table Full Enable 0: Interrupt disabled. 1: Interrupt Enabled.	R/W ^{*1}
1	—	This bit is read as 0. The write value should be 0.	R/W
2	MACTFE	MAC Table Full Enable 0: Interrupt disabled. 1: Interrupt Enabled.	R/W ^{*1}
3	VLANTFE	VLAN Table Full Enable 0: Interrupt disabled. 1: Interrupt Enabled.	R/W ^{*1}
16:4	—	These bits are read as 0. The write value should be 0.	R/W
17	MACADAE	MAC Address Deleted Aging Enable 0: Interrupt disabled. 1: Interrupt Enabled.	R/W ^{*1}
31:18	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Read value differs from written value.

LTHTFE bit (L3 Table Full Enable)

[Setting condition]

Writing 1 to this bit will set it.

[Clearing condition]

Writing 1 to FWMID0.LTHTFD register will clear this bit.

MACTFE bit (MAC Table Full Enable)

[Setting condition]

Writing 1 to this bit will set it.

[Clearing condition]

Writing 1 to FWMID0.MACTFD register will clear this bit.

VLANTFE bit (VLAN Table Full Enable)

[Setting condition]

Writing 1 to this bit will set it.

[Clearing condition]

Writing 1 to FWMID0.VLANTFD register will clear this bit.

MACADAE bit (MAC Address Deleted Aging Enable)

[Setting condition]

Writing 1 to this bit will set it.

[Clearing condition]

Writing 1 to FWMID0.MACADAD register will clear this bit.

30.3.3.51 FWMID0 : Monitoring Interrupt Disable Register 0

Base address: MFWD = 0x403C_0000
MFWD_NS = 0x503C_0000

Offset address: 0x7C08

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	MACA DAD	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	VLAN TFD	MACT FD	—	LTHTF D
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	LTHTFD	L3 Table Full Disable Writing 1 to this bit will clear FWMIE0.LTHTFE.	R/W
1	—	This bit is read as 0. The write value should be 0.	R/W
2	MACTFD	MAC Table Full Disable Writing 1 to this bit will clear FWMIE0.MACTFE.	R/W
3	VLANTFD	VLAN Table Full Disable Writing 1 to this bit will clear FWMIE0.VLANTFE.	R/W
16:4	—	These bits are read as 0. The write value should be 0.	R/W
17	MACADAD	MAC Address Deleted Aging Disable Writing 1 to this bit will clear FWMIE0.MACADAD.	R/W
31:18	—	These bits are read as 0. The write value should be 0.	R/W

30.4 Register Utilization

30.4.1 Software Flows

- Note:
- SW: Forwarding engine registers can only be written using the flows specified in this section
 - SW: All the software flows described in this section can only be used sequentially.

30.4.1.1 Software Flow Legend

Figure 30.2 shows the software flow legend.

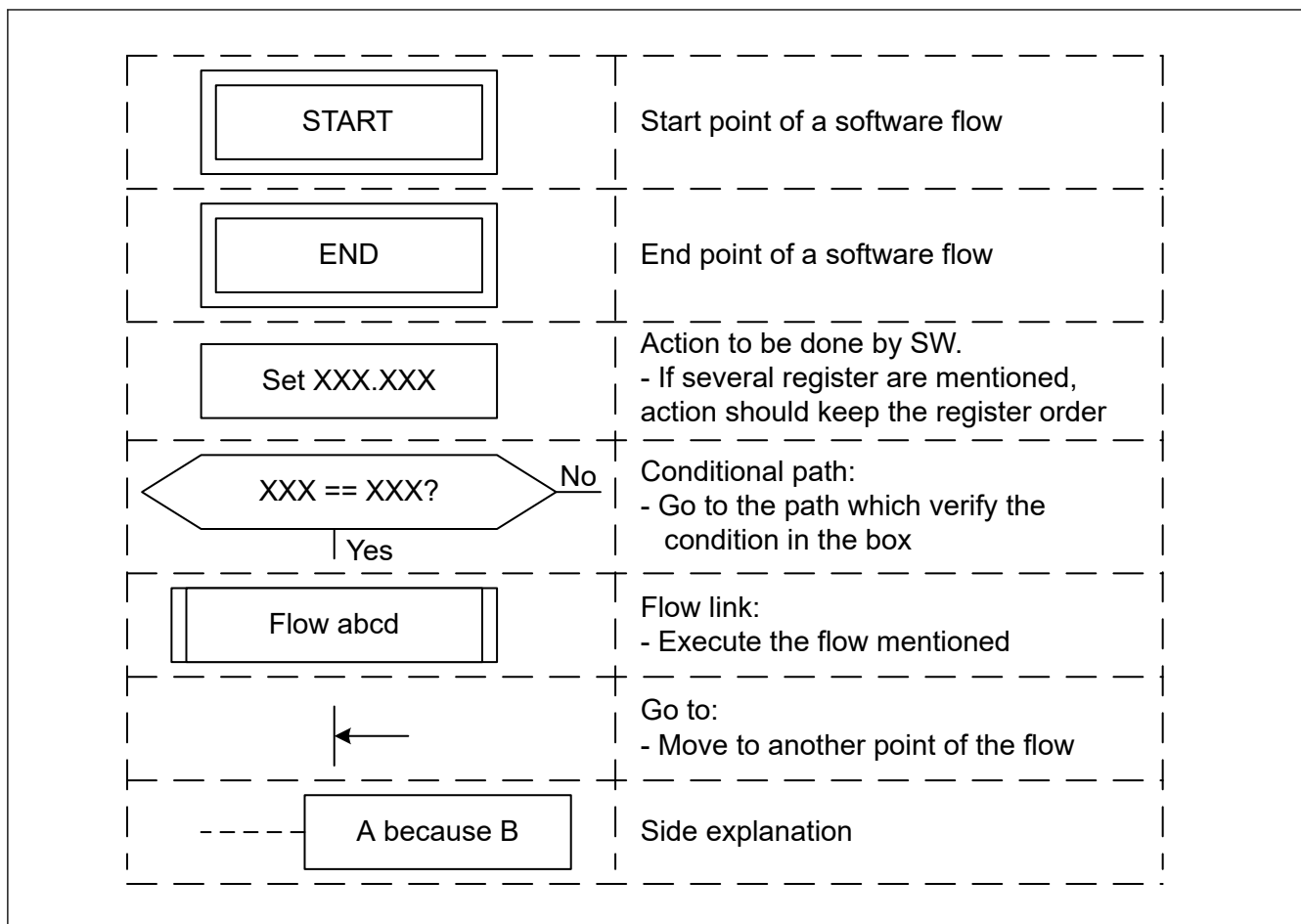


Figure 30.2 Software flow legend

30.4.1.2 Interrupt Handling Flow

Interrupt handling flow can be used in any mode except RESET mode.

Figure 30.3 describes the interrupt handling flow.

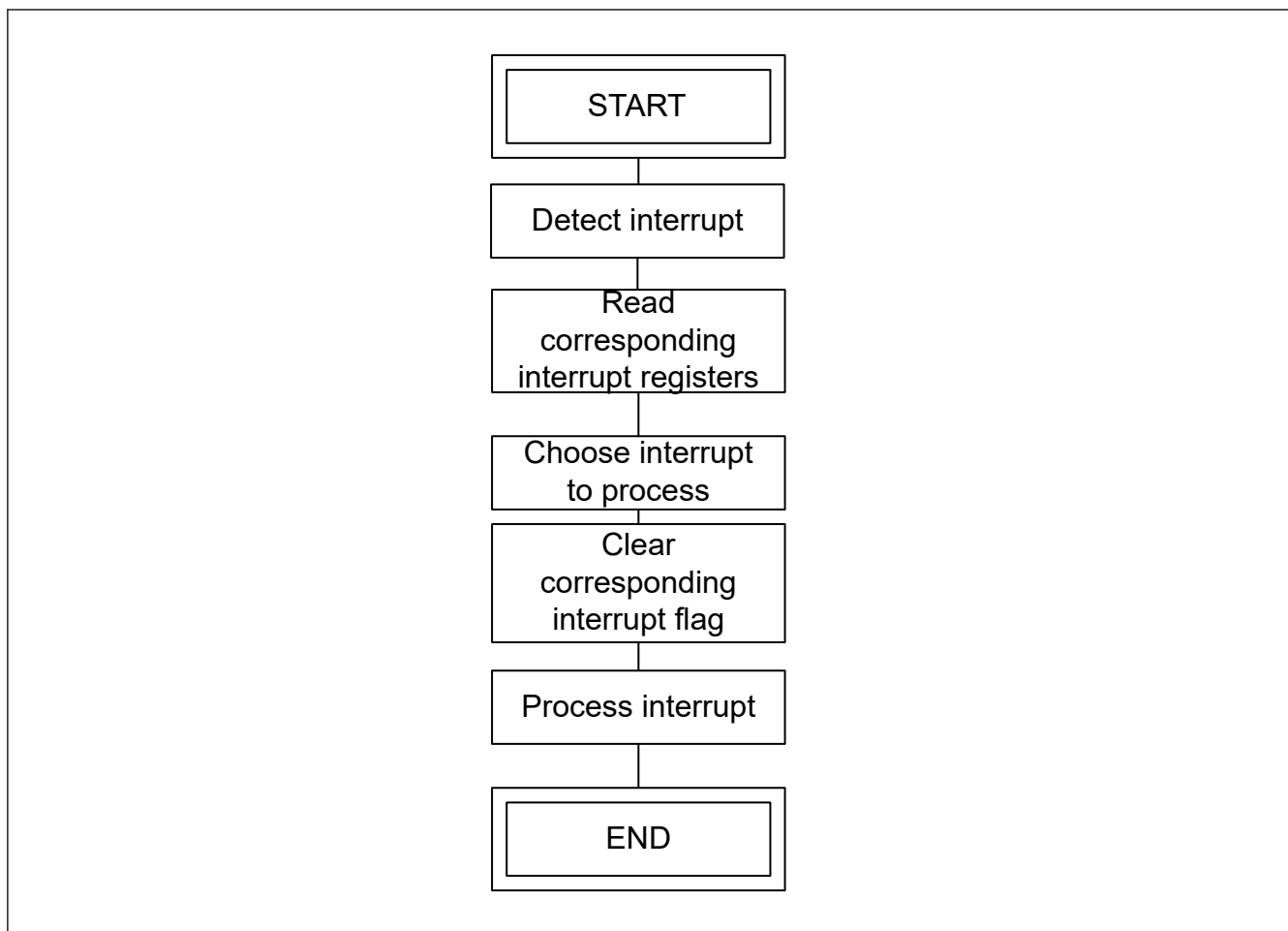


Figure 30.3 Interrupt handling flow

30.4.1.3 Port i Setting Flow

Port i setting flow is described in [Figure 30.4](#).

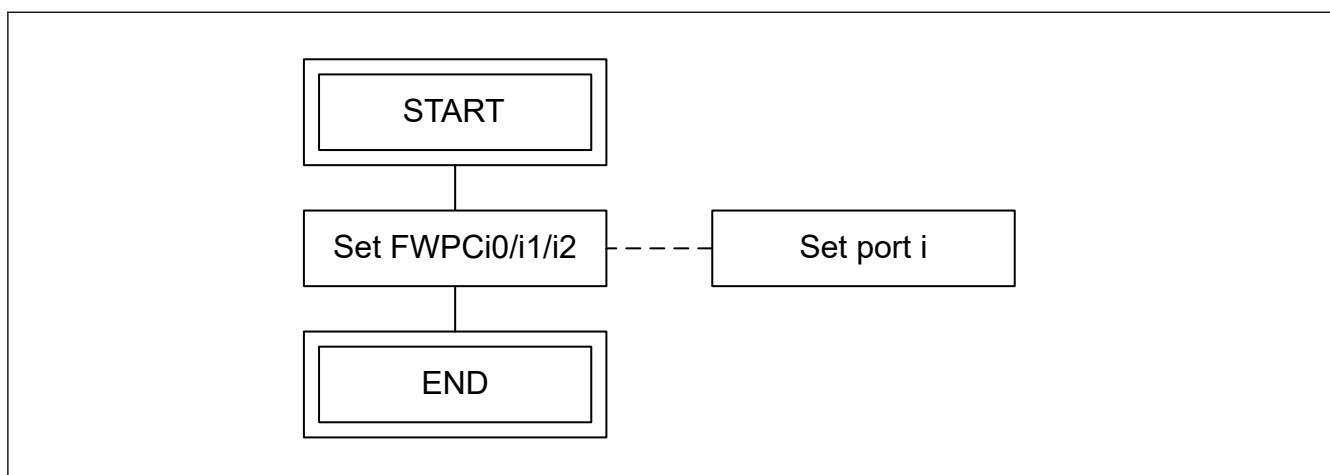


Figure 30.4 Port i setting flow

30.4.1.4 Cut-through Forwarding Flows

30.4.1.4.1 Cut-through Rule i Setting Flow

Cut-through rule i setting flow is described in [Figure 30.5](#).

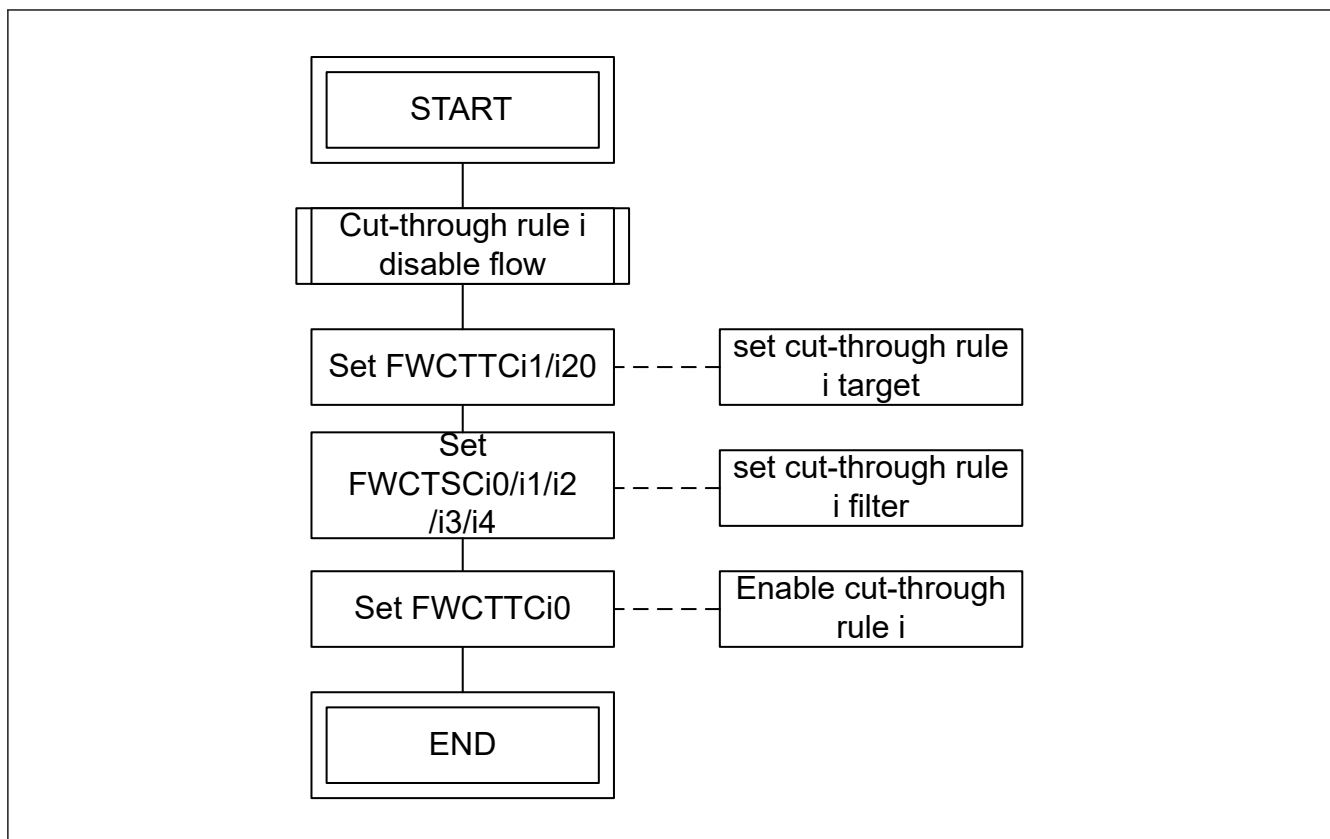


Figure 30.5 Cut-through rule i setting flow

30.4.1.4.2 Cut-through Rule i Disable Flow

Cut-through rule i disable flow is described in [Figure 30.6](#).

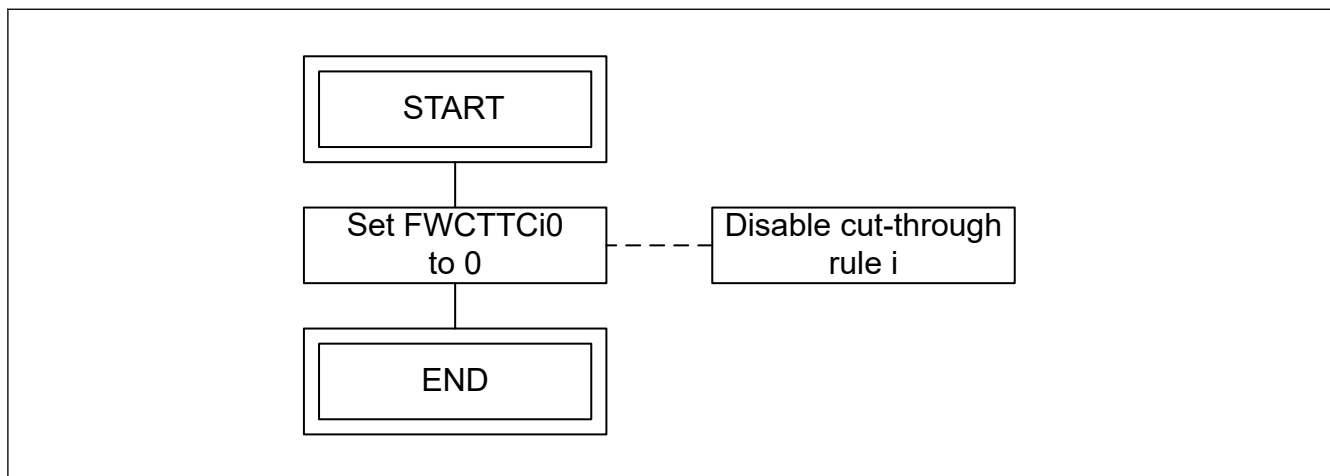


Figure 30.6 Cut-through rule i disable flow

30.4.1.5 Layer 3 Forwarding/Routing/Filtering Flows

30.4.1.5.1 Perfect Filter i Setting Flow

[Figure 30.7](#) describes the setting flow of perfect filter i.

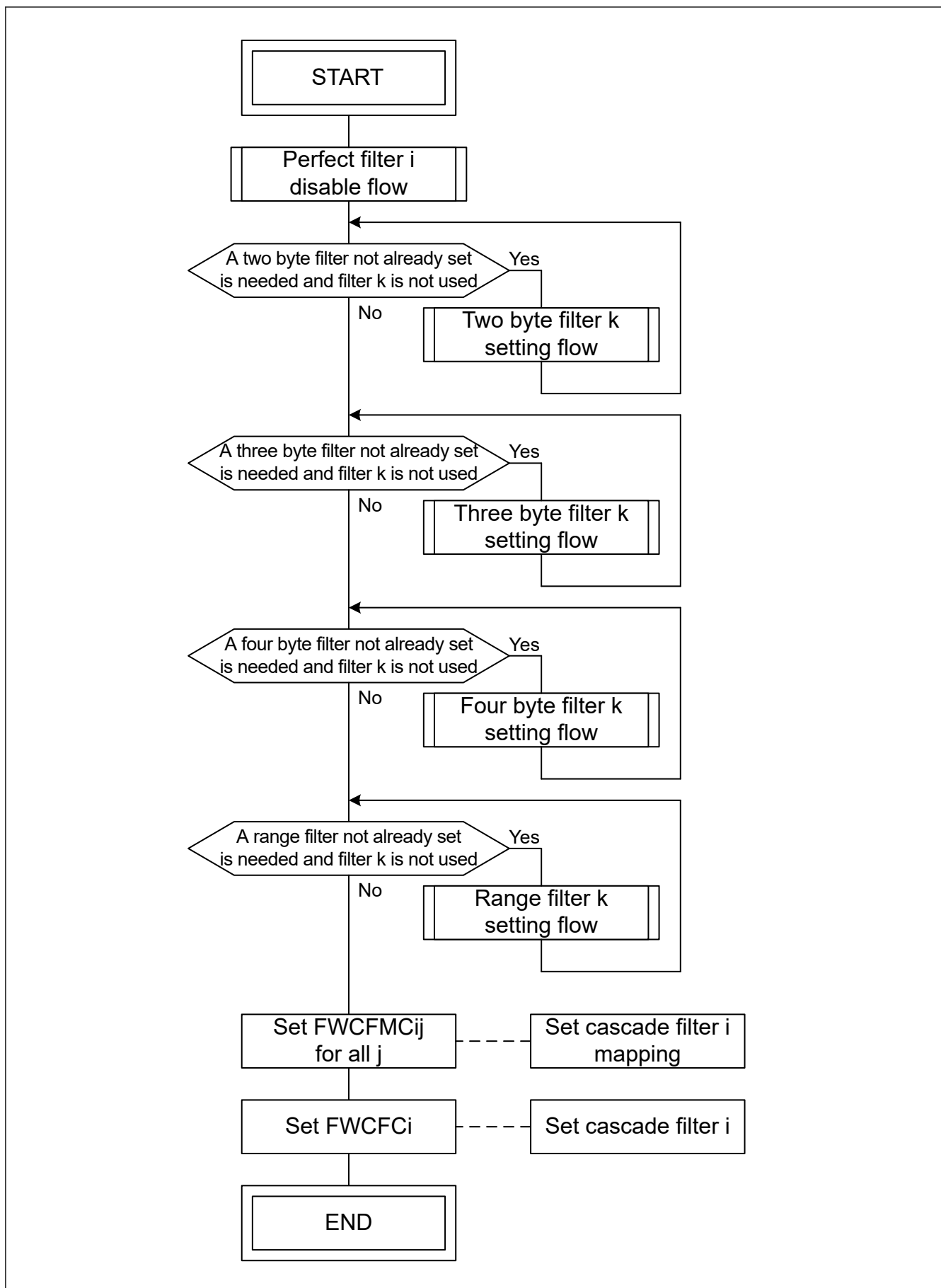


Figure 30.7 Setting flow of perfect filter i

30.4.1.5.2 Perfect Filter i Disable Flow

Figure 30.8 describes the disable flow of perfect filter i.

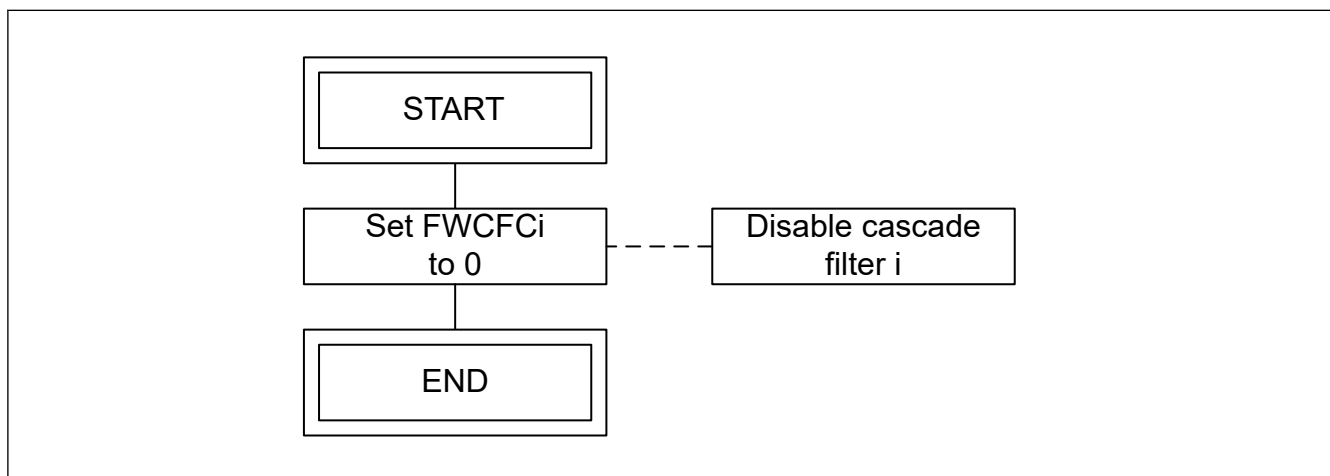


Figure 30.8 Disable flow of perfect filter i

30.4.1.5.3 Perfect Filter Priority Change Flow

Figure 30.9 describes the change flow of perfect filter priority.

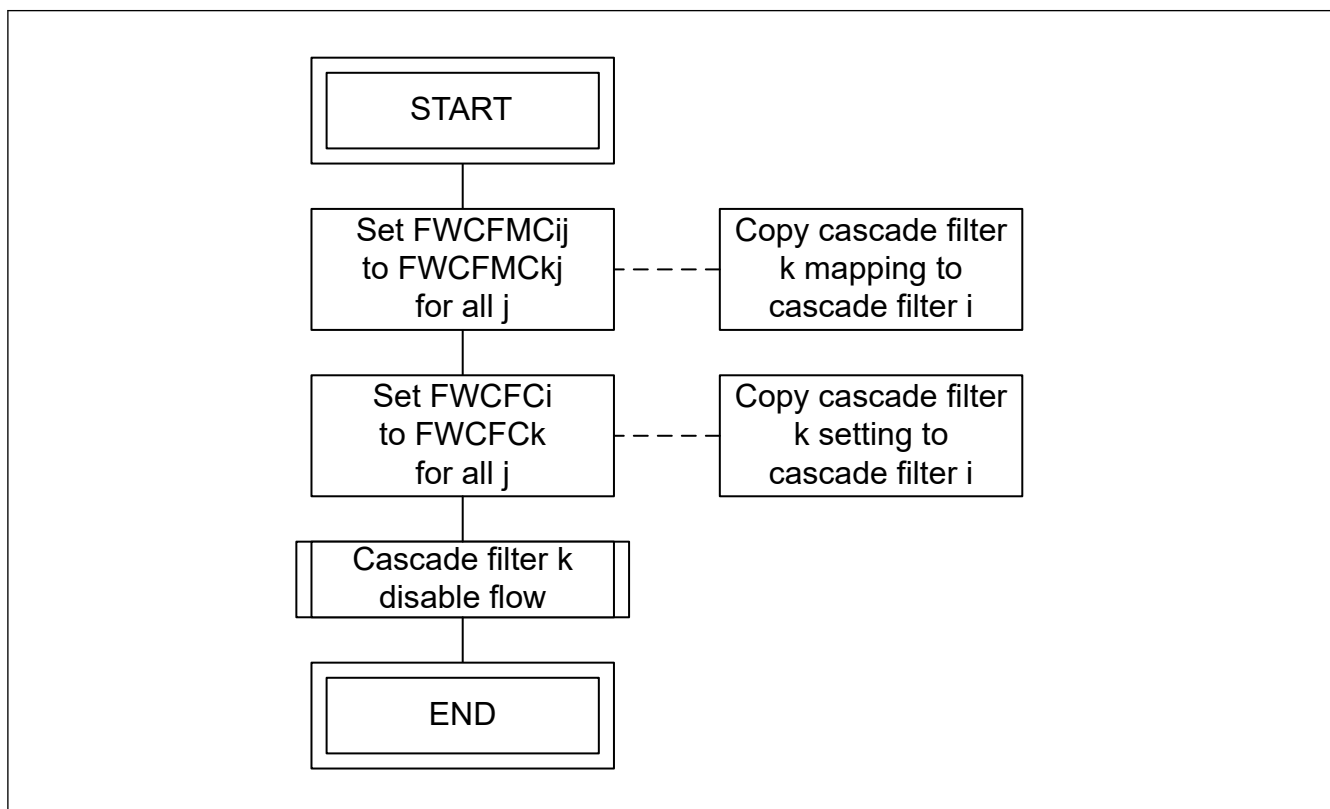


Figure 30.9 Change flow of perfect filter priority (from k to i)

30.4.1.5.4 Hash Hardware Calculation Flow (for IPv4/IPv6 Stream Creation)

Hash hardware calculation flow is described in Figure 30.10.

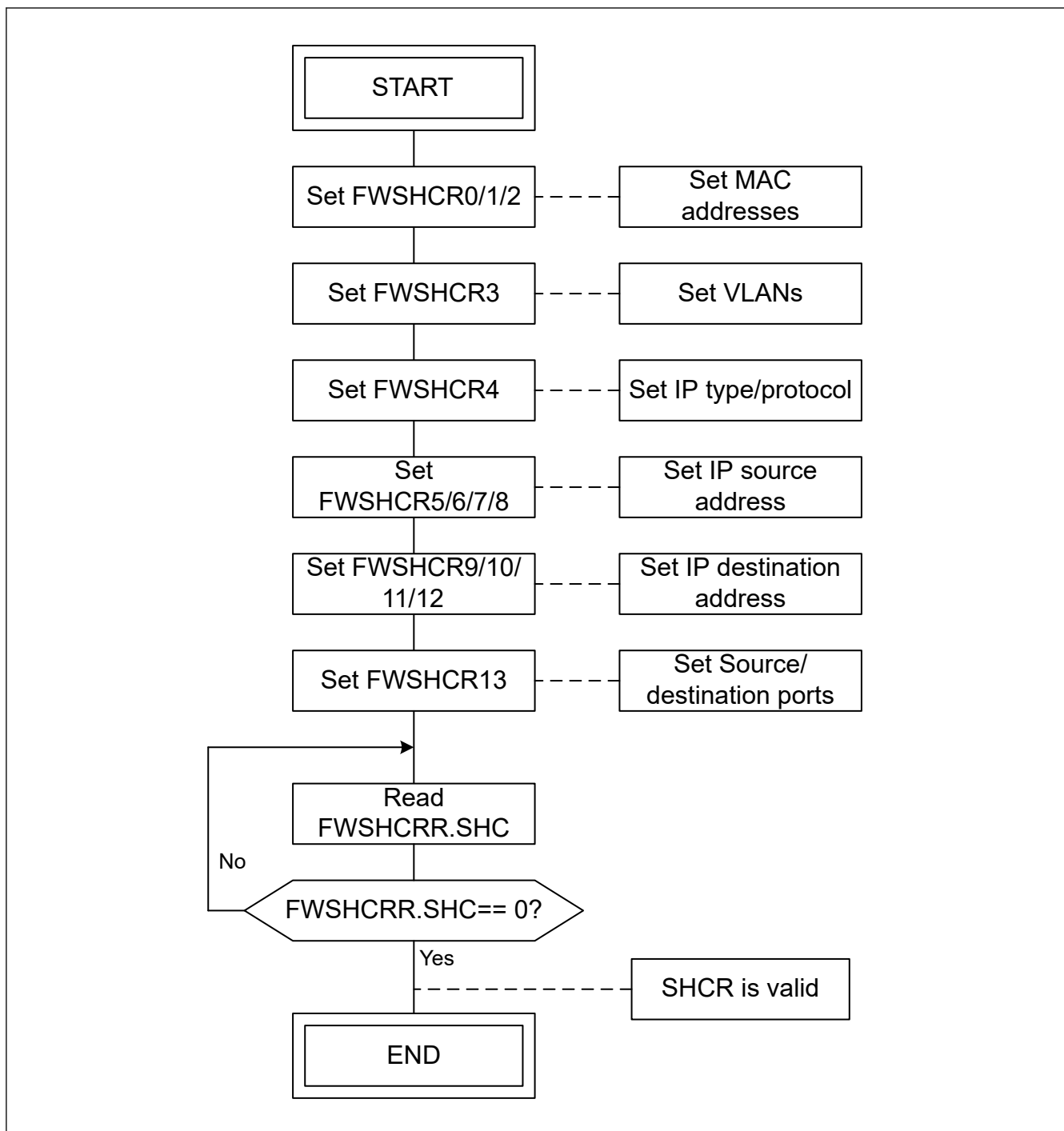


Figure 30.10 Hash hardware calculation flow

30.4.1.5.5 Layer 3 Table Reset Flow

Figure 30.11 describes the reset flow of Layer 3 table.

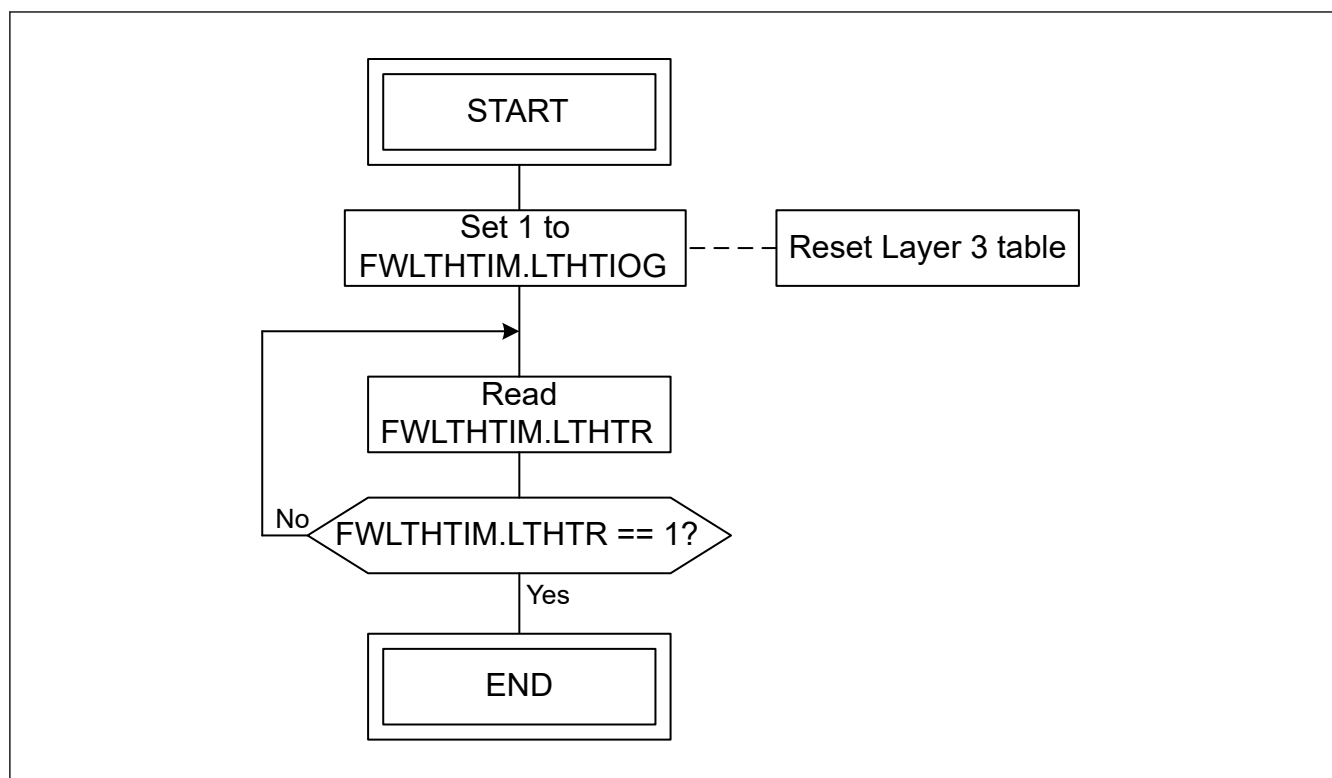


Figure 30.11 Reset flow of Layer 3 table

30.4.1.5.6 Layer 3 Entry Learn Flow

Layer 3 entry learn flow is described in [Figure 30.12](#).

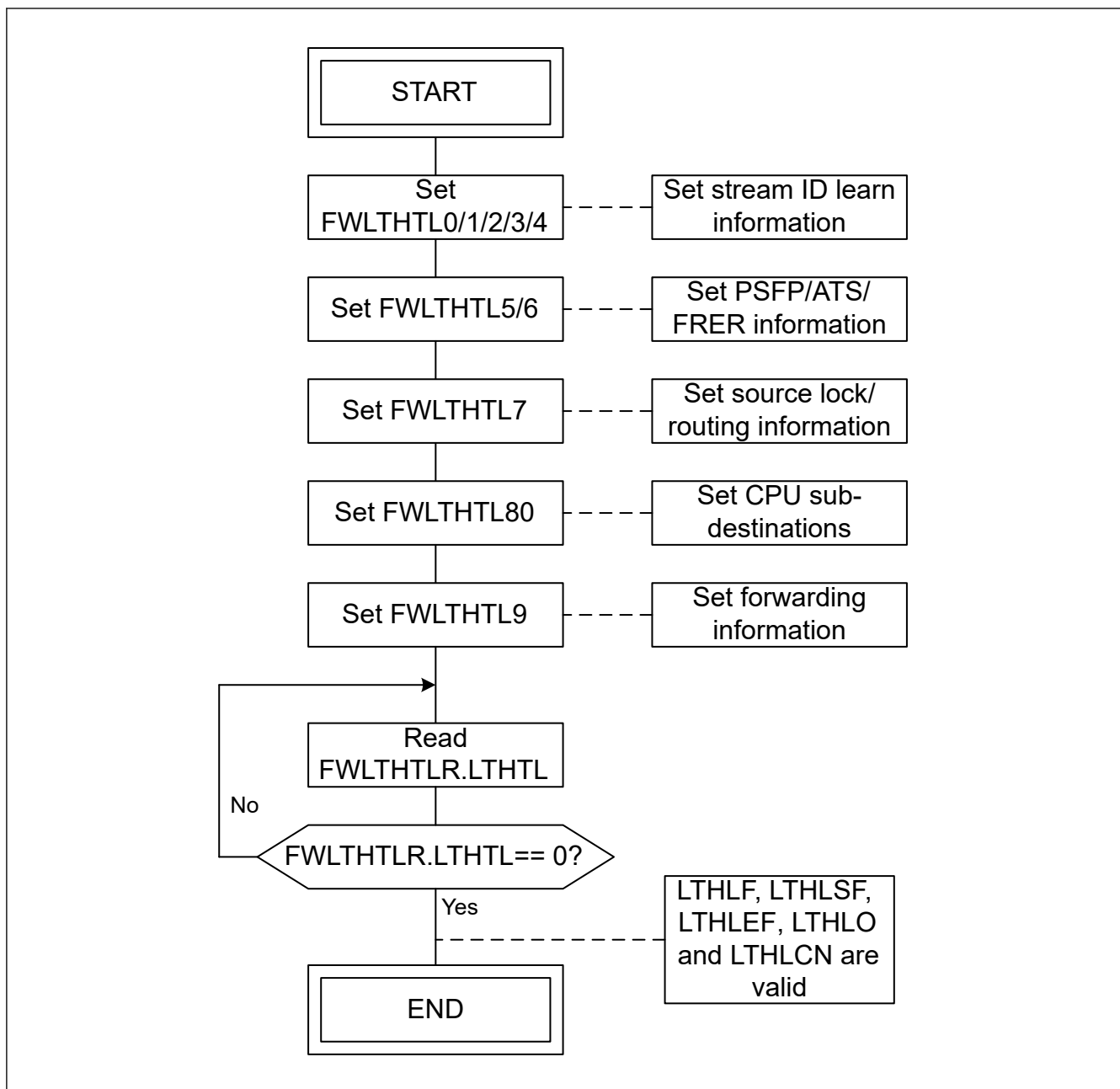


Figure 30.12 Layer 3 entry learn flow

30.4.1.5.7 Layer 3 Entry Search Flow

Layer 3 entry search flow is described in [Figure 30.13](#).

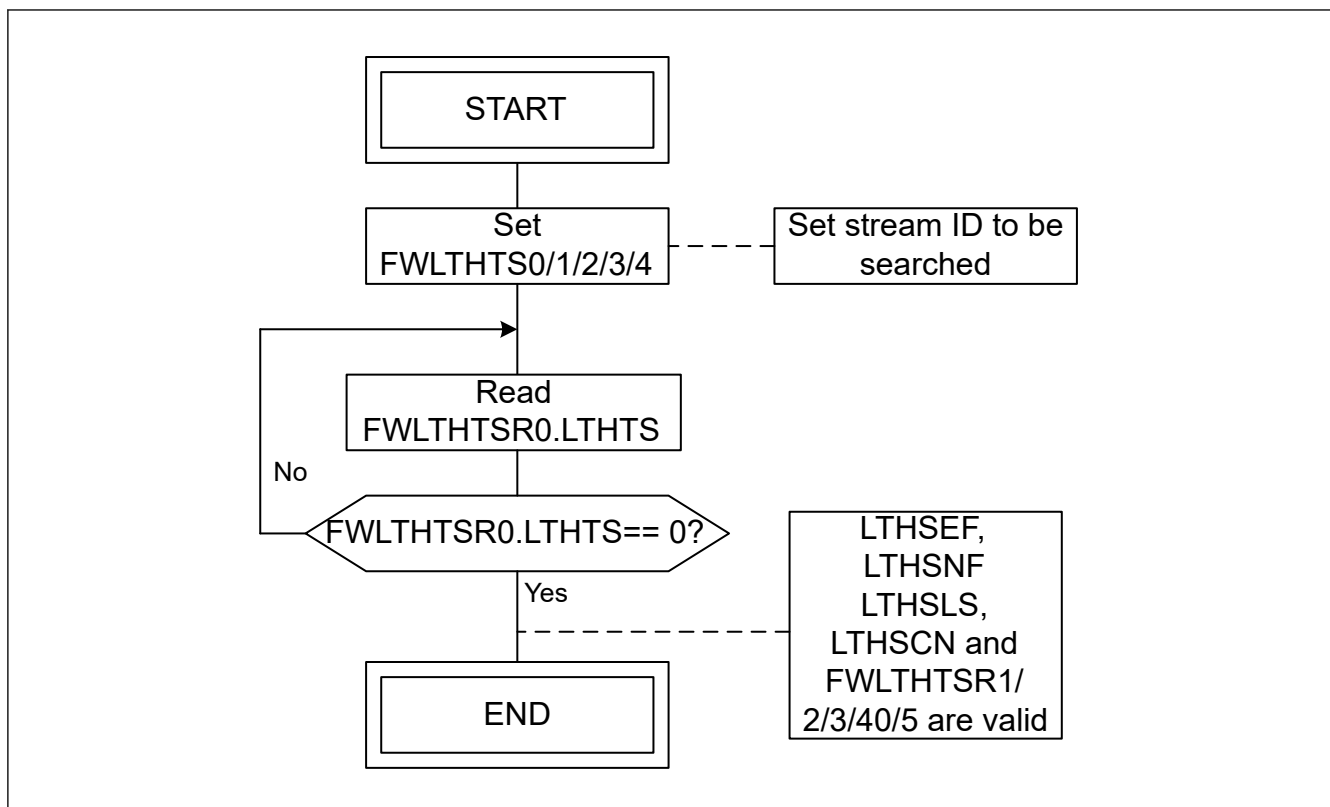


Figure 30.13 Layer 3 entry search flow

30.4.1.5.8 Layer 3 Entry Read Flow

Layer 3 entry read flow is described in [Figure 30.14](#).

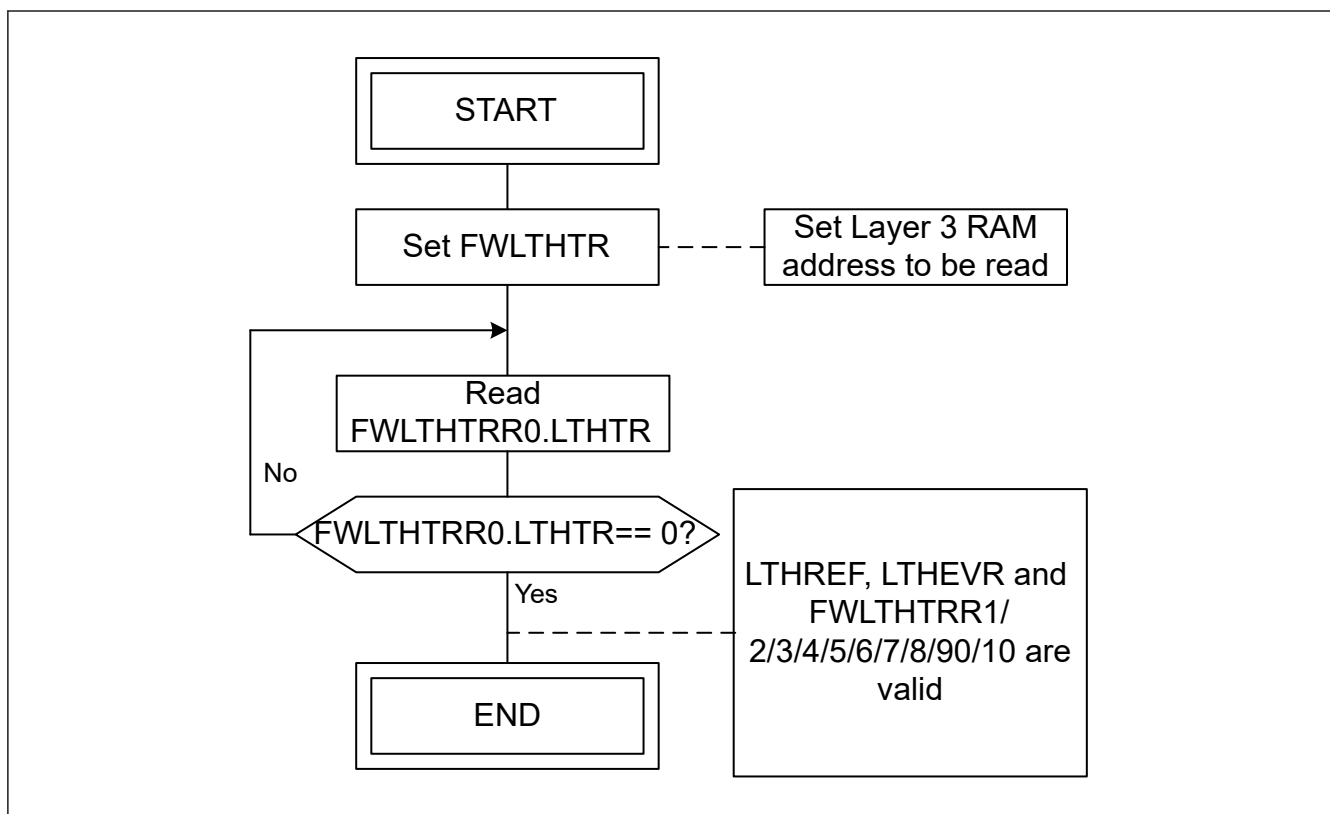


Figure 30.14 Layer 3 entry read flow

30.4.1.6 Layer 2 Forwarding Flows

30.4.1.6.1 MAC Table Reset Flow

Figure 30.15 describes the reset flow of MAC table.

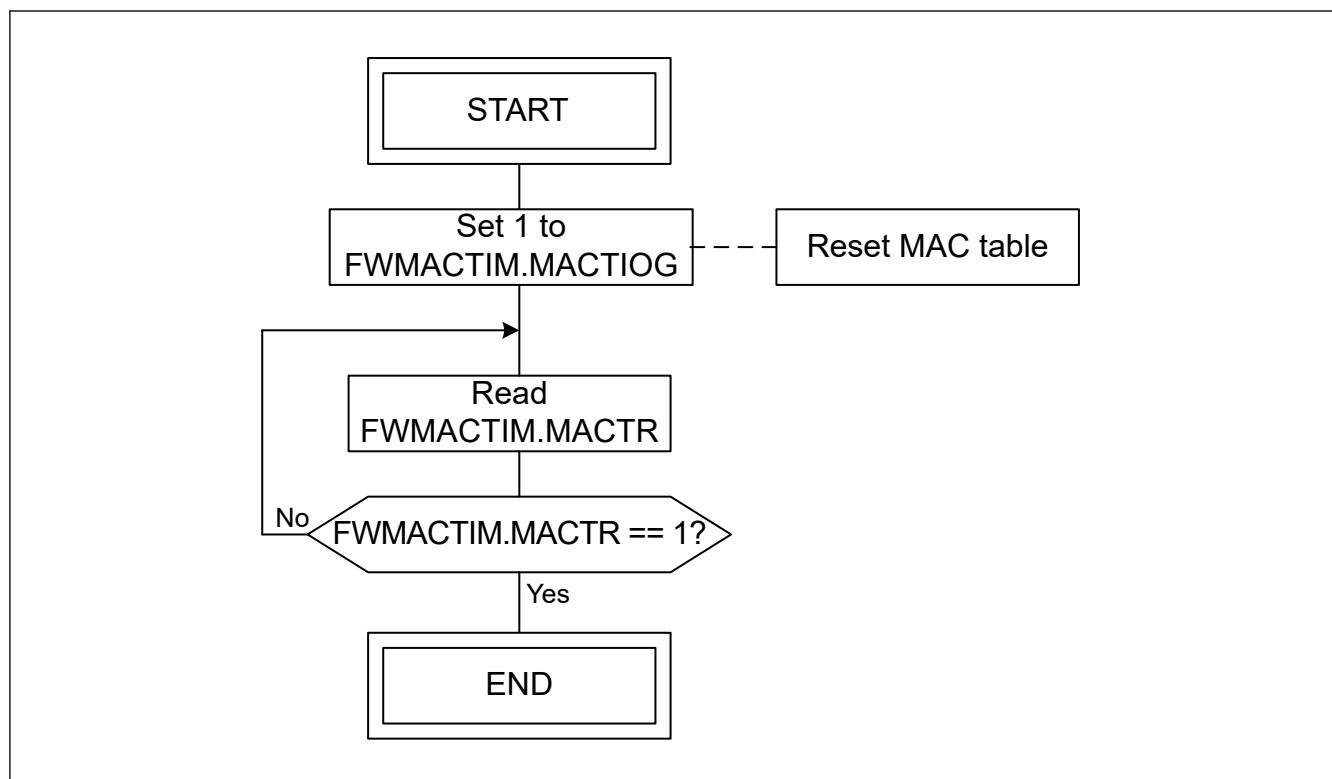


Figure 30.15 Reset flow of MAC table

30.4.1.6.2 VLAN Table Reset Flow

Figure 30.16 describes the reset flow of VLAN table.

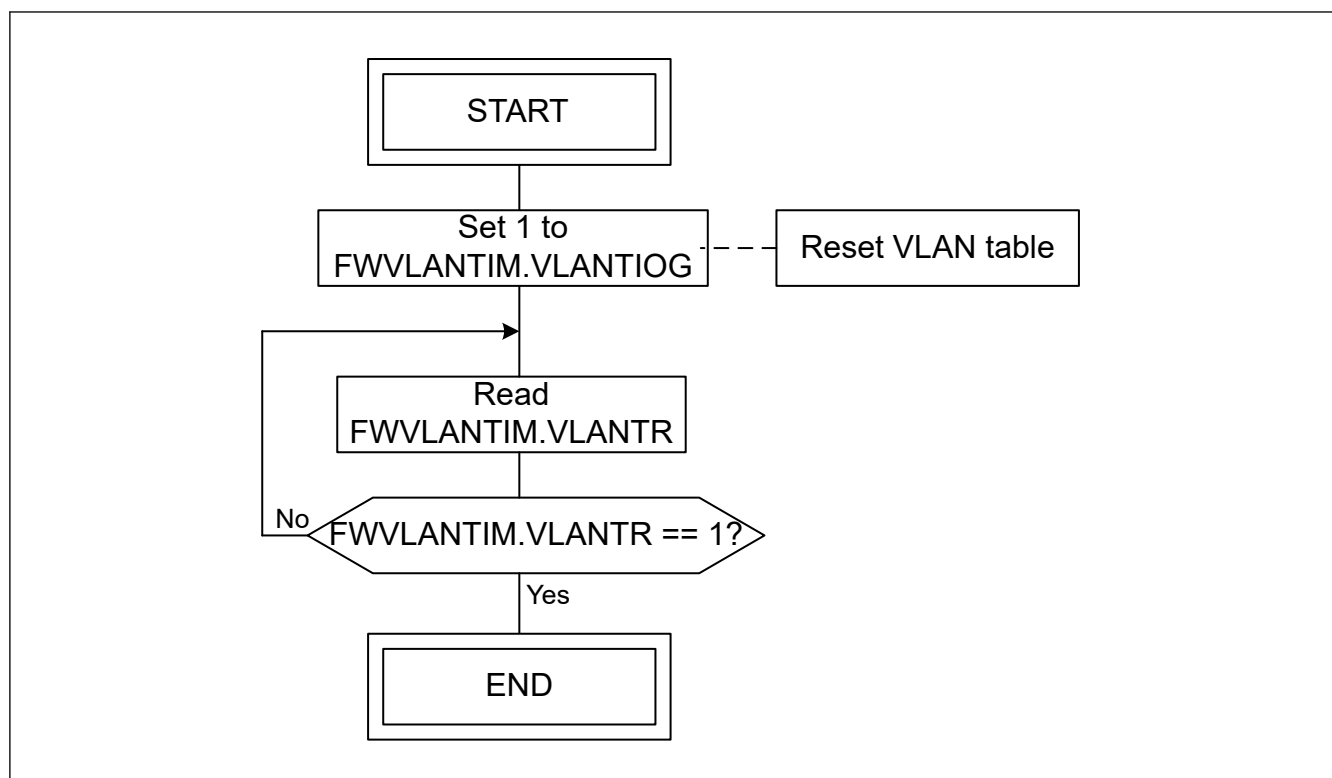


Figure 30.16 Reset flow of VLAN table

30.4.1.6.3 MAC Entry Learn Flow

MAC entry learn flow is described in [Figure 30.17](#).

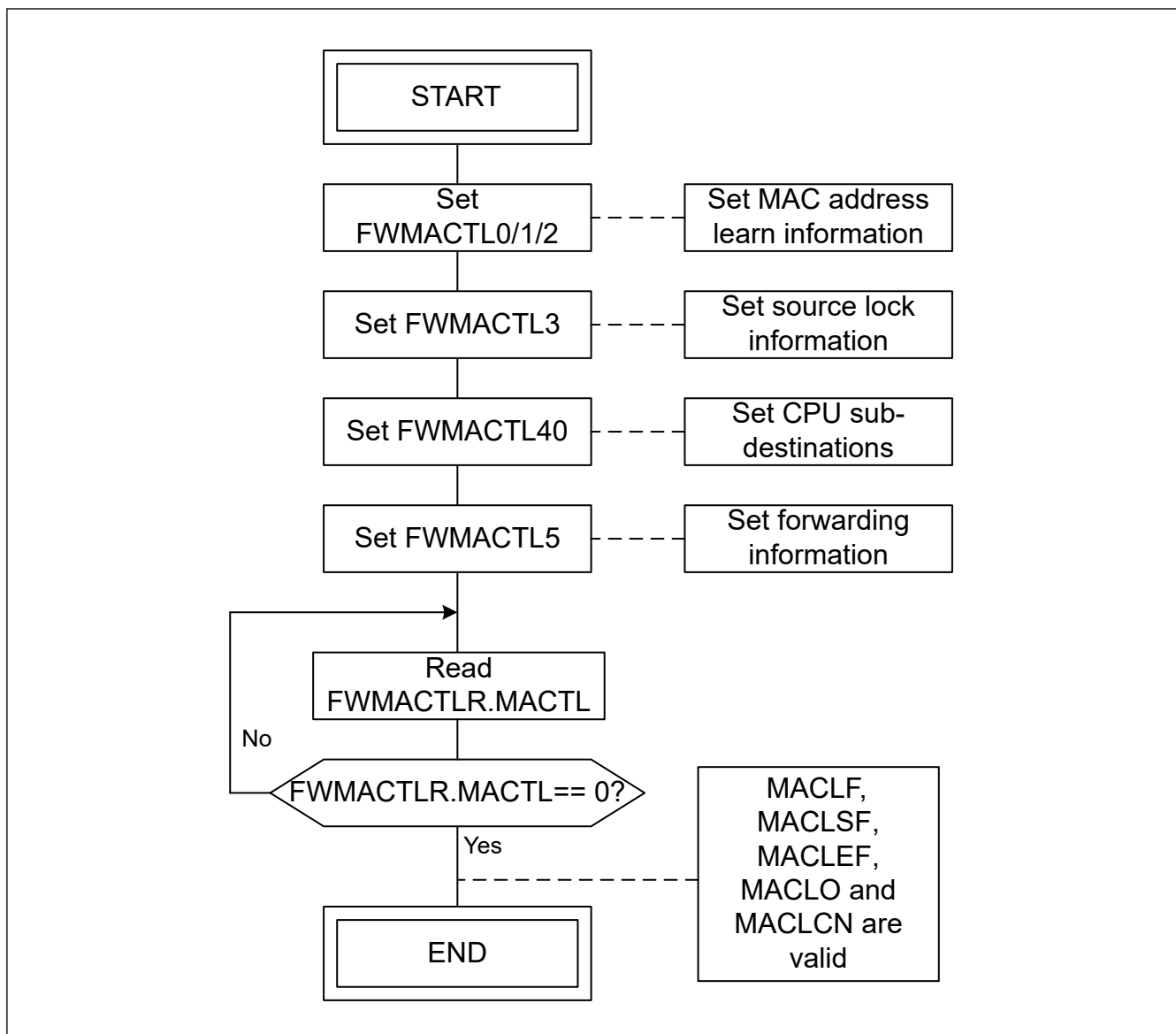


Figure 30.17 MAC entry learn flow

30.4.1.6.4 VLAN Entry Learn Flow

VLAN entry learn flow is described in [Figure 30.18](#).

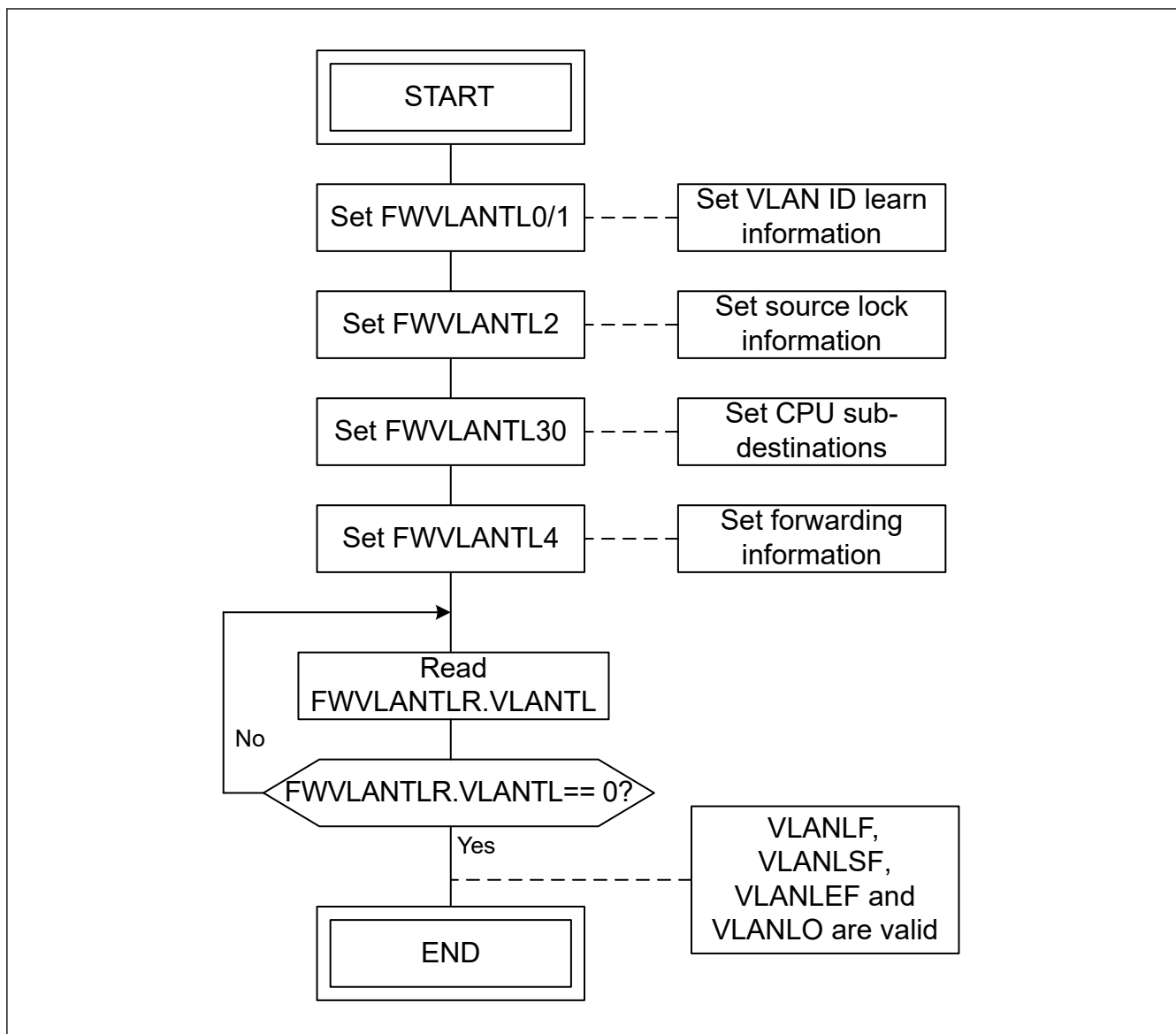


Figure 30.18 VLAN entry learn flow

30.4.1.6.5 MAC Entry Search Flow

MAC entry search flow is described in [Figure 30.19](#).

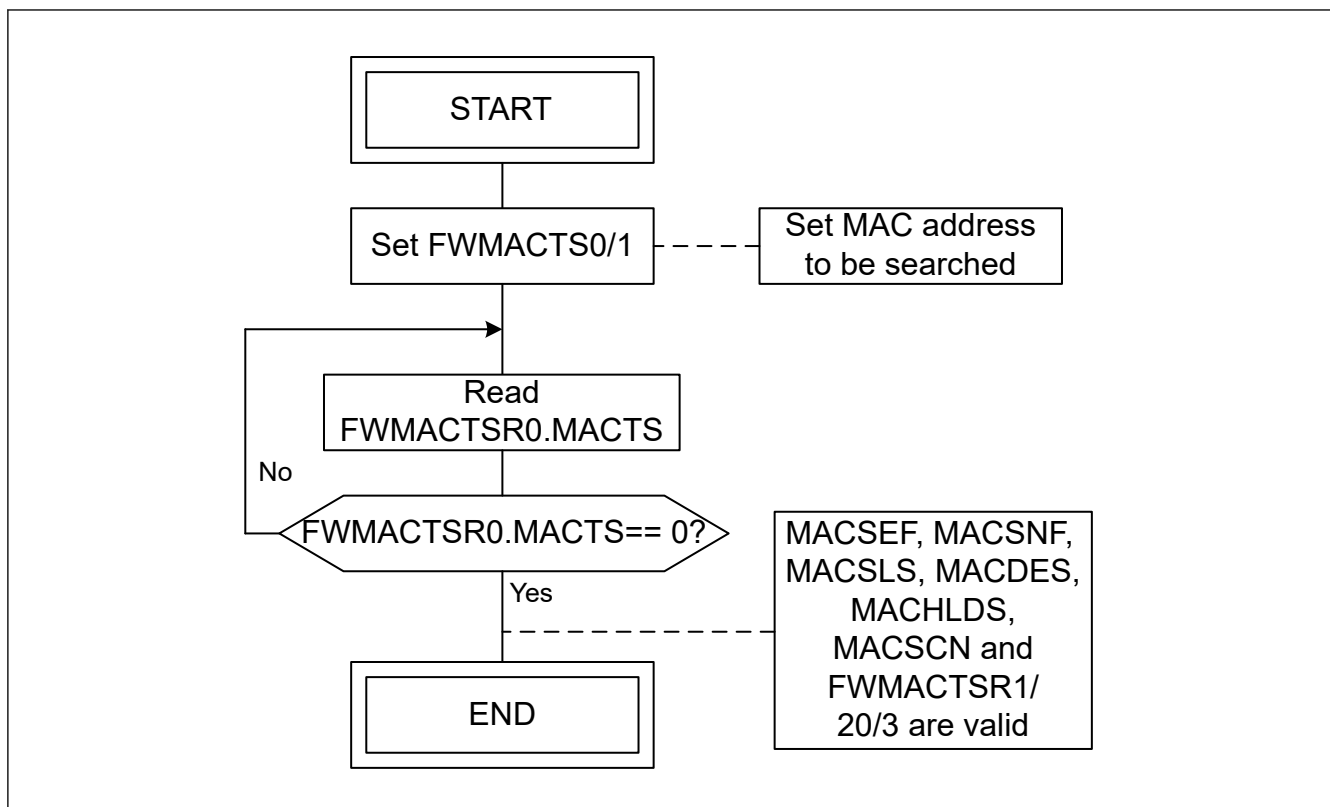


Figure 30.19 MAC entry search flow

30.4.1.6.6 VLAN Entry Search Flow

VLAN entry search flow is described in [Figure 30.20](#).

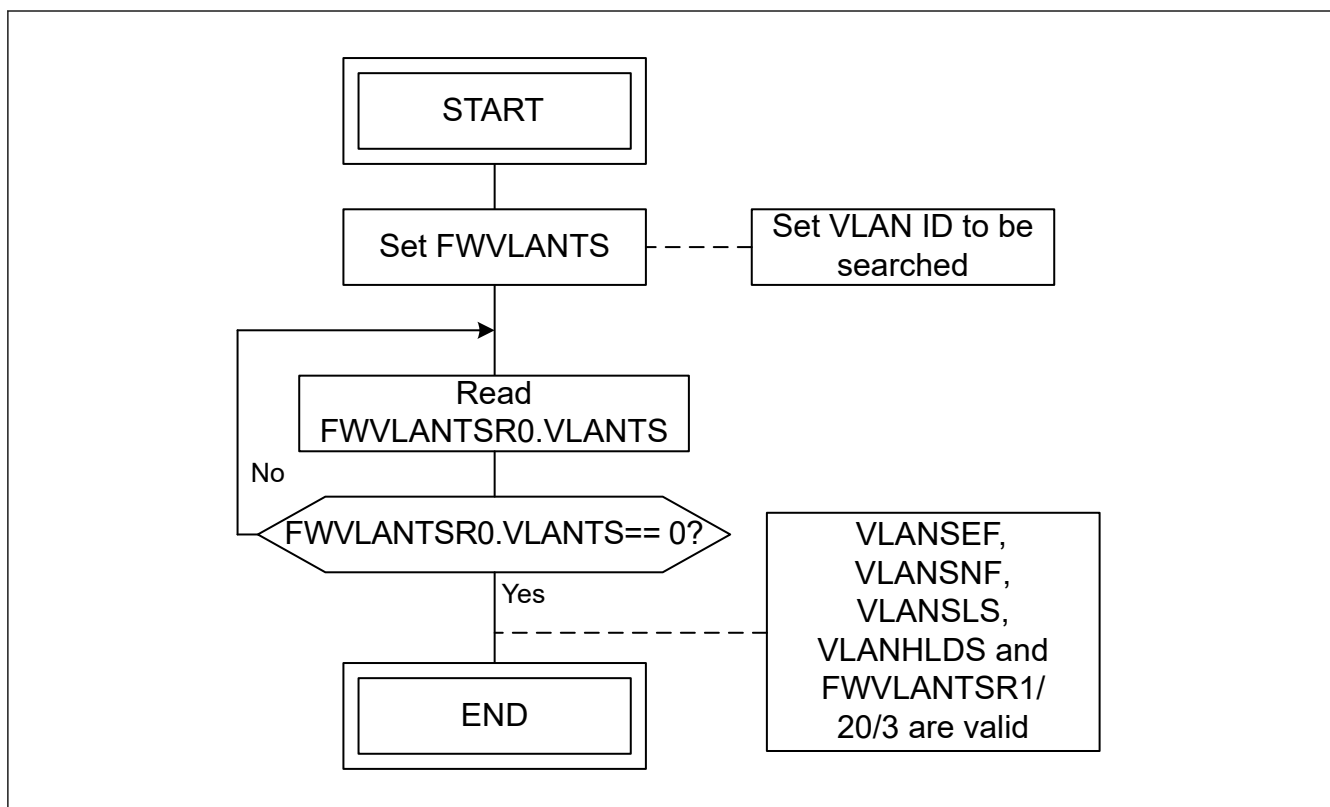


Figure 30.20 VLAN entry search flow

30.4.1.6.7 MAC Entry Read Flow

MAC entry read flow is described in [Figure 30.21](#).

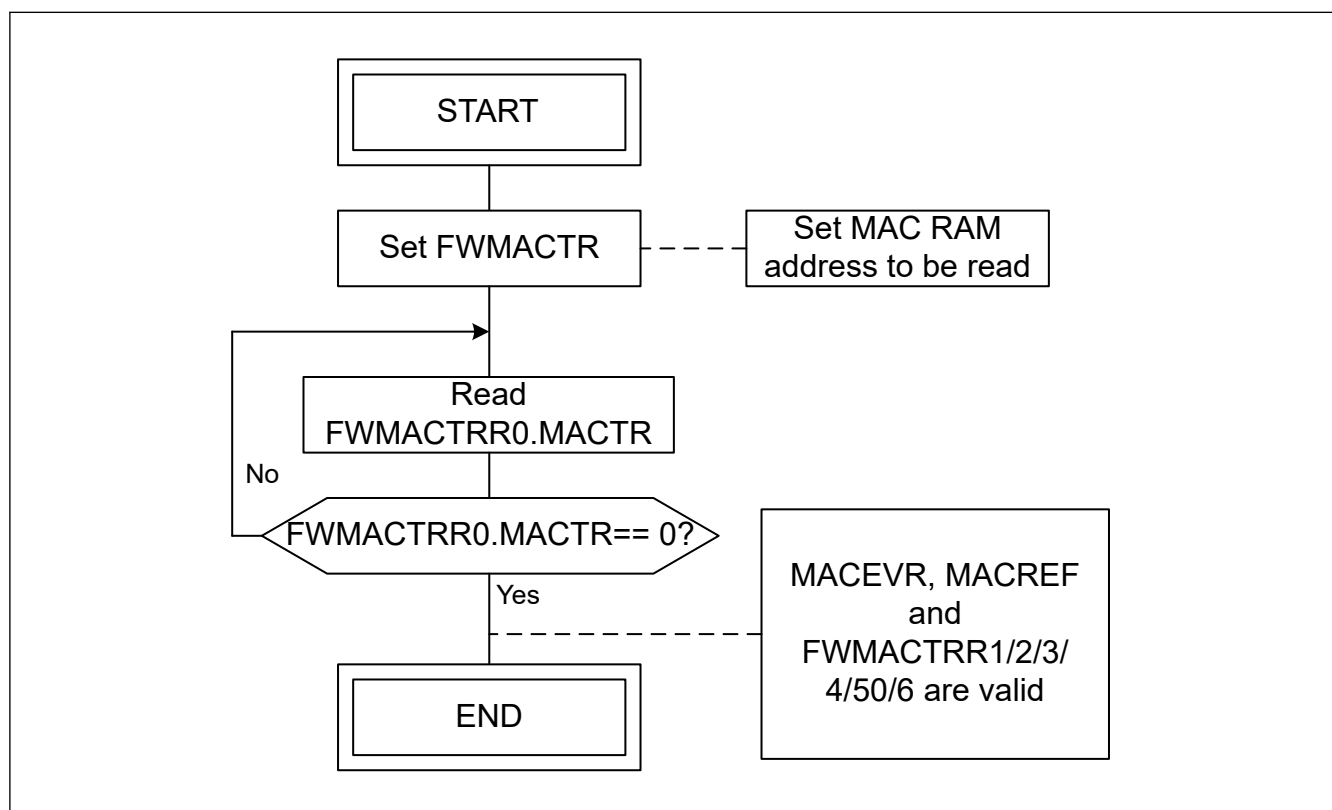


Figure 30.21 MAC entry read flow

30.4.1.7 Port based forwarding flows

30.4.1.7.1 Port i Port-based Setting Flow

Port i port-based setting flow is described in [Figure 30.22](#).

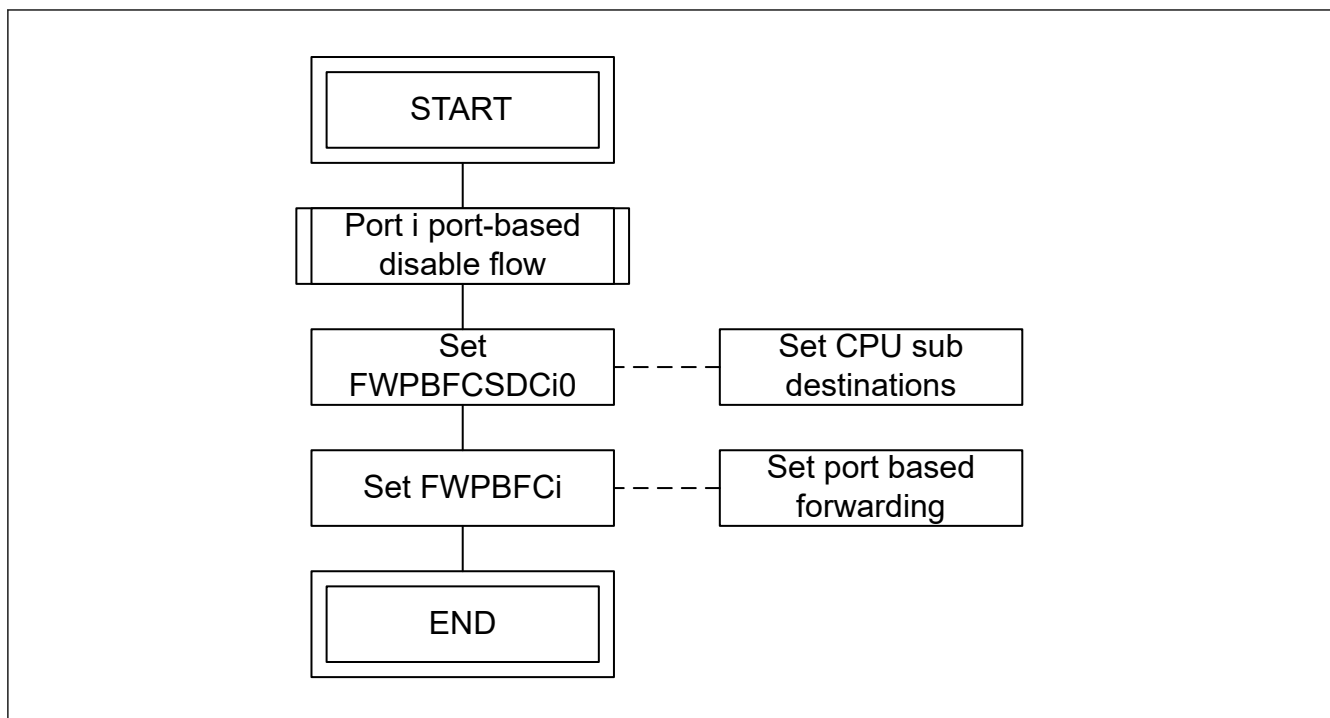


Figure 30.22 Port i port-based setting flow

30.4.1.7.2 Port i Port-based Disable Flow

Port i port-based disable flow is described in [Figure 30.23](#).

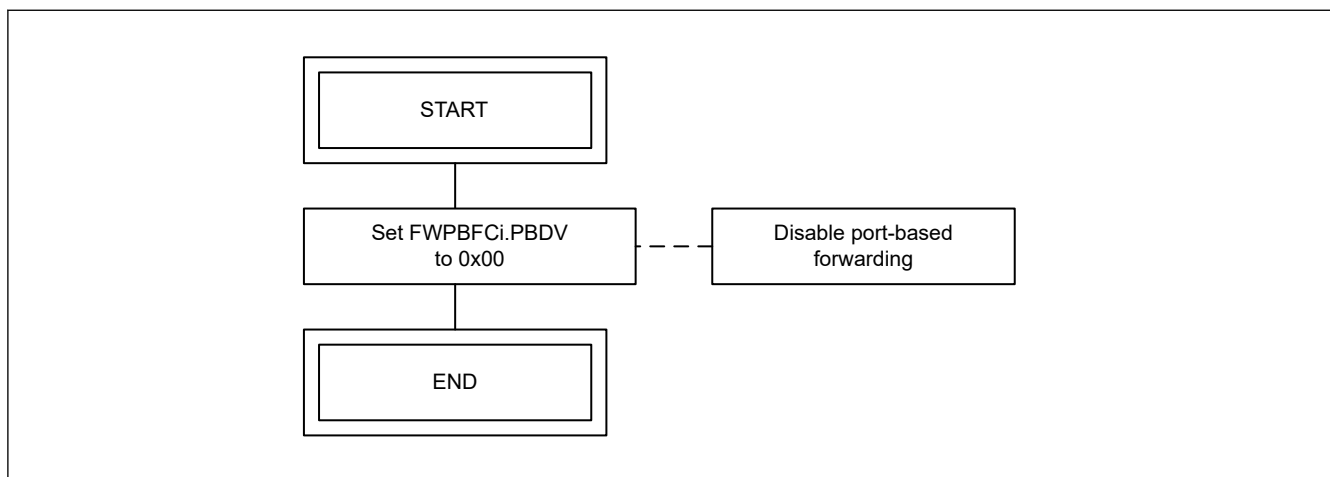


Figure 30.23 Port i port-based disable flow

30.4.1.8 Layer 2/Layer 3 Update Flows

30.4.1.8.1 Layer 2/Layer 3 Update Table Reset Flow

[Figure 30.24](#) describes the reset flow of Layer 2/Layer 3 Update table.

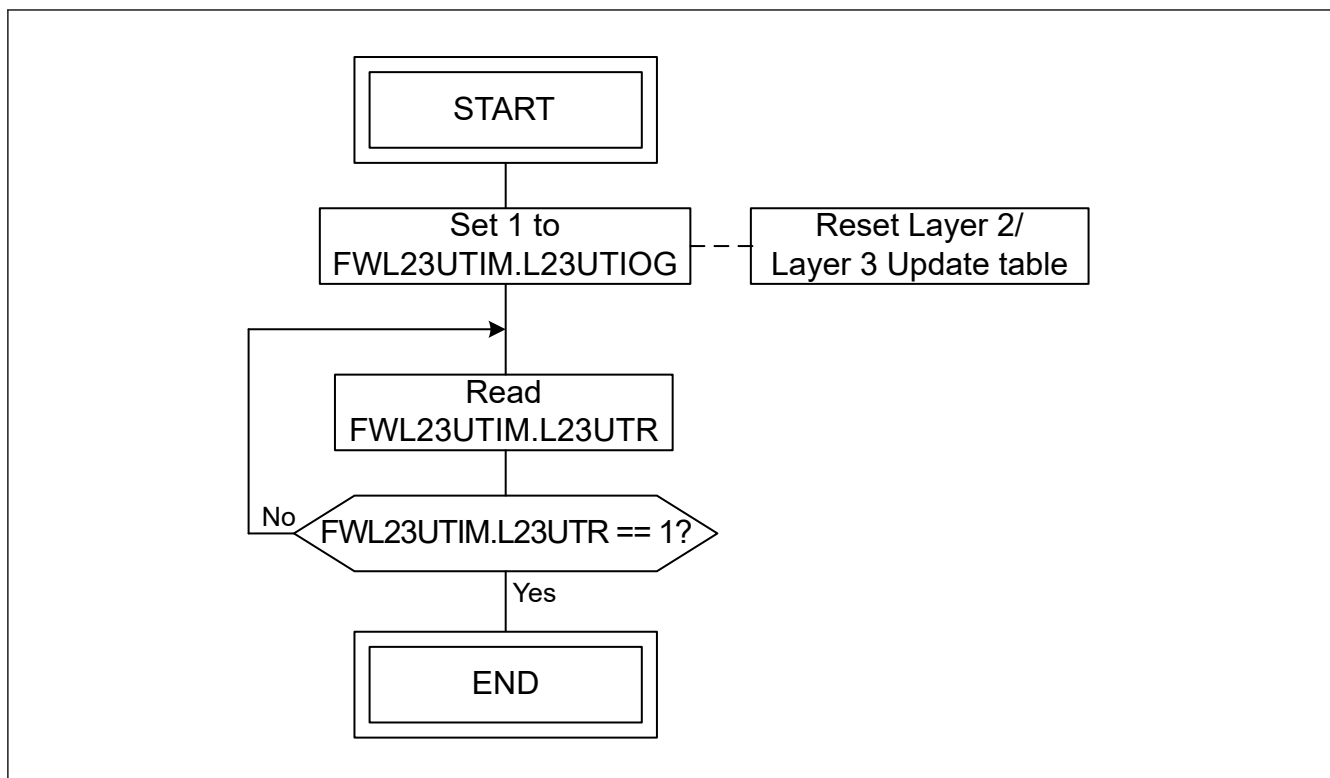


Figure 30.24 Reset flow of Layer 2/Layer 3 Update table

30.4.1.8.2 Layer 2/Layer 3 Update Rule Learn Flow

Figure 30.25 describes the learn flow of Layer 2/Layer 3 update rule.

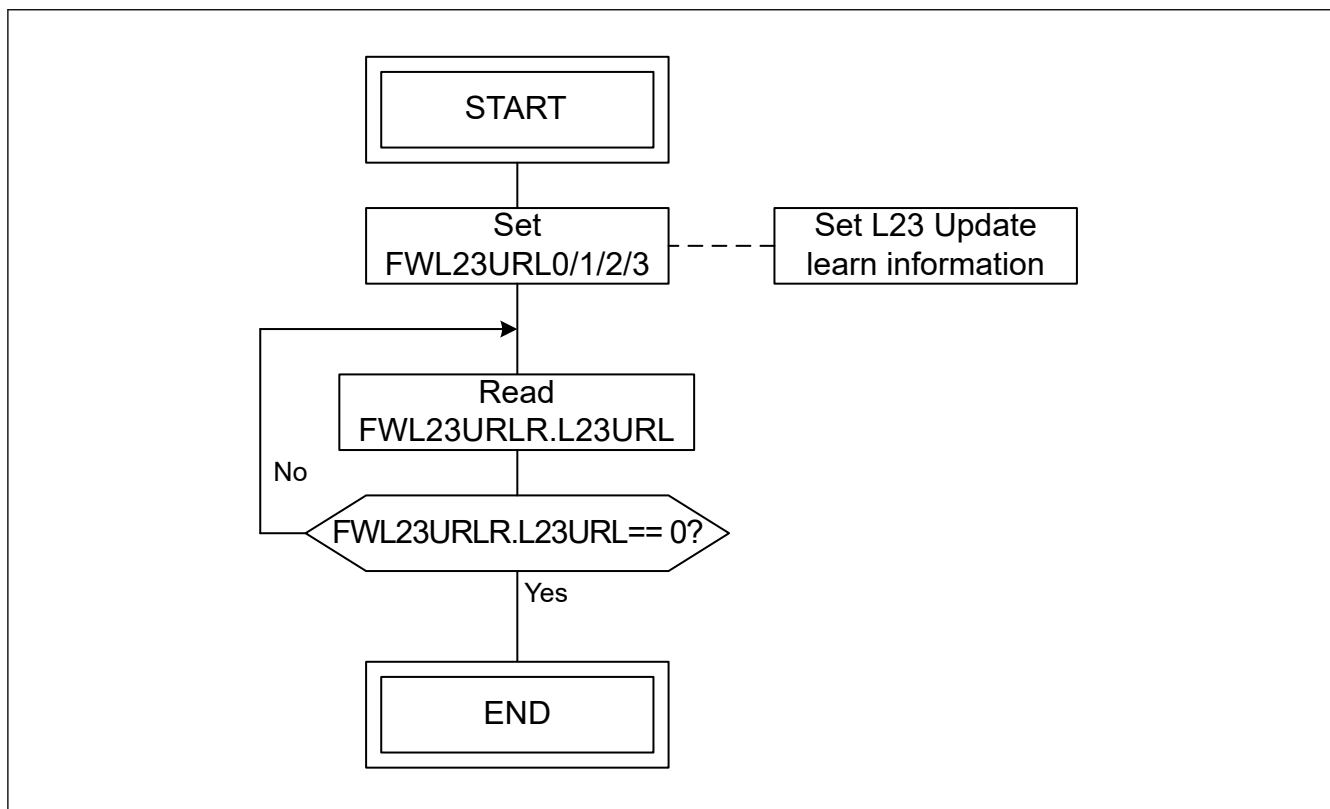


Figure 30.25 Learn flow of Layer 2/Layer 3 update rule

30.4.1.8.3 Layer 2/Layer 3 Update Rule Read Flow

Figure 30.26 describes the read flow of Layer 2/Layer 3 Update rule.

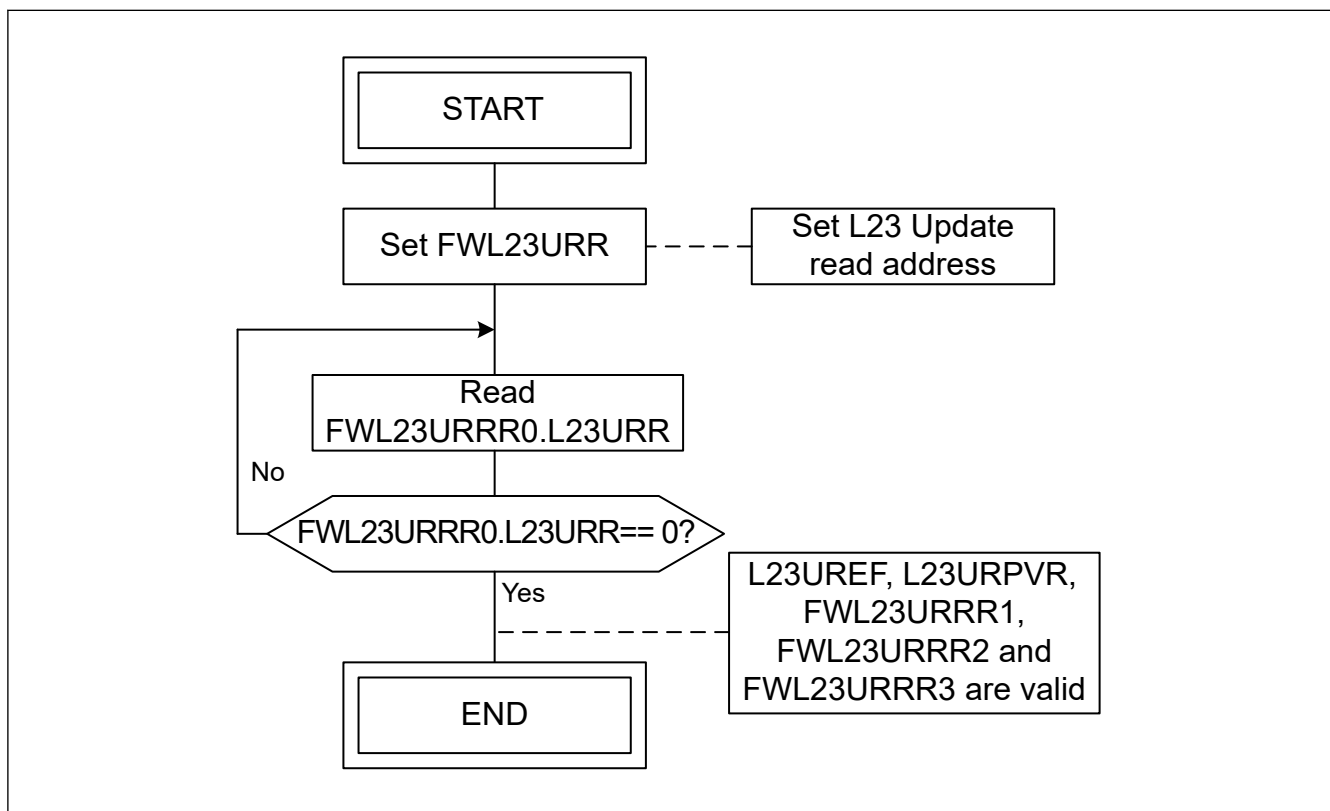


Figure 30.26 Read flow of Layer 2/Layer 3 Update rule

30.4.1.9 PSFP Flows

30.4.1.9.1 MSDU Filter i Setting Flow

Figure 30.27 describes the setting flow of MSDU filter i.

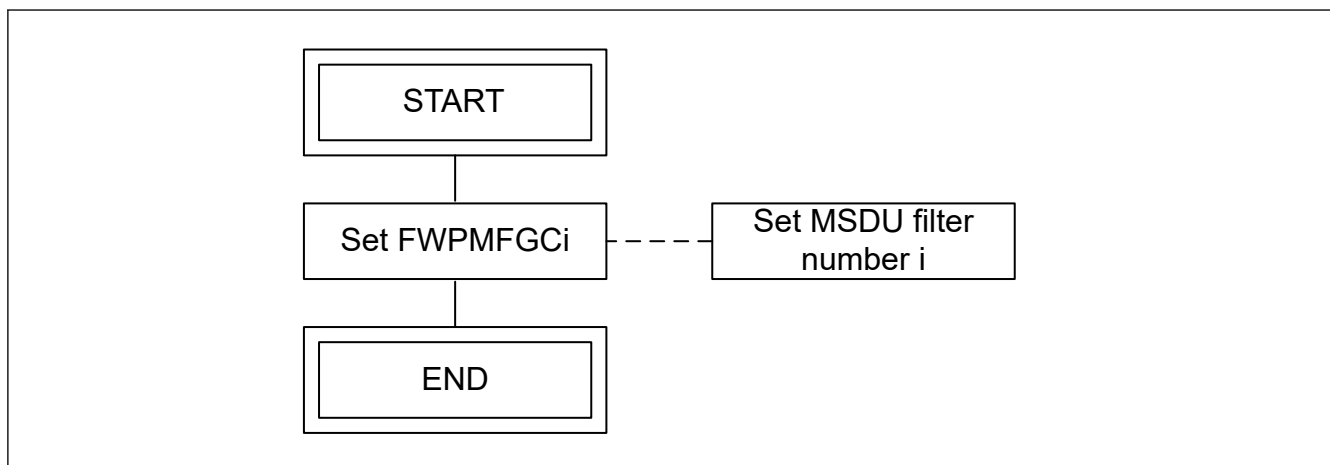


Figure 30.27 Setting flow of MSDU filter i

30.4.1.9.2 Meter i Setting Flow

Meter i setting flow is described in Figure 30.28.

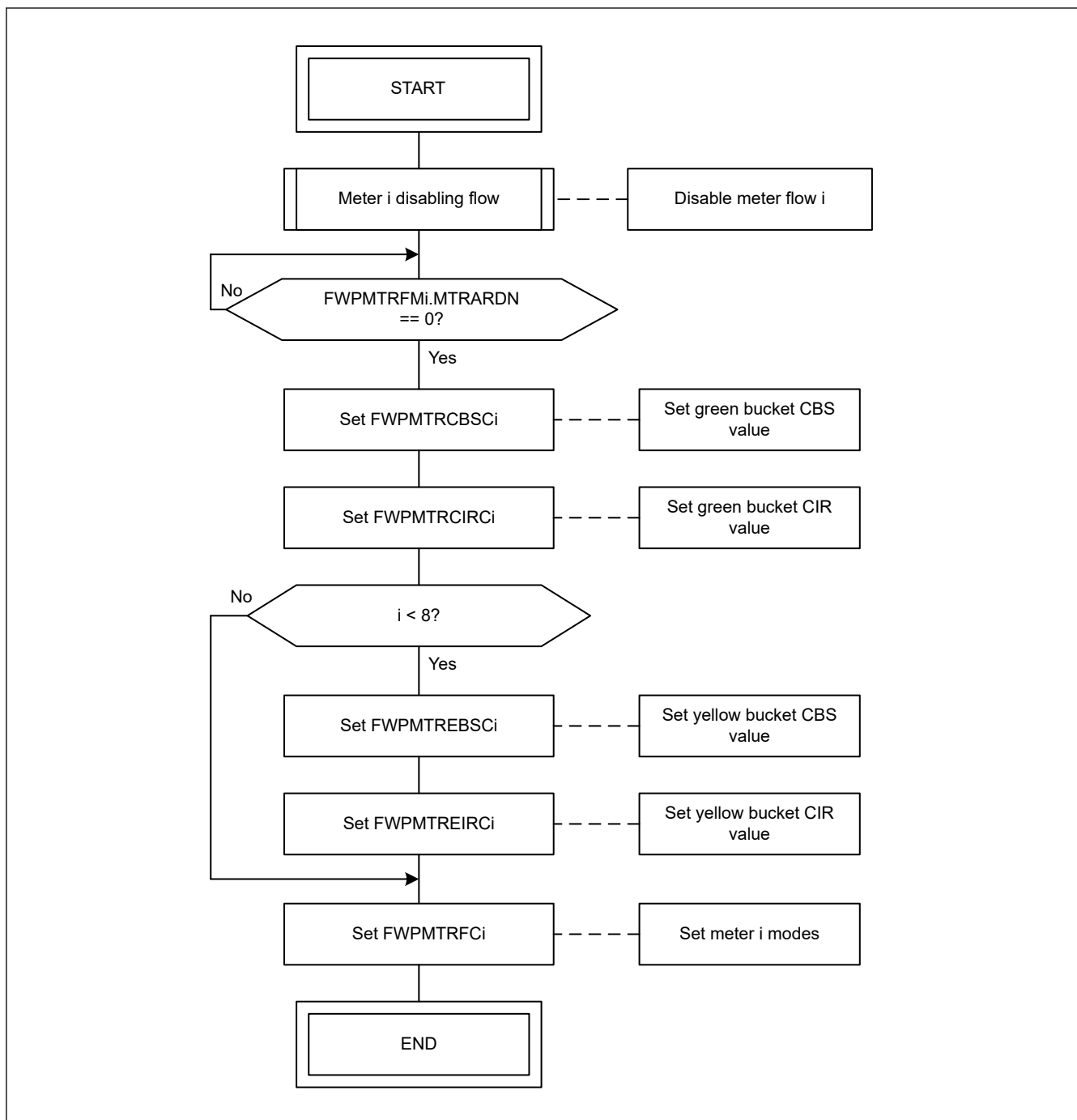


Figure 30.28 Meter i setting flow

30.4.1.9.3 Meter i Disabling Flow

Figure 30.29 describes the disabling flow of meter *i*.

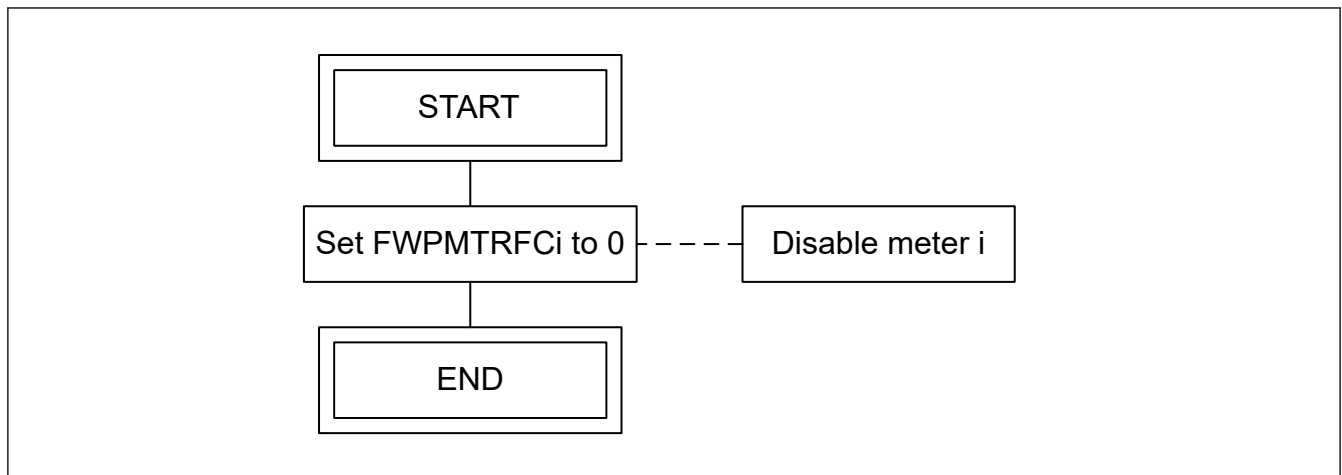


Figure 30.29 Disabling flow of meter *i*

30.4.1.10 FRER Flows

30.4.1.10.1 FRER Table Reset Flow

Figure 30.30 describes the reset flow of FRER table.

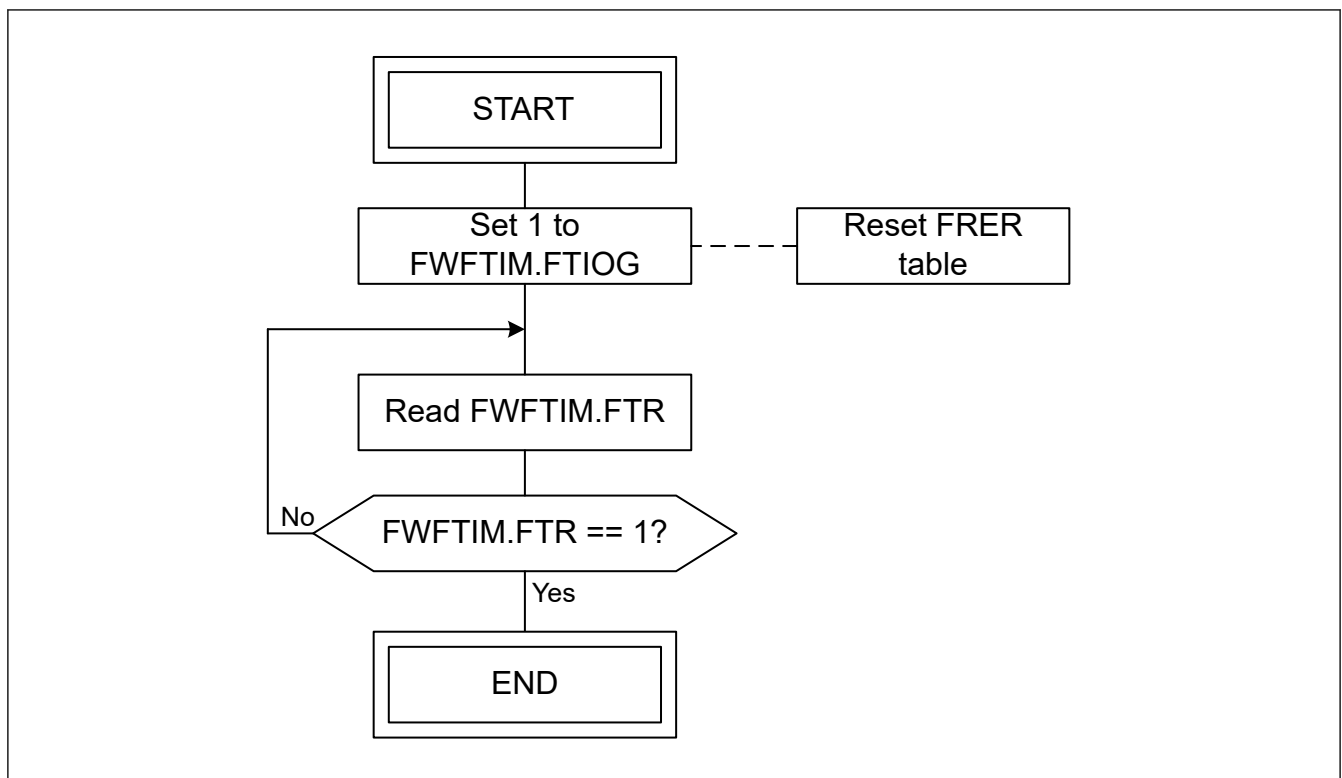


Figure 30.30 Reset flow of FRER table

30.4.1.10.2 FRER Entry *i* Learn Flow

Figure 30.31 describes the learn flow of FRER entry *i*.

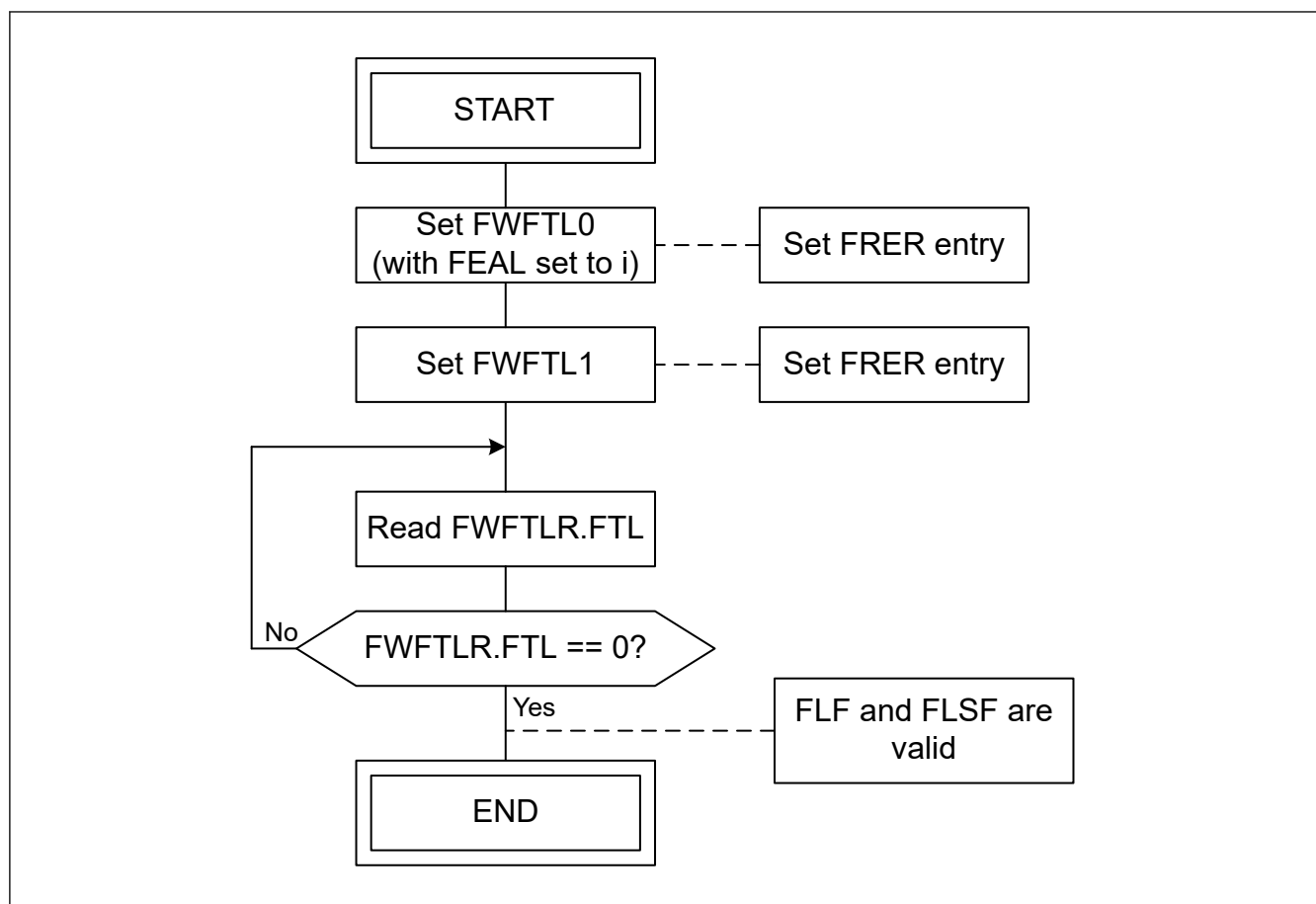


Figure 30.31 Learn flow of FRER entry i

30.4.1.10.3 FRER Entry i Read Flow

Figure 30.32 describes the read flow of FRER entry i.

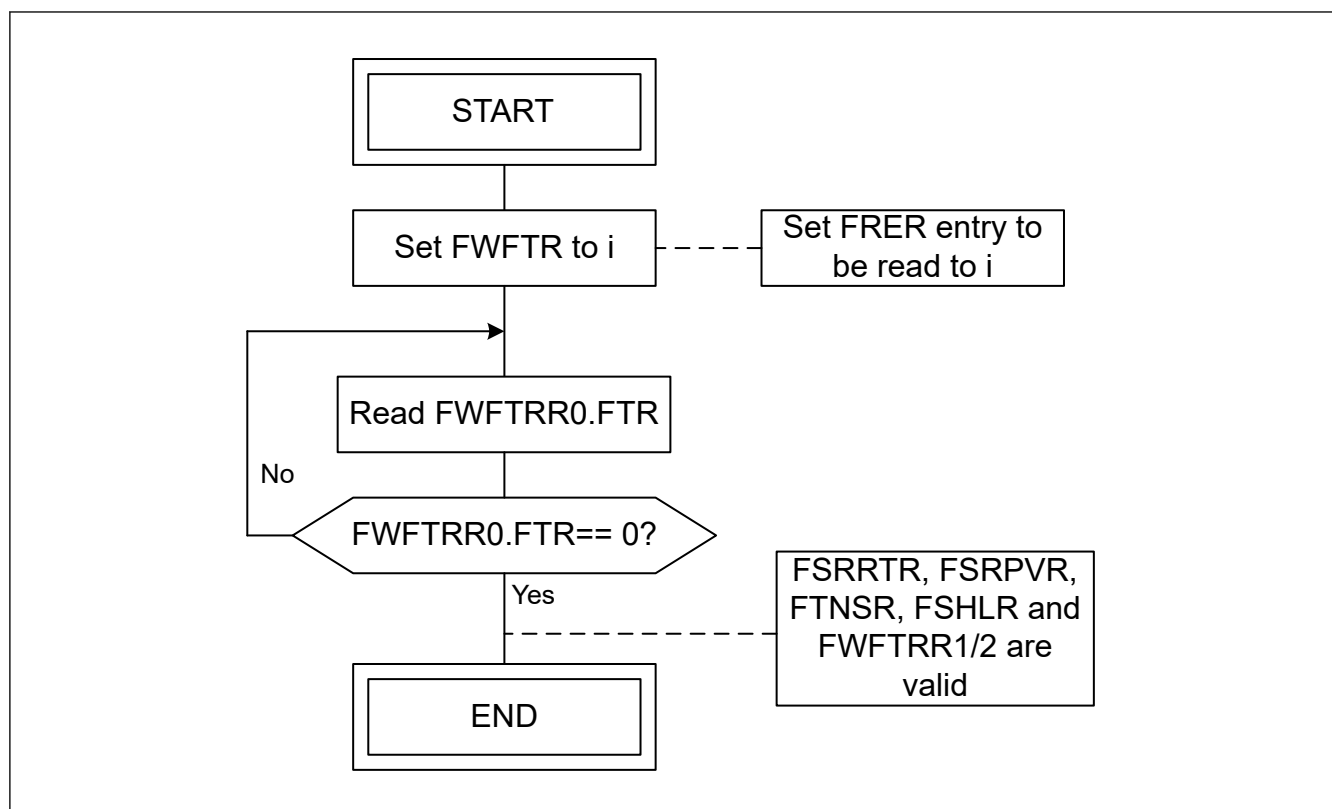


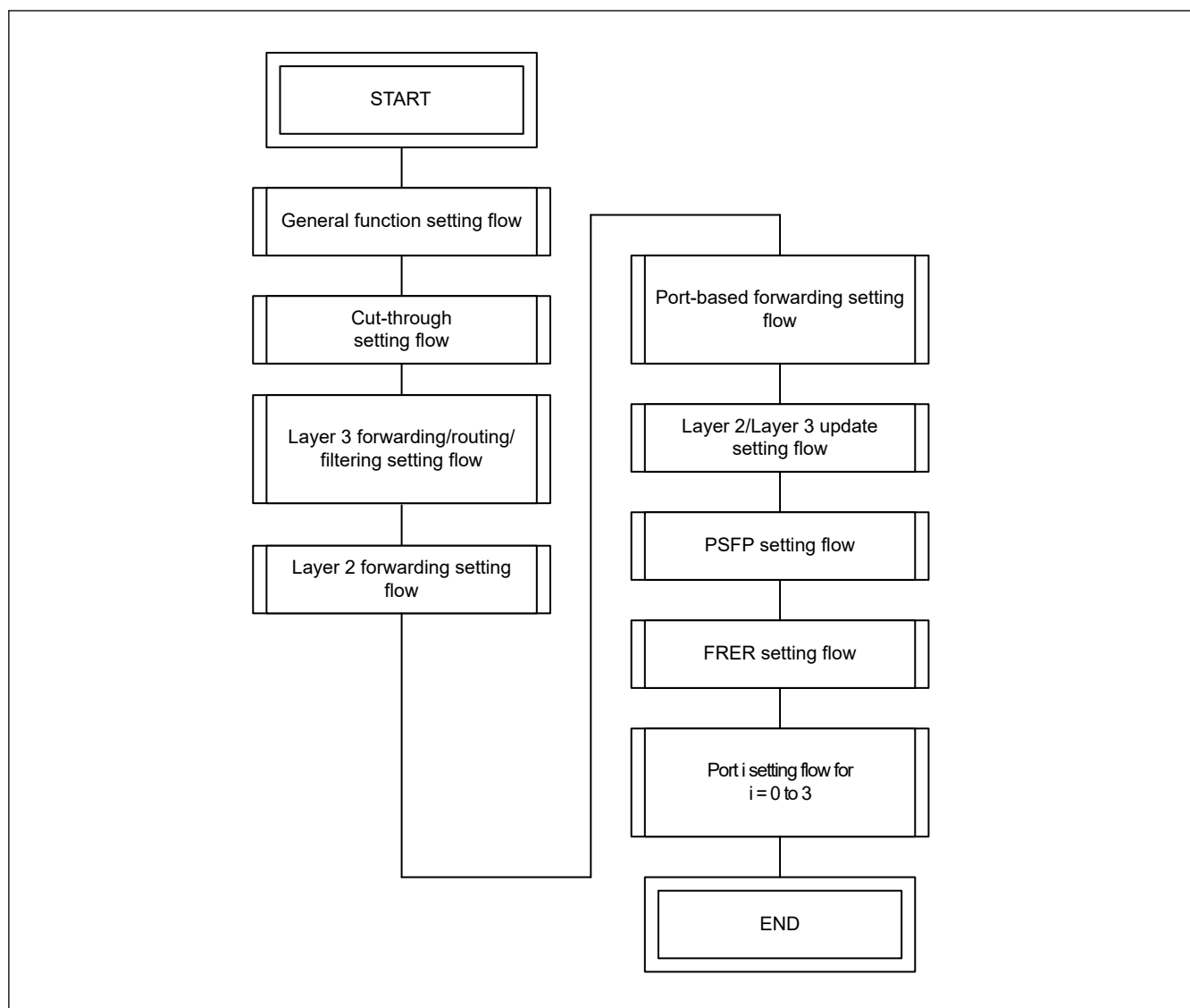
Figure 30.32 Read flow of FRER entry i

30.4.1.11 Called Software Flows

The flows described in this section can only be called from other flows (as shown in the “flow link” box in [Figure 30.2](#)) and cannot be used alone.

30.4.1.11.1 Full Setting Flow

Full setting flow is described in [Figure 30.33](#).

**Figure 30.33 Full setting flow**

30.4.1.11.2 General Function Setting Flow

[Figure 30.34](#) describes the setting flow of general function.

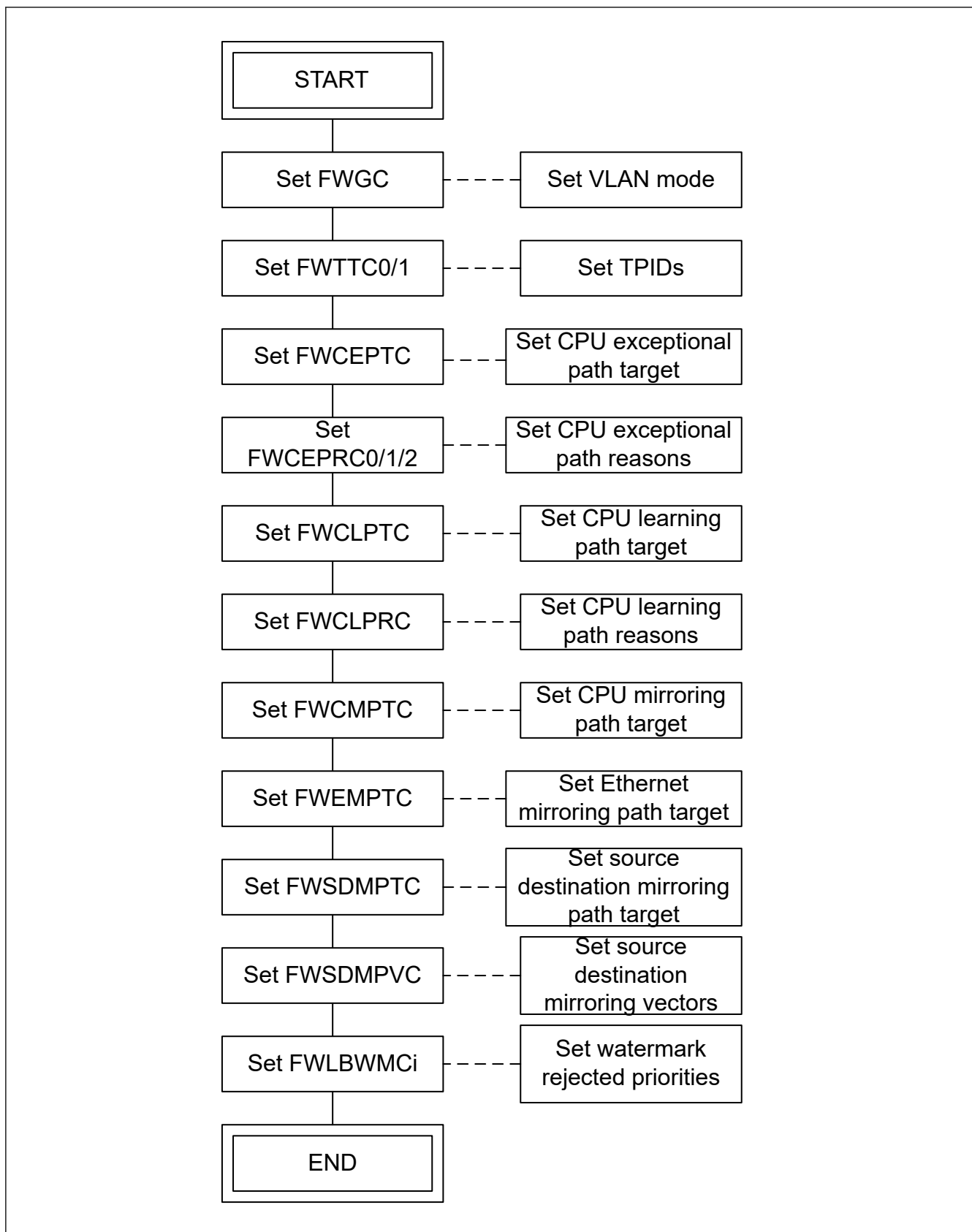


Figure 30.34 Setting flow of general function

30.4.1.11.3 Cut-through Setting Flow

Figure 30.35 describes the setting flow of cut-through.

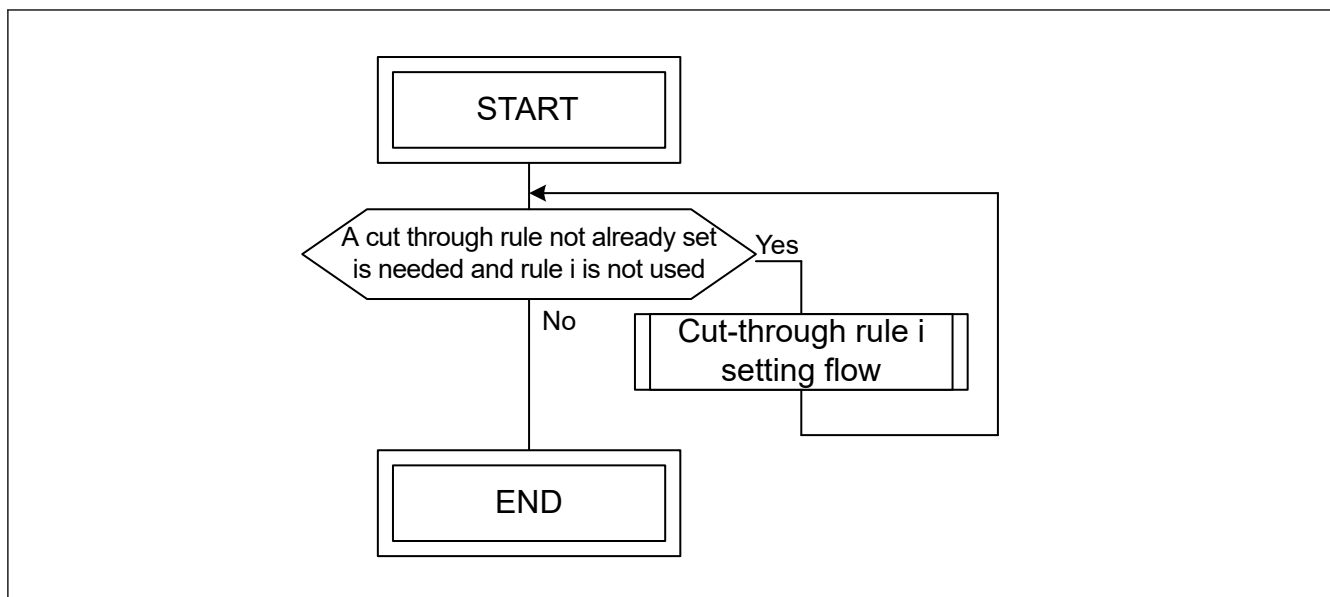


Figure 30.35 Setting flow of cut-through

30.4.1.11.4 Two-byte Filter i Setting Flow

Figure 30.36 describes the setting flow of two-byte filter i.

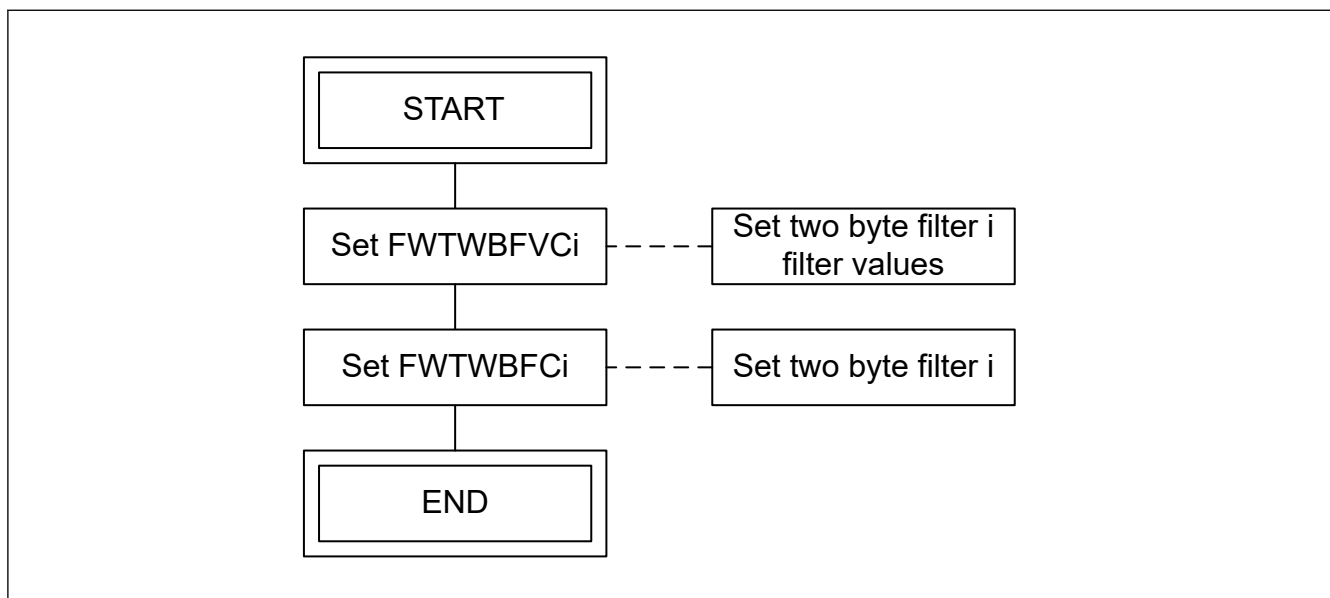


Figure 30.36 Setting flow of two-byte filter i

30.4.1.11.5 Three-byte Filter i Setting Flow

Figure 30.37 describes the setting flow of three-byte filter i.

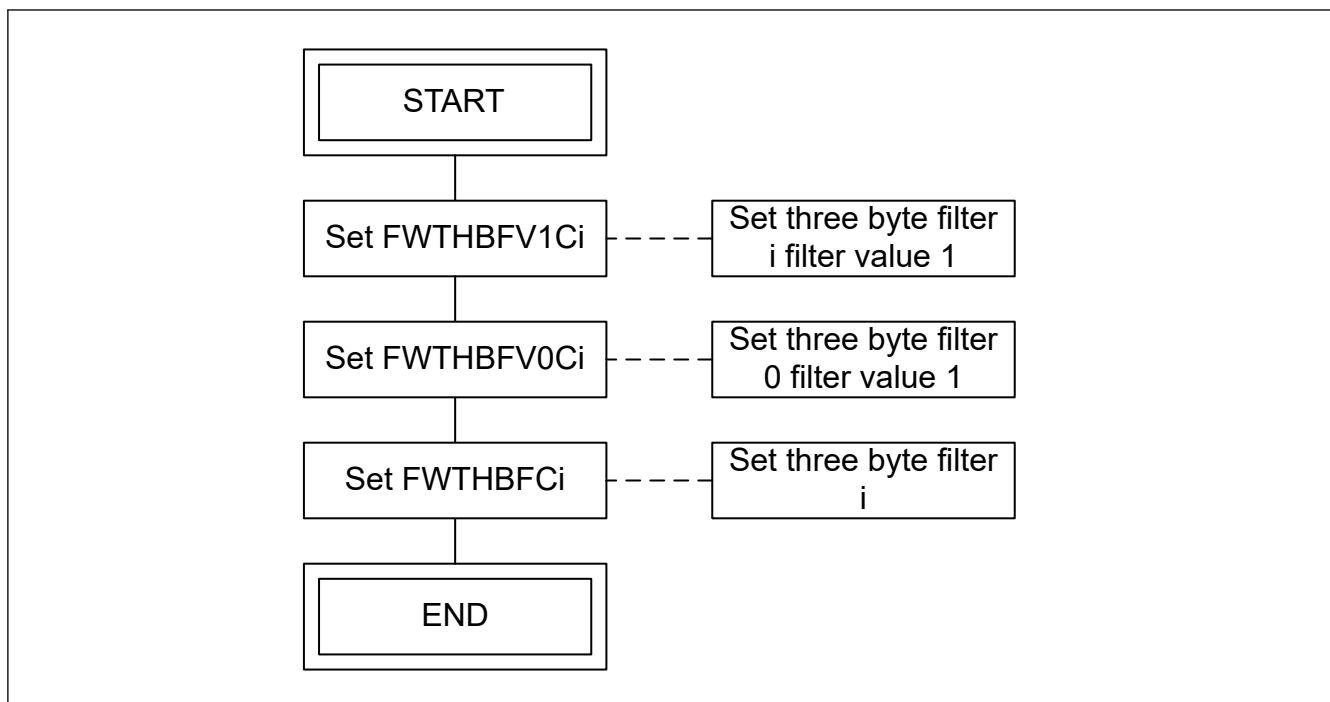


Figure 30.37 Setting flow of three-byte filter i

30.4.1.11.6 Four-byte Filter i Setting Flow

[Figure 30.38](#) describes the setting flow of four-byte filter i.

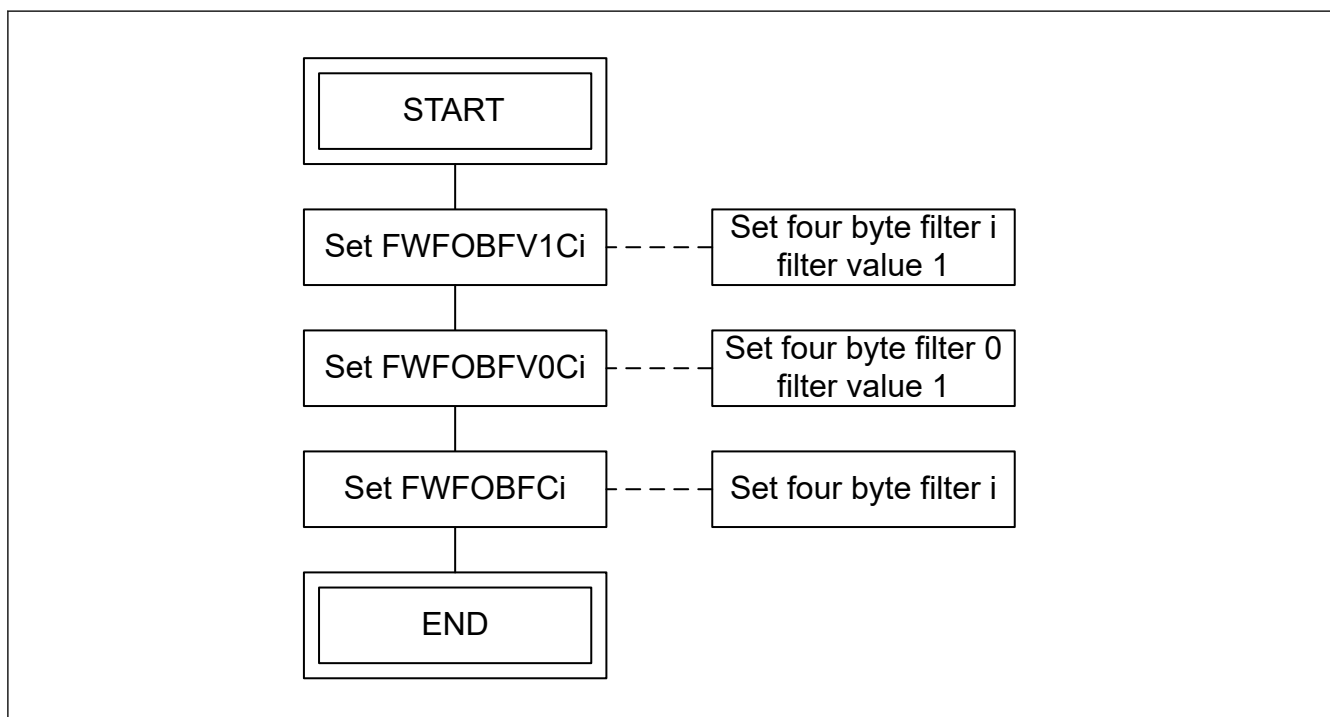


Figure 30.38 Setting flow of four-byte filter i

30.4.1.11.7 Range Filter i Setting Flow

[Figure 30.39](#) describes the setting flow of range filter i.

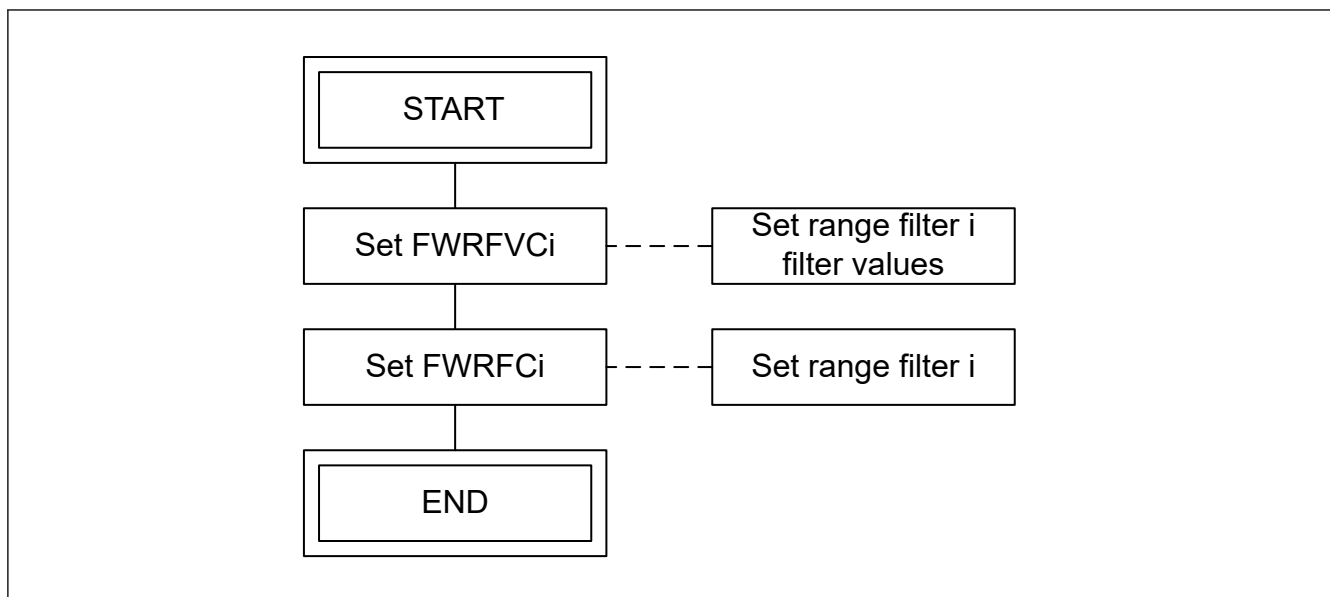


Figure 30.39 Setting flow of range filter i

30.4.1.11.8 IPv4/IPv6/Layer 2 Stream Filter Setting Flow

Figure 30.40 describes the setting flow of IPv4/IPv6/Layer 2 stream filter.

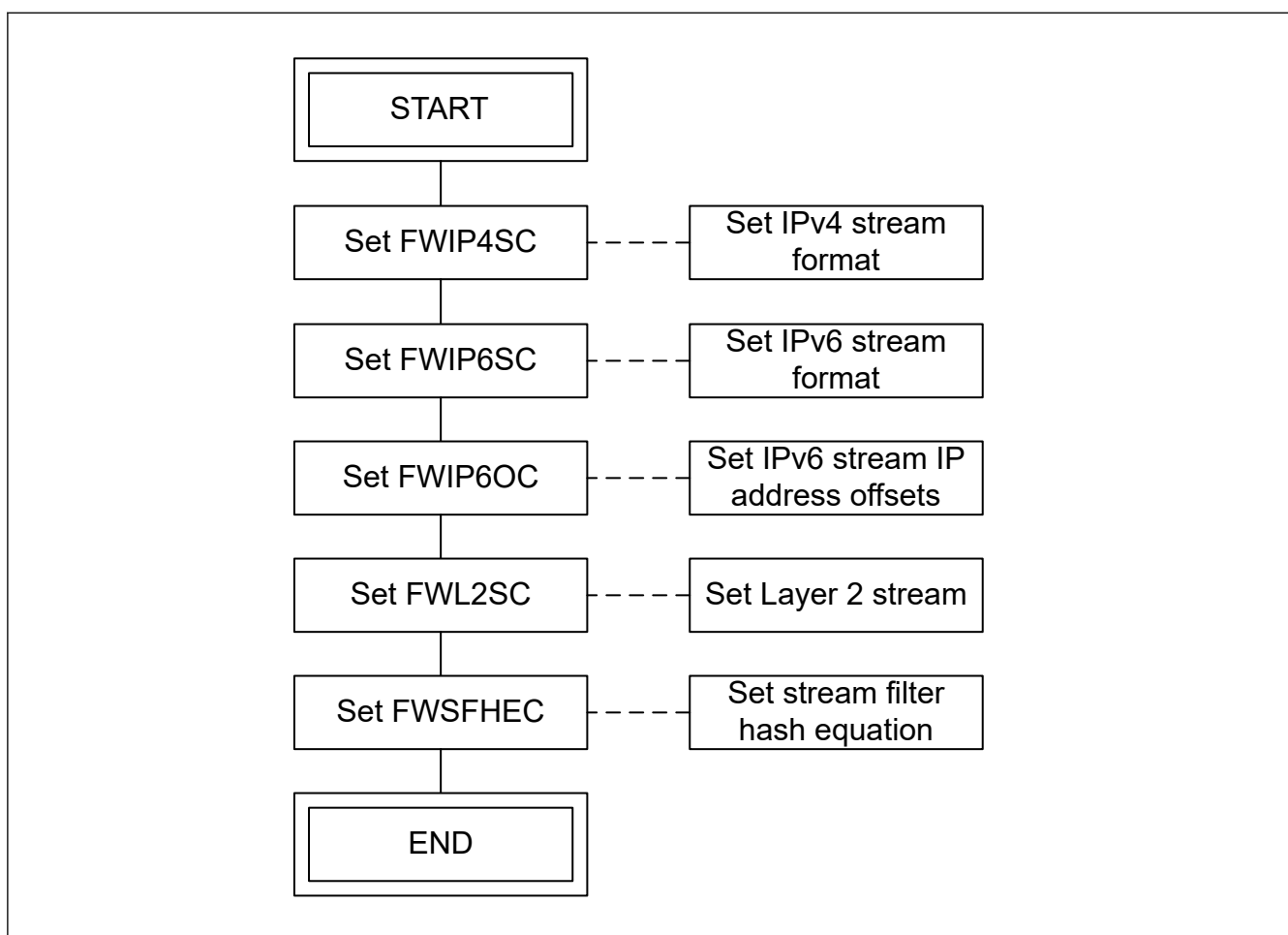


Figure 30.40 Setting flow of IPv4/IPv6/Layer 2 stream filter

30.4.1.11.9 Layer 3 Forwarding/Routing/Filtering Setting Flow

[Figure 30.41](#) describes the setting flow of Layer 3 forwarding/routing/filtering.

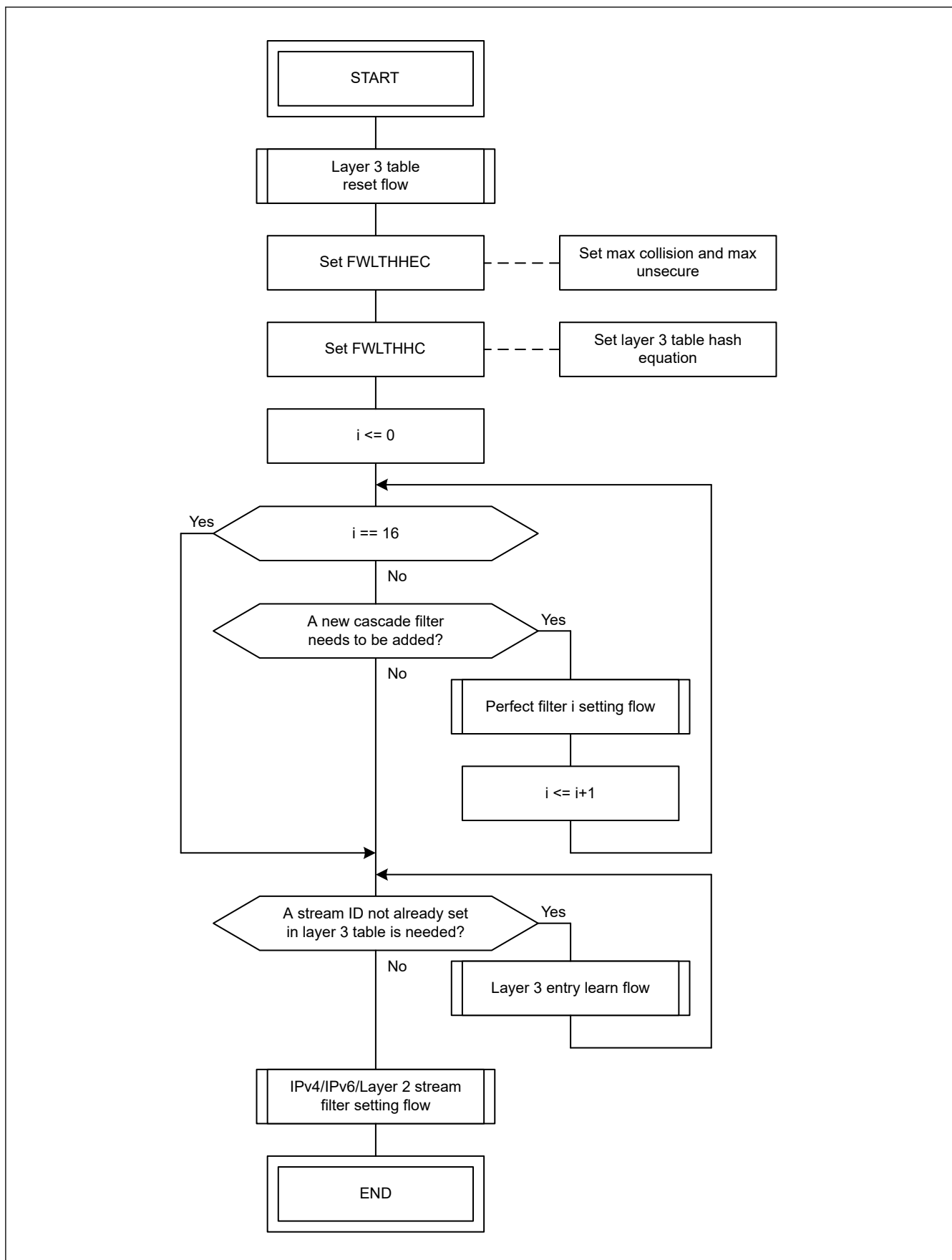


Figure 30.41 Setting flow of Layer 3 forwarding/routing/filtering

30.4.1.11.10 Layer 2 Forwarding Setting Flow

Figure 30.42 describes the setting flow of Layer 2 forwarding.

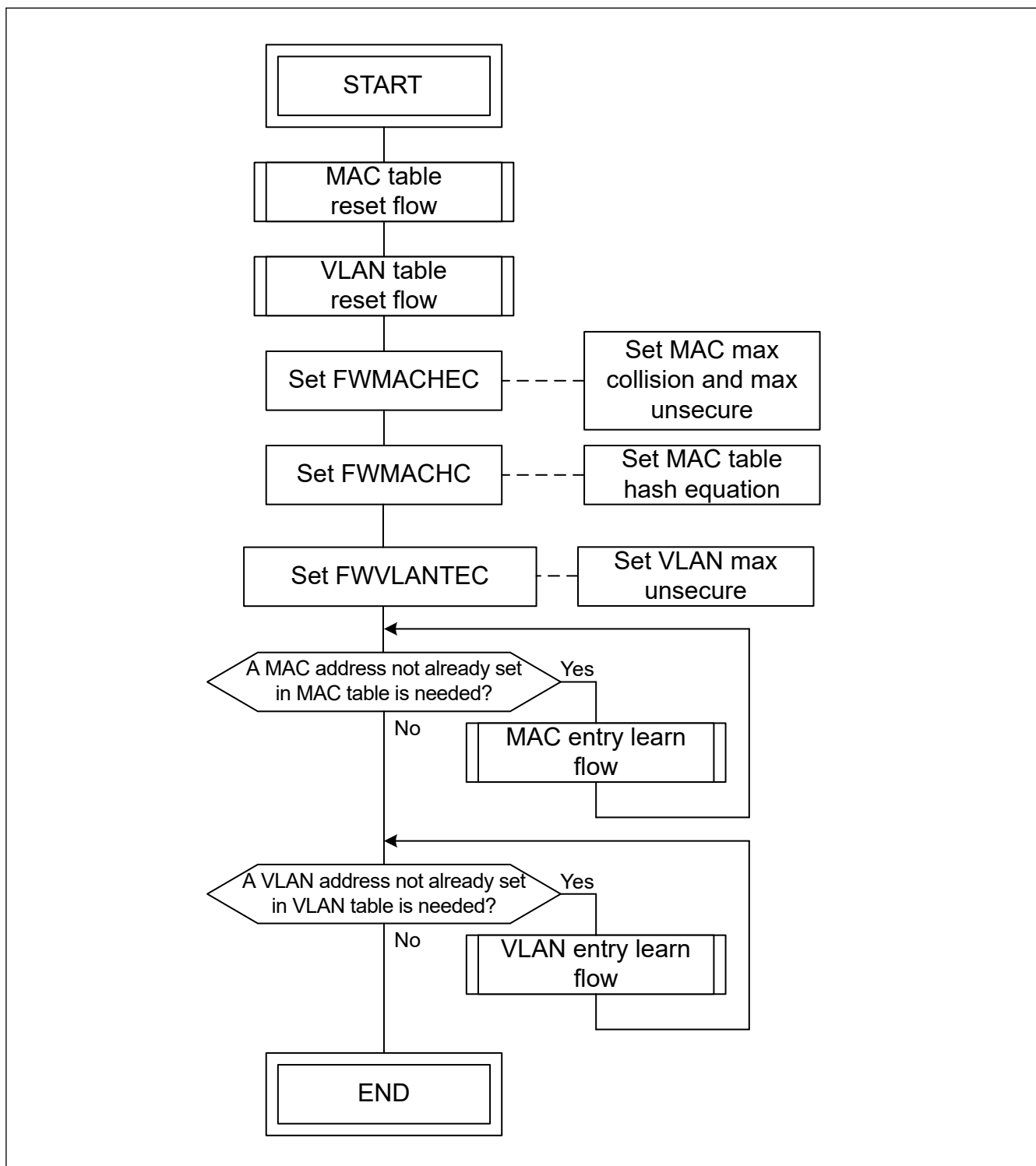


Figure 30.42 Setting flow of Layer 2 forwarding

30.4.1.11.11 Port-based Forwarding Setting Flow

Figure 30.43 describes the setting flow of port-based forwarding.

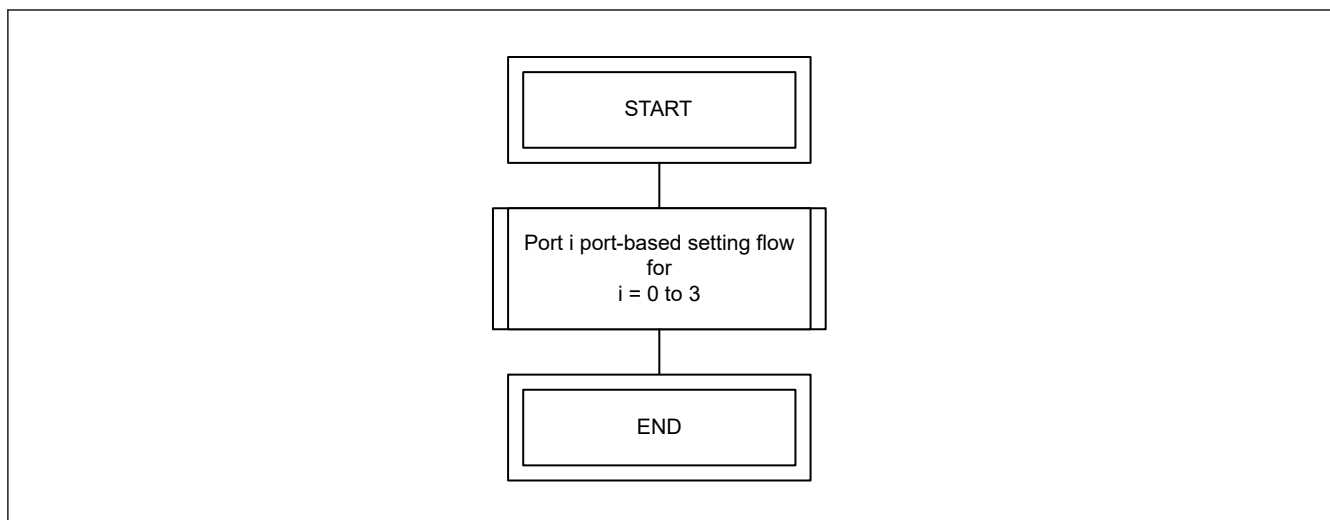


Figure 30.43 Setting flow of port-based forwarding

30.4.1.11.12 Layer 2/Layer 3 Update Setting Flow

Figure 30.44 describes the setting flow of Layer 2/Layer 3 update.

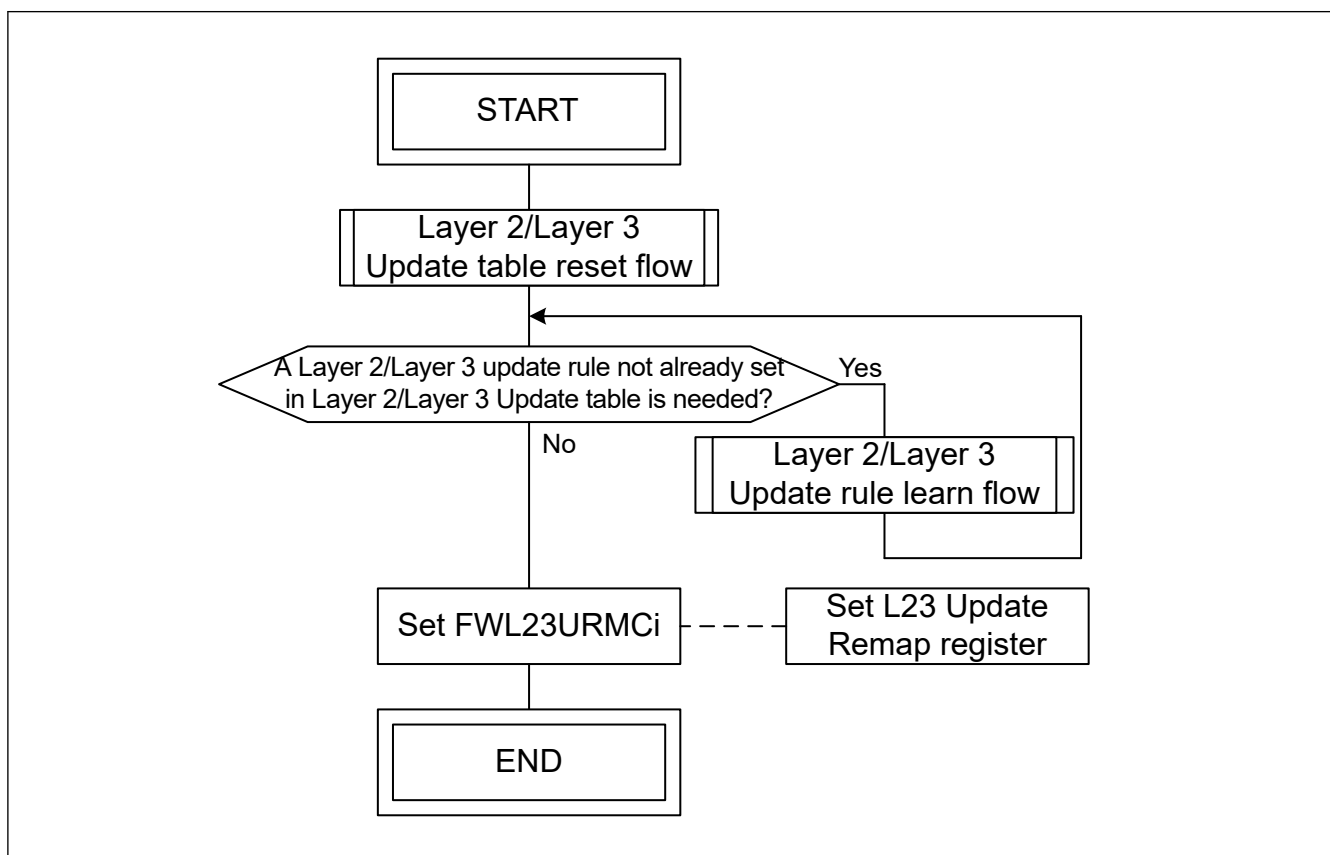


Figure 30.44 Setting flow of Layer 2/Layer 3 update

30.4.1.11.13 PSFP Setting Flow

Figure 30.45 describes the PSFP setting flow.

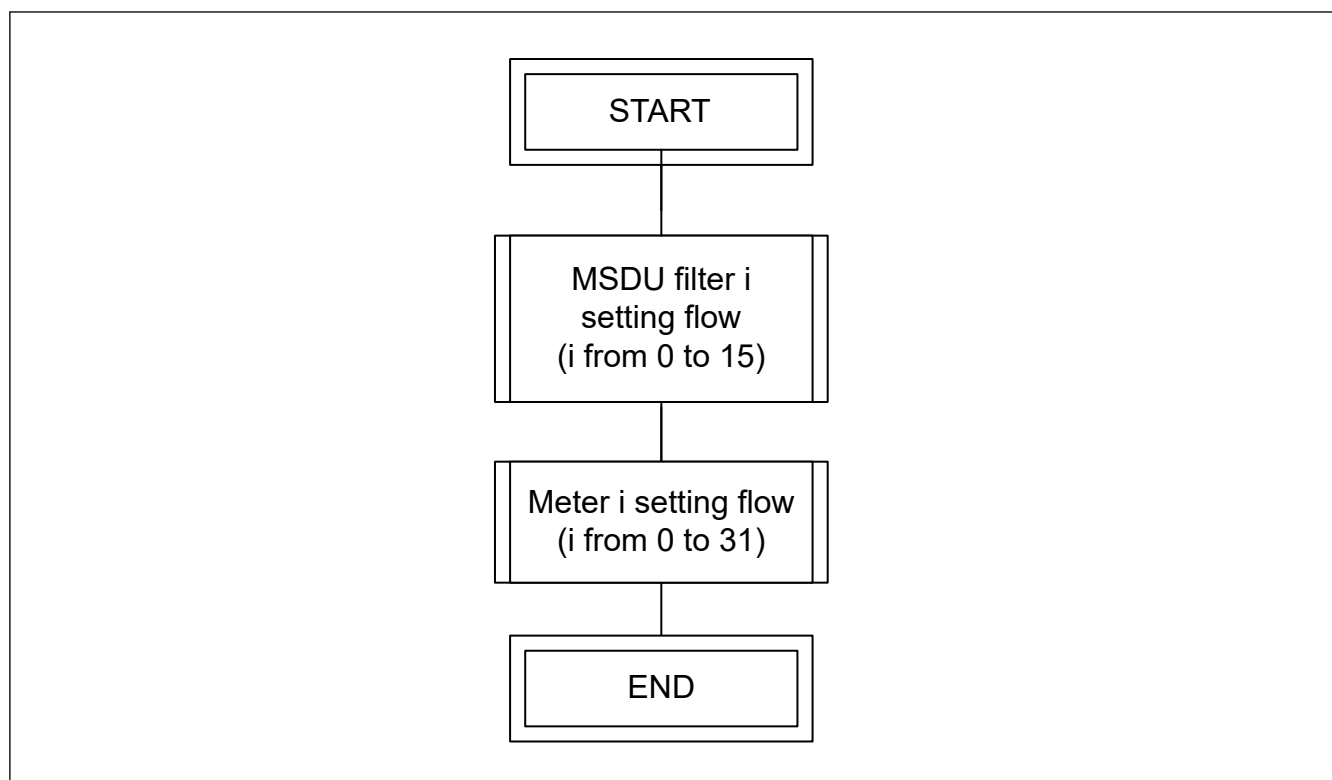


Figure 30.45 PSFP setting flow

30.4.1.11.14 FRER Setting Flow

FRER setting flow is described in [Figure 30.46](#).

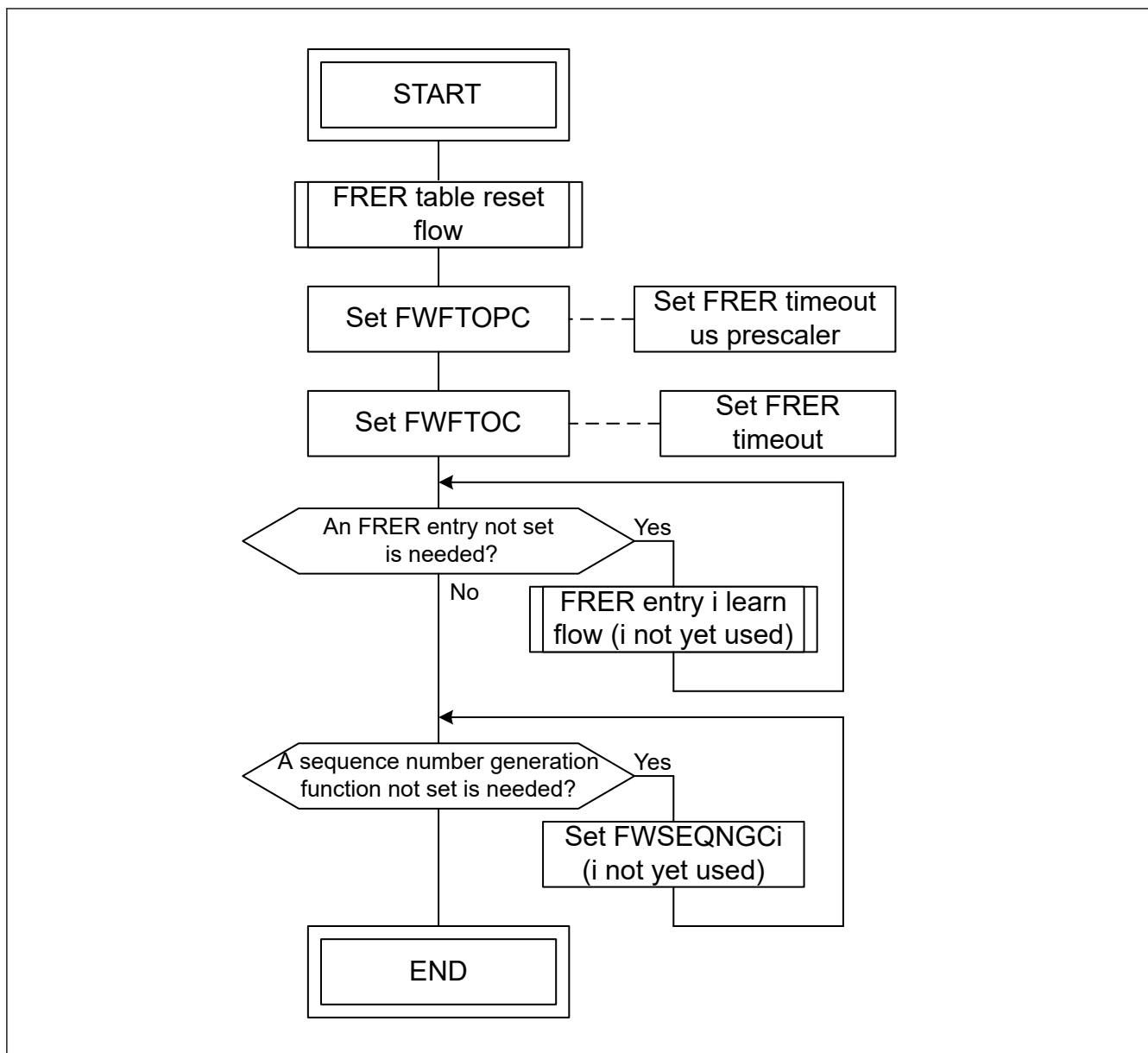


Figure 30.46 FRER setting flow

30.4.1.12 Register Writable Without Software Flow after Initialization

Table 30.4 describes the registers writable without software flow and which mode they can be written without software flow.

Table 30.4 Writable without software flow registers (1 of 2)

(✓ = writable, × = not writable)

Register name	Writable anytime
FWCEPTC	✓
FWCEPRC0/1/2	✓
FWCLPTC	✓
FWCLPRC	✓
FWCMPTC	✓
FWEMPTC	✓
FWSDMPTC	✓

Table 30.4 Writable without software flow registers (2 of 2)

(✓ = writable, × = not writable)

Register name	Writable anytime
FWSDMPVC	✓
FWLBWMCi	✓
FWPC0i/1i/2i	✓
FWL23URMCi	✓

30.5 Functional Details

30.5.1 General Functionalities

General functionalities are separated in several types:

- Descriptors
- ESWM modes
- ESWM TPIDs
- ESWM debug functions
- ESWM watermark

30.5.1.1 Descriptors

To forward a frame to one or several port, the forwarding emits descriptors through the descriptor bus. All descriptor emitted by the forwarding engine shall follow the descriptor format described in [Table 30.5](#). In following sections when a function emits a descriptor it will be explained using the field name. Only fields that are not determined by the input local descriptor or the input frame will be explained.

Table 30.5 Descriptor field (1 of 2)

Field name FDESCR	Bit width	Field description	Values
DV	3	Values: • Bit i set to 0: Descriptor not valid for port i • Bit i set to 1: Descriptor valid for port i	Depends on descriptor. Will be explained per descriptor in following sections.
SEC	1	Values: • 0: Descriptor is unsecure • 1: Descriptor is secure	Depends on descriptor. Will be explained per descriptor in following sections.
CSD	6	Bits [5:0] (written as FDESCR.CSDi) define the CPU sub-destination for GWCA number i (CPU sub-destination is equivalent to AXI descriptor queue number)	Depends on descriptor. Will be explained per descriptor in following sections.
IPV	3	Internal priority value before agent internal priority remapping	Depends on descriptor. Will be explained per descriptor in following sections.
SPN	2	Port number from which the frame came from	Set to the source agent port number given by mfw_spn_fwd pin
TPL	16	Input frame total payload length.	Set to the TPL value given by agent through local descriptors
MINFO	24	Meta-information associated to a frame for CPU forwarding	Depends on descriptor. Will be explained per descriptor in following sections.
RV	1	Routing valid. Informs agents that routing rules should be fetched to forwarding engine through L23 update bus to update frame information	Depends on descriptor. Will be explained per descriptor in following sections.
RN	8	Routing number. Informs agents that routing number should be used when FDESCR.RV is set.	Depends on descriptor. Will be explained per descriptor in following sections.
VCTRL	3	VLAN control information	Set to the VCTRL value given by agent through local descriptors

Table 30.5 Descriptor field (2 of 2)

Field name FDESCR	Bit width	Field description	Values
RTGI	1	R-TAG in	Set to the RTGI value given by agent through local descriptors
SEQN	16	Frame sequence number	Depends on descriptor. Will be explained per descriptor in following sections.
FI	1	Values: 0: Input frame does not contain its FCS 1: Input frame contains its FCS	Set to the FI value given by agent through local descriptors
FW	1	Values: 0: If input frame contains its FCS, it was correct when given to the switch 1: If input frame contains its FCS, it was incorrect when given to the switch	Set to the FW value given by agent through local descriptors

30.5.1.2 ESWM Modes

ESWM modes define the general behavior of the switch using FWGC registers. The functions of ESWM modes are as follows:

- FWGC register sets the VLAN format that is used by the forwarding engine. Depending on this setting, all the agents should convert ethernet frames in the required format (for format conversion, see GWCA specification document [GWCA]).

30.5.1.3 ESWM TPIDs

ESWM TPIDs for C-TAG, S-TAG and R-TAG [801.2Q] [801.2CB] can be set by the user using FW TTC0 and FW TTC1 registers.

Functions:

- FW TTC0 register sets the S-TAG and C-TAG TPIDs [801.2Q].
- FW TTC1 register sets the R-TAG TPID [801.2CB].
- FW TTC0 and FW TTC1 register reset values are as per defined in IEEE standards [801.2Q] [801.2CB].

30.5.1.4 ESWM Frame Input Priority

Any incoming frame has an input priority given by forwarding engine depending on the switch mode FWGC.SVM, and the incoming priority settings in FW PBFCi register including the frame format. [Figure 30.47](#) describes the frame input priority assignment for a frame coming from port i (the ToS DSCP field decoding is fixed in hardware and described in [Table 30.6](#)).

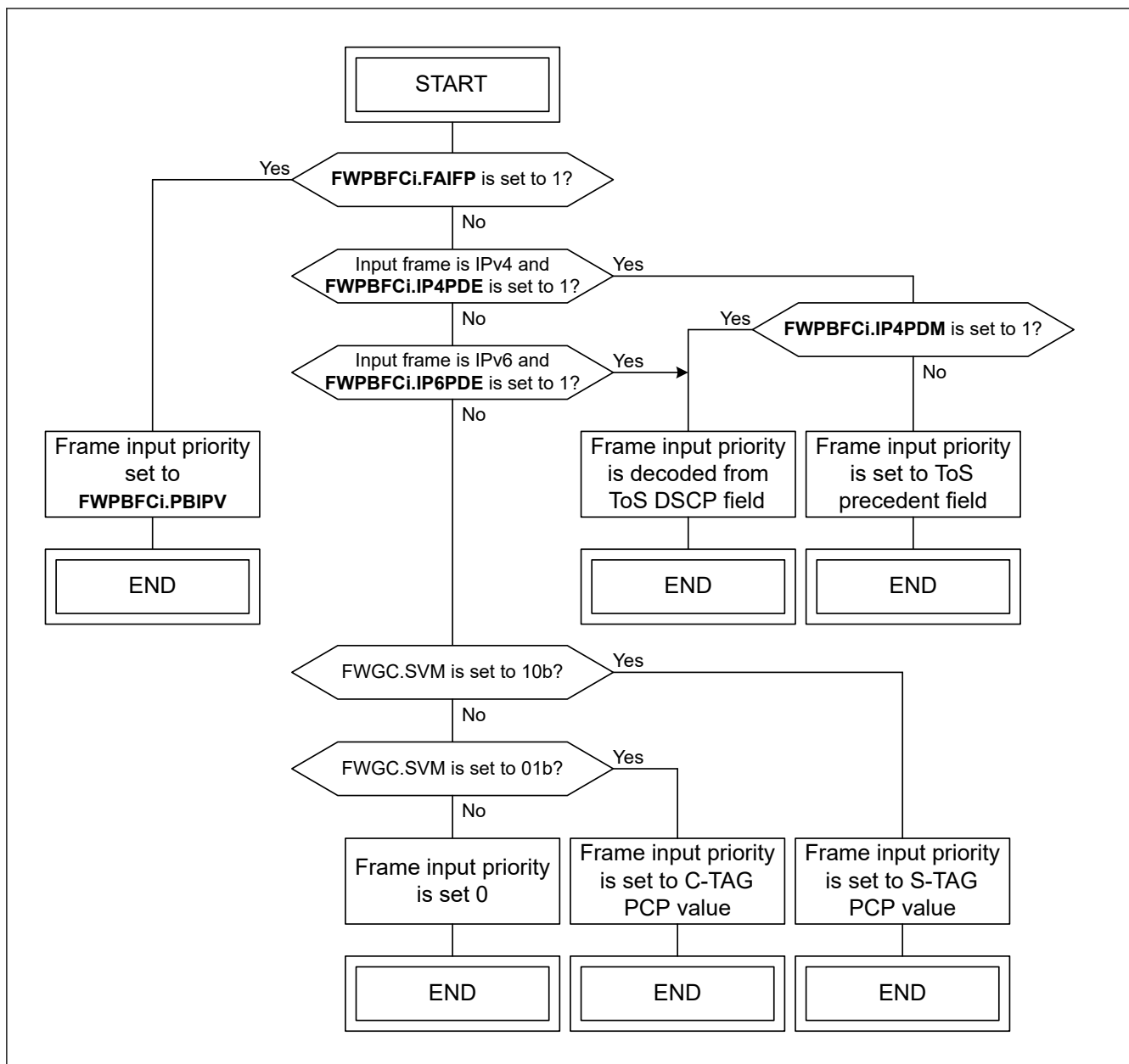


Figure 30.47 Priority assignment of frame input for port i

Note: HW: For cut-through forwarding, ToS and DSCP are not considered as frame input priority. Frame input priority is always extracted from S-TAG, S-TAG or set to 0.

Table 30.6 DSCP to priority mapping

DSCP values in decimal	Frame input priority
56	7
48	6
40, 46	5
32, 34, 36, 38	4
24, 26, 28, 30	3
16, 18, 20, 22	2
8, 10, 12, 14	1
Others	0

30.5.1.5 ESWM Debug Functions

30.5.1.5.1 CPU Exceptional Path

Exceptional path is used to forward faulty frames to CPU for diagnosis using FWCEPTC and FWCEPRC0/1/2 registers. When a frame is selected by exceptional path, an exceptional descriptor is emitted. The functions of exceptional path are as follows:

- FWCEPTC register sets the target to which type of faulty frames is forwarded
- FWCEPRC0/1/2 registers set which error frame is forwarded to the exceptional path.

Note: For exceptional path descriptor formats and associated error details, see [section 30.5.2.3.3. Direct Exceptional Descriptor](#), [\(4\) Layer 3 exceptional descriptor](#), [\(3\) Layer 2 exceptional descriptor](#), [section 30.5.2.6.3. Port Based Exceptional Descriptor](#) and [\(5\) Filter exceptional descriptor](#).

30.5.1.5.2 CPU Learning Path

Learning path is used to forward frames containing an unknown field to CPU for learning by using the FWCLPTC and FWCLPRC registers for the CPU to learn it. Learning can also be done by hardware using the FWPCi0 register and can be monitored using the FWEIS0i interrupt register. [Table 30.7](#) sums-up the field that can be learned in software (corresponding frames will be forwarded to learning path) and [Table 30.8](#). Any frame sent to learning path will be sent using learning descriptor described in [\(1\) Learning descriptor](#).

Note: HW: Any frame rejected by Watermark is not analyzed for learning. As a result, learning path is deactivated and hardware learning does not happen.

Functions:

- FWCLPTC register sets the target to which unknown frames will be forwarded.
- FWCLPRC registers set which field should be unknown for a frame to be forwarded to the learning path.
- FWPCi0 registers sets for which fields hardware learning should happen.
- FWEIS0i interrupt register notifies that a field could not be learnt in hardware.

Table 30.7 Software learning summary table

Field to learn	Learning type	Learning detection conditions for a frame coming from port i	Learning enable bit in FWCLPRC register	Hardware learning possible
Stream ID	Learning	FWPCi0.LHTA is set. At least a stream ID is valid. Stream ID is not set in the layer 3 table.	USIDLF	No
Destination MAC	Learning	FWPCi0.MACDSA is set. Destination MAC is not set in the MAC table.	UDMACLF	No
Source MAC	Learning	FWPCi0.MACSSA is set. Source MAC is a unicast address. Source MAC is not set in the MAC table.	USMACLF	Yes See Table 30.8 .
Source MAC port	Migration	FWPCi0.MACSSA is set. Source MAC is a unicast address. Source MAC is set in the MAC table. MAC.DV[i] is not set.	UPSMACLF	Yes See Table 30.8 .
VLAN	Learning	FWPCi0.VLANSA is set. VLAN is valid. VLAN ID is not set in the VLAN table.	UVLANLF	No

Table 30.8 Hardware learning summary table

Field to learn	Learning type	Hardware learning enable bit in FWPCi0 register for port i	Hardware learning fail interrupt in FWEIS0i register for port i	Hardware learning condition	Hardware learning disable condition (for certain frames HW learning can be disabled)
Source MAC	Learning	MACHLA	SMHLFS	Same as SW learning. See Table 30.7 .	FWPCi0.MACDSA is set. Destination MAC is set in the MAC table. MAC.HLD is set. Or FWPCi0.VLANSA is set. VLAN is valid. VLAN ID is set in the VLAN table. VLAN.HLD is set.
Source MAC port	Migration	MACHMA	SMHMFS	Same as SW learning. See Table 30.7 .	FWPCi0.MACDSA is set. Destination MAC is set in the MAC table. MAC.HLD is set. Or FWPCi0.VLANSA is set. VLAN is valid. VLAN ID is set in the VLAN table. VLAN.HLD is set.

Note: A multicast MAC address is an address with bit 40 set to 1.

(1) Learning descriptor

Learning descriptor fields are described in [Table 30.9](#).

Table 30.9 Learning descriptor fields

Field name FDESCR.	Values
DV	Learning descriptors are forwarded to GWCA number FWCLPTC.LPCS (port number 2 + FWCLPTC.LPCS).
SEC	FWCLPTC.LPSL
CSDj	FWCLPTC.LPCSD
IPV	FWCLPTC.LPIPV
MINFO	See Figure 30.48 and Table 30.10 .
RV	0
RN	ALL0
SEQN	Set to corresponding normal descriptor sequence number.

Byte	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0	—				FWDC[3:0]			
1	VL	SMPL	SML	DML	—			SIDL
2	—				SMHMF	SMHLF	SMHM	SMHL

Figure 30.48 Learning descriptor MINFO format**Table 30.10 Learning descriptor MINFO field explanation (1 of 2)**

Field name FDESCR.	Bit width	Field explanation	Values
FWDC	4	ForWarDing Code Present in all descriptors, allowing the type of forwarding to be identified	Fixed to 13 for Learning descriptors

Table 30.10 Learning descriptor MINFO field explanation (2 of 2)

Field name FDESCR.	Bit width	Field explanation	Values
SIDL	1	Stream ID Learn	See section 30.5.1.5.2. CPU Learning Path.
DML	1	Destination MAC Learn	See section 30.5.1.5.2. CPU Learning Path.
SML	1	Source MAC Learn	See section 30.5.1.5.2. CPU Learning Path.
SMPL	1	Source MAC Port Learn (migration)	See section 30.5.1.5.2. CPU Learning Path.
VL	1	VLAN Learn	See section 30.5.1.5.2. CPU Learning Path.
SMHL	1	Source MAC Hardware Learn* ¹	See section 30.5.1.5.2. CPU Learning Path.
SMHM	1	Source MAC Hardware Migration* ¹	See section 30.5.1.5.2. CPU Learning Path.
SMHLF	1	Source MAC Hardware Learn Fail* ¹	See section 30.5.1.5.2. CPU Learning Path.
SMHMF	1	Source MAC Hardware Migration Fail* ¹	See section 30.5.1.5.2. CPU Learning Path.

Note 1. "Hardware Learn/Migration" will set regardless of "Hardware Learn/Migration Fail". (If "Hardware Learn Fail" is set, "Hardware Learn" is always set.)

30.5.1.5.3 Mirroring Paths

Forwarding engine contains 3 mirroring paths:

- CPU mirroring path
- Ethernet mirroring path
- Source-destination mirroring path

(1) CPU mirroring path

CPU Mirroring path is used to copy frames to a CPU for monitoring purpose using FWCMPTC register. [Table 30.11](#) describes CPU mirroring selection conditions depending on the forwarding mechanism selected by forwarding flow. When a frame is selected for CPU mirroring, a mirroring descriptor is emitted ((4) [Mirroring descriptor](#)).

Functions:

- FWCMPTC register sets the target to which mirrored frames will be forwarded.

Table 30.11 CPU mirroring selection conditions

Descriptor selected by forwarding flow	Mirroring selection conditions
Cut-through normal descriptor (Figure 30.53)	Cut-through rule corresponding FWCTTCi1.CTCME is set.
Layer 3 normal descriptor (Figure 30.85 , Figure 30.86)	L3.CME is set for stream ID
Layer 2 normal descriptor (Figure 30.95 , Figure 30.96)	If {MAC.CME is set for destination MAC or VLAN.CME is set} when MAC destination address and VLAN ID are both found and MAC.SL = VLAN.SL. If MAC.CME is set for destination MAC when MAC destination only is found or if MAC destination address and VLAN ID are both found and MAC.SL & !VLAN.SL. If VLAN.CME is set when VLAN ID only is found or if MAC destination address and VLAN ID are both found and !MAC.SL & VLAN.SL.
Port based normal descriptor (Figure 30.99 , Figure 30.100)	If FWPBFCi.PBCME is set for incoming frame source port.

(2) Ethernet mirroring path

Ethernet Mirroring path is used to copy frames to an ethernet agent for monitoring purpose using FWEMPTC register. [Table 30.12](#) describes Ethernet mirroring selection conditions depending on the forwarding mechanism selected by forwarding flow. When a frame is selected for Ethernet mirroring, a mirroring descriptor is emitted ([\(4\) Mirroring descriptor](#)).

Functions:

- FWEMPTC register sets the target to which mirrored frames will be forwarded.

Table 30.12 Ethernet mirroring selection conditions

Descriptor selected by forwarding flow	Mirroring selection conditions
Cut-through normal descriptor (Figure 30.53)	Cut-through rule corresponding FWCTTCi1.CTEME is set.
Layer 3 normal descriptor (Figure 30.85 , Figure 30.86)	L3.EME is set for stream ID
Layer 2 normal descriptor (Figure 30.95 , Figure 30.96)	If {MAC.EME is set for destination MAC or VLAN.EME is set} when MAC destination address and VLAN ID are both found and MAC.SL = VLAN.SL. If MAC.EME is set for destination MAC when MAC destination only is found or if MAC destination address and VLAN ID are both found and MAC.SL & !VLAN.SL. If VLAN.EME is set when VLAN ID only is found or if MAC destination address and VLAN ID are both found and !MAC.SL & VLAN.SL.
Port based normal descriptor (Figure 30.99 , Figure 30.100)	If FWPBFCi.PBEME is set for incoming frame source port.

(3) Source-destination Mirroring path

Mirroring path is used to copy frames to an Ethernet port or a CPU for monitoring purpose using FWSDMPTC and FWSDMPVC register. When a frame is selected for Source-destination mirroring, a mirroring descriptor is emitted ([\(4\) Mirroring descriptor](#)).

Functions:

- FWSDMPTC register sets the target to which mirrored frames will be forwarded.
- FWSDMPVC register sets the source from which should come and the destination where should go a frame to be mirrored.
- A frame is selected for source-destination mirroring if the descriptor selected by forwarding flow is {Layer 3 normal descriptor ([Figure 30.85](#), [Figure 30.86](#)) or IP normal descriptor ([Figure 30.95](#), [Figure 30.96](#)) or Layer 2 normal descriptor ([Figure 30.95](#), [Figure 30.96](#)) or Port based normal descriptor ([Figure 30.99](#), [Figure 30.100](#))}, if the frame source port is set in FWSDMPVC.SDMSV register and if FDESCR.DV & FWSDMPVC.SDMDV is not null.

(4) Mirroring descriptor

Mirroring descriptor fields are described in [Table 30.13](#).

Table 30.13 Mirroring descriptor fields (1 of 2)

Field name FDESCR.	Values for CPU mirroring	Values for Ethernet mirroring	Values for Source-destination mirroring
DV	Mirroring descriptors are forwarded to GWCA number FWCMPVC.CMPCS (port number 2 + FWCMPVC.CMPCS)	Mirroring descriptors are forwarded to ETHA number FWEMPTC.EMPPS (port number FWEMPTC.EMPPS)	Mirroring descriptors are forwarded to port number FWSDMPTC.SDMPPS
SEC	FWCMPVC.CMPSL	FWEMPTC.EMPSL	FWSDMPTC.SDMPSL
CSDj	FWCMPVC.CMPCSD	All0	All0
IPV	If FWCMPVC.CMPIPU is set, FWCMPVC.CMPIPV If FWCMPVC.CMPIPU is not set, same IPV as normal descriptor	If FWEMPTC.EMPIPU is set, FWEMPTC.EMPIPV If FWEMPTC.EMPIPU is not set, same IPV as normal descriptor	If FWSDMPTC.SDMPIPU is set, FWSDMPTC.SDMPIPV If FWSDMPTC.SDMPIPU is not set, same IPV as normal descriptor
MINFO	See Figure 30.49 and Table 30.14 .	See Figure 30.49 and Table 30.14 .	See Figure 30.49 and Table 30.14 .
RV	0	0	0

Table 30.13 Mirroring descriptor fields (2 of 2)

Field name FDESCR.	Values for CPU mirroring	Values for Ethernet mirroring	Values for Source-destination mirroring
RN	ALL0	ALL0	ALL0
SEQN	Set to corresponding normal descriptor sequence number	Set to corresponding normal descriptor sequence number	Set to corresponding normal descriptor sequence number

Byte	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0	—				FWDC[3:0]			
1	NDFB[7:0]							
2	—							

Figure 30.49 Mirroring descriptor MINFO format**Table 30.14 Mirroring descriptor MINFO field explanation**

Field name FDESCR.	Bit width	Field explanation	Values
FWDC	4	ForWarDing Code Present in all descriptors, it allows identifying the type of forwarding.	Fixed to 14 for Mirroring descriptors
NDFB	8	Normal Descriptor First Byte	Set to mirrored frame corresponding normal descriptor FDESCR.MINFO[7:0] value

30.5.1.6 Watermark

ESWM watermark defines the switch behavior in case of overflow using FWLBWMCi and FWCEPRC2 registers and can be monitored using FWEIS0i interrupt register.

Functions:

- FWLBWMCi is used to set which frame should be discarded by level-based watermark [COMA].
- FWCEPRC2 register enables exceptional path for error descriptors.
- FWEIS0i interrupt register notifies that a frame coming from port i has been rejected because of watermark.
- Watermark uses has frame priority the IPV value given by forwarding mechanism before filtering. This correspond to the FDESCR.IPV in the normal descriptor emitted by the selected forwarding.
- Watermark uses as DEI value the value extracted from the input frame which is the S-TAG DEI value when FWGC.SVM is set to 10b, the C-TAG DEI value when FWGC.SVM is set to 01b, and 0 when FWGC.SVM is set to 00b.

Note:

- SW: The watermark is an indispensable functionality for switch QoS behavior definition, it is not advised to disable it.
- Error frames are not going through watermark.

(1) Watermark errors

Table 30.15 describes the Watermark errors. Instead of being discarded by forwarding engine, an error frame can be forwarded to exceptional path using Watermark exceptional descriptor described in (2) [Watermark exceptional descriptor](#) by setting corresponding error exceptional bit. For error set conditions, refer to corresponding error interrupt register explanations.

Table 30.15 Watermark errors (1 of 2)

Error	Error name	Error interrupt	Error exceptional bit
WMCf	Watermark Critical Filtering	FWEIS0i.WMCFS	FWCEPRC2.FWMFEF

Table 30.15 Watermark errors (2 of 2)

Error	Error name	Error interrupt	Error exceptional bit
WMFF	Watermark Flush Filtering	FWEIS0i.WMFFS	FWCEPRC2.FWMFEF
WMISF	Watermark IPV Secure Filtering	FWEIS0i.WMISFS	FWCEPRC2.FWMFEF
WMIUF	Watermark IPV Unsecure Filtering	FWEIS0i.WMIUFS	FWCEPRC2.FWMFEF

(2) Watermark exceptional descriptor

Watermark exceptional descriptor fields are described in [Table 30.16](#).

Table 30.16 Watermark exceptional descriptor fields

Field name FDESCR.	Values
DV	Exceptional descriptors are forwarded to GWCA number FWCEPTC.EPCS (port number 2 + FWCEPTC.EPCS)
SEC	FWCEPTC.EPSL
CSDj	FWCEPTC.EPCSD
IPV	FWCEPTC.EPIPV
MINFO	See Figure 30.50 and Table 30.17 .
RV	0
RN	ALL0
SEQN	Set to frame input sequence number

Byte	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0	—				FWDC[3:0]			
1	—				WMIUF	WMISF	WMFF	WMCF
2	—							

Figure 30.50 Watermark exceptional descriptor MINFO format**Table 30.17 Watermark exceptional descriptor MINFO field explanation**

Field name FDESCR.	Bit width	Field explanation	Values
FWDC	4	ForWarDing Code Present in all descriptors, it allows identifying the type of forwarding.	Fixed to 4 for Watermark exceptional descriptors
WMCF	1	Watermark Critical Filtering	See (1) Watermark errors .
WMFF	1	Watermark Flush Filtering	See (1) Watermark errors .
WMISF	1	Watermark IPV Secure Filtering	See (1) Watermark errors .
WMIUF	1	Watermark IPV Unsecure Filtering	See (1) Watermark errors .

30.5.2 Forwarding/Routing

Forwarding engine contains different types of forwarding which are:

- Cut-through forwarding
- Direct descriptor forwarding
- L3 forwarding/routing
- L2 forwarding
- Port based forwarding

30.5.2.1 Cut-through Forwarding

Cut-through forwarding is used to forward high priority frames without storing them in the local memory to suppress the frame reception time impact on the latency. Under the condition of ESWM insure best latency (no busy traffic on MFWD and no TAG frame) of

$0.7 \mu s + \text{"PHY clock 4 cycle"}(ns)$, $4.6 \mu s + \text{"PHY clock 4 cycle"}(ns)$ and $46 \mu s + \text{"PHY clock 4 cycle"}(ns)$ at respectively 10/5/2.5/1 Gbps, 100 Mbps and 10 Mbps [ETHA].

Depending on the setting conditions, the value may be smaller than these.

Generally, "Cut-through" only checks 6 bytes (Destination MAC Address) of a frame head and starts frame forwarding after this check immediately. Since internal data bus width in this switch is wider than 6 bytes, it needs more data bytes to start frame forwarding. Because of this logic, "Cut-through" forwarding time of this switch needs above times.

The Cut-through forwarding block diagram is represented in [Figure 30.51](#).

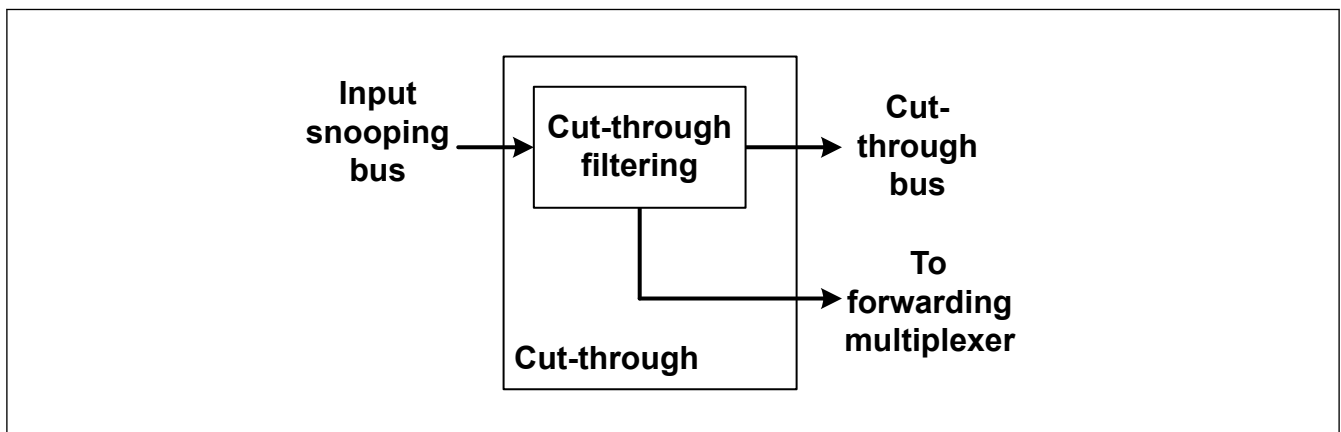


Figure 30.51 Cut-through forwarding block diagram

The Cut-through forwarding function has one functional block:

- **Cut-through filtering:** This function filters incoming frame first beat of data to select set frames for cut-through forwarding, creates the cut-through bus protocol destined to time agents [ETHA] and send required information to forwarding multiplexer to apply store and forward to cut-through frames.

- Note:**
- Forwarding multiplexer contains the forwarding flow. The forwarding flow is explained per forwarding type. For Cut-through, related Forwarding flow is described in [Figure 30.53](#)
 - HW: The usage "From-one-to-one between same PHY speed" is recommended. So in case of forwarding engine back pressure, it can happen that cut-through frames are cut in the middle and lost.
 - HW: P-frames cannot be forwarded by cut-through.
 - HW: Cut-through forwarding should not be set for ingress ports in Port-based VLAN mode because cut-through can only forward frame as they were received. ([ETHA] EAVCC.VIM set to 0)
 - HW: Watermark is not applied on Cut-through frames.
 - HW: Cut-through frames can be ingress tagging by FWCTGCI0.CTVCRCL but not egress tagging.
 - HW: L23 update is not applied on Cut-through frames.
 - HW: Data padding applied on Cut-through frames.
 - HW: Cut-through frames can be shaped using TAS but not CBS.
 - HW: Cut-through frames will be forward in cut-through mode as there are received and will not be updated.
 - HW: For cut-through, only mirroring set in FWCTTCI1/i2 register is available. All other debugging and mirroring are disabled.
 - HW: Cut-through frames will not be filtered by [RMAC] MAC Address filtering on Reception. Because [ETHA] and [GWCA] reception filter information are ignored.
For example, a frame which is having a nibble alignment error ([ETHA] reception filter information) is ignored by cut through forwarding. And a frame is transmitted out by padding bits of data to complete a byte.

30.5.2.1.1 Cut-through Filtering

Cut-through filtering is used to identify cut-through frames using FWCTGCI0/i1 and FWCTSCI0/i1/i2/i3/i4 registers. Cut-through filtering and register utilization are summarized in [Figure 30.52](#).

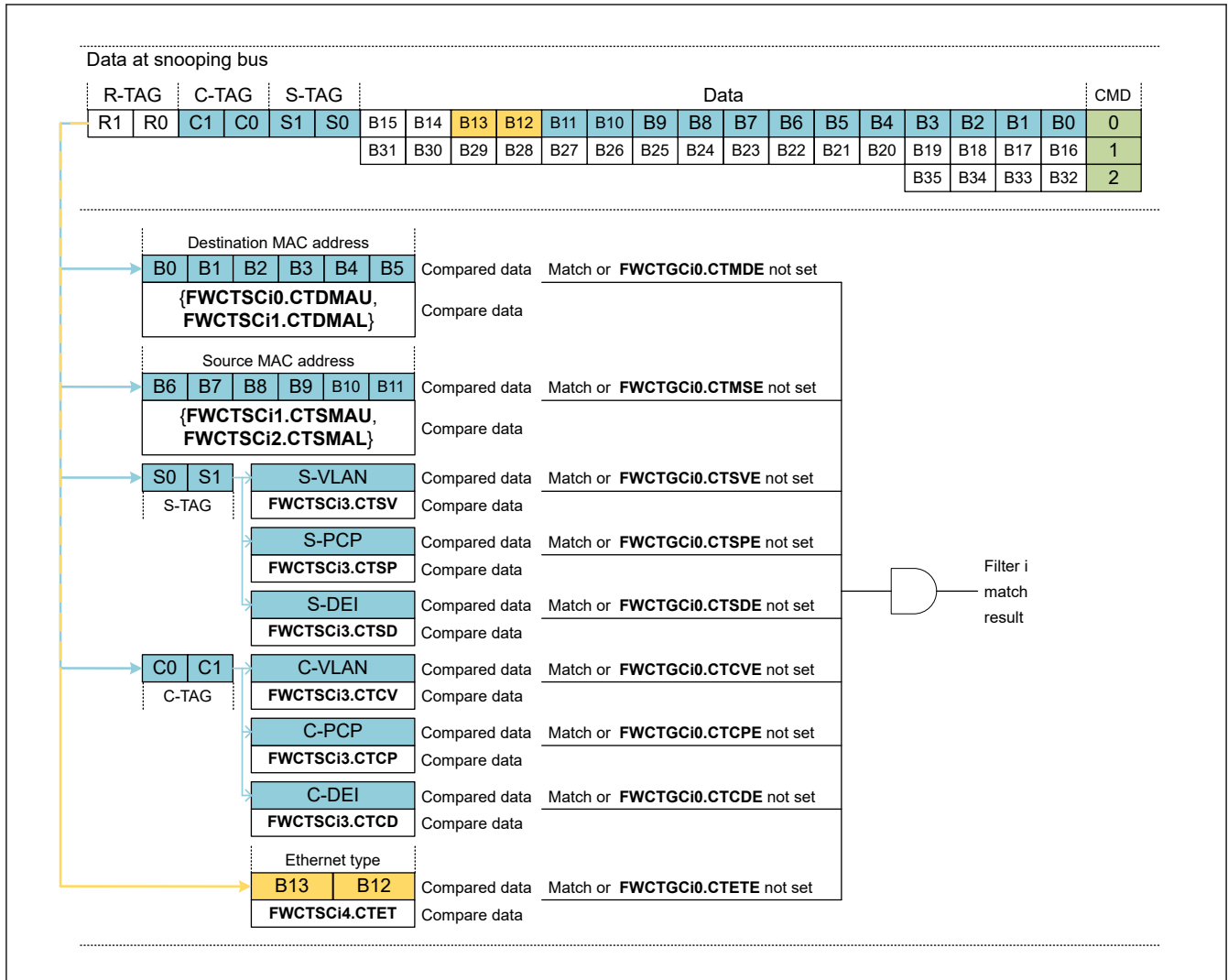


Figure 30.52 Cut-through filtering

Functions:

- A cut-through rule can only match frames from corresponding source port FWCTSCi4.CTSPN.
- If several Cut-through rules match the same frame, the Cut-through rules with the highest number will be selected.
- A cut-through rule is only valid when corresponding FWCTTCi0.CTDV is different from All0.
- A cut-through matching frames is forwarded by cut-through and store and forward independently of any error that could have happened in the ingress agent.
- For register FWCTGCI0 bit 0-8 set to 0 (all fields compare match is disabled), Cut through becomes “port based cut through” by FWCTTCi0.CTDV.

30.5.2.1.2 Cut-through Forwarding

Cut-through forwarding function creates the cut-through bus protocol destined to time agents [ETHA] using FWCTTCi0, FWCTGCI0/i1 registers. It is used to send frames while receiving them to reduce latency though cut-through bus.

Functions:

- FWCTTCi0 register is to determine the target port for cut-through. All ports set in FWCTTCi0.CTDV[PORT_TIME_N-1:0] and not set in FWCTTCi0.CTDFM are targeted for rule i cut-through forwarding. Any frame received from an agent with an error will still be forwarded/mirrored by cut-through forwarding.
- FWCTGCI1.CTMT, FWCTGCI0.CTVCTRL, and FWCTGCI0.CTFI registers are respectively used as Maximum Time on the PHY, VLAN control and FCS in for the matching cut-through frame.

- Note:
- HW: Ethernet timestamping functionalities are not available during cut-through forwarding [ETHA]. E.g. while forwarding frames to RMAC [RMAC], the ethernet agent will fix "Timestamp capture", "Insert egress timestamp" and "Calculate residence time" TX interface bit to 0.
 - HW: For cut-through forwarding, all (From & To) PHY speeds have to be the same [ETHA] [RMAC].
 - SW: FWCTGCi0.CTRTGI and FWCTGCi0.CTVCTRL should always correspond to the ingress frame format. If then do no correspond, errors can happen.

30.5.2.1.3 Store and Forward Forwarding

Store and forward forwarding function apply store and forward to cut-through frames using FWCTTCi0/i1/i2. Cut-through forwarding flow for Store and forward forwarding is described in [Figure 30.53](#).

Functions:

- FWCTTCi0 register is to determine the target port for store and forward. All ports set in FWCTTCi0.CTDV[2:0] and set in FWCTTCi0.CTDFM are targeted for rule i store and forward forwarding, so are ports set in FWCTTCi0.CTDV[3:2]. Any frame received from an agent with an error will still be forwarded/mirrored in store and forward mode.
- FWCTTCi1/i2 registers sets the forwarding information.

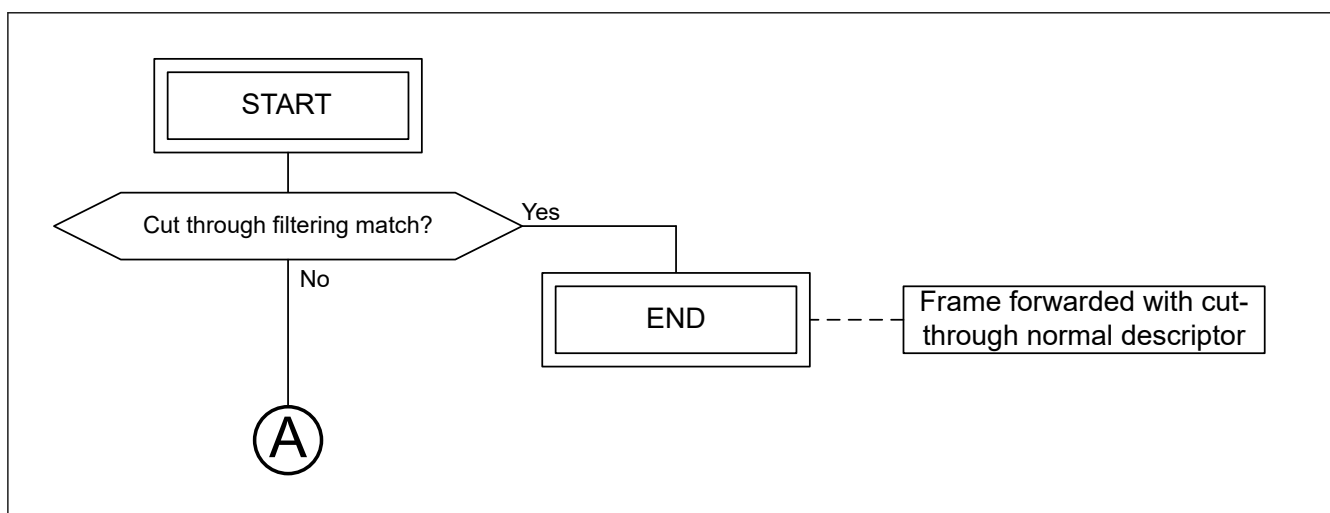


Figure 30.53 Cut-through forwarding flow

Note: Link A links to Figure 2.

(1) Cut-through normal descriptor

Cut-through normal descriptor fields for a frame matching cut-through rule number i are described in [Table 30.18](#).

Table 30.18 Cut-through normal descriptor fields

Field name FDESCR.	Values
DV	{FWCTTCi0.CTDV[3:2], {FWCTTCi0.CTDV[1] & FWCTTCi0.CTDFM}}
SEC	1
CSDj	FWCTTC2ji.CTCSD
IPV	If FWCTTCi1.CTIPU is set, set to FWCTTCi1.CTIPV If FWCTTCi1.CTIPU not set and FWGC.SVM is set to 10b, set to frame input S-TAG PCP If FWCTTCi1.CTIPU not set and FWGC.SVM is set to 01b, set to frame input C-TAG PCP If FWCTTCi1.CTIPU not set and FWGC.SVM is set to 00b, set to frame input 0
MINFO	See Figure 30.54 and Table 30.19
RV	0
RN	ALL0
SEQN	Set to frame input sequence number

Byte	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0	—				FWDC[3:0]			
1	—					CTRN[2:0]		
2	—							

Figure 30.54 Cut-through normal descriptor MINFO format

Table 30.19 Cut-through normal descriptor MINFO field explanation

Field name FDESCR.	Bit width	Field explanation	Values
FWDC	4	ForWarDing Code Present in all descriptors, allowing the type of forwarding to be identified	Fixed to 0 for Cut-through normal descriptors
CTRN	3	Cut-through rule number	Set to the cut-through rule number which forwarded the frame
—	- -	Reserved field	Set to 0

30.5.2.2 Integrity Check

Integrity check is used to check if a frame can be forwarded or if it contains an error. The Integrity check block diagram is represented in [Figure 30.55](#). Integrity check flow is described in [Figure 30.56](#).

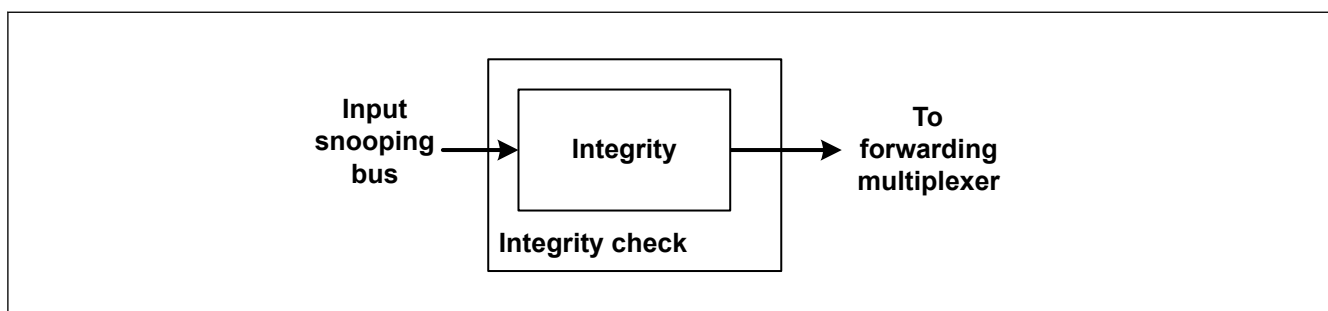


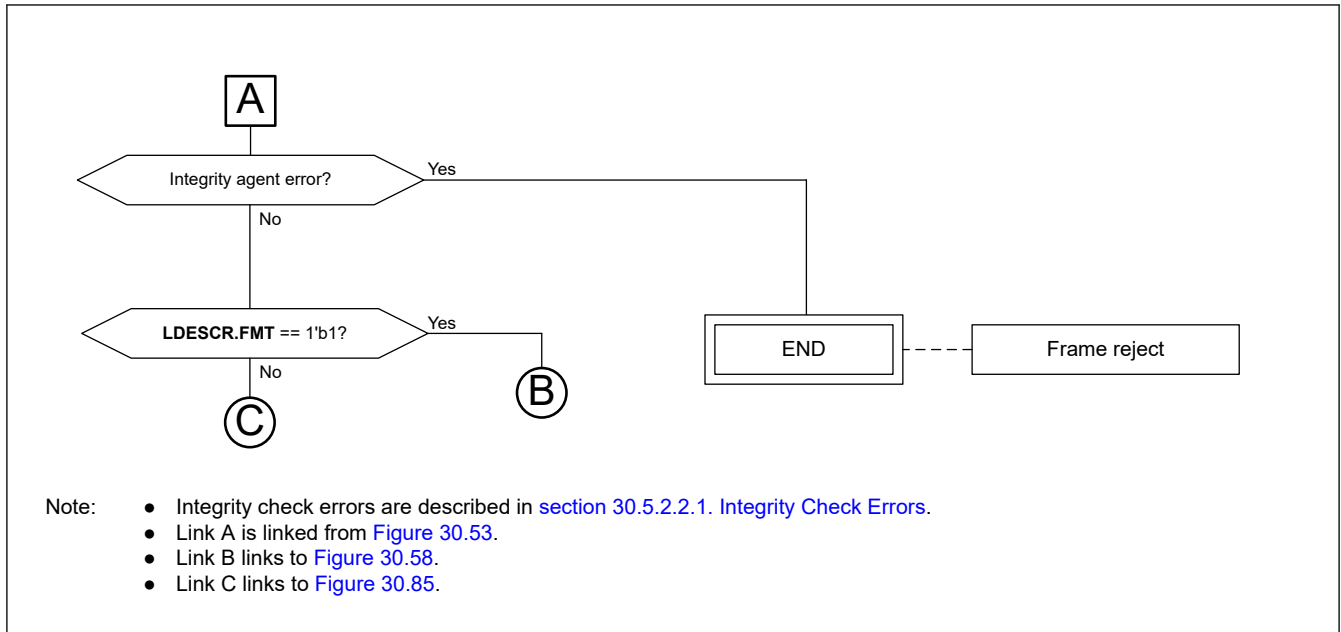
Figure 30.55 Integrity check block diagram

The Integrity check function has one functional block:

- Integrity: The ether type 0x8100, 0x88A8 and 0xF1C1 are ignored and the next ether type field are extracted. However, invalid frames (exp CC-TAG) are judged to have no header, and the first ether type field is the extract.
- Integrity: This functions check if an error has been received from the input agent or if incoming frames don't respect the required format.
- Integrity check for both IPv4 filtering and IPv6 filtering will happen based on ether type field. i.e. Ether type is only used for parsing frame type in MFWD integrity check and is irrespective of header length or TPL size of the required frame type.

Note:

- Forwarding multiplexer contains the forwarding flow. The forwarding flow is explained per forwarding type. For integrity check, related Forwarding flow is described in [Figure 30.56](#).

**Figure 30.56 Integrity check flow**

30.5.2.2.1 Integrity Check Errors

[Table 30.20](#) describes the Integrity check errors. Instead of being discarded by forwarding engine, an error frame can be forwarded to exceptional path using Integrity exceptional descriptor described in [section 30.5.2.2.2. Integrity Exceptional Descriptor](#) by setting its corresponding error exceptional bit.

Table 30.20 Integrity check errors (1 of 2)

Error	Error name	Error interrupt/Explanation	Error exceptional bit
Integrity agent errors			
GTFE	GWCA TAG filtering error	Happens when a frame is received from a GWCA and LDESCR.SAEF[0] is set [GWCA]	FWCEPRC0.GTFEEF
GAXE	GWCA AXI Error	Happens when a frame is received from a GWCA and LDESCR.SAEF[3] is set [GWCA]	FWCEPRC0.GAXEEF
GSEGE	GWCA Sequence Error	Happens when a frame is received from a GWCA and LDESCR.SAEF[4] is set [GWCA]	FWCEPRC0.GSEQEEF
GDNE	GWCA Descriptor Number Error	Happens when a frame is received from a GWCA and LDESCR.SAEF[6] is set [GWCA]	FWCEPRC0.GDNEEF
ETFE	Ethernet TAG filtering Error	Happens when a frame is received from an Ethernet agent and LDESCR.SAEF[0] is set [ETHA]	FWCEPRC0.ETFEF
EPHYE	Ethernet TAG PHY Error	Happens when a frame is received from an Ethernet agent and LDESCR.SAEF.EC0 is set to 4'd1 [ETHA]	FWCEPRC0.EPHYEEF
EPCRCE	Ethernet PCH CRC Error	Happens when a frame is received from an Ethernet agent and LDESCR.SAEF.EC0 is set to 4'd2 [ETHA]	FWCEPRC0.EPCRCEEF
ENIBE	Ethernet Nibble Error	Happens when a frame is received from an Ethernet agent and LDESCR.SAEF.EC0 is set to 4'd3 [ETHA]	FWCEPRC0.ENIBEEF
EFCSE	Ethernet FCS Error	Happens when a frame is received from an Ethernet agent and LDESCR.SAEF.EC0 is set to 4'd4 [ETHA]	FWCEPRC0.EFCSEEF
EFFME	Ethernet Final Fragment Missing Error	Happens when a frame is received from an Ethernet agent and LDESCR.SAEF.EC0 is set to 4'd5 [ETHA]	FWCEPRC0.EFFMEEF
ECFSE	Ethernet C-Fragment SMD Error	Happens when a frame is received from an Ethernet agent and LDESCR.SAEF.EC0 is set to 4'd6 [ETHA]	FWCEPRC0.ECFSEEF
ECFFCE	Ethernet C-Fragment FRAG_COUNT Error	Happens when a frame is received from an Ethernet agent and LDESCR.SAEF.EC0 is set to 4'd7 [ETHA]	FWCEPRC0.ECFFCEEF

Table 30.20 Integrity check errors (2 of 2)

Error	Error name	Error interrupt/Explanation	Error exceptional bit
ERFF	Ethernet RMAC Frame Filtered	Happens when a frame is received from an Ethernet agent and LDESCR.SAEF.EC0 is set to 4'd8 [ETHA]	FWCEPRC0.ERFFEF
ERPOO	Ethernet Reception Partially Out of Operation	Happens when a frame is received from an Ethernet agent and LDESCR.SAEF.EC0 is set to 4'd9 [ETHA]	FWCEPRC0.ERPOOEF
EBOE	Ethernet Buffer Overflow Error	Happens when a frame is received from an Ethernet agent and LDESCR.SAEF.EC1 is set to 2'd1 [ETHA]	FWCEPRC0.EBOEEF
EUE	Ethernet Undersize Error	Happens when a frame is received from an Ethernet agent and LDESCR.SAEF.EC1 is set to 2'd2 [ETHA]	FWCEPRC0.EUEEF
EOE	Ethernet Oversize Error	Happens when a frame is received from an Ethernet agent and LDESCR.SAEF.EC1 is set to 2'd3 [ETHA]	FWCEPRC0.EOEEF

30.5.2.2.2 Integrity Exceptional Descriptor

Integrity exceptional descriptor fields are described in [Table 30.21](#).

Table 30.21 Integrity exceptional descriptor fields

Field name FDESCR.	Values
DV	Exceptional descriptors are forwarded to GWCA number FWCEPTC.EPCS (port number PORT_TIME_N+ FWCEPTC.EPCS)
SEC	FWCEPTC.EPSL
CSDj	FWCEPTC.EPCSD
IPV	FWCEPTC.EPIPV
MINFO	Refer to Figure 30.57 and Table 30.22
RV	1'b0
RN	ALL0
SEQN	Set to frame input sequence number

Byte	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0	—				FWDC[3:0]			
1	—				L4F	IP6F	IP4F	TYPEF
2	—							

Figure 30.57 Integrity exceptional descriptor MINFO format**Table 30.22 Integrity exceptional descriptor MINFO field explanation (1 of 2)**

Field name FDESCR.	Bit width	Field explanation	Values
FWDC	4	ForWarDing Code Present in all descriptors, it allows identifying the type of forwarding.	Fixed to 1 for Integrity exceptional descriptors
TYPEF	1	Type filtering	Refer to section 30.5.2.2.1. Integrity Check Errors
IP4F	1	IPv4 filtering	Refer to section 30.5.2.2.1. Integrity Check Errors
IP6F	1	IPv6 filtering	Refer to section 30.5.2.2.1. Integrity Check Errors
L4F	1	Layer 4 filtering	Refer to section 30.5.2.2.1. Integrity Check Errors

Table 30.22 Integrity exceptional descriptor MINFO field explanation (2 of 2)

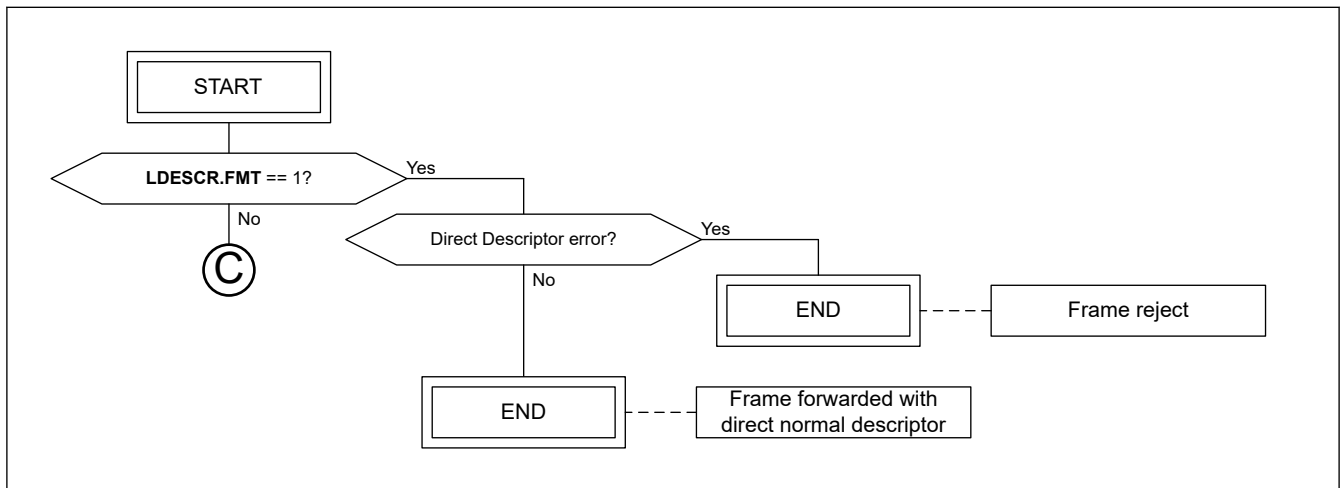
Field name FDESCR.	Bit width	Field explanation	Values
—	—	Reserved field	Set to 0

30.5.2.3 Direct Descriptor Forwarding

Direct descriptor forwarding is used to forward local direct descriptors (LDESCR.FMT [GWCA] set to 1) received from a slow interface [GWCA] using FWPCi1 and FWCEPRC0 register and can be monitored using FWEIS0i interrupt register. Direct descriptor forwarding flow is described in [Figure 30.58](#).

Functions:

- FWPCi1.DDE register is used to enable direct descriptor reception from port i.
- FWPCi1.DDSL register is used to enable secure direct descriptor reception from port i when direct descriptor reception from port I is enabled.
- FWCEPRC0 register enables exceptional path for error descriptors.
- FWEIS0i interrupt register notifies that a direct descriptor has been rejected.

**Figure 30.58 Direct descriptor forwarding flow**

- Note:
- Direct descriptor errors are described in [section 30.5.2.3.1. Direct Descriptor Forwarding Errors](#).
 - Direct normal descriptor format is described in [section 30.5.2.3.2. Direct Normal Descriptor](#).
 - Link C links to [Figure 30.85](#).

30.5.2.3.1 Direct Descriptor Forwarding Errors

[Table 30.23](#) describes the Direct descriptor forwarding errors. Instead of being discarded by forwarding engine, an error frame can be forwarded to exceptional path using Direct exceptional descriptor described in [section 30.5.2.3.3. Direct Exceptional Descriptor](#) by setting corresponding error exceptional bit. For error set conditions, refer to corresponding error interrupt register explanations.

Table 30.23 Direct descriptor errors

Error	Error name	Error interrupt	Error exceptional bit
DDE	Direct Descriptor Error	FWEIS0i.DDES	FWCEPRC0.DDEEF
DDSF	Direct Descriptor Security Filtering	FWEIS0i.DDSES	FWCEPRC0.DDFSFEF
NTF	No Target Filtering	FWEIS0i.DDNTFS	FWCEPRC2.DDNTFEF

30.5.2.3.2 Direct Normal Descriptor

[Table 30.24](#) describes fields of direct normal descriptor. For LDESCR explanation, see GWCA specification [GWCA].

Table 30.24 Direct normal descriptor fields

Field name FDESCR.	Values
DV	LDESCR.DV
SEC	LDESCR.SEC
CSD	LDESCR.CSD
IPV	LDESCR.IPV
MINFO	Refer to Figure 30.59 and Table 30.25
RV	LDESCR.RV
RN	If LDESCR.RV is set, set to LDESCR.RN If LDESCR.RV is not set, set to 0
SEQN	Set to frame input sequence number

Byte	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0	—				FWDC[3:0]			
1	—							
2								

Figure 30.59 MINFO format of direct normal descriptor**Table 30.25 MINFO field explanation of direct normal descriptor**

Field name FDESCR.	Bit width	Field explanation	Values
FWDC	4	Forwarding Code Present in all descriptors, it allows identifying the type of forwarding.	Fixed to 2 for direct normal descriptors
—	--	Reserved field	Set to 0

30.5.2.3.3 Direct Exceptional Descriptor

Direct exceptional descriptor fields are described in [Table 30.26](#).

Table 30.26 Direct exceptional descriptor fields

Field name FDESCR.	Values
DV	Exceptional descriptors are forwarded to GWCA number FWCEPTC.EPCS (port number 2 + FWCEPTC.EPCS)
SEC	FWCEPTC.EPSL
CSDj	FWCEPTC.EPCSD
IPV	FWCEPTC.EPIPV
MINFO	Refer to Figure 30.60 and Table 30.27
RV	0
RN	ALL0
SEQN	Set to frame input sequence number

Byte	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0	—				FWDC[3:0]			
1	—				NTF	DDFF	DDSF	DDF
2	—							

Figure 30.60 Direct exceptional descriptor MINFO format

Table 30.27 Direct exceptional descriptor MINFO field explanation

Field name FDESCR.	Bit width	Field explanation	Values
FWDC	4	ForWarDing Code Present in all descriptors, it allows identifying the type of forwarding.	Fixed to 3 for Direct exceptional descriptors
DDF	1	Direct Descriptor Filtered	See section 30.5.2.3.1. Direct Descriptor Forwarding Errors.
DDSF	1	Direct Descriptor Security Filtered	See section 30.5.2.3.1. Direct Descriptor Forwarding Errors.
DDFF	1	Direct Descriptor Format Filtered	See section 30.5.2.3.1. Direct Descriptor Forwarding Errors.
NTF	1	No Target Filtered	See section 30.5.2.3.1. Direct Descriptor Forwarding Errors.

30.5.2.4 Layer 3 Forwarding/Routing/Filtering

Figure 30.61 shows the block diagram of Layer 3 forwarding/routing/filtering. This block is denoted by L3, but it contains all the stream-based functions of the forwarding engine, even the Layer 2 streams.

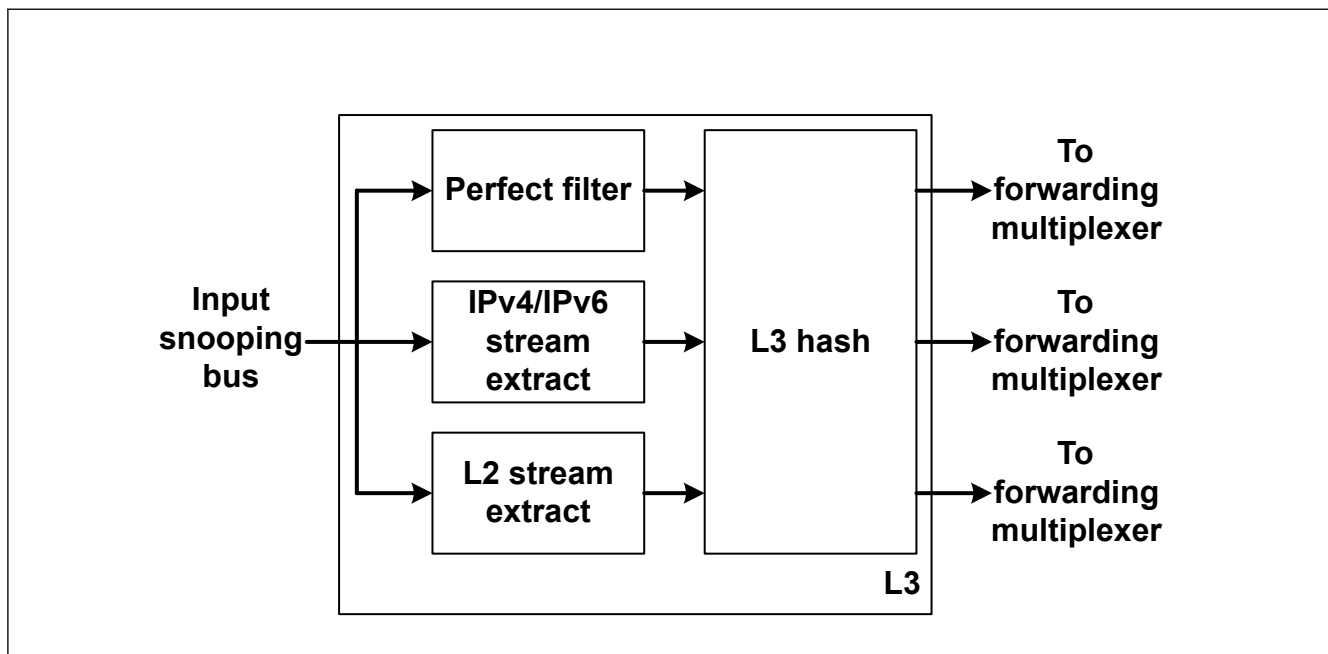


Figure 30.61 Block diagram of Layer 3 forwarding/routing/filtering

The Layer 3 forwarding/routing/filtering function is separated in four functional blocks:

- Perfect filter: This block snoops all the required information from the Forwarding engine snooping bus and create a user-defined stream.
- IPv4/IPv6 stream extract: This block snoops all the required information from the Forwarding engine snooping bus and create an IPv4/IPv6 stream.

- L2 stream extract: This block snoops all the required information from the Forwarding engine snooping bus and create a L2 stream.
- L3 hash: This block handles the stream meta information such as forwarding, routing and filtering information using a hash function.

Note: Forwarding multiplexer contains the forwarding flow. The forwarding flow is explained by forwarding type. For Layer 3 forwarding/routing/filtering, related forwarding flows are described in [Figure 30.85](#), [Figure 30.86](#) and [Figure 30.87](#).

30.5.2.4.1 Perfect Filter

The perfect filter is used to identify streams. This filter has the advantage to be flexible and can identify a stream without any statistic error but, because of its high cost, is available in a limited number (16 streams available).

The Perfect filter block diagram is represented in [Figure 30.62](#).

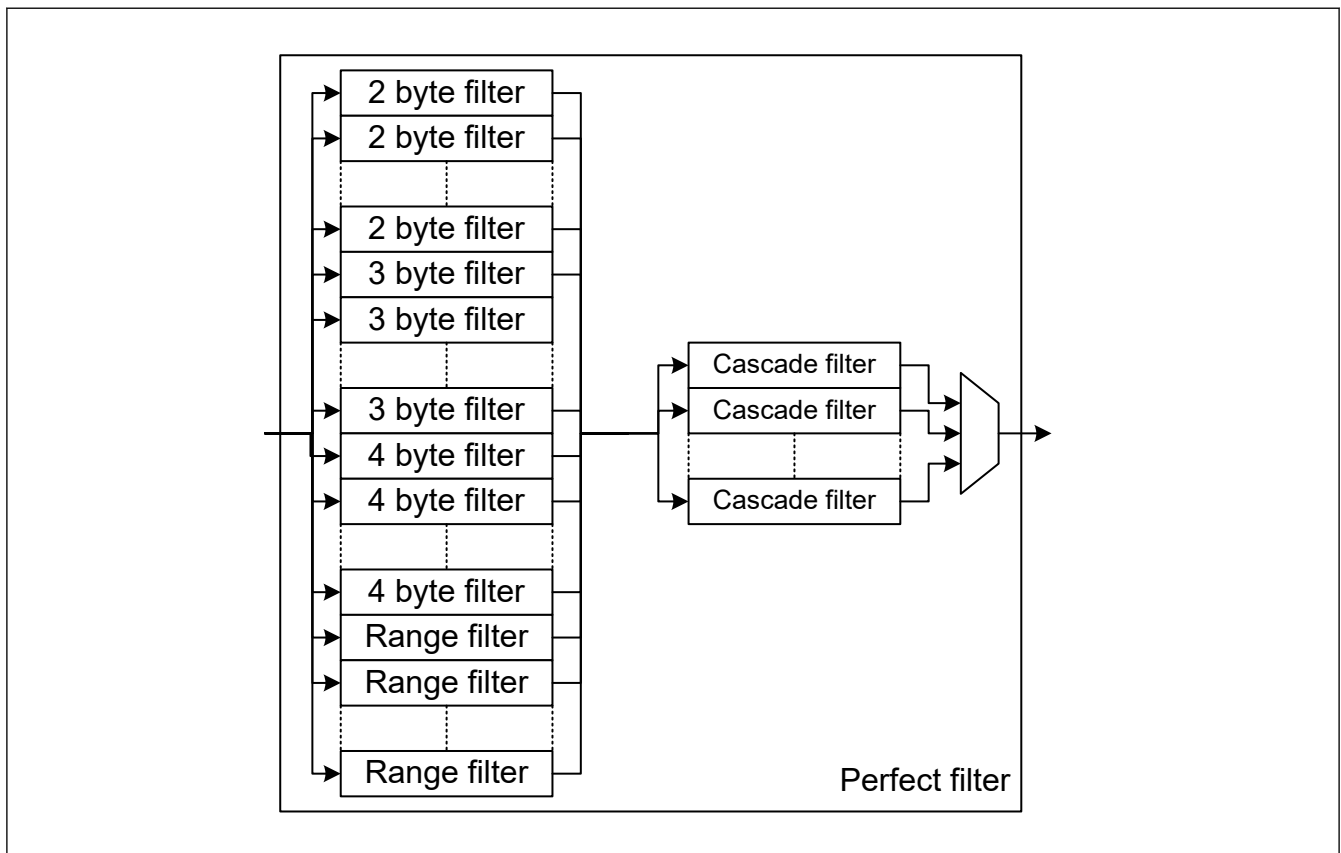


Figure 30.62 Perfect filter block diagram

The perfect filtering happens in two steps:

- The first step is done by 2-byte, 3-byte, 4-byte and range filters. Each one of these filters is independent of others but only able to filter a precise part of a frame.
- The second step is done by Cascade filters. Each one of the cascade filters can associate 7 filtering results from the first step to create a stream ID.

Note: HW: All frames smaller than 33 bytes (received with no command 1 from snooping bus, the 33 bytes are not including the TAG saved in the TAG RAM) will be considered has unknown frames by the perfect filter.

(1) 2-byte filters

A 2-byte filter aims at filtering a specific a portion of a frame and happens in two steps, the data extraction and the filtering.

(a) Data extraction for filtering

A 2-byte filter has two modes to extract the data from a frame. [Figure 30.63](#) describes these modes using examples.

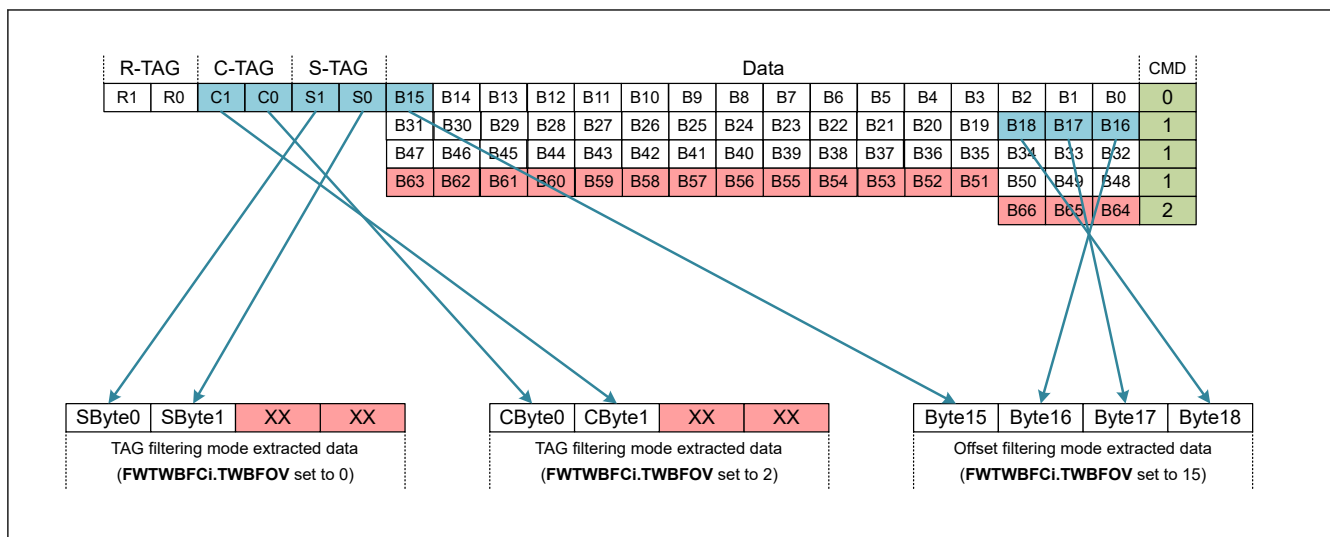


Figure 30.63 2-byte filter data extraction for filtering

In Offset filtering mode (FWTWBFCi.TWBFM == 0):

- Data bytes from byte FWTWBFCi.TWBFOV to byte FWTWBFCi.TWBFOV+3 are extracted for filtering.

In TAG filtering mode (FWTWBFCi.TWBFM == 1):

- If FWTWBFCi.TWBFOV is set to 0, VLAN S-TAG is extracted for filtering.
- If FWTWBFCi.TWBFOV is set to 2, VLAN C-TAG is extracted for filtering.

- Note:
- HW: The data for filtering in 2-byte filters can only be extracted until the first three bytes in the penultimate beat of data (first three bytes of the last time CMD is equal to 1. In Figure 30.63, the bytes colored in red are the none-extractible bytes for filtering). I.e., the offset filtering will only be effective if the offset value is set until the last byte of the antepenultimate beat of data (the last before last time CMD is equal to 1. Byte 47 in Figure 30.63).
 - HW: A filter with a setting that should use non-extractible data for filtering will always unmatched.
 - HW: In TAG filtering mode, if S-TAG or C-TAG is not present in the frame for filtering, the value 0 will be extracted by the filter.

(b) Extracted data filtering

A 2-byte filter has three modes to filter extracted data. Figure 30.64 describes these modes.

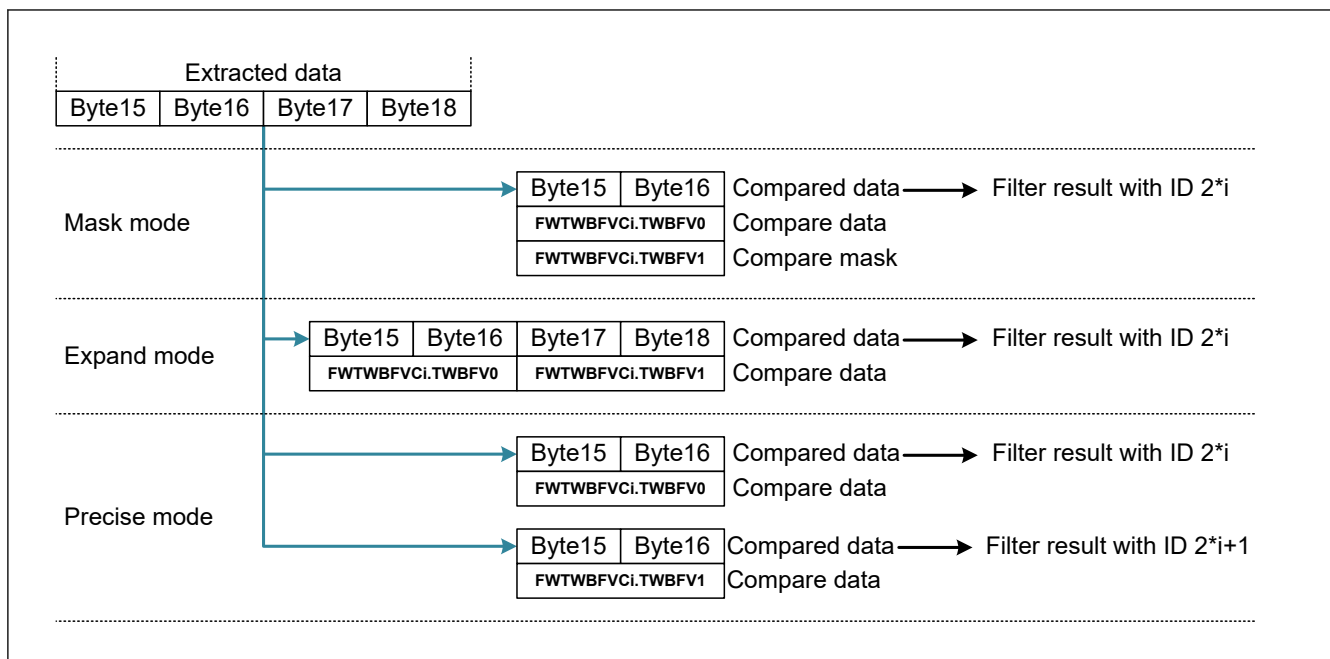


Figure 30.64 2-byte filter data extraction for filtering

In mask mode (FWTWBFVCI.TWBFUM == 00b):

- The first two bytes of the extracted data are compared to FWTWBFVCI.TWBFV0 using FWTWBFVCI.TWBFV1 as a mask (when a bit is set to 1, the corresponding bit is not used for comparison). The result is associated to $2*i$ filter ID.

In Expand mode (FWTWBFVCI.TWBFUM == 01b):

- The four-byte extracted data is compared to {FWTWBFVCI.TWBFV0, FWTWBFVCI.TWBFV1}. The result is associated to $2*i$ filter ID.

In Precise mode (FWTWBFVCI.TWBFUM == 10b):

- The first two bytes of the extracted data are compared to FWTWBFVCI.TWBFV0. The result is associated to $2*i$ filter ID.
- The first two bytes of the extracted data are compared to FWTWBFVCI.TWBFV1. The result is associated to $2*i+1$ filter ID.

(2) 3-byte filters

A 3-byte filter aims at filtering a specific a portion of a frame and happens in two steps, the data extraction and the filtering.

(a) Data extraction for filtering

A 3-byte filter has one mode to extract the data from a frame. [Figure 30.65](#) describes this mode using an example.

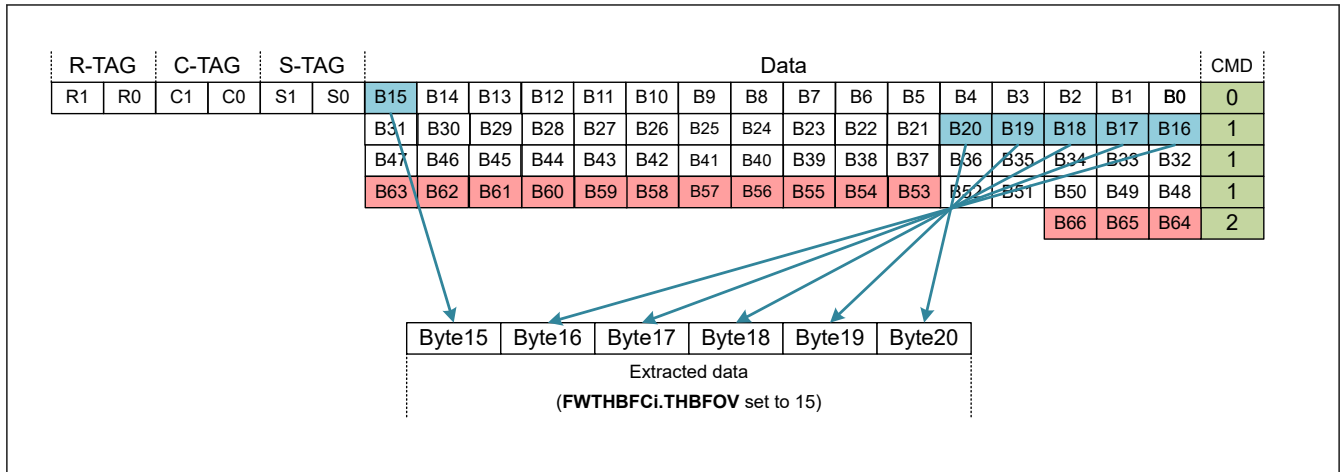


Figure 30.65 3-byte filter data extraction for filtering

For 3-byte filters:

- Data bytes from byte FWTHBFCi.THBFOV to byte FWTHBFCi.THBFOV+5 are extracted for filtering.

Note:

- HW: The data for filtering in 3-byte filters can only be extracted until the first five bytes in the penultimate beat of data (first five bytes of the last time CMD is equal to 1. In Figure 30.65, the bytes colored in red are the none-extractable bytes for filtering). I.e. the offset filtering will only be effective if the offset value is set until the last byte of the antepenultimate beat of data (the last before last time CMD is equal to 1. Byte 47 in Figure 30.65).
- HW: A filter with a setting that should use non-extractible data for filtering will always unmatched.

(b) Extracted data filtering

A 3-byte filter has three modes to filter extracted data. Figure 30.66 describes these modes.

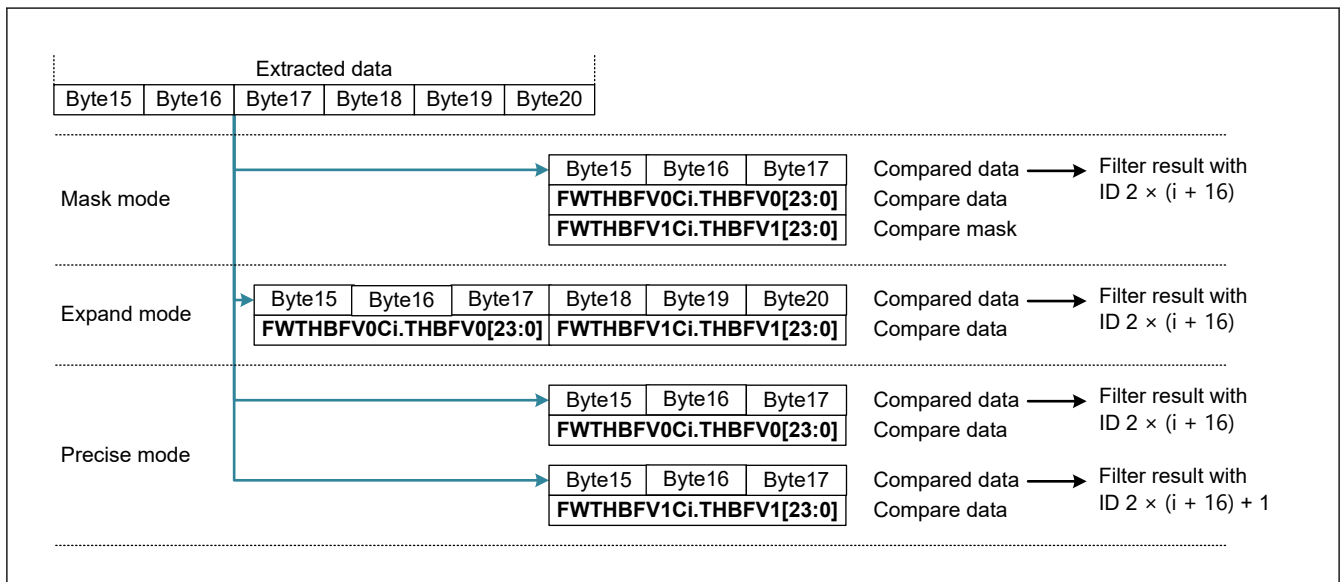


Figure 30.66 3-byte filter data extraction for filtering

In mask mode (FWTHBFCi.THBFUM[1:0] == 00b):

- The first three bytes of the extracted data are compared to FWTHBFV0Ci.THBV0[23:0] using FWTHBFV1Ci.THBV1[23:0] as a mask (when a bit is set to 1, the corresponding bit is not used for comparison). The result is associated to $2 \times (i + 16)$ filter ID.

In Expand mode (FWTHBFCi.THBFUM[1:0] == 01b):

- The six-byte extracted data is compared to {FWTHBFV0Ci.THBV0[23:0], FWTHBFV1Ci.THBV1[23:0]}. The result is associated to $2 \times (i + 16)$ filter ID.

In Precise mode (FWTHBFCi.THBFUM[1:0] == 10b):

- The first three bytes of the extracted data are compared to FWTHBFV0Ci.THBFV0[23:0]. The result is associated to $2 \times (i + 16)$ filter ID.
- The first three bytes of the extracted data are compared to FWTHBFV1Ci.THBFV1[23:0]. The result is associated to $2 \times (i + 16) + 1$ filter ID.

(3) 4-byte filters

A 4-byte filter aims at filtering a specific a portion of a frame and happens in two steps, the data extraction and the filtering.

(a) Data extraction for filtering

A 4-byte filter has one mode to extract the data from a frame. Figure 30.67 describes this mode using an example.

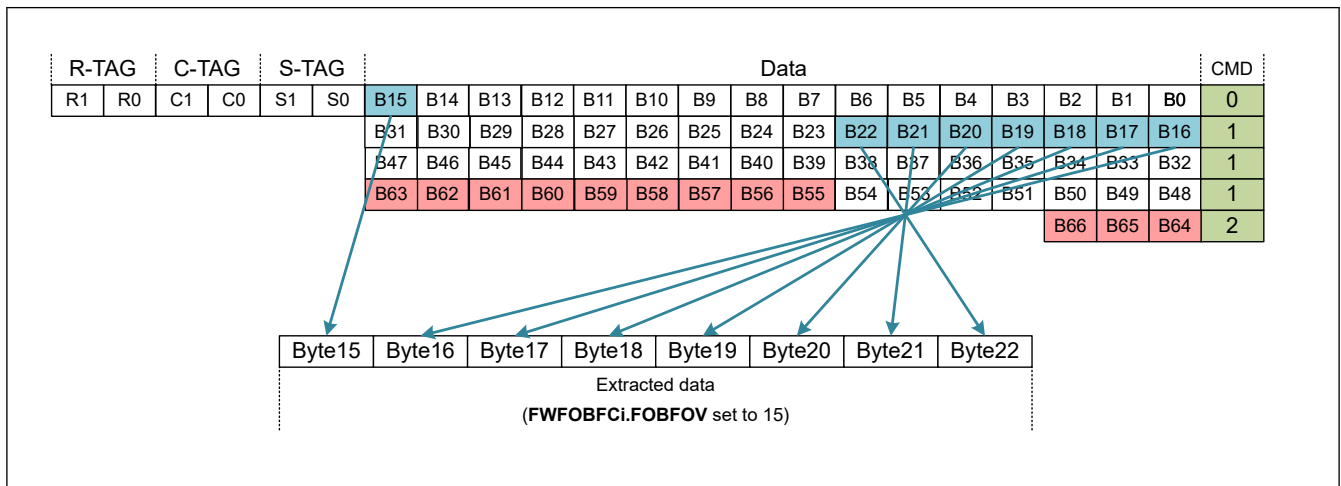


Figure 30.67 4-byte filter data extraction for filtering

For 4-byte filters:

- Data bytes from byte FWFOBFCi.FOBFOV to byte FWFOBFCi.FOBFOV+7 are extracted for filtering.

- Note:
- HW: The data for filtering in 4-byte filters can only be extracted until the first seven bytes in the penultimate beat of data (first seven bytes of the last time CMD is equal to 1. In Figure 30.67, the bytes colored in red are the none-extractible bytes for filtering). I.e. the offset filtering will only be effective if the offset value is set until the last byte of the antepenultimate beat of data (the last before last time CMD is equal to 1. Byte 47 in Figure 30.67).
 - HW: A filter with a setting that should use non-extractible data for filtering will always unmatched.

(b) Extracted data filtering

A 4-byte filter has three modes to filter extracted data. Figure 30.68 describes these modes.

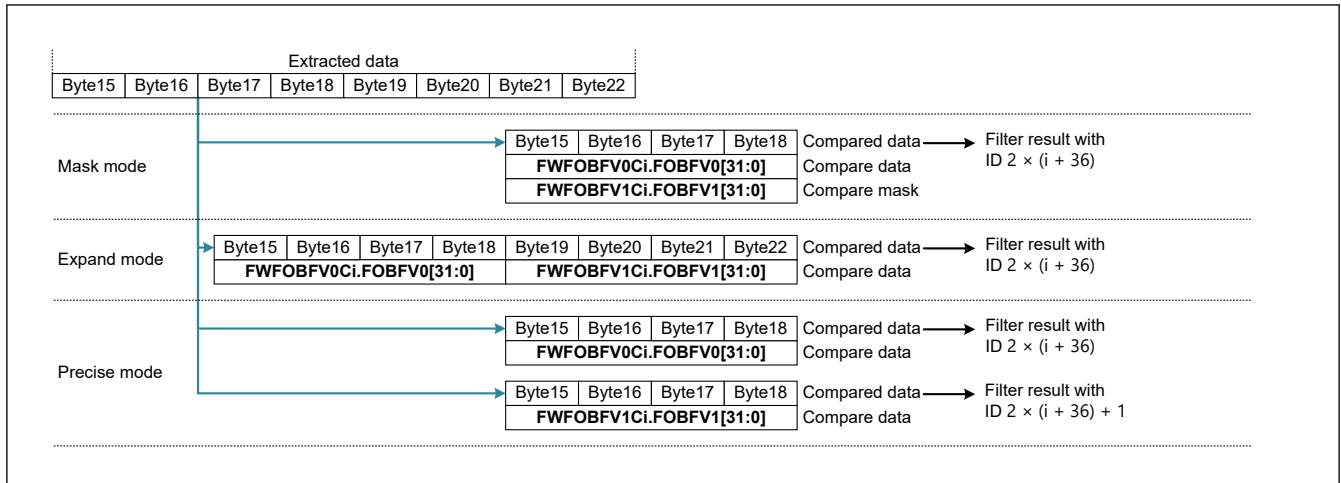


Figure 30.68 4-byte filter data extraction for filtering

In mask mode (FWFOBF0Ci.FOBFUM[1:0] == 00b):

- The first four bytes of the extracted data are compared to FWFOBFV0Ci.FOBFV0[31:0] using FWFOBFV1Ci.FOBFV1[31:0] as a mask (when a bit is set to 1, the corresponding bit is not used for comparison). The result is associated to $2 \times (i + 36)$ filter ID.

In Expand mode (FWFOBF0Ci.FOBFUM[1:0] == 01b):

- The eight-byte extracted data is compared to {FWFOBFV0Ci.FOBFV0[31:0], FWFOBFV1Ci.FOBFV1[31:0]}. The result is associated to $2 \times (i + 36)$ filter ID.

In Precise mode (FWFOBF0Ci.FOBFUM[1:0] == 10b):

- The first three bytes of the extracted data are compared to FWFOBFV0Ci.FOBFV0[31:0]. The result is associated to $2 \times (i + 36)$ filter ID.
- The first three bytes of the extracted data are compared to FWFOBFV1Ci.FOBFV1[31:0]. The result is associated to $2 \times (i + 36) + 1$ filter ID.

(4) Range filters

A Range filter aims at filtering a specific portion of a frame and happens in two steps, the data extraction and the filtering.

(a) Data extraction for filtering

A Range filter has two modes to extract the data from a frame. Figure 30.69 describes these modes using examples.

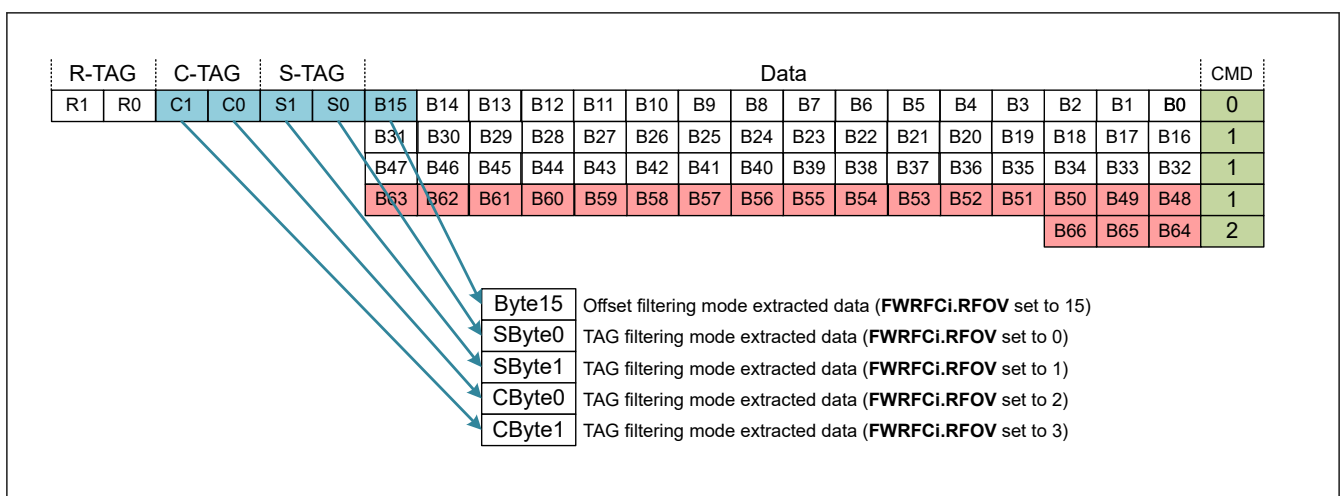


Figure 30.69 Range filter data extraction for filtering

In Offset filtering mode (FWRFCi.RFM == 0):

- Data byte FWRFCi.RFOV is extracted for filtering.

In TAG filtering mode (FWRFCi.RFM == 1):

- If FWRFCi.RFOV is set to 0, VLAN S-TAG VID upper 4 bits are extracted for filtering (PCP and DEI are fixed to 0).
- If FWRFCi.RFOV is set to 1, VLAN S-TAG VID lower 8 bits are extracted for filtering.
- If FWRFCi.RFOV is set to 2, VLAN C-TAG VID upper 4 bits are extracted for filtering (PCP and DEI are fixed to 0).
- If FWRFCi.RFOV is set to 3, VLAN C-TAG VID lower 8 bits are extracted for filtering.

Restrictions:

- Note:
- HW: PCP and DEI values in C-TAGs and S-TAGs are ignored by the range filter (Set to 0).
 - HW: The data for filtering in range filters can only be extracted until the antepenultimate beat of data (the last before last time CMD is equal to 1. In [Figure 30.69](#), the bytes colored in red are the none-extractible bytes for filtering). I.e. the offset filtering will only be effective if the offset value is set until the last byte of the antepenultimate beat of data (the last before last time CMD is equal to 1. Byte 47 in [Figure 30.69](#)).
 - HW: A filter with a setting that should use non-extractible data for filtering will always unmatched.

(b) Extracted data filtering

A Range filter has one mode to filter extracted data. [Figure 30.70](#) describes this mode.

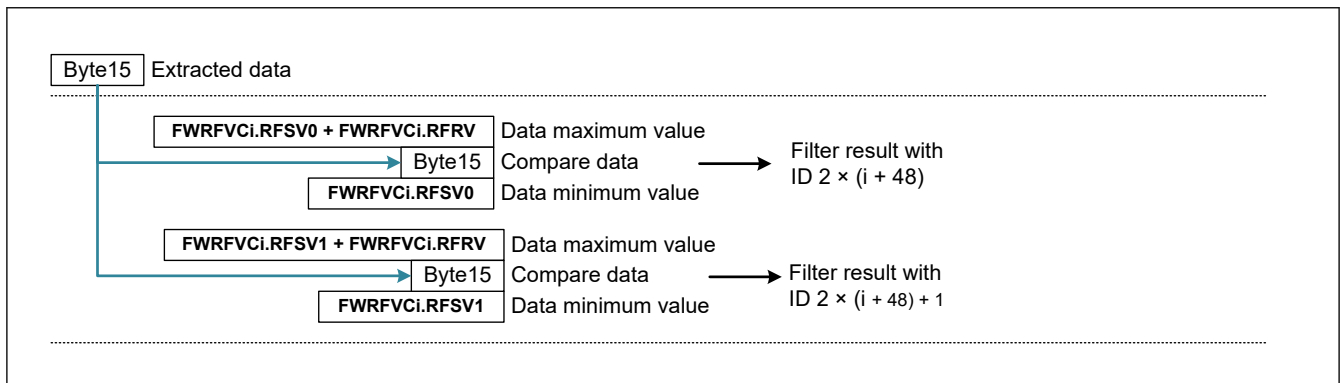


Figure 30.70 Range filter data extraction for filtering

For Range filters:

- The one-byte extracted data is checked to be included in [FWRFCi.RFSV0: FWRFCi.RFSV0 + FWRFCi.RFRV]. The result is associated to $2 \times (i + 48)$ filter ID.
- The one-byte extracted data is checked to be included in [FWRFCi.RFSV1: FWRFCi.RFSV1 + FWRFCi.RFRV]. The result is associated to $2 \times (i + 48) + 1$ filter ID.

(5) Cascade filters

A Cascade filter aims at gathering several filter results (from 2-byte, 3-byte, 4byte and Range filters) to regroup them to create a stream ID. The number of cascade filters defines the number of possible stream IDs in the perfect filter 16.

Functions:

- FWCFMCij register is used to map until 7 filters results to cascade filter i.
- FWCFCi register is used to select from which port the stream is expected and with which type of frame (p or e)
- If several cascade filters match the same frame, the cascade filter with the highest number will be selected.
- If a cascade filter matches a frame, a frame format code of 3 bits equal to 0 is associated to the frame.

Cascade filter stream ID:

- When cascade filter i matches a frame, the stream ID associated to this frame will be a 131-bit stream ID equal to {frame format code, 128'di}.

(6) Perfect filter utilization example

Figure 30.71 contains an example of perfect filter utilization. In this example, the user wants to filter a frame which:

- Has a stream ID based on MAC destination address, and VLAN C-TAG.
- Has a MAC destination address equal to 'hAABBCCDDEEFF'.
- Has a VLAN C-TAG equal to ABCD.
- Is known to be and IPv4 frame (IPv4 type is added in filtering for security).

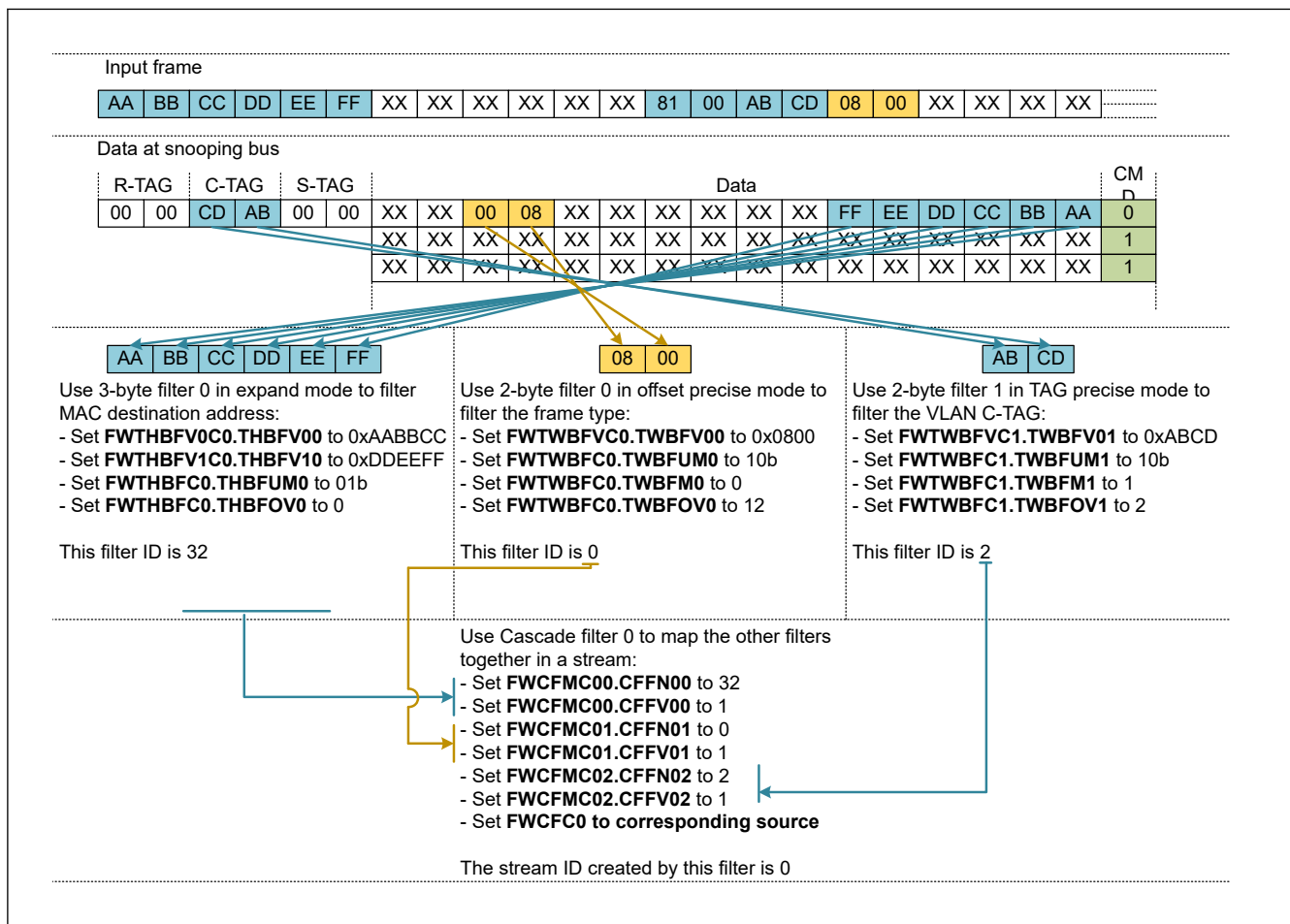


Figure 30.71 Perfect filter utilization example

30.5.2.4.2 IPv4/IPv6 Hash Stream Filter

The Hash filter is used to identify streams. This filter has the advantage to be able to generate a big number of streams but, because of the use of hash, can have some stream ID collision problem (that should be resolved using the perfect filter).

The Hash filter block diagram is represented in Figure 30.72.

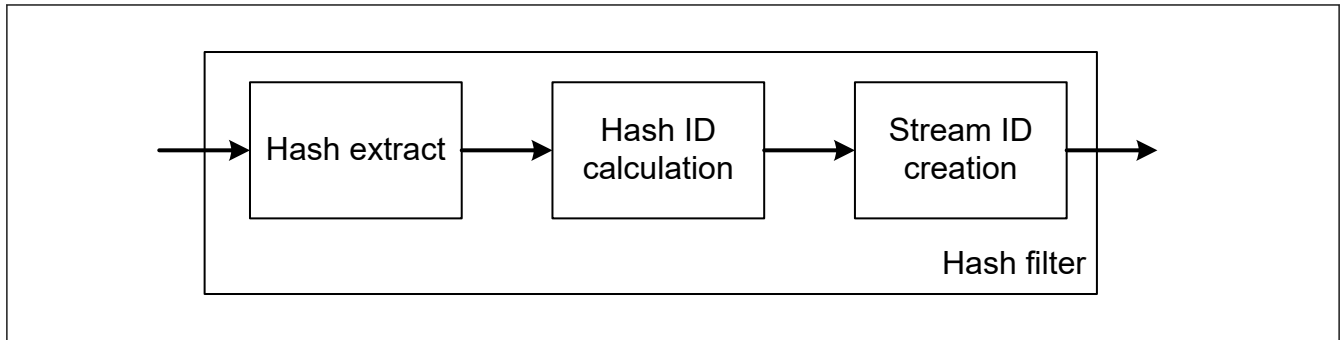


Figure 30.72 Hash filter block diagram

The Hash filtering happens in Three steps:

- The first step is done by Hash extract. It consists in extracting all the possible field for the IPv4/IPv6 stream ID creation.
- The second step is done by Hash ID calculation. It consists in calculating the hash ID associated to each frame depending on the extracted data and the fields chosen by user for Hash.
- The third step is done by Stream ID creation. It consists in the stream ID assembling depending on the fields chosen by user.

(1) Hash extract

The Hash extract is used to extract the Stream IDs from IPv4 and IPv6 frames to create their associated stream ID. The extraction happens on six formats which are IPv4, IPv4/TCP, IPv4/UDP, IPv6, IPv6/TCP and IPv6/UDP.

Note: • HW: Hash extract will ignore any data received from Fabric [MFAB] with command 2.

(a) IPv4 detection and data extraction

[Figure 30.73](#) describes how IPv4 detection and data extraction happens. This data extraction happens only if the frame did not match a cascade filter, if the frame format decode condition are satisfied and if FWPCi0.IP4OE is set for the corresponding source port. In this case, a frame format code of 3 bits equal to 1 is associated to the frame.

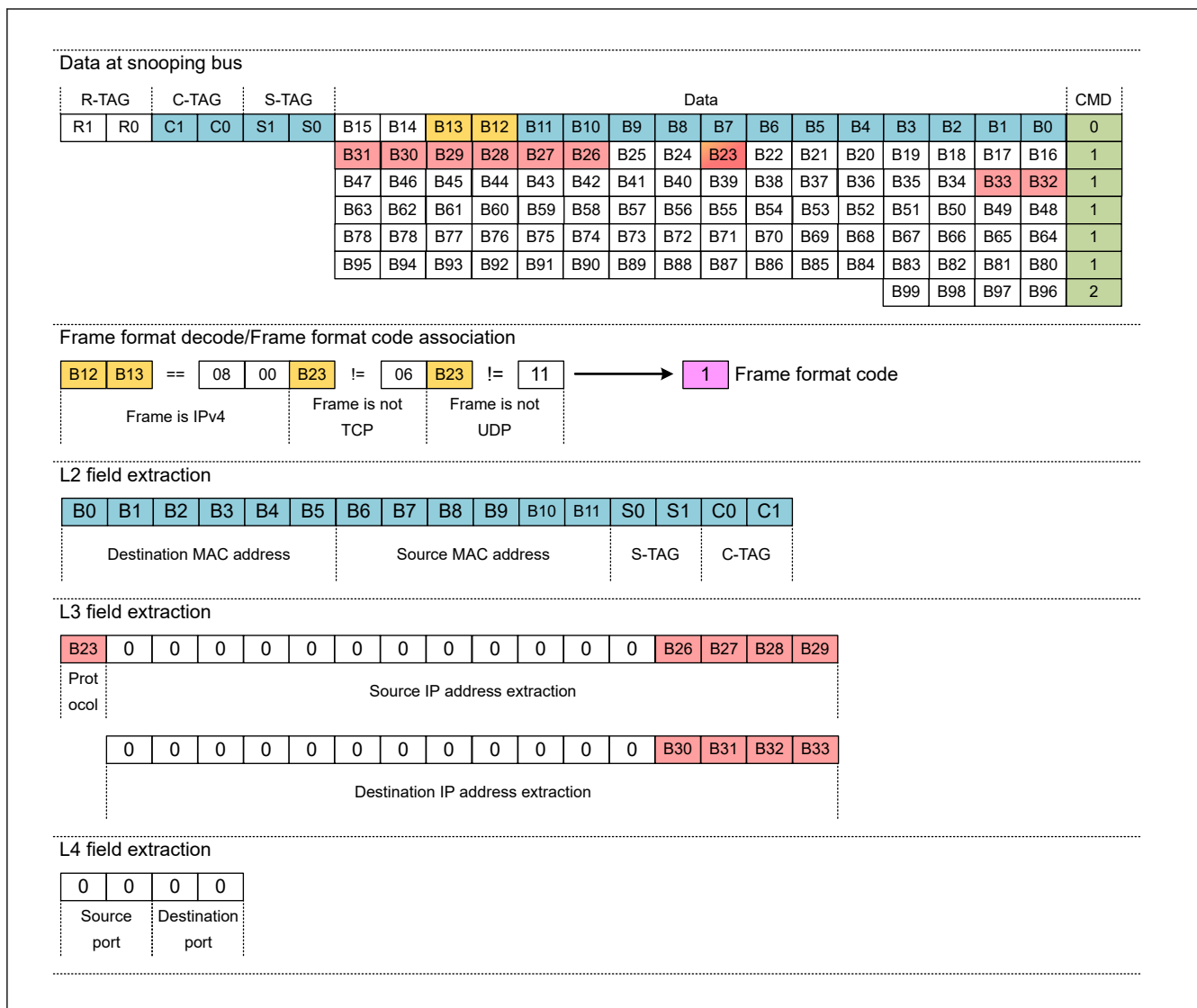
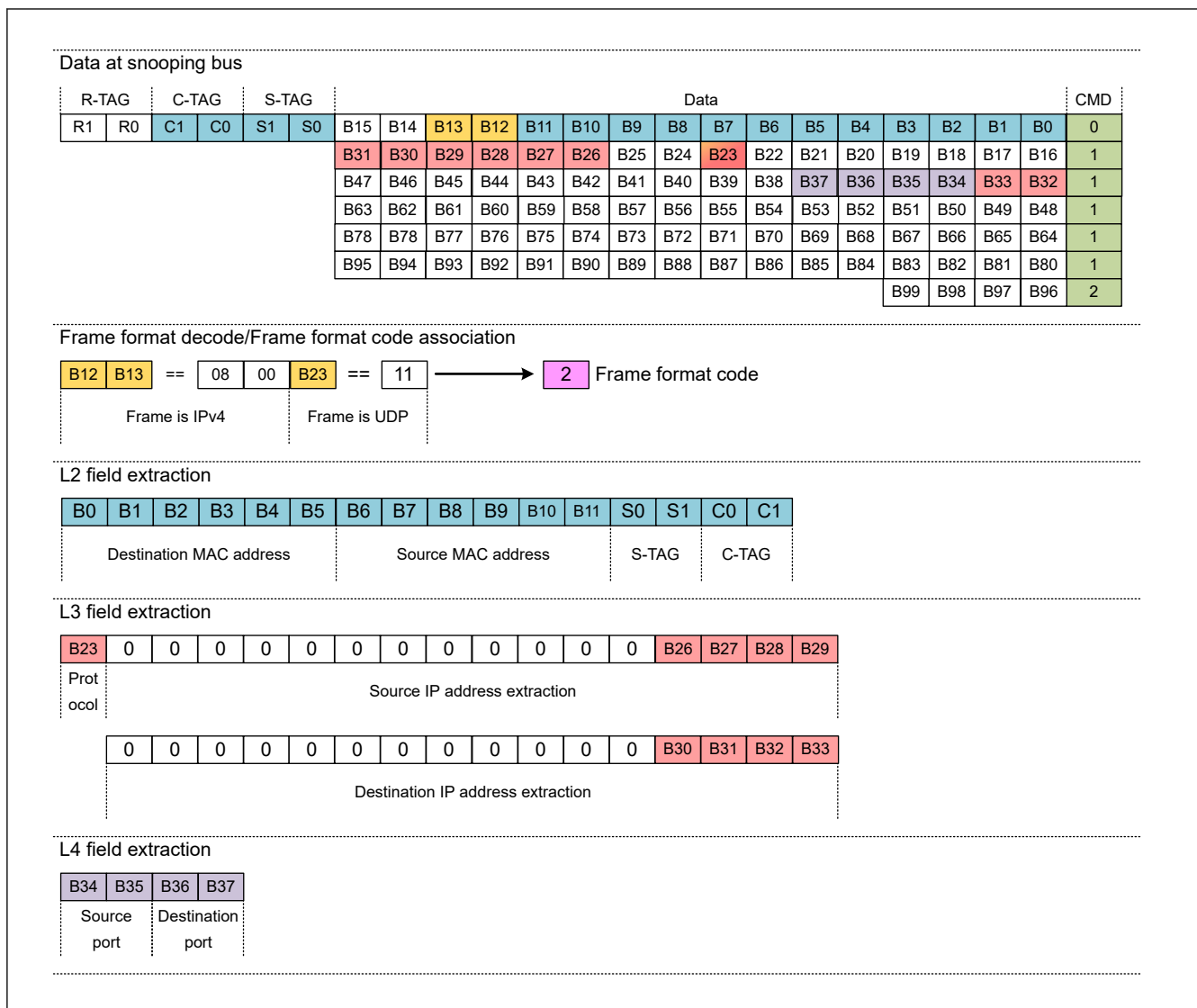


Figure 30.73 IPv4 detection and data extraction

(b) IPv4/UDP detection and data extraction

Figure 30.74 describes how IPv4/UDP detection and data extraction happens. This data extraction happens only if the frame did not match a cascade filter, if the frame format decode condition are satisfied and if FWPCi0.IP4UE is set for the corresponding source port. In this case, a frame format code of 3 bits equal to 2 is associated to the frame.

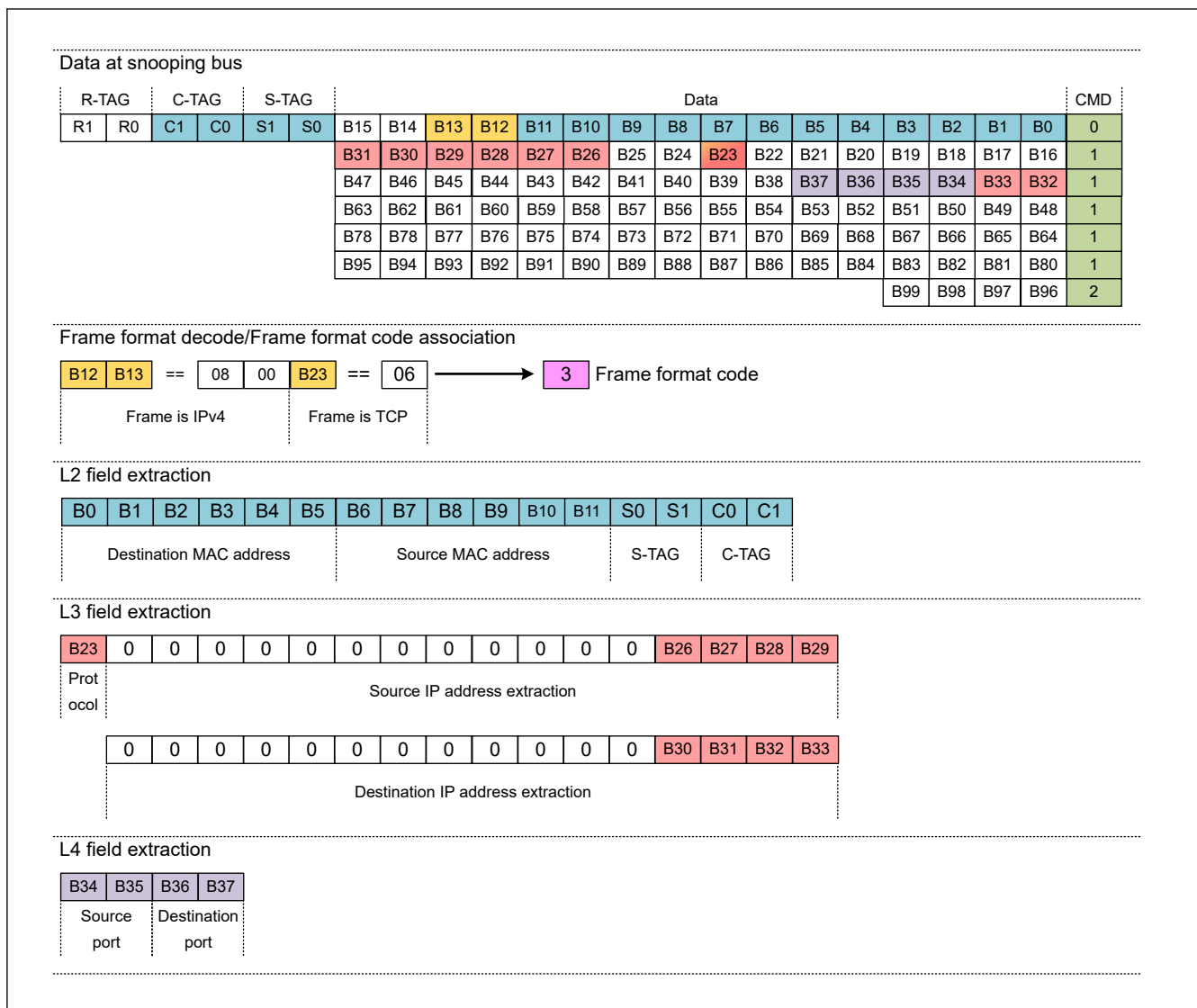
**Figure 30.74 IPv4/UDP detection and data extraction**

Function:

- The L4 extraction function takes in account the IPv4 HeaderLength field. Figure 30.74 only describes the case where HeaderLength is equal to 5 (default value).

(c) IPv4/TCP detection and data extraction

Figure 30.75 describes how IPv4/TCP detection and data extraction happens. This data extraction happens only if the frame did not match a cascade filter, if the frame format decode condition are satisfied and if FWPCi0.IP4TE is set for the corresponding source port. In this case, a frame format code of 3 bits equal to 3 is associated to the frame.

**Figure 30.75 IPv4/TCP detection and data extraction**

Function:

- The L4 extraction function takes in account the IPv4 HeaderLength field. [Figure 30.75](#) only describes the case where HeaderLength is equal to 5 (default value).

(d) IPv6 detection and data extraction

[Figure 30.76](#) describes how IPv6 detection and data extraction happens. This data extraction happens only if the frame did not match a cascade filter, if the frame format decode condition are satisfied and if FWPCi0.IP6OE is set for the corresponding source port. In this case, a frame format code of 3 bits equal to 4 is associated to the frame.

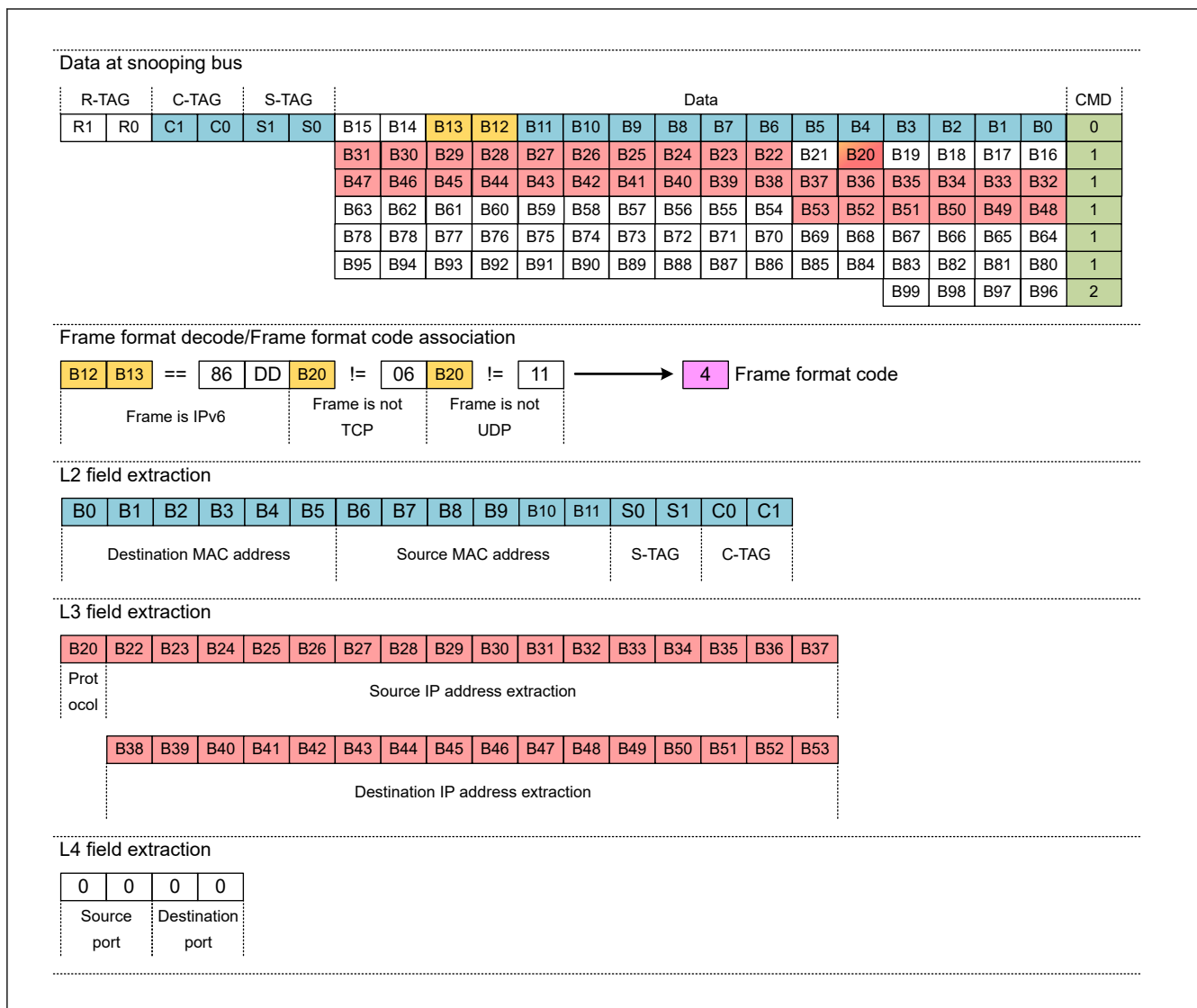


Figure 30.76 IPv6 detection and data extraction

(e) IPv6/UDP detection and data extraction

Figure 30.77 describes how IPv6/UDP detection and data extraction happens. This data extraction happens only if the frame did not match a cascade filter, if the frame format decode condition are satisfied and if FWPCi0.IP6UE is set for the corresponding source port. In this case, a frame format code of 3 bits equal to 5 is associated to the frame.

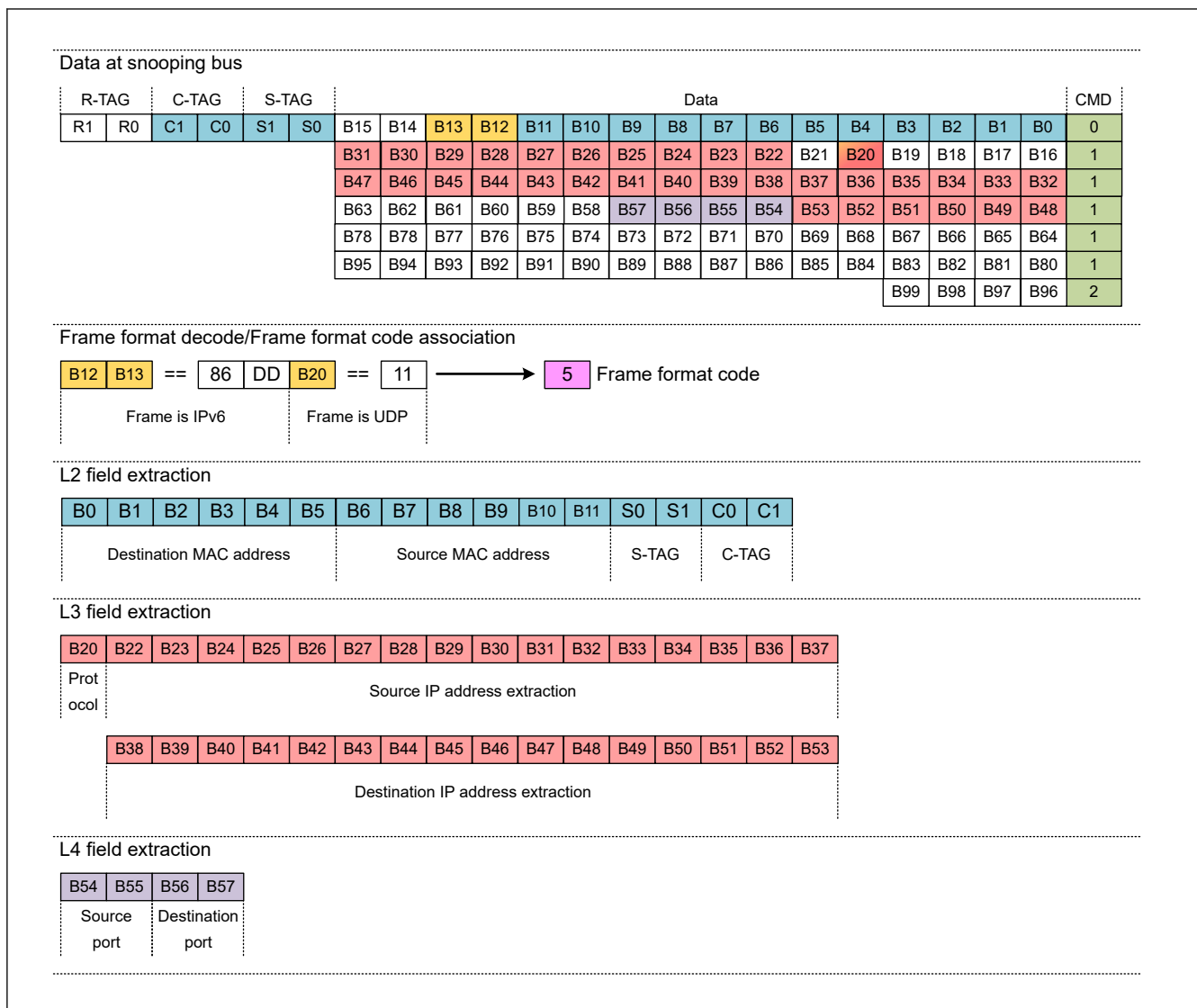


Figure 30.77 IPv6/UDP detection and data extraction

(f) IPv6/TCP detection and data extraction

Figure 30.78 describes how IPv6/TCP detection and data extraction happens. This data extraction happens only if the frame did not match a cascade filter, if the frame format decode condition are satisfied and if FWPCi0.IP6TE is set for the corresponding source port. In this case, a frame format code of 3 bits equal to 6 is associated to the frame.

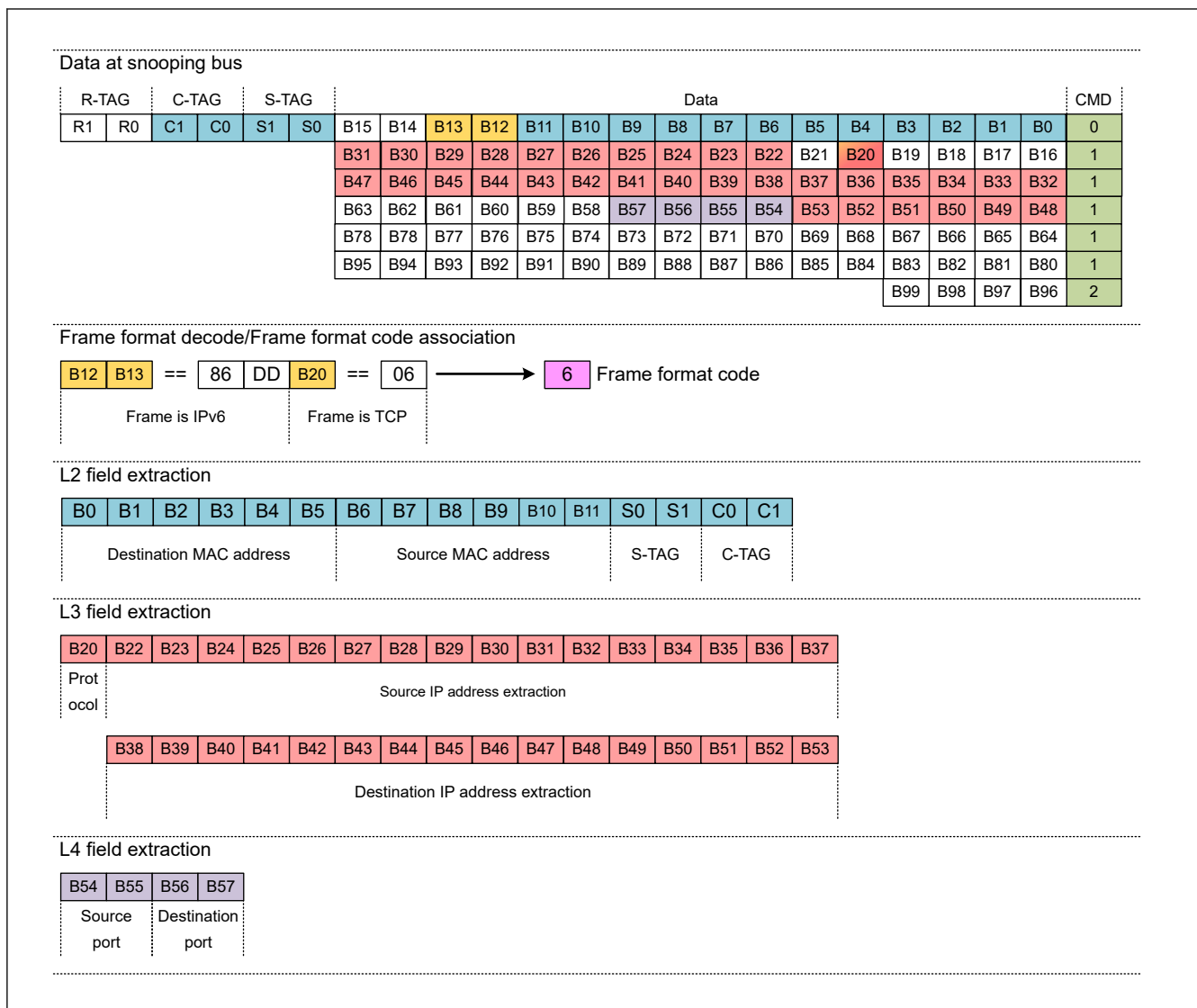


Figure 30.78 IPv6/TCP detection and data extraction

(2) Hash ID calculation

The Hash ID calculation is used to select the interesting fields for the Hash ID (security purpose) and to calculate the hash ID.

(a) Hash equation

Refer to FWSFHEC register explanation.

(b) IPv4 Hash ID calculation

The IPv4 Hash ID calculation is done depending on extracted fields in Hash extract. The calculation is described in [Figure 30.79](#).

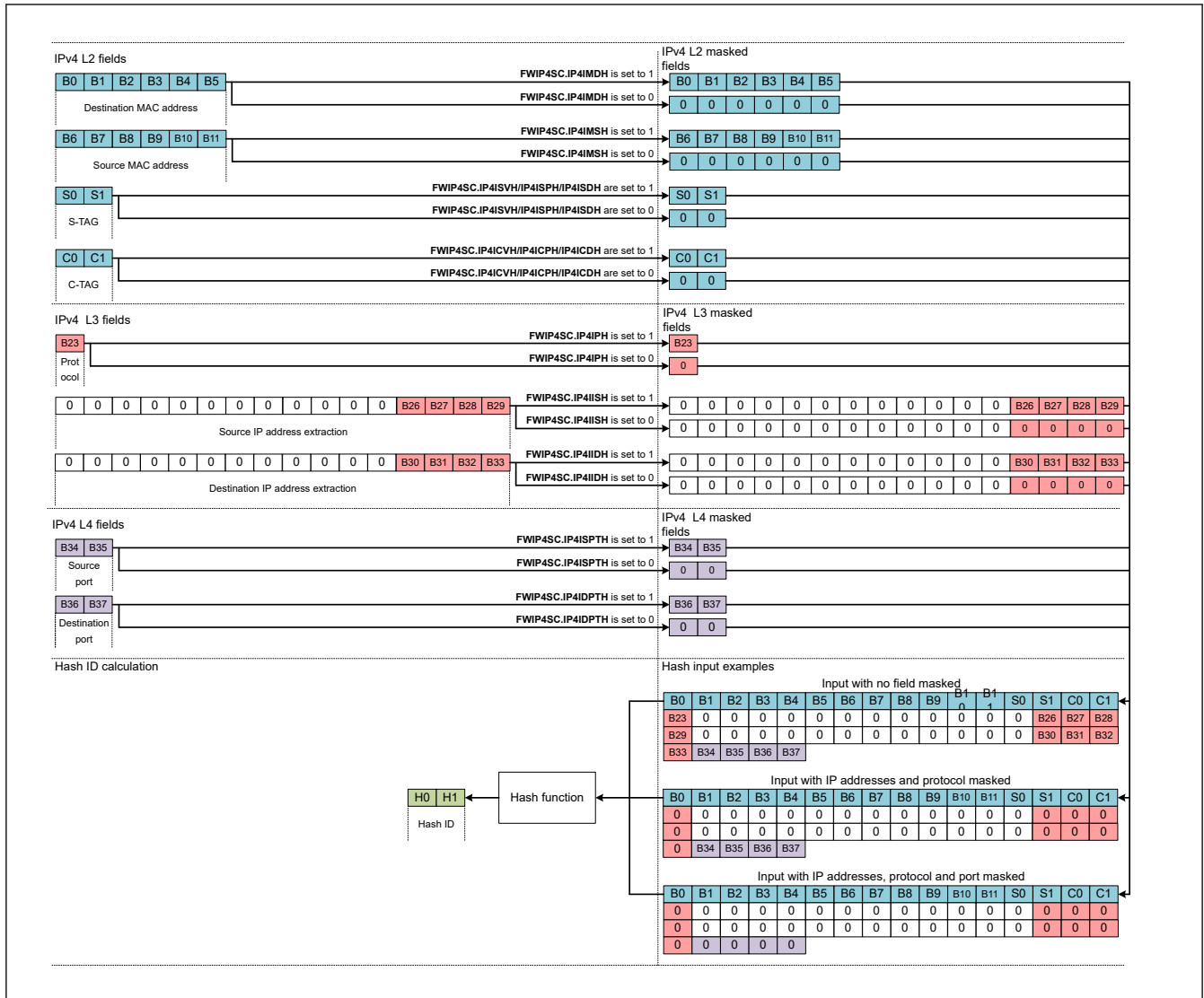


Figure 30.79 IPv4 Hash ID calculation

Functions:

- In Figure 30.79, The C-TAG and S-TAG masking is represented done at once, but masking is possible independently on PCP, DEI and VID.
- Setting all the field inclusion bits for Hash to 0 will fix the Hash ID to 0 in the stream (FWIP4SC.IP4IDPTH/IP4ISPTH/IP4IPH/IP4IIDH/IP4IISH/IP4ICDH/IP4ICPH/IP4ICVH/IP4ISDH/IP4ISPH/IP4ISVH/IP4IMSH/IP4IMDH).
- Setting the hash equation to 0 will fix the Hash ID to 0 in the stream (FWSFHEC.IP4HE).

(c) IPv6 Hash ID calculation

The IPv6 Hash ID calculation is done depending on extracted fields in Hash extract. The calculation is described in Figure 30.80.

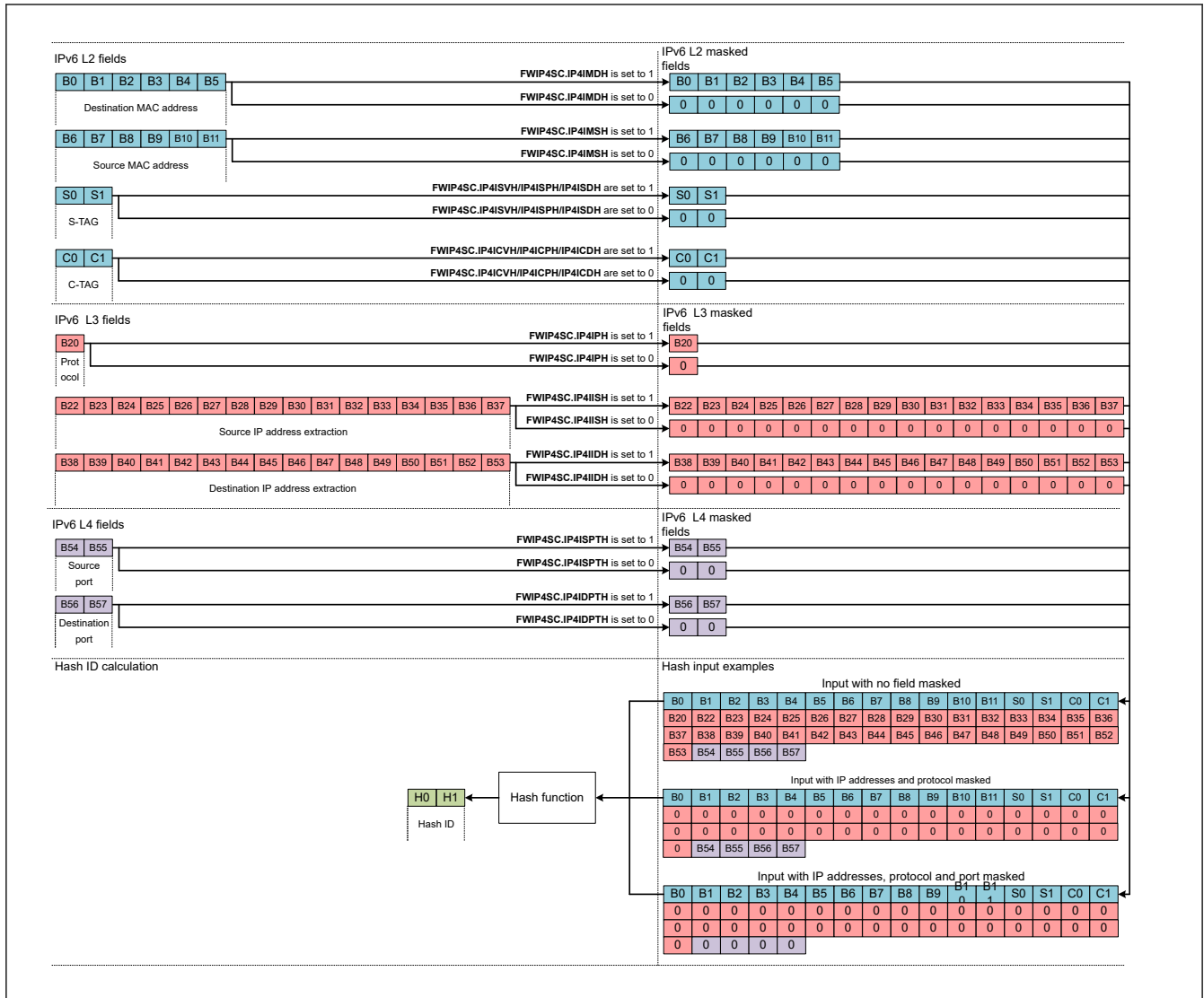


Figure 30.80 IPv6 Hash ID calculation

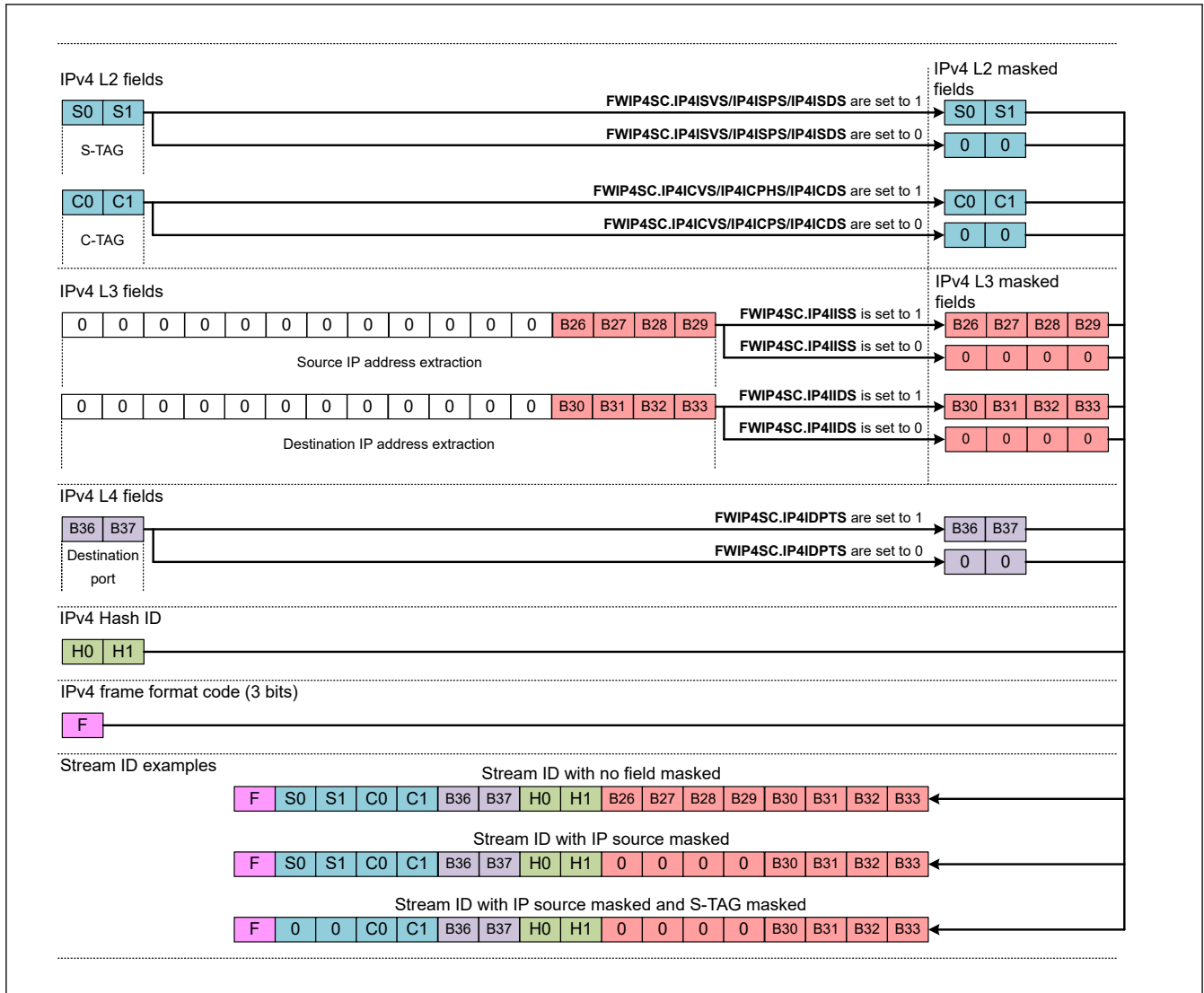
Functions:

- In [Figure 30.80](#), The C-TAG and S-TAG masking is represented done at once, but masking is possible independently on PCP, DEI and VID.
- Setting all the field inclusion bits for Hash to 0 will fix the Hash ID to 0 in the stream (FWIP6SC.IP6IDPTH/IP6ISPTH/IP6IPH/IP6IIDH/IP6IISH/IP6ICDH/IP6ICPH/IP6ICVH/IP6ISDH/IP6ISPH/IP6ISVH/IP6IMSH/IP6IMDH)
- Setting the hash equation to 0 will fix the Hash ID to 0 in the stream (FWSFHEC.IP6HE)

(3) Stream ID creation

(a) IPv4 Stream ID creation

The IPv4 Stream ID is assembled from some of the extracted fields in Hash extract, the Hash ID calculated in Hash ID calculation and the frame format code. The stream ID creation is described in [Figure 30.81](#).

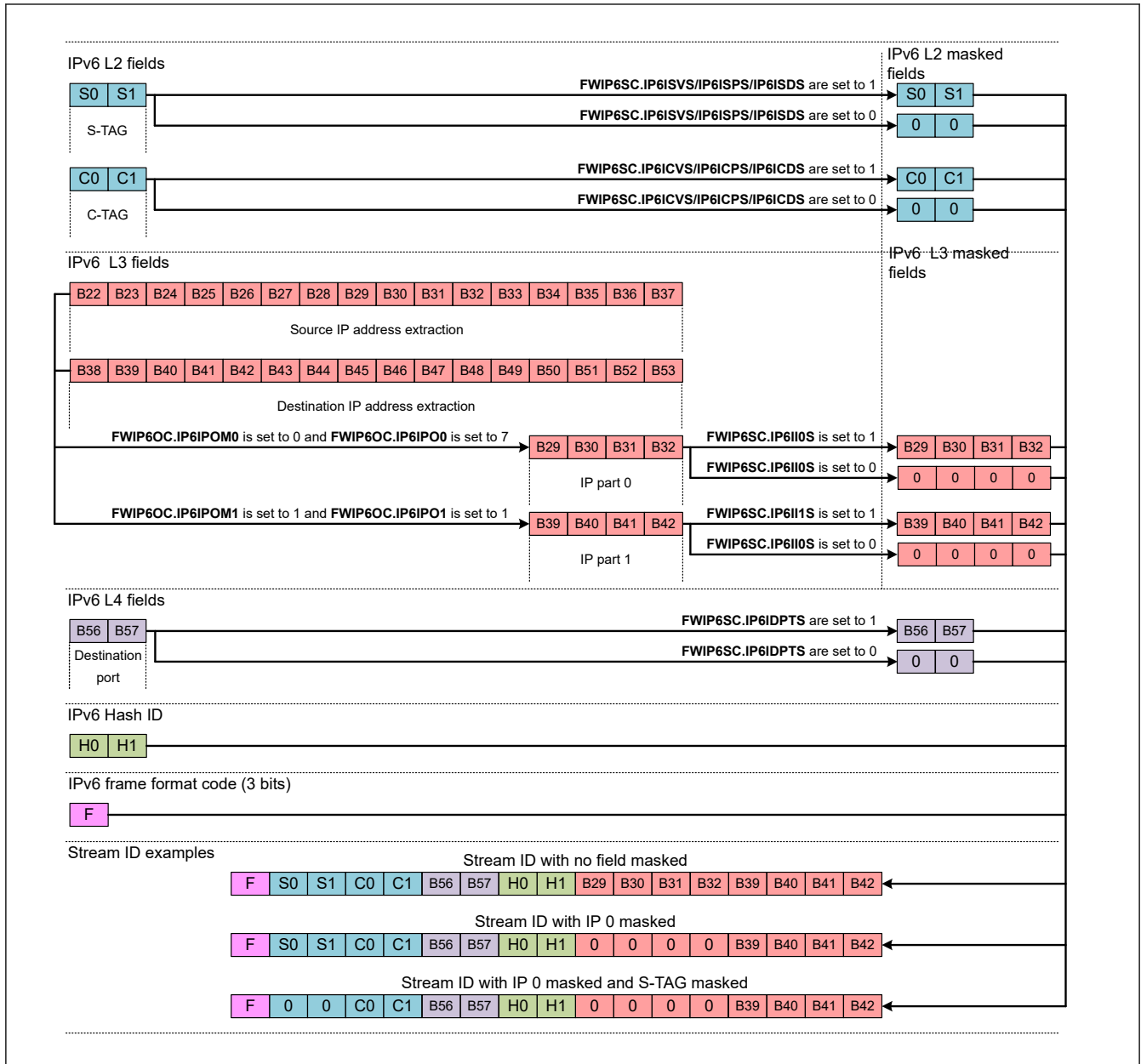
**Figure 30.81 IPv4 Stream ID creation**

Functions:

- In [Figure 30.81](#), The C-TAG and S-TAG masking is represented done at once, but masking is possible independently on PCP, DEI and VID.
- The Stream ID field (FWLTHTL1-4) should be configured as 0 related to disabled part (bits) with “Stream in (FWIP4SC.IP4IDPTS/IP4IIDS/IP4IIS/IP4ICDS/IP4ICPS/IP4ICVTS/IP4ISDS/IP4ISPS/IP4ISVS) = 0x0”

(b) IPv6 Stream ID creation

The IPv6 Stream ID is assembled from some of the extracted fields in Hash extract, the Hash ID calculated in Hash ID calculation and the frame format code. The stream ID creation is described in [Figure 30.82](#).

**Figure 30.82 IPv6 Stream ID creation**

Functions:

- In Figure 30.82, The C-TAG and S-TAG masking is represented done at once, but masking is possible independently on PCP, DEI and VID.
- In Figure 30.82, IP 0 is extracted from IPv6 source IP address and IP 1 is extracted from destination IP address, but it both can extract their 4 bytes from any of the IP addresses. This setting is done through FWIP6OC register.
- The Stream ID field (FWLTHTL1-4) should be configured as 0 related to disabled part (bits) with “Stream in (FWIP6SC.IP6IDPTS/IP6IIS/IIP6IIO/IIP6ICDS/IP6ICPS/IP6ICVS/IP6ISDS/IP6ISPS/IP6ISVS) = 0x0”

(4) Software Hash ID calculation

Because hash ID calculation is complex to implement in software, a hardware accelerator is available for hash ID calculation. Figure 30.83 gives software hash ID calculation examples.

IPv4 Data at snooping bus

R-TAG		C-TAG		S-TAG		Data																CMD	
R1	R0	C1	C0	S1	S0	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0		
						B31	B30	B29	B28	B27	B26	B25	B24	B23	B22	B21	B20	B19	B18	B17	B16	0	
						B47	B46	B45	B44	B43	B42	B41	B40	B39	B38	B37	B36	B35	B34	B33	B32	1	
						B63	B62	B61	B60	B59	B58	B57	B56	B55	B54	B53	B52	B51	B50	B49	B48	1	
						B78	B77	B76	B75	B74	B73	B72	B71	B70	B69	B68	B67	B66	B65	B64		1	
						B95	B94	B93	B92	B91	B90	B89	B88	B87	B86	B85	B84	B83	B82	B81	B80	1	
																	B99	B98	B97	B96		2	

SFR setting for IPv4

B0	B1	B2	B3	FWSHCR0
B4	B5	B6	B7	FWSHCR1
B8	B9	B10	B11	FWSHCR2
S0	S1	C0	C1	FWSHCR3
0	0	0	B23	FWSHCR4
0	0	0	0	FWSHCR5
0	0	0	0	FWSHCR6
0	0	0	0	FWSHCR7
B26	B27	B28	B29	FWSHCR8
0	0	0	0	FWSHCR9
0	0	0	0	FWSHCR10
0	0	0	0	FWSHCR11
B30	B31	B32	B33	FWSHCR12
0	0	0	0	FWSHCR13

IPv6/TCP data at snooping bus

R-TAG		C-TAG		S-TAG		Data																CMD	
R1	R0	C1	C0	S1	S0	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0		
						B31	B30	B29	B28	B27	B26	B25	B24	B23	B22	B21	B20	B19	B18	B17	B16	1	
						B47	B46	B45	B44	B43	B42	B41	B40	B39	B38	B37	B36	B35	B34	B33	B32	1	
						B63	B62	B61	B60	B59	B58	B57	B56	B55	B54	B53	B52	B51	B50	B49	B48	1	
						B78	B77	B76	B75	B74	B73	B72	B71	B70	B69	B68	B67	B66	B65	B64		1	
						B95	B94	B93	B92	B91	B90	B89	B88	B87	B86	B85	B84	B83	B82	B81	B80	1	
																	B99	B98	B97	B96		2	

SFR setting for IPv6/TCP

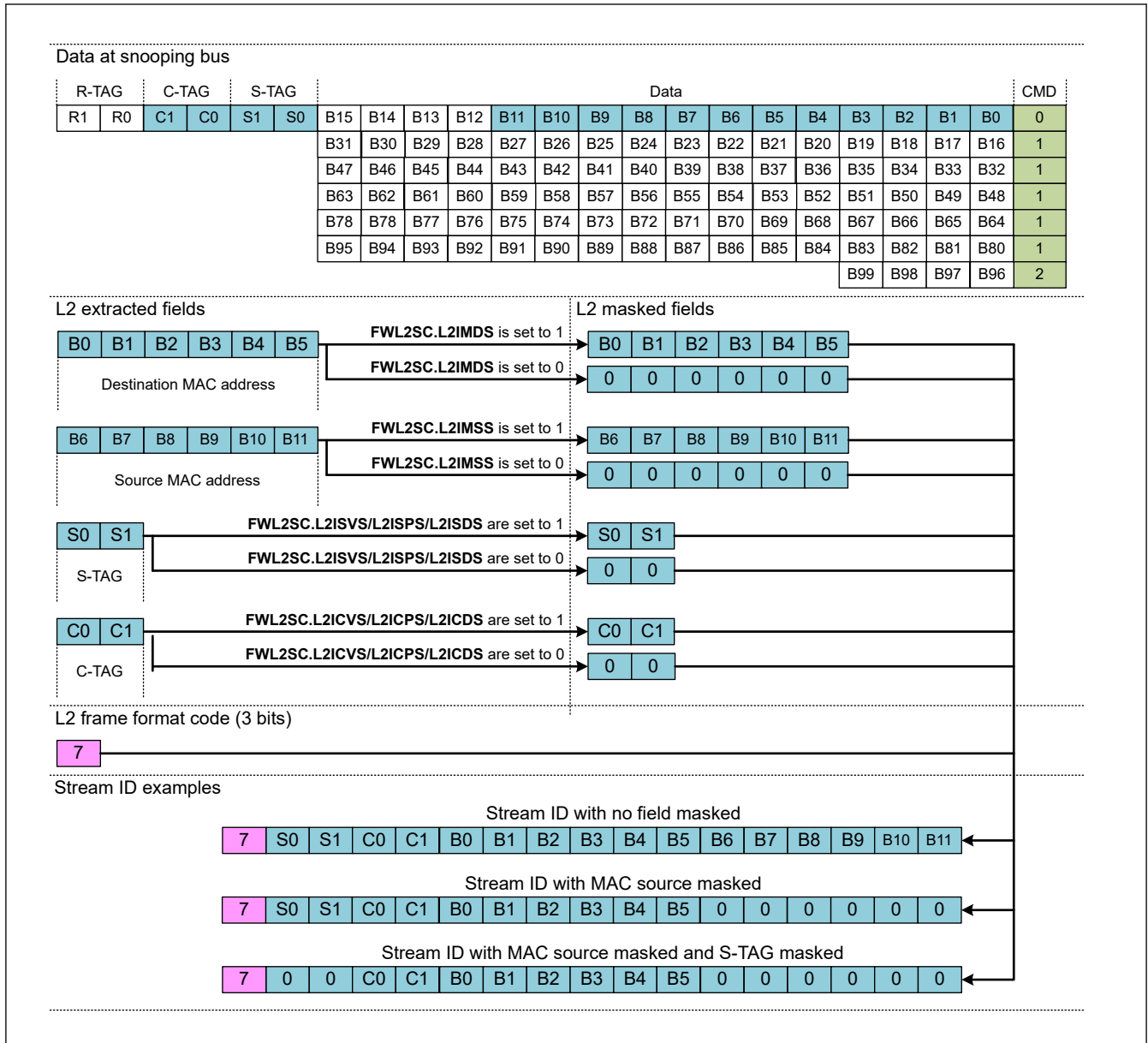
B0	B1	B2	B3	FWSHCR0
B4	B5	B6	B7	FWSHCR1
B8	B9	B10	B11	FWSHCR2
S0	S1	C0	C1	FWSHCR3
0	1	0	B20	FWSHCR4
B22	B23	B24	B25	FWSHCR5
B26	B27	B28	B29	FWSHCR6
B30	B31	B32	B33	FWSHCR7
B34	B35	B36	B37	FWSHCR8
B38	B39	B40	B41	FWSHCR9
B42	B43	B44	B45	FWSHCR10
B46	B47	B48	B49	FWSHCR11
B50	B51	B52	B53	FWSHCR12
B54	B55	B56	B57	FWSHCR13

Figure 30.83 Software hash ID calculation example

Note: SW: For software hash ID calculation, the software should set to 0 the fields that are masked by the Hash ID calculation function.

30.5.2.4.3 L2 Stream Filter

The L2 stream filter is used to create a stream based on L2 fields. This stream creating happens only if the frame did not match a cascade filter or an IPv4/IPv6 hash stream filter and if FWPCi0.L2SE is set for the corresponding source port. In this case, a frame format code of 3 bits equal to 7 is associated to the frame. Figure 30.84 describes how the L2 stream ID is created.

**Figure 30.84 L2 stream ID creation**

Functions:

- In Figure 30.84, The C-TAG and S-TAG masking is represented done at once, but masking is possible independently on PCP, DEI and VID.
- A frame which did not match a cascade filter, an IPv4/IPv6 hash stream filter or the L2 stream filter is considered an unknown frame.
- The Stream ID field (FWLTHTL1-4) should be configured as 0 related to disabled part (bits) with “Stream in (FWL2SC.L2ICDS/L2ICPS/L2ICVS/L2ISDS/L2ISPS/L2ISVS/L2IMSS/L2IMDS) = 0x0”.

30.5.2.4.4 L3 Hash

L3 hash aims at searching/learning/reading L3 rules thanks to a programmable hash. The hash algorithm is able to solve all the collisions in hardware, so it does not need any software intervention to operate.

L3 hash is divided in the following items:

- Rule format: Determines what information will be store along with the stream ID in the table.
- SW Learning: Function to add/suppress an entry from the L3 table by software.

- Hash maximum collision number: Avoids hash search to slow down the forwarding pipeline and apply back pressure on the ingress ports.
- Hash equation: Function to set hash calculation to minimize collisions.
- SW Searching: Function to find the forwarding/routing/filtering information related to a stream by software.
- SW Reading: Function to read the entries in the L3 table by software.
- HW searching: Function to find the forwarding/routing/filtering information related to a stream by forwarding mechanism.

(1) Rule format

The L3 table is used to store L3 rules. A L3 rule contains the information the frame forwarding, routing and filtering. All the fields in this table, if quoted, will be written L3.{Field name}. [Table 30.28](#) describes fields contained in a L3 rule.

Table 30.28 L3 rule format (1 of 2)

Field name	Field size (bit)	Field Explanation
EV	1	Entry Valid Values: • 0: Entry is not valid • 1: Entry is valid
SID	131	Stream ID of the entry. The stream ID is created in perfect filter, IPv4/IPv6 hash stream filter or L2 stream filter (respectively refer to section (5) Cascade filters , section (3) Stream ID creation and section section 30.5.2.4.3. L2 Stream Filter)
SL	1	Security Level Values: • 0: Entry is unsecure • 1: Entry is secure
MSDUV	1	MSDU Valid Values: • 0: Frames matching this entry will be ignored by PSFP MSDU filter. • 1: Frames matching this entry will be processed by PSFP MSDU filter.
MSDUN	4	MSDU Number Functions: • HW: If corresponding entry L3.MSDUV value is set to 1, frames matching this entry will be processed by PSFP MSDU filter number L3.MSDUN.
MTRV	1	Meter Valid Values: • 0: Frames matching this entry will be ignored by PSFP Meter filter. • 1: Frames matching this entry will be processed by PSFP Meter filter.
MTRN	5	Meter Number Functions: • HW: If corresponding entry L3.MTRV value is set to 1, frames matching this entry will be processed by PSFP MSDU filter number L3.MTRN.
FRERV	1	FRER Valid Values: • 0: Frames matching this entry will be ignored by FRER individual recovery. LINK • 1: Frames matching this entry will be processed by FRER individual recovery. LINK
FRERN	7	FRER Number Functions: • HW: If corresponding entry L3.FRERV value is set to 1, frames matching this entry will be processed by FRER entry number L3.FRERN for individual recovery.
RV	1	Routing Valid Values: • 0: Frames matching this entry will be ignored by routing. • 1: Frames matching this entry will be processed by routing.
RN	8	Routing Number Functions: • HW: If corresponding entry L3.RV value is set to 1, frames matching this entry will be processed by routing rule number L3.RN.

Table 30.28 L3 rule format (2 of 2)

Field name	Field size (bit)	Field Explanation
SLV	4	Source Lock Vector Values: - Bit i set to 0: Frames matching this entry coming from port i are rejected by source port lock error. - Bit i set to 1: Frames matching this entry coming from port i are forwarded/routed.
DV	4	Destination vector Functions: HW: Ports to where the frame should be sent.
CSD	6	CPU Sub-Destinations Functions: HW: CPU Sub-Destinations the frame should be sent.
CME	1	CPU Mirroring Enable Values: 0: CPU mirroring is disabled. 1: CPU mirroring is enabled.
EME	1	Ethernet Mirroring Enable Values: 0: Ethernet mirroring is disabled. 1: Ethernet mirroring is enabled.
IPU	1	Internal Priority Update Values: 0: Frames matching this entry will not have their internal priority updated. 1: Frames matching this entry will have their internal priority updated.
IPV	3	Internal Priority Value Functions: HW: If corresponding entry L3.IPU value is set to 1, frames matching this entry will have their priority updated to L3.IPV.

(2) SW Learning

Learning is used to add/remove an entry in the L3 table. [Table 30.29](#) describes register used to learn a stream ID in the Layer 3 table. [Table 30.30](#) describes the learn results.

Table 30.29 L3 learn registers (1 of 2)

Register name	Field name/corresponding field in I3 table	Field explanation
FWLTHTL0.LTHED	Entry delete	Not present in L3 table, used to select between learning an entry and deleting an entry
FWLTHTL0.LTHSLL	L3.SL	See (1) Rule format .
FWLTHTL0.LTHSLP0/ FWLTHTL1.LTHSLP1/ FWLTHTL2.LTHSLP2/ FWLTHTL3.LTHSLP3/ FWLTHTL4.LTHSLP4	L3.SID	See (1) Rule format .
FWLTHTL5.LTHMSDUVL	L3.MSDUV	See (1) Rule format .
FWLTHTL5.LTHMSDUNL	L3.MSDUN	See (1) Rule format .
FWLTHTL6.LTHMTRVL	L3.MTRV	See (1) Rule format .
FWLTHTL6.LTHMTRNL	L3.MTRN	See (1) Rule format .
FWLTHTL6.LTHFRERV	L3.FRERV	See (1) Rule format .
FWLTHTL6.LTHFRERNL	L3.FRERN	See (1) Rule format .
FWLTHTL7.LTHRVL	L3.RV	See (1) Rule format .
FWLTHTL7.LTHRNL	L3.RN	See (1) Rule format .
FWLTHTL7.LTHSLVL	L3.SLV	See (1) Rule format .
FWLTHTL8.LTHCSDL	L3.CSD[5:0]	See (1) Rule format .
FWLTHTL9.LTHDVL	L3.DV	See (1) Rule format .

Table 30.29 L3 learn registers (2 of 2)

Register name	Field name/corresponding field in I3 table	Field explanation
FWLTHTL9.LTHCMEL	L3.CME	See (1) Rule format.
FWLTHTL9.LTHEMEL	L3.EME	See (1) Rule format.
FWLTHTL9.LTHIPUL	L3.IPU	See (1) Rule format.
FWLTHTL9.LTHIPVL	L3.IPV	See (1) Rule format.

Table 30.30 L3 learn result

Register name	Field name/corresponding field in I3 table	Field explanation
FWLTHTLR.LTHLF	Learning fail	Learning fail with happen in one of the following conditions: - The learn entry is not yet in the table and the table is already full (FWLTHTEM.LHTEN == 256). - The Layer 3 table is not ready (FWLTHTIM.LHTTR is not set) - The learning function is used to delete an entry (FWLTHTL0.LTHED set for learning) and the entry is not found in the Layer 3 table
FWLTHTLR.LTHLSF	Learning security fail	Security fail with happen in one of the following conditions: - The learn entry is an unsecure entry (FWLTHTL0.LTHSLL not set for learning) and the unsecure entry number FWLTHTEM.LHTUEN has reached the maximum unsecure entry number FWLTHHEC.LTHHMUE. - The learn action is done by the APB which tries to learn an unsecure entry when the entry is already in the table, is secure (L3.SL is set) and when FWLTHTEM.LHTUEN has reached the maximum unsecure entry number FWLTHHEC.LTHHMUE.
FWLTHTLR.LTHLO	Learning overwrite	Overwrite with happen in one of the following conditions: The learn stream ID is already present in the table, and no error happened during learning.
FWLTHTLR.LTHLCN	Learning collision number	Give the number of collisions that needs to be resolved to access the learnt entry if the lean succeeded. If FWLTHTLR.LTHLCN value comes out bigger than FWLTHHEC.LTHHMC register value, the newly learnt entry will be ignored by forwarding engine during forwarding. A switch reconfiguration with a different L3 Hash equation should be considered.

Note: HW: The fuller the table is, the slower Learning gets.

(3) Hash maximum collision number

Hash maximum number avoids the ESWM to have back pressure due to a slow forwarding mechanism.

Equation (1) shows how to calculate this number. This number should be set in FWLTHHEC.LTHHMC register.

$$(1) \text{FWLTHHEC.LTHHMC} = (\text{clk_freq}[\text{Hz}] * \text{Average_frame_size}[\text{bit}] / \text{Incoming_throughput}[\text{bps}] - 4) / 3$$

Where:

- clk_freq is the clock frequency of ESWCLK.
- $\text{Average_frame_size}$ is the average frame size expected on switch inputs
- $\text{Incoming_throughput}$ is the total throughput expected on switch inputs.

For example, if clk_freq is 150 MHz, $\text{Average_frame_size}$ is 128 bytes and $\text{Incoming_throughput}$ is 3 Gbps (1 Gbps of data on 2 GMII and 1 Gbps of data on 1 GWCA), the value 15 should be set to FWLTHHEC.LTHHMC register.

(4) Hash equation

The hash equation is controlled by register FWLTHHC.LTHHE.

The initial value of the equation is $1 + x^8$.

If there are too many collisions in the table ($\text{FWLTHTLR.LTHLCN} > \text{FWLTHHEC.LTHHMC}$ during learning), this hash equation can be changed to minimize the number of collisions by:

- SW: Modelling the hash equations and find an equation which have a reasonable number of collisions.

- SW: Trying randomly some equations until finding an equation which have a reasonable number of collisions.

(5) SW Searching

Searching is used to find the information related to a stream ID. [Table 30.31](#) describes register used to search a stream ID in the Layer 3 table. [Table 30.32](#) describes the search results.

Table 30.31 L3 search registers

Register name	Field name/corresponding field in I3 table	Field explanation
FWLTHTS0.LTHSSP0/ FWLTHTS1.LTHSSP1/ FWLTHTS2.LTHSSP2/ FWLTHTS3.LTHSSP3/ FWLTHTS4.LTHSSP4	L3.SID	See (1) Rule format .

Table 30.32 L3 search result

Register name	Field name/corresponding field in I3 table	Field explanation
FWLTHTSR0.LTHSNF	Search not found.	Search failed because the searched stream ID is not in the table
FWLTHTSR0.LTHSCN	Search collision number	Give the number of collisions occurred to find the entry.
FWLTHTSR0.LTHSLS	L3.SL	See (1) Rule format .
FWLTHTSR1.LTHMSDUVS	L3.MSDUV	See (1) Rule format .
FWLTHTSR1.LTHMSDUNS	L3.MSDUN	See (1) Rule format .
FWLTHTSR2.LTHMTRVS	L3.MTRV	See (1) Rule format .
FWLTHTSR2.LTHMTRNS	L3.MTRN	See (1) Rule format .
FWLTHTSR3.LTHFRERVS	L3.FRERV	See (1) Rule format .
FWLTHTSR3.LTHFRERN	L3.FRERN	See (1) Rule format .
FWLTHTSR3.LTHSLVS	L3.SLV	See (1) Rule format .
FWLTHTSR3.LTHRVS	L3.RV	See (1) Rule format .
FWLTHTSR3.LTHRNS	L3.RN	See (1) Rule format .
FWLTHTSR40.LTHCSDS	L3.CSD[5:0]	See (1) Rule format .
FWLTHTSR5.LTHDVS	L3.DV	See (1) Rule format .
FWLTHTSR5.LTHCMES	L3.CME	See (1) Rule format .
FWLTHTSR5.LTHEMES	L3.EME	See (1) Rule format .
FWLTHTSR5.LTHIPUS	L3.IPU	See (1) Rule format .
FWLTHTSR5.LTHIPVS	L3.IPV	See (1) Rule format .

(6) SW Reading

Reading is used to dump the Layer 3 table content. This function cannot be used to read a specific entry because it is not possible for the software to know at which address an entry can be found in the Layer 3 RAM. [Table 30.33](#) describes register used to read an entry in the Layer 3 table. [Table 30.34](#) describes the read results.

Table 30.33 L3 read registers

Register name	Field name/corresponding field in I3 table	Field explanation
FWLTHTR.LTHAR	Read address	Address that will be used to read an entry in the Layer 3 table

Table 30.34 L3 read result

Register name	Field name/corresponding field in I3 table	Field explanation
FWLTHTRR0.LTHEVR	L3.EV	The read address holds a valid entry Only if this bit is set when reading an address, the corresponding entry and the following registers in this table are valid. See (1) Rule format
FWLTHTRR1.LTHSRP0/ FWLTHTRR2.LTHSRP1/ FWLTHTRR3.LTHSRP2/ FWLTHTRR4.LTHSRP3/ FWLTHTRR5.LTHSRP4	L3.SID	See (1) Rule format.
FWLTHTRR1.LTHSLR	L3.SL	See (1) Rule format.
FWLTHTRR6.LTHMSDUVR	L3.MSDUV	See (1) Rule format.
FWLTHTRR6.LTHMSDUNR	L3.MSDUN	See (1) Rule format.
FWLTHTRR7.LTHMTRVR	L3.MTRV	See (1) Rule format.
FWLTHTRR7.LTHMTRNR	L3.MTRN	See (1) Rule format.
FWLTHTRR7.LTHFRERV	L3.FRERV	See (1) Rule format.
FWLTHTRR7.LTHFRERNR	L3.FRERN	See (1) Rule format.
FWLTHTRR8.LTHSLVR	L3.SLV	See (1) Rule format.
FWLTHTRR8.LTHRVR	L3.RV	See (1) Rule format.
FWLTHTRR8.LTHRNR	L3.RN	See (1) Rule format.
FWLTHTRR90.LTHCSDR	L3.CSD[5:0]	See (1) Rule format.
FWLTHTRR10.LTHDVR	L3.DV	See (1) Rule format.
FWLTHTRR10.LTHCMER	L3.CME	See (1) Rule format.
FWLTHTRR10.LTHEMER	L3.EME	See (1) Rule format.
FWLTHTRR10.LTHIPUR	L3.IPU	See (1) Rule format.
FWLTHTRR10.LTHIPVR	L3.IPV	See (1) Rule format.

30.5.2.4.5 Layer 3 Forwarding/Routing/Filtering

Layer 3 forwarding/routing/filtering is used to forward local ethernet descriptors (LDESCR.FMT [GWCA] set to 1'b0) received from agents using FWPCi0/FWPCi1 and FWCEPRC2 register and can be monitored using FWEIS0i interrupt register. Layer 3 forwarding/routing/filtering searches received valid stream IDs (section (5) Cascade filters, section (a) IPv4 Stream ID creation and section (b) IPv6 Stream ID creation) in the layer 3 table to extract corresponding entry fields (section (1) Rule format). Depending on the entry fields, Layer 3 forwarding/routing/filtering will forward/route/filter frames following Layer 3 forwarding/routing secure flow in Figure 30.85, Layer 3 forwarding/routing unsecure flow in Figure 30.86 and Layer 3 filtering flow in Figure 30.87.

Functions:

- FWPCi0.LHTA register is used to activate layer 3 table for frames received from port i.
- FWPCi0.LTHRUS register is used to reject unknown streams received from port i.
- FWPCi0.LTHRUS register is used to reject secure unknown streams received from port i.
- FWPCi1.LTHFM register is used to set to which ports frames received from port i can be forwarded.
- FWCEPRC2 register enables exceptional path for error descriptors.
- FWEIS0i interrupt register notifies that an ethernet descriptor has been rejected because of an error.

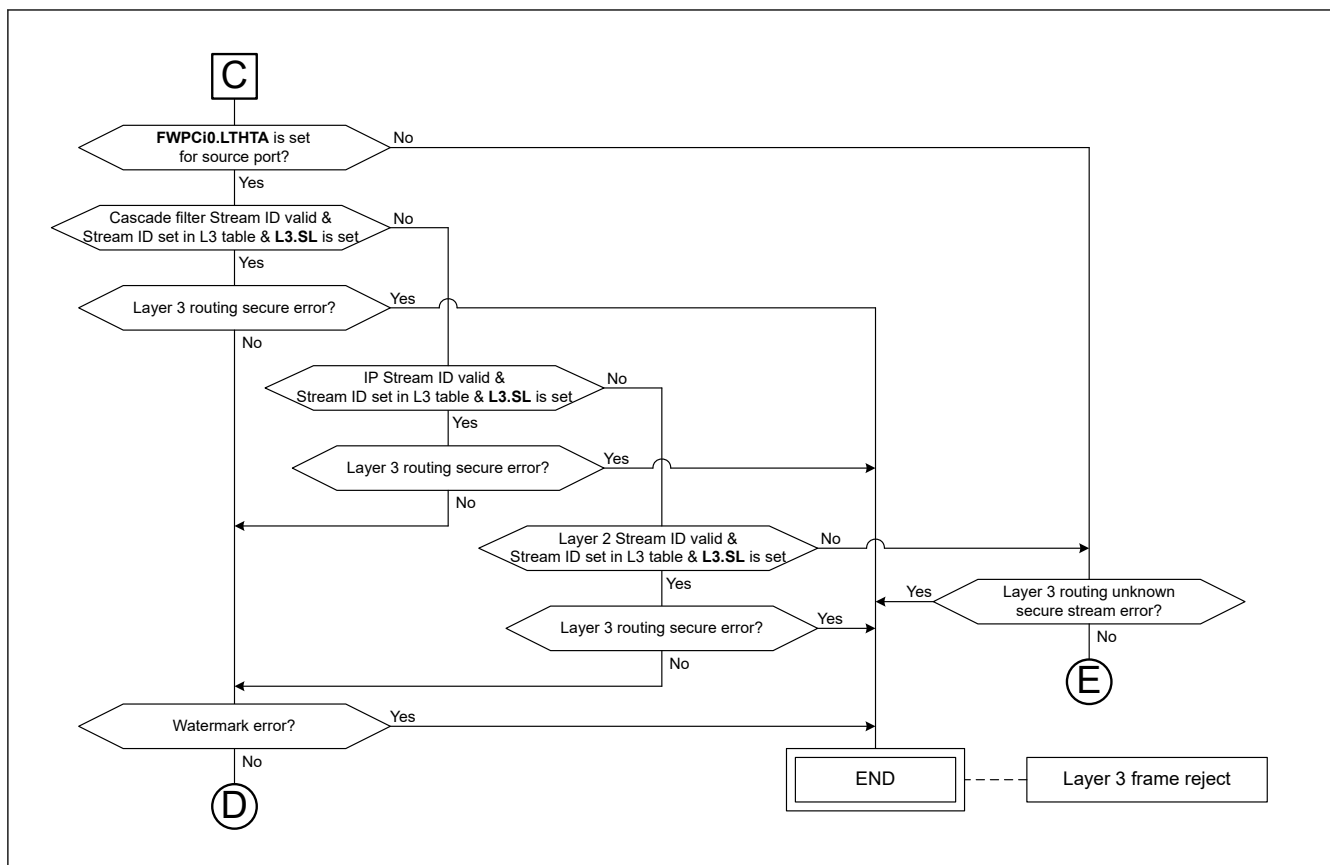


Figure 30.85 Layer 3 forwarding/routing secure flow

- Note:
- Layer 3 routing secure errors and Layer 3 routing unknown secure stream errors are described in section (1) [Layer 3 routing errors](#).
 - Watermark errors are described in section (1) [Watermark errors](#).
 - Link C is linked from Figure 2.
 - Link D links to [Figure 30.87](#).
 - Link E links to [Figure 30.95](#)

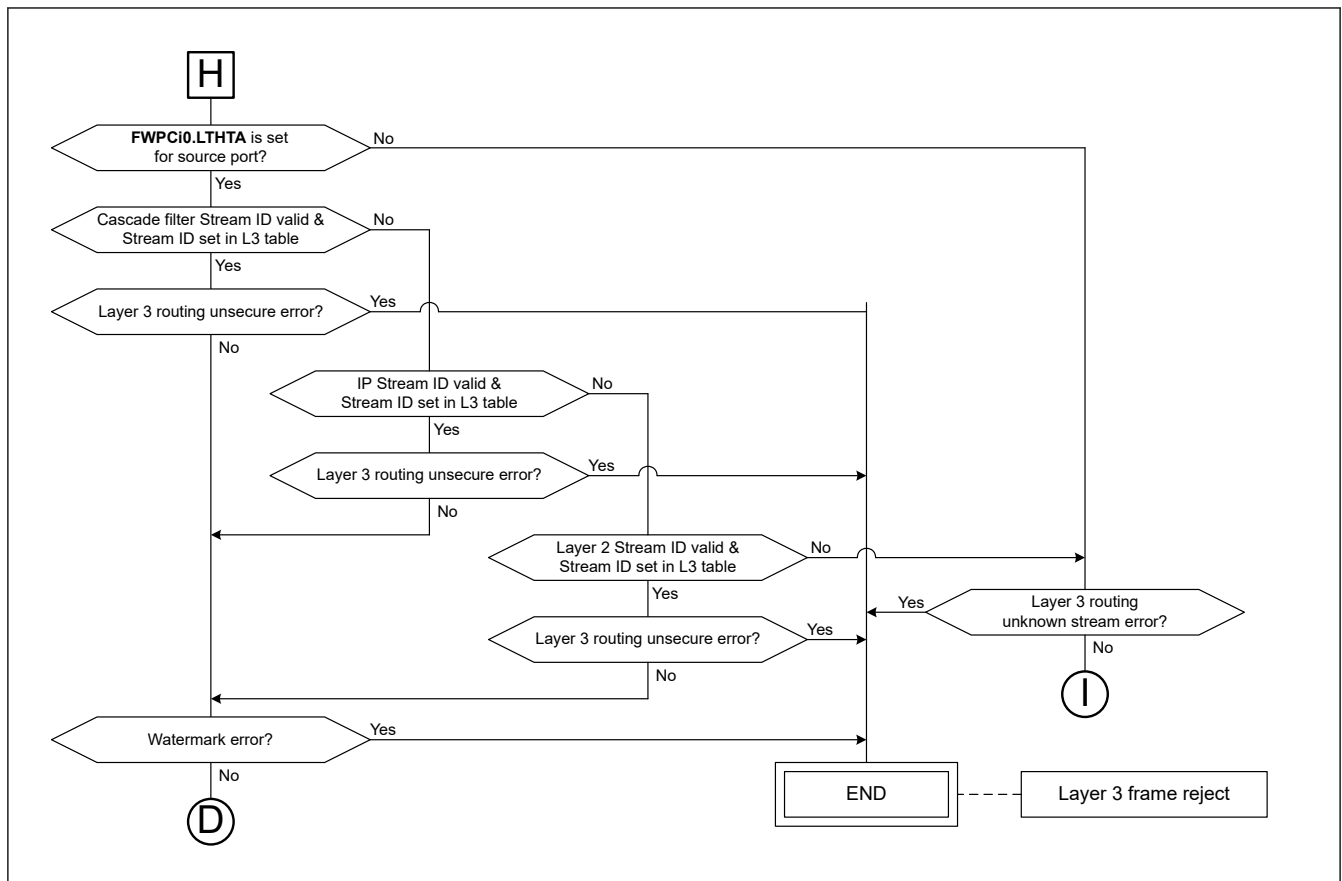


Figure 30.86 Layer 3 forwarding/routing unsecure flow

- Note:
- Layer 3 routing errors and Layer 3 routing unknown stream errors are described in section (1) [Layer 3 routing errors](#).
 - Watermark errors are described in section (1) [Watermark errors](#).
 - Link H is linked from [Figure 30.99](#).
 - Link D links to [Figure 30.87](#).
 - Link I links to [Figure 30.96](#).

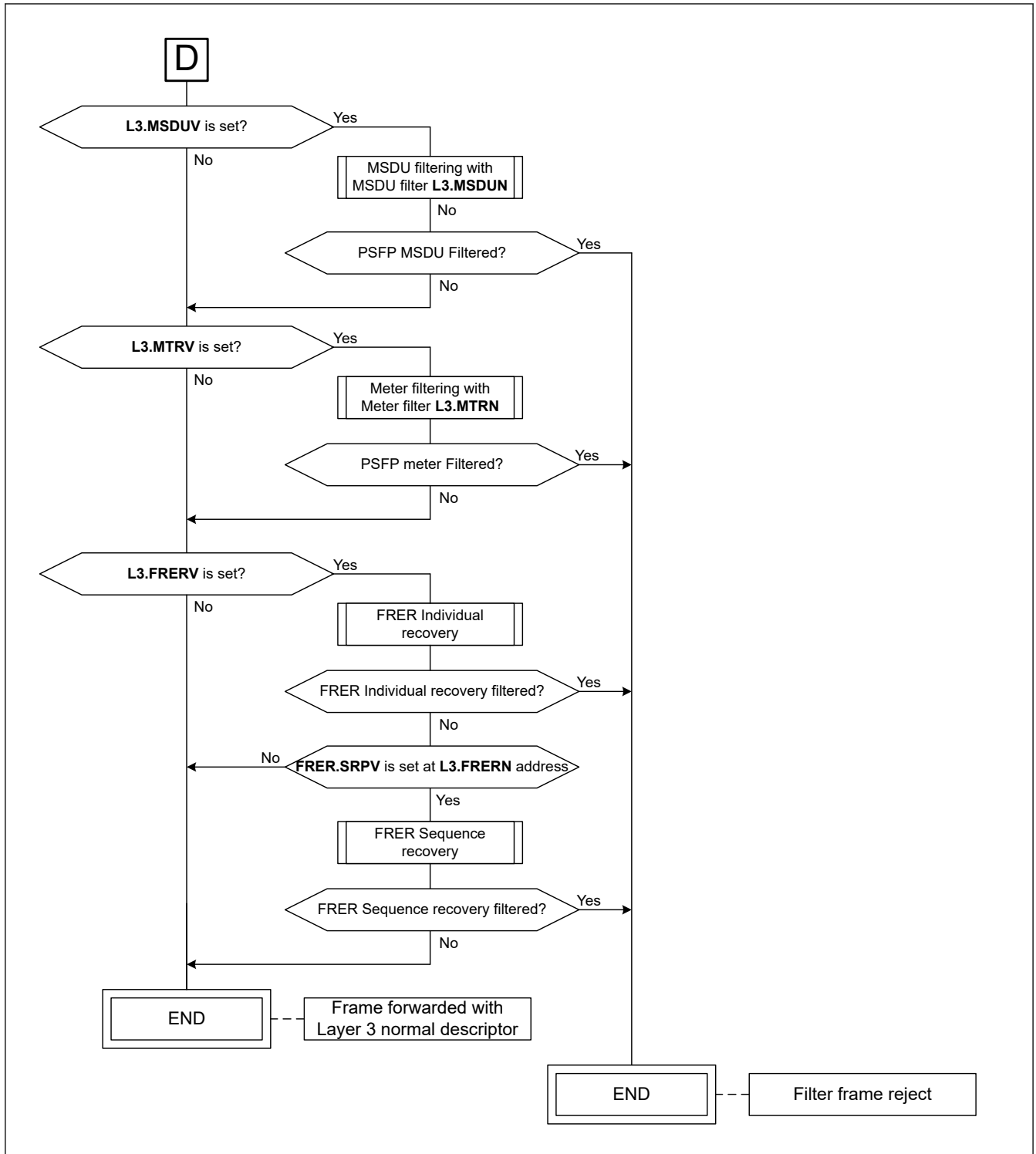


Figure 30.87 Layer 3 filtering flow

- Note:
- Link D is linked from [Figure 30.85](#) and [Figure 30.86](#).
 - Filter frame reject is described in section (2) [Filtering errors](#).
 - MSDU filtering is described in [Figure 30.103](#).
 - Meter filtering is described in [Figure 30.104](#).
 - Individual recovery is described in section (e) [Individual recovery algorithms](#).
 - Sequence recovery is described in section (f) [Sequence recovery algorithms](#).

(1) Layer 3 routing errors

Table 30.35 describes the Layer 3 routing errors. Instead of being discarded by forwarding engine, an error frame can be forwarded to exceptional path using Layer 3 exceptional descriptor described in section (4) [Layer 3 exceptional descriptor](#) by setting corresponding error exceptional bit. For error set conditions, refer to corresponding error interrupt register explanations.

Table 30.35 Layer 3 routing errors

Error	Error name	Error interrupt	Error exceptional bit
Layer 3 routing secure errors			
SNTF	Secure No Target Filtering	FWEIS0i.LTHNTFS	FWCEPRC2.FLTHNTFEF
SSPF	Secure Source Port Filtering	FWEIS0i.LTHSPFS	FWCEPRC2.FLTHSLFEF
Layer 3 routing unknown secure stream errors			
SUF	Secure Unknown filtering	FWEIS0i.LTHUFS	FWCEPRC2.FLTHUFEF
Layer 3 routing errors			
NTF	No Target Filtering	FWEIS0i.LTHNTFS	FWCEPRC2.FLTHNTFEF
SPF	Source Port Filtering	FWEIS0i.LTHSPFS	FWCEPRC2.FLTHSLFEF
Layer 3 routing unknown stream errors			
UF	Unknown filtering	FWEIS0i.LTHUFS	FWCEPRC2.FLTHUFEF

(2) Filtering errors

Table 30.36 describes the Filter errors. Instead of being discarded by forwarding engine, an error frame can be forwarded to exceptional path using Filter exceptional descriptor described in section (5) [Filter exceptional descriptor](#) by setting corresponding error exceptional bit. For error set conditions, refer to corresponding error interrupt register explanations.

Table 30.36 Filtering errors

Error	Error name	Error interrupt	Error exceptional bit
MSDUF	MSDU filtered	FWEIS2.PMF5	FWCEPRC1.FMSDUFEF
MTRF	Meter filtered	FWEIS5.PMRFS	FWCEPRC1.FMTRFEF
IFRERF	Individual FRER Filtered	FWEIS60/61/62/63.FFS for no sequence error of stuck error FWEIS70/71/72/73.FOORS for out of range error	FWCEPRC1.FIFFEF
SFRERF	Sequence FRER Filtered	FWEIS60/61/62/63.FFS for no sequence error FWEIS70/71/72/73.FOORS for out of range error No interrupt if a frame is discarded because it was a duplicate	FWCEPRC1.FSFFEF

(3) Layer 3 normal descriptor

Layer 3 normal descriptor fields are described in Table 30.37.

Table 30.37 Layer 3 normal descriptor fields (1 of 2)

Field name FDESCR.	Values
DV	L3.DV & !FWPCi1.LTHFM
SEC	L3.SL
CSD	L3.CSD
IPV	If L3.IPU is set, set to L3.IPV. If L3.IPU is not set, set to the frame input priority. (See section 30.5.1.4. ESWM Frame Input Priority.)
MINFO	See Figure 30.88 and Table 30.38 .

Table 30.37 Layer 3 normal descriptor fields (2 of 2)

Field name FDESCR.	Values
RV	L3.RV
RN	L3.RN
SEQN	Set to frame input sequence number if L3.RN is not mapped to a sequence number generation function (refer to section section 30.5.3.2.2. Sequence Number Generation)

Byte	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0	FCLR[2:0]		FFC[2:0]		FWDC[3:0]			
1	FID[15:0]							
2								

Figure 30.88 Layer 3 normal descriptor MINFO format**Table 30.38 Layer 3 normal descriptor MINFO field explanation**

Field name FDESCR.	Bit width	Field explanation	Values
FWDC	4	ForWarDing Code Present in all descriptors, allowing the type of forwarding to be identified	Fixed to 5 for Layer 3 normal descriptors
FFC	2	Frame Format Code	Set to 0 for perfect filter forwarding Set to 1 for IPv4 forwarding Set to 2 for IPv6 forwarding Set to 3 for Layer 2 stream forwarding
FCLR	2	Frame CoLoR	Set to 0 when frame has no color Set to 1 when frame is green Set to 2 when frame is yellow Set to 3 when frame is red Refer to section section 30.5.3.1.2. Meter Filter/ATS for color attribution
FID	16	Frame IDentifier	For Cascade filter stream ID, set to the cascade filter number that matched. For IPV stream ID, set to the hash ID inserted in the stream ID. For L2 stream, set to All0.

(4) Layer 3 exceptional descriptor

Layer 3 exceptional descriptor fields are described in [Table 30.39](#).

Table 30.39 Layer 3 exceptional descriptor fields

Field name FDESCR.	Values
DV	Exceptional descriptors are forwarded to GWCA number FWCEPTC.EPCS (port number 2)
SEC	FWCEPTC.EPSL
CSDj	FWCEPTC.EPCSD
IPV	FWCEPTC.EPIPV
MINFO	See Figure 30.89 and Table 30.40 .
RV	0
RN	ALL0
SEQN	Set to frame input sequence number

Byte	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0	FCLR[1:0]		FFC[1:0]		FWDC[3:0]			
1	—					MTRF	GATEF	MSDUF
2	—							

Figure 30.89 Layer 3 exceptional descriptor MINFO format

Table 30.40 Layer 3 exceptional descriptor MINFO field explanation

Field name FDESCR.	Bit width	Field explanation	Values
FWDC	4	ForWarDing Code Present in all descriptors, allowing the type of forwarding to be identified	Fixed to 6 for Layer 3 exceptional descriptors
FFC	2	Frame Format Code	Set to 0 for perfect filter forwarding Set to 1 for IPv4 forwarding Set to 2 for IPv6 forwarding Set to 3 for Layer 2 stream forwarding Restrictions: HW: This field is not valid when Secure Unknown filtering or Unknown filtering happened.
SNTF	1	Secure No Target Filtering	See (1) Layer 3 routing errors.
SSPF	1	Secure Source Port Filtering	See (1) Layer 3 routing errors.
SUF	1	Secure Unknown filtering	See (1) Layer 3 routing errors.
NTF	1	No Target Filtering	See (1) Layer 3 routing errors.
SPF	1	Source Port Filtering	See (1) Layer 3 routing errors.
UF	1	Unknown filtering	See (1) Layer 3 routing errors.

(5) Filter exceptional descriptor

Filter exceptional descriptor fields are described in [Table 30.41](#).

Table 30.41 Filter exceptional descriptor fields

Field name FDESCR.	Values
DV	Exceptional descriptors are forwarded to GWCA number FWCEPTC.EPCS (port number 2)
SEC	FWCEPTC.EPSL
CSDj	FWCEPTC.EPCSD
IPV	FWCEPTC.EPIPV
MINFO	Refer to Figure 30.90 and Table 30.42
RV	0
RN	ALL0
SEQN	Set to frame input sequence number

Byte	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0	FCLR[1:0]		FFC[1:0]		FWDC[3:0]			
1	—					MTRF	GATEF	MSDUF
2	—							

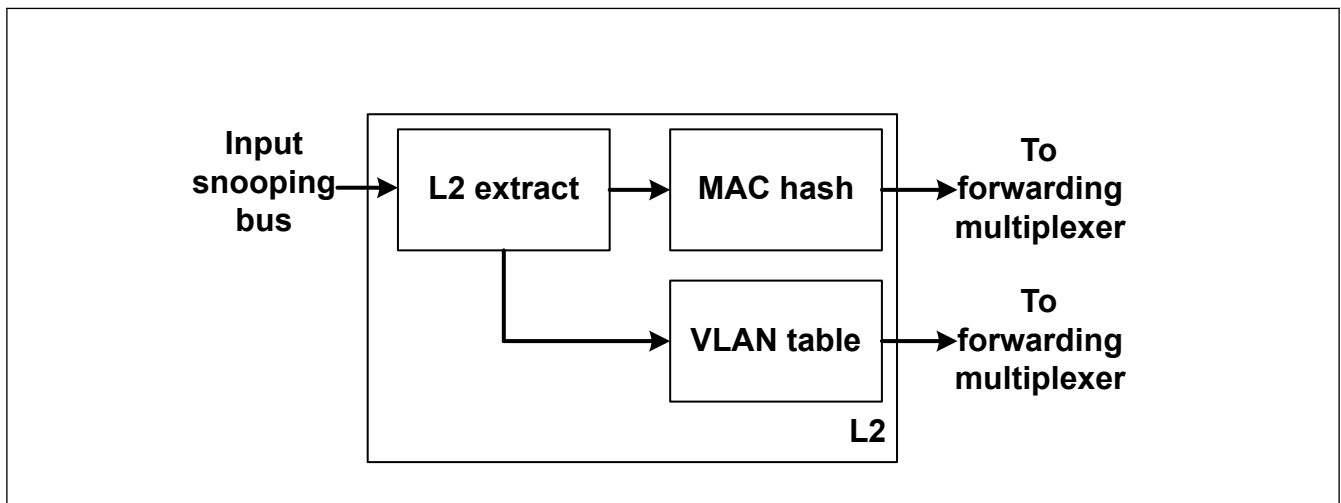
Figure 30.90 Filter exceptional descriptor MINFO format

Table 30.42 Filter exceptional descriptor MINFO field explanation

Field name FDESCR.	Bit width	Field explanation	Values
FWDC	4	ForWarDing Code Present in all descriptors, allowing the type of forwarding to be identified	Fixed to 15 for Layer 3 exceptional descriptors
FFC	2	Frame Format Code	Set to 0 for perfect filter forwarding Set to 1 for IPv4 forwarding Set to 2 for IPv6 forwarding Set to 3 for Layer 2 stream forwarding
FCLR	2	Frame CoLoR	Set to 0 when frame has no color Set to 1 when frame is green Set to 2 when frame is yellow Set to 3 when frame is red Refer to section section 30.5.3.1.2. Meter Filter/ATS for color attribution
MSDUF	1	MSDU Filtered	See (2) Filtering errors .
MTRF	1	Meter Filtered	See (2) Filtering errors .
IFRERF	1	Individual FRER Filtered	See (2) Filtering errors .
IFNOSEQ	1	Individual FRER NO SEQUENCE	See (2) Filtering errors .
IFOOR	1	Individual FRER Out Of Range (For match algorithm, this flag is set for Out of order error)	See (2) Filtering errors .
FSTUCK	1	FRER Stuck	See (2) Filtering errors .
SFRERF	1	Sequence FRER Filtered	See (2) Filtering errors .
SFNOSEQ	1	Sequence FRER NO SEQUENCE	See (2) Filtering errors .
SFOOR	1	Sequence FRER Out Of Range (For match algorithm, this flag is set for Out of order error)	See (2) Filtering errors .

30.5.2.5 Layer 2 Forwarding

[Figure 30.91](#) describes the block diagram of Layer 2 forwarding.

**Figure 30.91 Block diagram of Layer 2 forwarding**

The Layer 2 forwarding function is separated in three functional blocks:

- **L2 extract:** This block snoops all the required information from the forwarding engine snooping bus and extracts the MAC destination, MAC source addresses and the VLAN ID from frames.
- **MAC hash:** This block handles the MAC addresses meta information such as forwarding and routing information using a hash function.

- VLAN table: This block handles the VLAN ID meta information such as forwarding and routing information.

Note: Forwarding multiplexer contains the forwarding flow. The forwarding flow is explained by forwarding type. For Layer 3 forwarding/routing/filtering, related forwarding flows are described in [Figure 30.95](#) and [Figure 30.96](#).

30.5.2.5.1 L2 Extract

The L2 extract is used to extract the source and destination MAC addresses and the VLAN ID all frames.

[Figure 30.92](#) describes MAC addresses and VLAN ID extraction. MAC addresses extraction always occurs. VLAN extraction only occurs when FWGC.SVM is different from 00b. If VLAN ID extraction does not occur, all VLANs are considered as unknown.

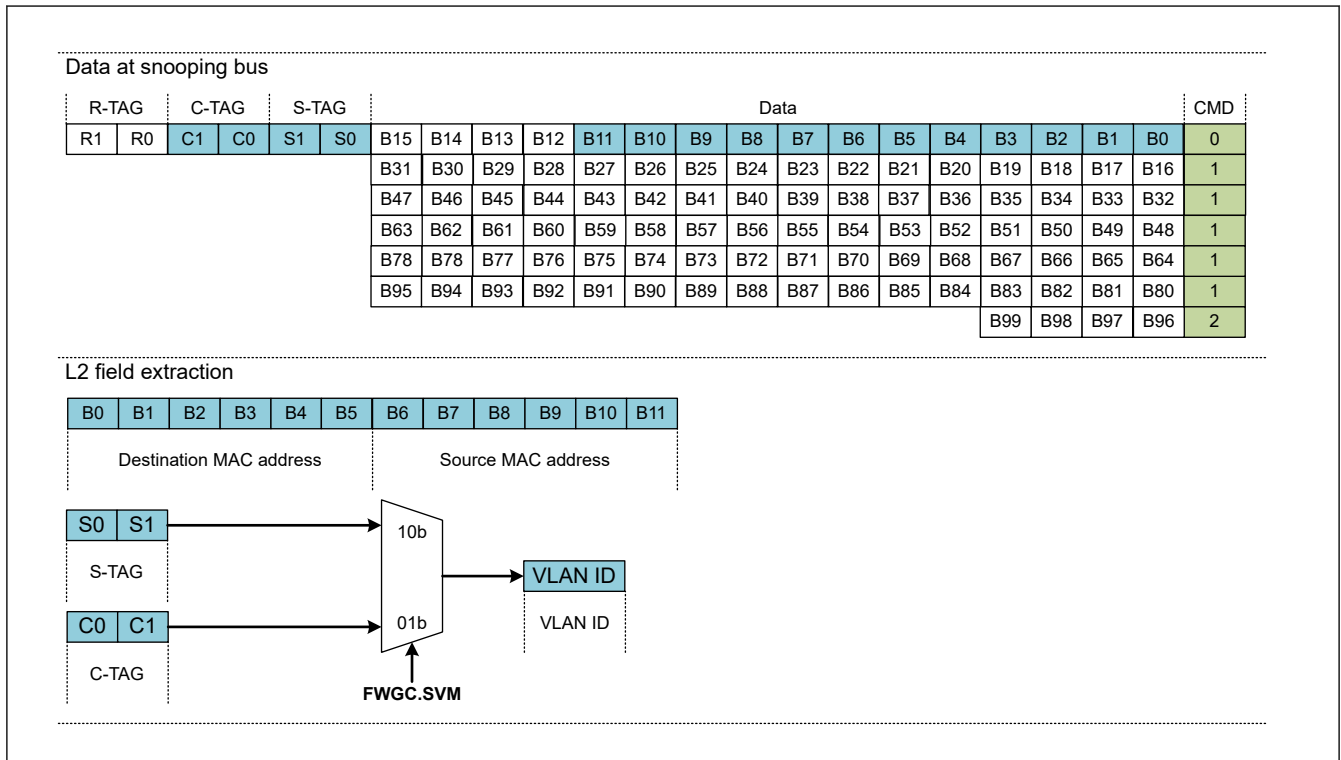


Figure 30.92 MAC address and VLAN ID extraction

30.5.2.5.2 MAC Hash

MAC hash aims at searching/learning/reading MAC rules thanks to a programmable hash. The hash algorithm is able to solve all the collisions in hardware, so it does not need any software intervention to operate.

MAC hash is divided in the following items:

- Rule format: Determines what information will be store along with the MAC address in the table.
- SW Learning: Function to add/suppress an entry from the MAC table by software.
- Hash maximum collision number: Avoids hash search to slow down the forwarding pipeline and apply back pressure on the ingress ports.
- Hash equation: Function to set hash calculation to minimize collisions.
- SW Searching: Function to find the forwarding information related to a MAC address by software.
- SW Reading: Function to read the entries in the MAC table by software.
- HW Searching: Function to find the forwarding information related to a MAC address by forwarding mechanism.
- HW Learning: Function to learn unicast MAC source addresses by forwarding mechanism.
- HW aging: Function used to suppress unused dynamic MAC addresses.

(1) Rule format

The MAC table is used to store MAC rules. A MAC rule contains the information for the frame forwarding. All the fields in this table, if quoted, will be written MAC.{Field name}. [Table 30.43](#) describes fields contained in a MAC rule.

Table 30.43 MAC rule format (1 of 2)

Field name	Field size (bit)	Field Explanation
EV	1	Entry Valid Values: 0: Entry is not valid 1: Entry is valid
SL	1	Security Level Values: 0: Entry is unsecure 1: Entry is secure
MA	48	MAC Address of the entry. The MAC address is extracted by L2 extract. (See section 30.5.2.5.1. L2 Extract) This address is in big endian.
HLD	1	Hardware Learn Disable Values: 0: Source MAC hardware learning is enabled for frames containing MAC.MAC as destination MAC address. 1: Source MAC hardware learning is disabled for frames containing MAC.MAC as destination MAC address.
DSLVL	4	Destination Source Lock Vector Values: - Bit i set to 0: Frames with a destination MAC matching this entry coming from port i are rejected by source port lock error. - Bit i set to 1: Frames with a destination MAC matching this entry coming from port i are forwarded/routed.
SSLVL	4	Source Source Lock Vector Values: - Bit i set to 0: Frames with a source MAC matching this entry coming from port i are rejected by source port lock error. - Bit i set to 1: Frames with a source MAC matching this entry coming from port i are forwarded/routed.
DV	4	Destination vector Functions: HW: Ports to where the frame should be sent.
CSD	6	CPU Sub-Destinations Functions: HW: CPU Sub-Destinations the frame should be sent.
CME	1	CPU Mirroring Enable Values: 0: CPU mirroring is disabled. 1: CPU mirroring is enabled.
EME	1	Ethernet Mirroring Enable Values: 0: Ethernet mirroring is disabled. 1: Ethernet mirroring is enabled.
IPU	1	Internal Priority Update Values: 0: Frames matching this entry will not have their internal priority updated. 1: Frames matching this entry will have their internal priority updated.
IPV	3	Internal Priority Value Functions: HW: If corresponding entry MAC.IPU value is set to 1, frames matching this entry will have their priority updated to MAC.IPV.
DE	1	Dynamic Entry Values: 0: Entry is static. Migration and aging cannot be applied to the entry. 1: Entry is dynamic. Migration and aging can be applied to the entry.

Table 30.43 MAC rule format (2 of 2)

Field name	Field size (bit)	Field Explanation
AB	1	Aging Bit: Values: 0: Entry has been searched as a MAC source address by forwarding mechanism since previous aging. The entry should not be deleted during next aging. 1: Entry has not been searched as a MAC source address by forwarding mechanism since previous aging. The entry should be deleted during next aging if it is not searched as a MAC source address by forwarding mechanism before.

(2) SW Learning

Learning is used to add/remove an entry in the MAC table. [Table 30.44](#) describes register used to learn an entry in the MAC table. [Table 30.45](#) describes the learn results.

Table 30.44 MAC learn registers

Register name	Field name/corresponding field in MAC table	Field explanation
FWMACTL0.MACED	Entry delete	Not present in L3 table, used to select between learning an entry and deleting an entry
FWMACTL0.MACSL	MAC.SL	See (1) Rule format.
{FWMACTL1.MACMALP0, FWMACTL2.MACMALP1}	MAC.MA	See (1) Rule format.
FWMACTL0.MACDEL	MAC.DE	See (1) Rule format.
FWMACTL0.MACHLDL	MAC.HLD	See (1) Rule format.
FWMACTL3.MACDSLVL	MAC.DSLV	See (1) Rule format.
FWMACTL3.MACSSLVL	MAC.SSLV	See (1) Rule format.
FWMACTL40.MACCSD	MAC.CSD[5:0]	See (1) Rule format.
FWMACTL5.MACDVL	MAC.DV	See (1) Rule format.
FWMACTL5.MACCMEL	MAC.CME	See (1) Rule format.
FWMACTL5.MACEMEL	MAC.EME	See (1) Rule format.
FWMACTL5.MACIPUL	MAC.IPU	See (1) Rule format.
FWMACTL5.MACIPVL	MAC.IPV	See (1) Rule format.

Table 30.45 MAC learn result (1 of 2)

Register name	Field name/corresponding field in MAC table	Field explanation
FWMACTLR.MACLF	Learning fail	Learning fail with happen in one of the following conditions: - The learn entry is not yet in the table and the table is already full (FWMACTEM.MACTEN == 2048). - The MAC table is not ready (FWMACTIM.MACTR is not set) - The learning function is used to delete an entry (FWMACTL0.MACED set for learning) and the entry is not found in the MAC table
FWMACTLR.MACLSF	Learning security fail	Security fail with happen in one of the following conditions: - The learn entry is an unsecure entry (FWMACTL0.MACSL not set for learning) and the unsecure entry number FWMACTEM.MACTUEN has reached the maximum unsecure entry number FWMACHEC.MACHMUE. - The learn action is done by the APB which tries to learn an unsecure entry when the entry is already in the table, is secure (MAC.SL is set) and when FWMACTEM.MACTUEN has reached the maximum unsecure entry number FWMACHEC.MACHMUE.
FWMACTLR.MACLO	Learning overwrite	Overwrite with happen in one of the following conditions: The learn MAC is already present in the table, and no error happened during learning.

Table 30.45 MAC learn result (2 of 2)

Register name	Field name/ corresponding field in MAC table	Field explanation
FWMACTLR.MACLCN	Learning collision number	Give the number of collisions that needs to be resolved to access the learnt entry if the lean succeeded. If FWMACTLR.MACLCN value comes out bigger than FWMACHEC.MACHMC register value, the newly learnt entry will be ignored by forwarding engine during forwarding. A switch reconfiguration with a different MAC Hash equation should be considered.

Note: HW: The fuller the table is, the slower Learning gets.

(3) Hash maximum collision number

Hash maximum number avoids the ESWM to have back pressure due to a slow forwarding mechanism.

Equation (2) shows how to calculate this number. This number should be set in FWMACHEC.MACHMC register.

$$(2) \text{ FWMACHEC.MACHMC} = (\text{clk_freq}[\text{Hz}] * \text{Average_frame_size}[\text{bit}] / \text{Incoming_throughput}[\text{bps}] - 4) / 3$$

Where:

- *clk_freq* is the clock frequency of ESWCLK.
- *Average_frame_size* is the average frame size expected on switch inputs
- *Incoming_throughput* is the total throughput expected on switch inputs.

For example, if *clk_freq* is 150 MHz, *Average_frame_size* is 128 bytes and *Incoming_throughput* is 3 Gbps (1 Gbps of data on 2 GMII and 1 Gbps of data on 1 GWCA), the value 15 should be set to FWLTHHEC.LTHHMC register.

(4) Hash equation

The hash equation is controlled by register FWMACHC.MACHE.

The initial value of the equation is $1 + x^{11}$.

If there are too many collisions in the table ($\text{FWMACTLR.MACLCN} > \text{FWMACHEC.MACHMC}$ during learning), this hash equation can be changed to minimize the number of collisions by:

- SW: Modelling the hash equations and find an equation which have a reasonable number of collisions.
- SW: Trying randomly some equations until finding an equation which have a reasonable number of collisions.

(5) SW Searching

Searching is used to find the information related to a MAC address. [Table 30.46](#) describes register used to search an MAC address in the MAC table. [Table 30.47](#) describes the search results.

Table 30.46 MAC search registers

Register name	Field name/corresponding field in MAC table	Field explanation
{FWMACTS0.MACMASP0, FWMACTS1.MACMASP1}	MAC.MA	See (1) Rule format .

Table 30.47 MAC search result (1 of 2)

Register name	Field name/corresponding field in MAC table	Field explanation
FWMACTSR0.MACSNF	Search not found.	Search failed because the searched MAC address is not in the table
FWMACTSR0.MACSCN	Search collision number	Give the number of collisions occurred to find the entry.
FWMACTSR0.MACSLS	MAC.SL	See (1) Rule format .
FWMACTSR0.MACDES	MAC.DE	See (1) Rule format .

Table 30.47 MAC search result (2 of 2)

Register name	Field name/corresponding field in MAC table	Field explanation
FWMACTSR0.MACHLDS	MAC.HLD	See (1) Rule format.
FWMACTSR1.MACDSLVS	MAC.DSLV	See (1) Rule format.
FWMACTSR1.MACSSLVS	MAC.SSLV	See (1) Rule format.
FWMACTSR20.MACCSDS	MAC.CSD[5:0]	See (1) Rule format.
FWMACTSR3.MACDVS	MAC.DV	See (1) Rule format.
FWMACTSR3.MACCMES	MAC.CME	See (1) Rule format.
FWMACTSR3.MACEMES	MAC.EME	See (1) Rule format.
FWMACTSR3.MACIPUS	MAC.IPU	See (1) Rule format.
FWMACTSR3.MACIPVS	MAC.IPV	See (1) Rule format.

(6) SW Reading

Reading is used to dump the MAC table content. This function cannot be used to read a specific entry because it is not possible for the software to know at which address an entry can be found in the MAC RAM.

Table 30.48 MAC read registers

Register name	Field name/corresponding field in MAC table	Field explanation
FWMACTR.MACAR	Read address	Address that will be used to read an entry in the Layer 3 table

Table 30.49 MAC read result

Register name	Field name/corresponding field in MAC table	Field explanation
FWMACTRR0.MACEVR	MAC.EV	The read address holds a valid entry Only if this bit is set when reading an address, the corresponding entry and the following registers in this table are valid. See (1) Rule format.
{FWMACTRR2.MACMARPO, FWMACTRR3.MACMARPI}	MAC.MA	See (1) Rule format.
FWMACTRR1.MACSLR	MAC.SL	See (1) Rule format.
FWMACTRR1.MACDER	MAC.DE	See (1) Rule format.
FWMACTRR1.MACHLDR	MAC.HLD	See (1) Rule format.
FWMACTRR1.MACAB	MAC.AB	See (1) Rule format.
FWMACTRR4.MACDSLVR	MAC.DSLV	See (1) Rule format.
FWMACTRR4.MACSSLVR	MAC.SSLV	See (1) Rule format.
FWMACTRR50.MACCSDR	MAC.CSD[5:0]	See (1) Rule format.
FWMACTRR6.MACDVR	MAC.DV	See (1) Rule format.
FWMACTRR6.MACCMER	MAC.CME	See (1) Rule format.
FWMACTRR6.MACEMER	MAC.EME	See (1) Rule format.
FWMACTRR6.MACIPUR	MAC.IPU	See (1) Rule format.
FWMACTRR6.MACIPVR	MAC.IPV	See (1) Rule format.

(7) HW learning/migration

MAC HW learning/migration conditions and settings are described in section [section 30.5.1.5.2. CPU Learning Path](#). [Table 30.50](#) describes to what value the MAC entry fields will be set while learning in hardware a frame coming from port i.

For Source MAC multicast address, HW learning/migration will invalid (ignored).

Table 30.50 MAC hardware learning values

Field name	Field Explanation
MAC.EV	1
MAC.SL	0
MAC.MA	Incoming frame MAC source
MAC.HLD	0
MAC.DSLV	0xF
MAC.SSLV	0xF
MAC.DV	001b << i
MAC.CSD	0x00
MAC.CME	0
MAC.EME	0
MAC.IPU	0
MAC.IPV	000b
MAC.DE	1
MAC.AB	0

(8) HW Aging

MAC HW aging function is used to remove the unused MAC addresses conditions using FWMACAGUSPC and FWMACAGC registers and can be monitored using FWMACAGM0/1 registers and FWMIS0.MACADAS interrupt register. MAC HW aging function thanks to two algorithms, the aging algorithm described in [Figure 30.93](#) and the MAC source address search algorithm described in [Figure 30.94](#).

Functions:

- FWMACAGUSPC register sets a prescaler to create a 1 Hz for aging.
- FWMACAGC register sets/enables the aging function
- FWMIS0.MACADAS and FWMACAGM0/1 can be used to keep track in SW of the aged MAC addresses. These registers should be used only in polling mode (FWMACAGC.MACAGPM set).

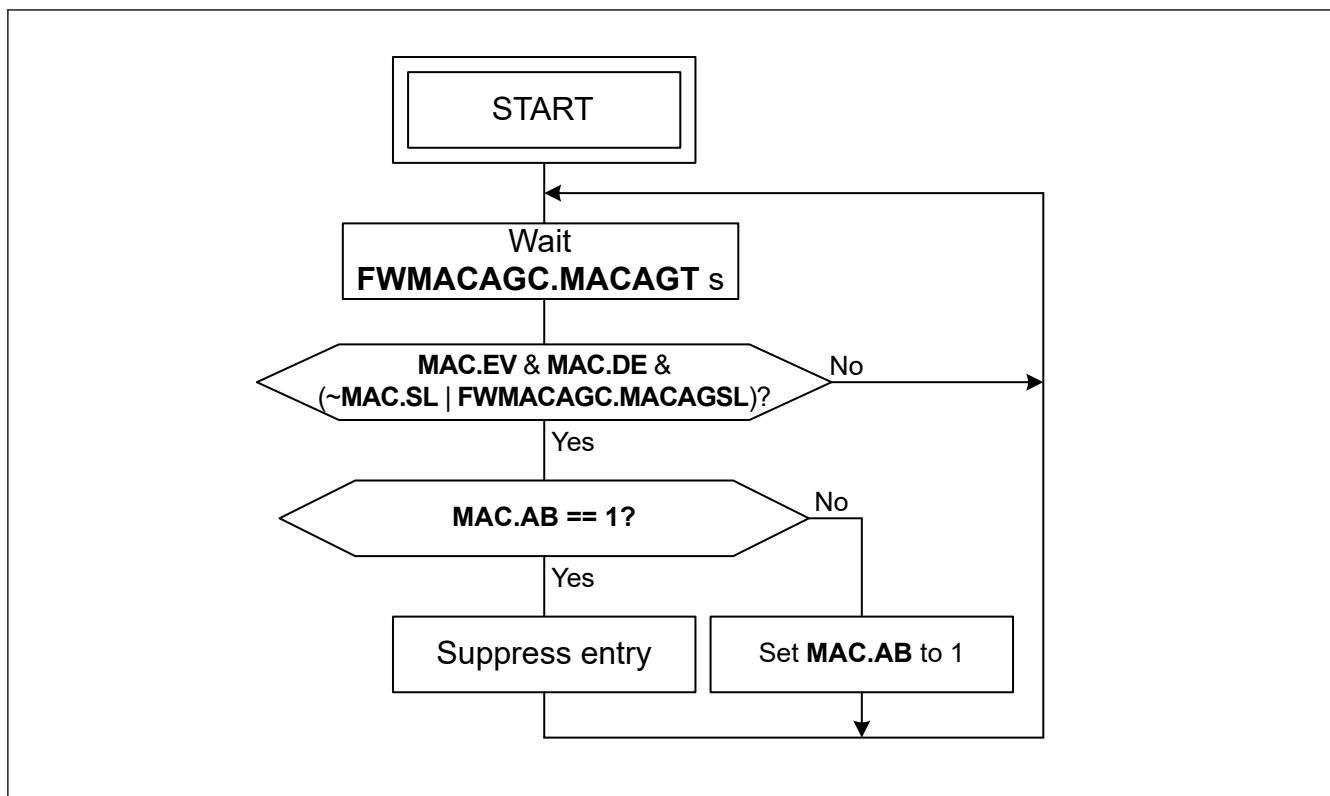


Figure 30.93 MAC aging algorithm

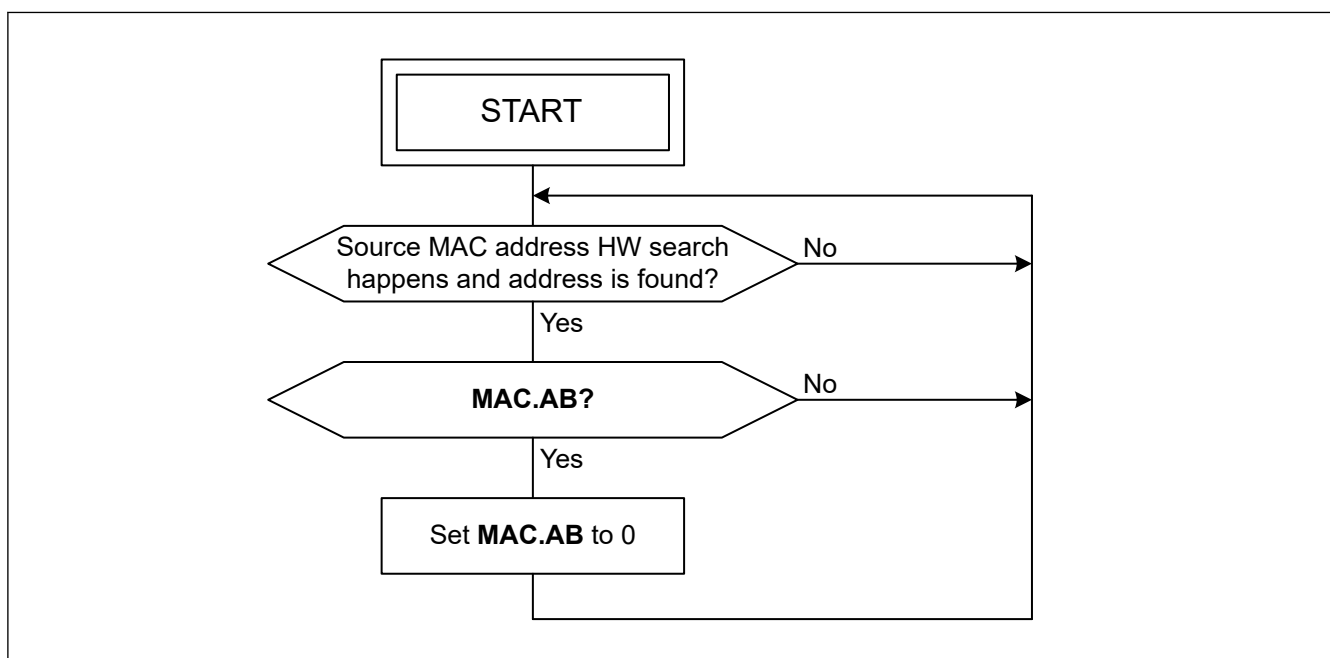


Figure 30.94 MAC source address search algorithm

30.5.2.5.3 VLAN Table

VLAN hash aims at searching/learning VLAN rules. The VLAN ID to VLAN RAM address mapping is a 1:1 mapping, so no hash is needed for VLAN handling.

VLAN table is divided in the following items:

- Rule format: Determines what information will be store in a VLAN rule. The VLAN ID is not stored a VLAN rule.
- SW Learning: Function to add/suppress an entry from the VLAN table by software.

- SW Searching: Function to find the forwarding information related information to a VLAN ID by software.
- HW Searching: Function to find the forwarding information related to a VLAN ID by forwarding mechanism.

(1) Rule format

The VLAN table is used to store VLAN rules. A VLAN rule contains the information for the frame forwarding. All the fields in this table, if quoted, will be written VLAN.{Field name}. [Table 30.51](#) describes fields contained in a VLAN rule.

Table 30.51 VLAN rule format

Field name	Field size (bit)	Field description
EV	1	Entry Valid Values: • 0: Entry is not valid • 1: Entry is valid
SL	1	Security Level Values: • 0: Entry is unsecure • 1: Entry is secure
HLD	1	Hardware Learn Disable Values: • 0: Source MAC hardware learning is enabled for frames containing corresponding VLAN ID. • 1: Source MAC hardware learning is disabled for frames containing corresponding VLAN ID.
SLV	4	Source Lock Vector Values: • Bit i set to 0: Frames with a VLAN ID matching this entry coming from port i are rejected by source port lock error. • Bit i set to 1: Frames with a VLAN ID matching this entry coming from port i are forwarded/routed.
DV	4	Destination vector Functions: • HW: Ports to where the frame should be sent.
CSD	6	CPU Sub-Destinations Functions: • HW: CPU Sub-Destinations the frame should be sent
CME	1	CPU Mirroring Enable Values: • 0: CPU mirroring is disabled. • 1: CPU mirroring is enabled.
EME	1	Ethernet Mirroring Enable Values: • 0: Ethernet mirroring is disabled. • 1: Ethernet mirroring is enabled.
IPU	1	Internal Priority Update Values: • 0: Frames matching this entry will not have their internal priority updated. • 1: Frames matching this entry will have their internal priority updated.
IPV	3	Internal Priority Value Functions: • HW: If corresponding entry VLAN.IPU value is set to 1, frames matching this entry will have their priority updated to VLAN.IPV.

(2) SW Learning

Learning is used to add/remove an entry in the VLAN table. [Table 30.52](#) describes register used to learn an entry in the VLAN table. [Table 30.53](#) describes the learn results.

Table 30.52 VLAN learn registers

Register name	Field name/corresponding field in VLAN table	Field explanation
FWVLANTL0.VLANED	Entry delete	Not present in L3 table, used to select between learning an entry and deleting an entry
FWVLANTL0.VLANSL	VLAN.SL	See (1) Rule format.
FWVLANTL1.VLANIDL	VLAN ID	Correspond to the address of the learnt entry.
FWVLANTL0.VLANHLDL	VLAN.HLD	See (1) Rule format.
FWVLANTL2.VLANSLVL	VLAN.SLV	See (1) Rule format.
FWVLANTL30.VLANCSDL	VLAN.CSD[5:0]	See (1) Rule format.
FWVLANTL4.VLANDVL	VLAN.DV	See (1) Rule format.
FWVLANTL4.VLANCME	VLAN.CME	See (1) Rule format.
FWVLANTL4.VLANEME	VLAN.EME	See (1) Rule format.
FWVLANTL4.VLANIPUL	VLAN.IPU	See (1) Rule format.
FWVLANTL4.VLANIPVL	VLAN.IPV	See (1) Rule format.

Table 30.53 VLAN learn result

Register name	Field name/corresponding field in VLAN table	Field explanation
FWVLANTLR.VLANLF	Learning fail	Learning fail with happen in one of the following conditions: - The VLAN table is not ready (FWVLANTIM.VLANTR is not set). - The learning function is used to delete an entry (FWVLANTL0.VLANED set for learning) and the entry is not VLAN table.
FWVLANTLR.VLANLSF	Learning security fail	Security fail with happen in one of the following conditions: - The learning entry is an unsecure entry (FWVLANTL0.VLANSL not set for learning) and the unsecure entry number FWVLANTEM.VLANTUEN has reached the maximum unsecure entry number FWVLANTEC.VLANTMUE. - The learning action is done by the APB which tries to learn an unsecure entry when the entry is already in the table, is secure (VLAN.SL is set) and when FWVLANTEM.VLANTUEN has reached the maximum unsecure entry number FWVLANTEC.VLANTMUE.
FWVLANTLR.VLANLO	Learning overwrite	Overwrite with happen in one of the following conditions: The learning VLAN is already present in the table, and no error happened during learning.

(3) SW Searching

Searching is used to find the information related to a VLAN address. Table 30.54 describes register used to search a VLAN address in the VLAN table. Table 30.55 describes the search results. This function is not actually searching the address but only reading the address corresponding to the VLAN ID in the VLAN table.

Table 30.54 VLAN search registers

Register name	Field name/corresponding field in VLAN table	Field explanation
FWVLANTS.VLANVIDS	VLAN ID	Address that should be read in VLAN table. Correspond to the VLAN ID

Table 30.55 VLAN search result (1 of 2)

Register name	Field name/corresponding field in VLAN table	Field explanation
FWVLANTSR0.VLANSNF	Search not found.	Search failed because the searched VLAN ID is not in the table
FWVLANTSR0.VLANSL	VLAN.SL	See (1) Rule format.
FWVLANTSR0.VLANHLD	VLAN.HLD	See (1) Rule format.

Table 30.55 VLAN search result (2 of 2)

Register name	Field name/corresponding field in VLAN table	Field explanation
FWVLANTSR1.VLANSLVS	VLAN.SLV	See (1) Rule format.
FWVLANTSR20.VLANCSDS	VLAN.CSD[5:0]	See (1) Rule format.
FWVLANTSR3.VLANDVS	VLAN.DV	See (1) Rule format.
FWVLANTSR3.VLANCMES	VLAN.CME	See (1) Rule format.
FWVLANTSR3.VLANEMES	VLAN.EME	See (1) Rule format.
FWVLANTSR3.VLANIPUS	VLAN.IPU	See (1) Rule format.
FWVLANTSR3.VLANIPVS	VLAN.IPV	See (1) Rule format.

30.5.2.5.4 Layer 2 Forwarding

Layer 2 forwarding is used to forward local ethernet descriptors (LDESCR.FMT [GWCA] set to 0) received from agents using FWPCi0 and FWCEPRC2 register and can be monitored using FWEIS0i interrupt register. Layer 2 forwarding searches received MAC addresses ([section 30.5.2.5.1. L2 Extract](#)) in the MAC table to extract corresponding entry fields ((1) Rule format) and searches valid VLAN IDs ([section 30.5.2.5.1. L2 Extract](#)) in the VLAN table to extract corresponding entry fields ((1) Rule format). Depending on the entry fields, Layer 2 forwarding will forward frames following L2 forwarding secure flow in [Figure 30.95](#) and L2 forwarding unsecure flow in [Figure 30.96](#).

Functions:

- FWPCi0.MACDSA register is used to activate MAC table for frames received from port i for destination MAC.
- FWPCi0.MACRUEDA register is used to reject unknown destination MAC received from port i.
- FWPCi0.MACRUEDA register is used to reject secure unknown destination MAC received from port i.
- FWPCi0.MACSSA register is used to activate MAC table for frames received from port i for source MAC.
- FWPCi0.MACRUUSA register is used to reject unknown source MAC received from port i.
- FWPCi0.MACRUSSA register is used to reject secure unknown source MAC received from port i.
- FWPCi0.VLANSA register is used to activate VLAN table for frames received from port i for source MAC.
- FWPCi0.VLANRU register is used to reject unknown VLAN IDs received from port i.
- FWPCi0.VLANRUS register is used to reject secure unknown VLAN IDs received from port i.
- FWPCi2.LTWFM register is used to set to which ports frames received from port i can be forwarded.
- FWCEPRC2 register enables exceptional path for error descriptors.
- FWEIS0i interrupt register notifies that an ethernet descriptor has been rejected because of an error.

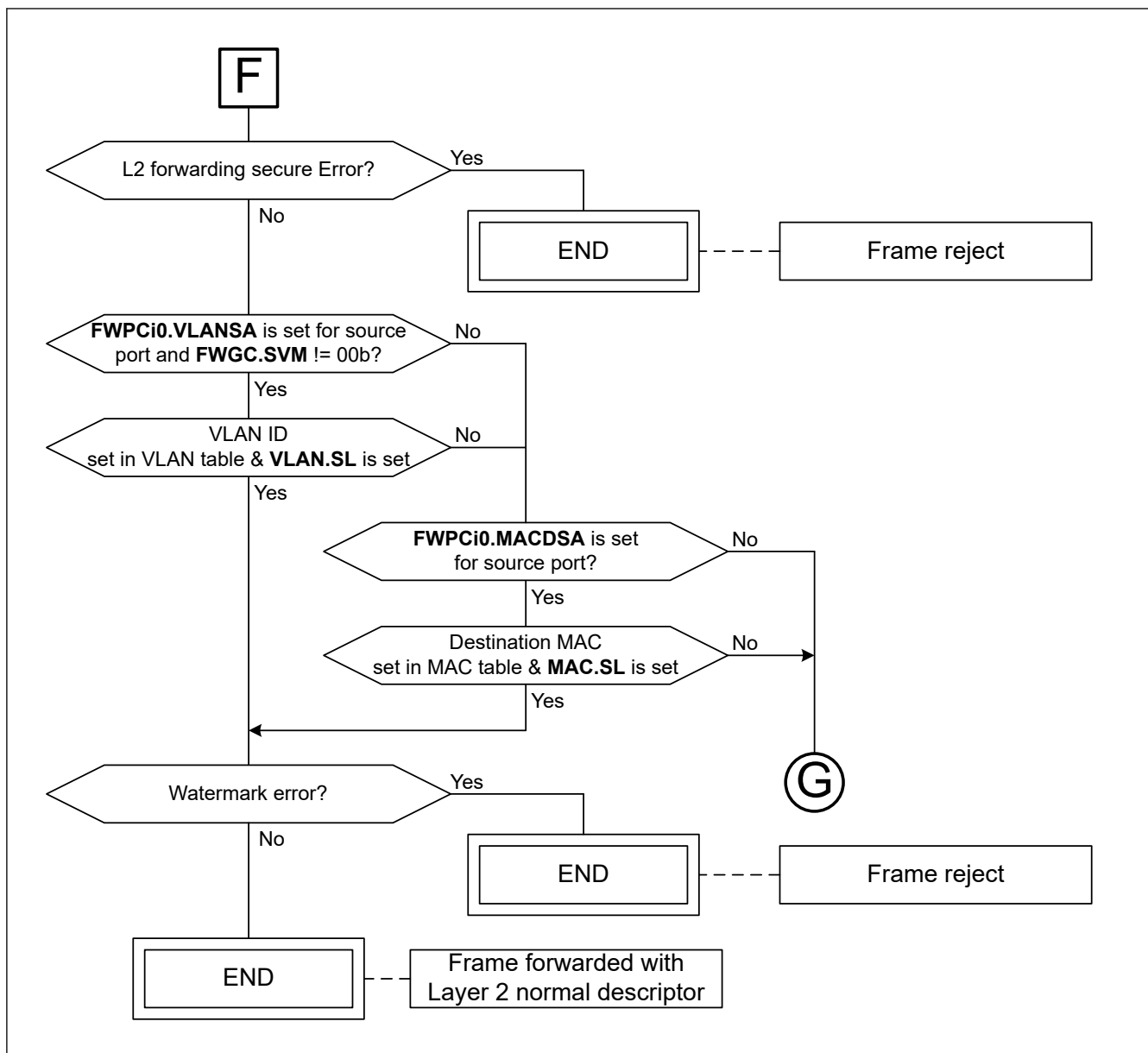


Figure 30.95 L2 forwarding secure flow

- Note:
- L2 forwarding secure errors are described in section (1) [L2 forwarding errors](#).
 - Watermark errors are described in section (1) [Watermark errors](#).
 - Link F is linked from [Figure 30.85](#) point E.
 - Link G links to [Figure 30.99](#).

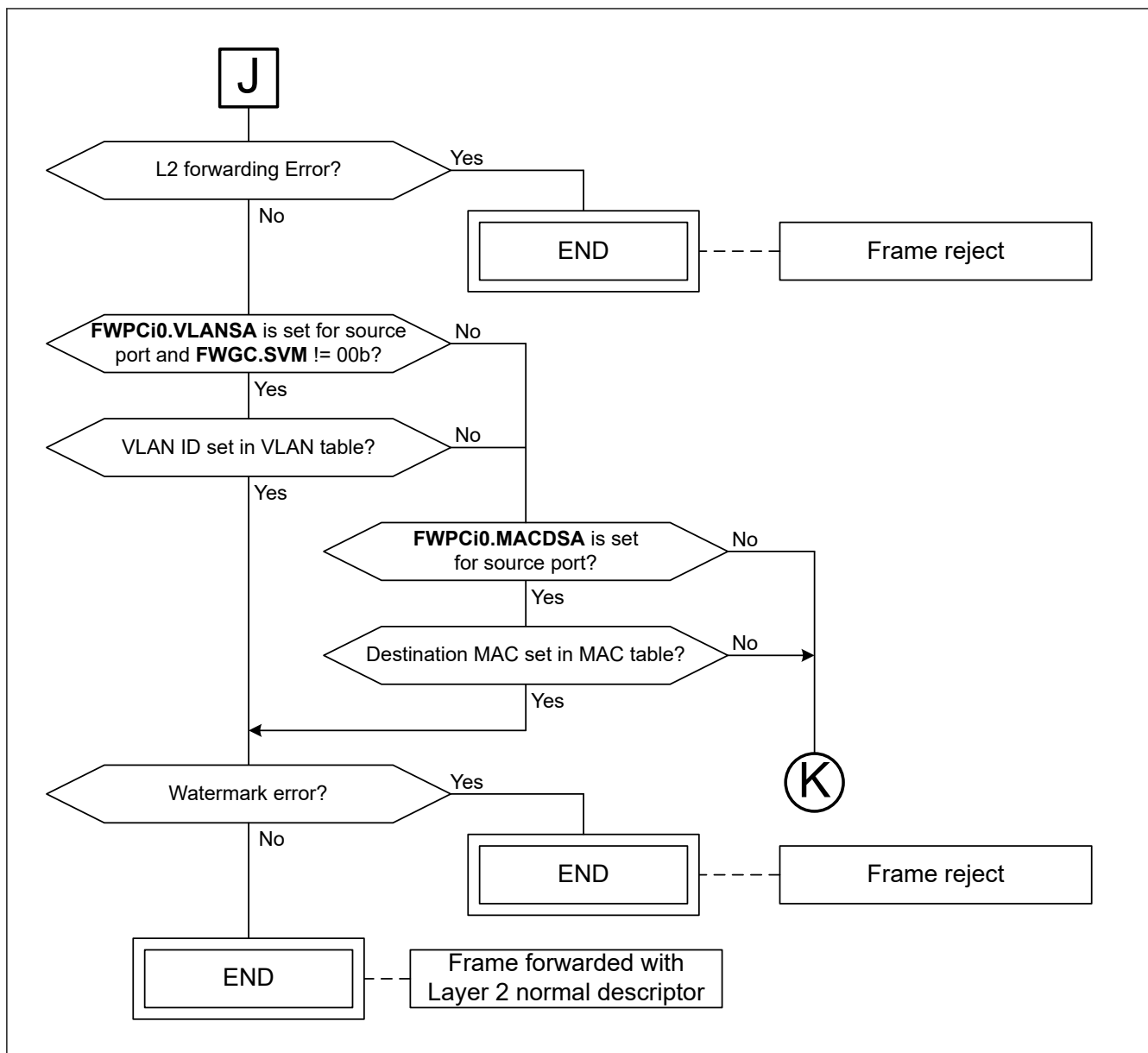


Figure 30.96 L2 forwarding unsecure flow

- Note:
- L2 forwarding errors are described in section (1) [L2 forwarding errors](#).
 - Watermark errors are described in section (1) [Watermark errors](#).
 - Link J is linked from [Figure 30.86](#) point I.
 - Link K links to [Figure 30.100](#).

(1) L2 forwarding errors

[Table 30.56](#) describes the L2 forwarding errors. Instead of being discarded by forwarding engine, an error frame can be forwarded to exceptional path using Layer 2 exceptional descriptor described in section (3) [Layer 2 exceptional descriptor](#) by setting corresponding error exceptional bit. For error set conditions, refer to corresponding error interrupt register explanations.

Table 30.56 L2 forwarding Errors (1 of 2)

Error	Error name	Error interrupt	Error exceptional bit
L2 forwarding secure errors			
SNTF	Secure No Target Filtering	FWEIS0i.LTWNTFS	FWCEPRC2.FLTWNTFEF
SDSPF	Secure Destination Source Port Filtering	FWEIS0i.LTWDSPFS	FWCEPRC2.FDMACSLFEF

Table 30.56 L2 forwarding Errors (2 of 2)

Error	Error name	Error interrupt	Error exceptional bit
SSSPF	Secure Source Source Port Filtering	FWEIS0i.LTWSSPFS	FWCEPRC2.FSMACSLFEF
SVSPF	Secure VLAN Source Port Filtering	FWEIS0i.LTWVSPFS	FWCEPRC2.FVLANSLEF
SDUF	Secure Destination Unknown filtering	FWEIS0i.LTWDUFS	FWCEPRC2.FDMACUFEF
SSUF	Secure Source Unknown filtering	FWEIS0i.LTWSUFS	FWCEPRC2.FSMACUFEF
SVUF	Secure VLAN Unknown filtering	FWEIS0i.LTWVUFS	FWCEPRC2.FVLANUFEF
L2 forwarding errors			
NTF	No Target Filtering	FWEIS0i.LTWNTFS	FWCEPRC2.FLTWNTFEF
DSPF	Destination Source Port Filtering	FWEIS0i.LTWDSPFS	FWCEPRC2.FDMACSLFEF
SSPF	Source Source Port Filtering	FWEIS0i.LTWSSPFS	FWCEPRC2.FSMACSLFEF
VSPF	VLAN Source Port Filtering	FWEIS0i.LTWVSPFS	FWCEPRC2.FVLANSLEF
DUF	Destination Unknown filtering	FWEIS0i.LTWDUFS	FWCEPRC2.FDMACUFEF
SUF	Source Unknown filtering	FWEIS0i.LTWSUFS	FWCEPRC2.FSMACUFEF
VUF	VLAN Unknown filtering	FWEIS0i.LTWVUFS	FWCEPRC2.FVLANUFEF

(2) Layer 2 normal descriptor

For Layer 2 normal descriptor creation, only MAC table fields read from MAC table thanks to the destination MAC and VLAN table fields read from VLAN table are used. Layer 2 normal descriptor fields are described in [Table 30.57](#).

Table 30.57 Layer 2 normal descriptor fields

Field name FDESCR.	Values if MAC destination address and VLAN ID are both found and MAC.SL = VLAN.SL	Values if MAC destination only is found or if MAC destination address and VLAN ID are both found and MAC.SL & !VLAN.SL	Values if VLAN ID only is found or if MAC destination address and VLAN ID are both found and !MAC.SL & VLAN.SL
DV	MAC.DV & VLAN.DV & !FWPCi2.LTWFM	MAC.DV & !FWPCi2.LTWFM	VLAN.DV & !FWPCi2.LTWFM
SEC	MAC.SL	MAC.SL	VLAN.SL
CSD	MAC.CSD	MAC.CSD	VLAN.CSD
IPV	If MAC.IPU is set, set to MAC.IPV If MAC.IPU is not set and VLAN.IPU is set, set to VLAN.IPV If MAC.IPU and VLAN.IPU both are not set, set to the frame input priority (section section 30.5.1.4. ESWM Frame Input Priority)	If MAC.IPU is set, set to MAC.IPV If MAC.IPU is not set, set to the frame input priority (section section 30.5.1.4. ESWM Frame Input Priority)	If VLAN.IPU is set, set to VLAN.IPV If VLAN.IPU is not set, set to the frame input priority (section section 30.5.1.4. ESWM Frame Input Priority)
MINFO	See Figure 30.97 and Table 30.58 .	See Figure 30.97 and Table 30.58 .	See Figure 30.97 and Table 30.58 .
RV	0	0	0
RN	ALL0	ALL0	ALL0
SEQN	Set to frame input sequence number	Set to frame input sequence number	Set to frame input sequence number

Byte	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0	—		FL2C[1:0]		FWDC[3:0]			
1	—							
2								

Figure 30.97 Layer 2 normal descriptor MINFO format

Table 30.58 Layer 2 normal descriptor MINFO field explanation

Field name FDESCR.	Bit width	Field explanation	Values
FWDC	4	ForWarDing Code Present in all descriptors, it allows identifying the type of forwarding.	Fixed to 9 for Layer 2 normal descriptors
FL2C	2	Forwarding layer 2 code	If MAC destination address and VLAN ID are both found and MAC.SL = VLAN.SL, set to 0. If MAC destination only is found or if MAC destination address and VLAN ID are both found and MAC.SL & !VLAN.SL, set to 1. If VLAN ID only is found or if MAC destination address and VLAN ID are both found and !MAC.SL & VLAN.SL, set to 2.
—	--	Reserved field	Set to 0

(3) Layer 2 exceptional descriptor

Layer 2 exceptional descriptor fields are described.

Table 30.59 Layer 2 exceptional descriptor fields

Field name FDESCR.	Values
DV	Exceptional descriptors are forwarded to GWCA number FWCEPTC.EPCS (port number 2 + FWCEPTC.EPCS)
SEC	FWCEPTC.EPSL
CSDj	FWCEPTC.EPCSD
IPV	FWCEPTC.EPIPV
MINFO	See Figure 30.98 and Table 30.60 .
RV	0
RN	ALL0
SEQN	Set to frame input sequence number

Byte	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0	—				FWDC[3:0]			
1	NTF	SVUF	SSUF	SDUF	SVSPF	SSSPF	SDSPF	SNTF
2	—		VUF	SUF	DUF	VSPF	SSPF	DSPF

Figure 30.98 Layer 2 exceptional descriptor MINFO format**Table 30.60 Layer 2 exceptional descriptor MINFO field explanation (1 of 2)**

Field name FDESCR.	Bit width	Field explanation	Values
FWDC	4	ForWarDing Code Present in all descriptors, it allows identifying the type of forwarding.	Fixed to 10 for Layer 2 exceptional descriptors
SNTF	1	Secure No Target Filtering	See (1) L2 forwarding errors .
SDSPF	1	Secure Destination Source Port Filtering	See (1) L2 forwarding errors .
SSSPF	1	Secure Source Source Port Filtering	See (1) L2 forwarding errors .
SVSPF	1	Secure VLAN Source Port Filtering	See (1) L2 forwarding errors .
SDUF	1	Secure Destination Unknown filtering	See (1) L2 forwarding errors .
SSUF	1	Secure Source Unknown filtering	See (1) L2 forwarding errors .
SVUF	1	Secure VLAN Unknown filtering	See (1) L2 forwarding errors .

Table 30.60 Layer 2 exceptional descriptor MINFO field explanation (2 of 2)

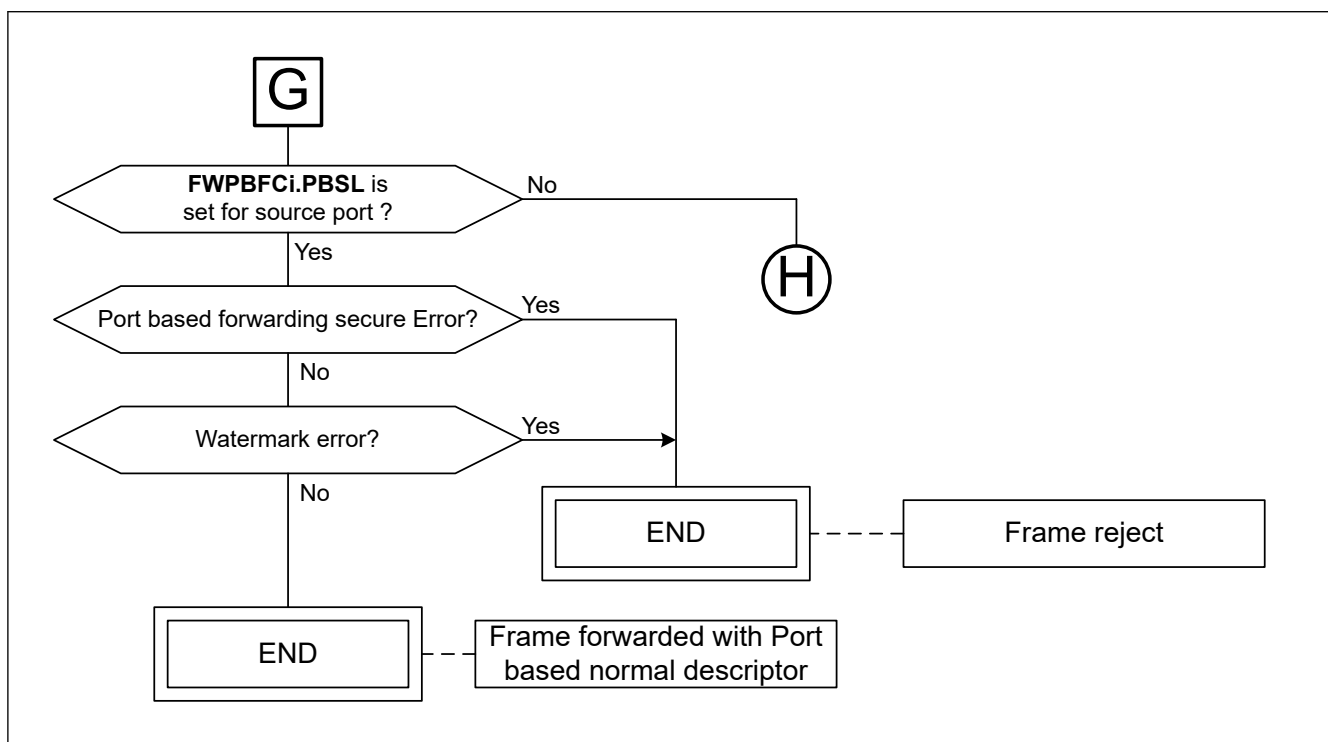
Field name FDESCR.	Bit width	Field explanation	Values
NTF	1	No Target Filtering	See (1) L2 forwarding errors.
DSPF	1	Destination Source Port Filtering	See (1) L2 forwarding errors.
SSPF	1	Source Source Port Filtering	See (1) L2 forwarding errors.
VSPF	1	VLAN Source Port Filtering	See (1) L2 forwarding errors.
DUF	1	Destination Unknown filtering	See (1) L2 forwarding errors.
SUF	1	Source Unknown filtering	See (1) L2 forwarding errors.
VUF	1	VLAN Unknown filtering	See (1) L2 forwarding errors.

30.5.2.6 Port Based Forwarding

Port based forwarding aims at forwarding all frames from a port to other ones. It used FWPBFCi and FWPBFCSDCi0 registers. Port based forwarding will forward frames following Port based forwarding secure flow in and Port based forwarding unsecure flow in .

Functions:

- FWPBFCi and FWPBFCSDCi0 registers are used to set forwarding information for frames coming from port i.

**Figure 30.99 Port based forwarding secure flow**

- Note:
- Port based forwarding secure errors are described in section.
 - Watermark errors are described in section (1) [Watermark errors](#).
 - Link G is linked from [Figure 30.95](#).
 - Link H links to [Figure 30.86](#).

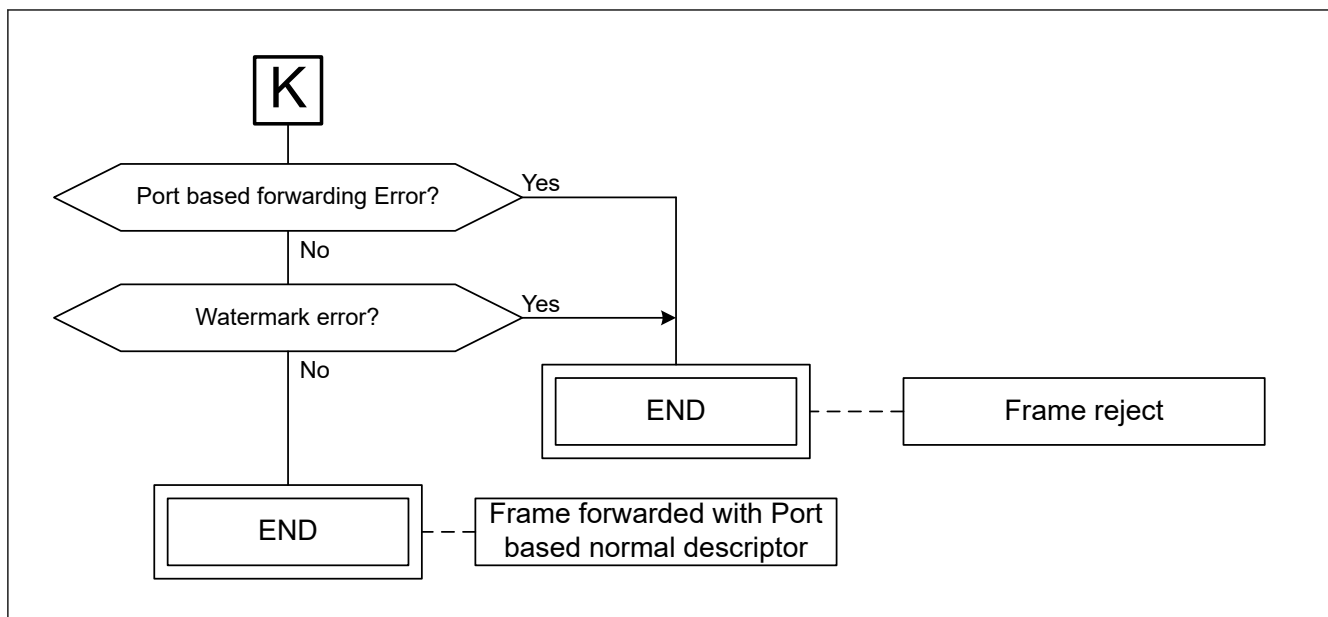


Figure 30.100 Port based forwarding unsecure flow

- Note:
- Port based forwarding errors are described in (1) [L2 forwarding errors](#).
 - Watermark errors are described in (1) [Watermark errors](#).
 - Link K is linked from [Figure 30.96](#).

30.5.2.6.1 Port based Forwarding Errors

[Table 30.61](#) describes the Port based forwarding errors. Instead of being discarded by forwarding engine, an error frame can be forwarded to exceptional path using Port based exceptional descriptor described in [section 30.5.2.6.3. Port Based Exceptional Descriptor](#) by setting corresponding error exceptional bit. For error set conditions, refer to corresponding error interrupt register explanations.

Table 30.61 Port based forwarding Errors

Error	Error name	Error interrupt	Error exceptional bit
Port based forwarding secure errors			
SNTF	Secure No Target Filtering	FWEIS0i.PBNTFS	FWCEPRC2.FPBNTFEF
Port based forwarding errors			
NTF	No Target Filtering	FWEIS0i.PBNTFS	FWCEPRC2.FPBNTFEF

30.5.2.6.2 Port based Normal Descriptor

Port based normal descriptor fields are described in [Table 30.62](#).

Table 30.62 Port based normal descriptor fields (1 of 2)

Field name FDESCR.	Values
DV	FWPBFCi.PBDV
SEC	FWPBFCi.PBSL
CSDj	FWPBFCSDCi0.PBCSD
IPV	If FWPBFCi.PBIPU is set, set to FWPBFCi.PBIPV If FWPBFCi.PBIPU is not set, set to the frame input priority (section 30.5.1.4. ESWM Frame Input Priority)
MINFO	Refer to Figure 30.101 and Table 30.63
RV	0
RN	ALL0

Table 30.62 Port based normal descriptor fields (2 of 2)

Field name FDESCR.	Values
SEQN	Set to frame input sequence number

Byte	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0	—				FWDC[3:0]			
1	—							
2								

Figure 30.101 Port based normal descriptor MINFO format**Table 30.63 Port based normal descriptor MINFO field explanation**

Field name FDESCR.	Bit width	Field explanation	Values
FWDC	4	ForWarDing Code Present in all descriptors, it allows identifying the type of forwarding.	Fixed to 11 for Port based normal descriptors
—	—	Reserved field	Set to 0

30.5.2.6.3 Port Based Exceptional Descriptor

Port based exceptional descriptor fields are described in [Table 30.64](#).

Table 30.64 Port based exceptional descriptor fields

Field name FDESCR.	Values
DV	Exceptional descriptors are forwarded to GWCA number FWCEPTC.EPCS (port number 2)
SEC	FWCEPTC.EPSL
CSDj	FWCEPTC.EPCSD
IPV	FWCEPTC.EPIPV
MINFO	See Figure 30.102 and Table 30.65 .
RV	0
RN	ALL0
SEQN	Set to frame input sequence number

Byte	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0	—				FWDC[3:0]			
1	—						NTF	SNTF
2	—							

Figure 30.102 Port based exceptional descriptor MINFO format**Table 30.65 Port based exceptional descriptor MINFO field explanation (1 of 2)**

Field name FDESCR.	Bit width	Field explanation	Values
FWDC	4	ForWarDing Code Present in all descriptors, it allows identifying the type of forwarding.	Fixed to 12 for Port based exceptional descriptors

Table 30.65 Port based exceptional descriptor MINFO field explanation (2 of 2)

Field name FDESCR.	Bit width	Field explanation	Values
SNTF	1	Secure No Target Filtering	See section 30.5.2.6.1. Port based Forwarding Errors.
NTF	1	No Target Filtering	See section 30.5.2.6.1. Port based Forwarding Errors.

30.5.2.7 Layer 2/Layer 3 Update

Layer 2/Layer 3 Update aims at updating frames before sending them. Agents will use the L23 update interface to fetch the update information from the forwarding engine and will automatically update the frames following the Layer 2/Layer 3 Update rule. Frames Layer 2/Layer 3 Update information fetching will happen for Layer 3 forwarding/routing/filtering frames with respectively L3.RV set.

Layer 2/Layer 3 Update is divided in the following items:

- Rule format: Determines what information will be store along with the stream ID in the table.
- Learning: Function to add/suppress a Layer 2/Layer 3 Update rule by software.
- Reading: Function to read a Layer 2/Layer 3 Update rule by software.
- Rule fetching: Function used by agents to read a Layer 2/Layer 3 Update rule.

30.5.2.7.1 Rule Format

The Layer 2/Layer 3 Update RAM is used to store Layer 2/Layer 3 Update rules. A Layer 2/Layer 3 Update rule contains the information for the frame routing. All the fields in this table, if quoted, will be written L23U.{Field name}. [Table 30.66](#) describes fields contained in a L3 rule.

Table 30.66 Layer 2/Layer 3 Update rule format (1 of 2)

Field name	Field size (bit)	Field Explanation
RPV	4	Routing Port Valid. Values: • Bit i set to 0: Routing rule won't apply for port i. • Bit i set to 1: Routing rule will apply for port i. Functions: • If set to all0, this field will disable the corresponding routing entry. In case RPV is 0 for an egress port, the frame is not updated and will be transmitted unmodified.
TTLU	1	TimeToLive update Functions: • If this bit is set and the corresponding frame is detected as an IPv4 frame, the TTL field will be decremented by 1 and the header checksum will be updated accordingly. • If this bit is set and the corresponding frame is detected as an IPv6 frame, the HopLimit field will be decremented by 1.
MDAU	1	MAC Destination Address Update Functions: • If this bit is set, the rule corresponding frame MAC address will be updated to L23U.MDA.
MSAU	1	MAC Source Address Update Functions: • If this bit is set, the rule corresponding frame MAC address will be updated to corresponding agent MAC address.
CVIDU	1	C-tag VLAN ID Update Functions: • If this bit is set, the rule corresponding frame C-tag VLAN ID will be updated to L23U.CVID.
CPCPU	1	C-tag PCP Update Functions: • If this bit is set, the rule corresponding frame C-tag PCP will be updated to L23U.CPCP.

Table 30.66 Layer 2/Layer 3 Update rule format (2 of 2)

Field name	Field size (bit)	Field Explanation
CDEIU	1	C-tag DEI Update Functions: If this bit is set, the rule corresponding frame C-tag DEI will be updated to L23U.CDEI.
SVIDU	1	S-tag VLAN ID Update Functions: If this bit is set, the rule corresponding frame S-tag VLAN ID will be updated to L23U.SVID.
SPCPU	1	S-tag PCP Update Functions: If this bit is set, the rule corresponding frame S-tag PCP will be updated to L23U.SPCP.
SDEIU	1	S-tag DEI Update Functions: If this bit is set, the rule corresponding frame S-tag DEI will be updated to L23U.SDEI.
RTU	2	R-TAG update Values: 00b: Frame received without R-TAG will be forwarded without R-TAG, Frame received with R-TAG will be forwarded with an R-TAG containing FDESCR.SEQN sequence number. 01b: All frames will be forwarded with an R-TAG containing FDESCR.SEQN sequence number (used for sequence number generation, refer to section 30.5.3.2.2. Sequence Number Generation). 10b: All frames will be forward without R-TAG (used to strip R-TAG after elimination, refer to section (1) Stream identification, FRER recovery, initial stream recovery).
MDA	48	MAC Destination Address Functions: Used to update MAC destination address.
CVID	12	C-tag VLAN ID Functions: Used to update C-tag VLAN ID.
CPCP	3	C-tag PCP Functions: Used to update C-tag PCP.
CDEI	1	C-tag DEI Functions: Used to update C-tag DEI.
SVID	12	S-tag VLAN ID Functions: Used to update S-tag VLAN ID.
SPCP	3	S-tag PCP Functions: Used to update S-tag PCP.
SDEI	1	S-tag DEI Functions: Used to update S-tag DEI.

30.5.2.7.2 SW Learning

Learning is used to add/remove an entry in the Layer 2/Layer 3 Update rule table. [Table 30.67](#) describes register used to learn an entry in the Layer 2/Layer 3 Update rule table. [Table 30.68](#) describes the learn results.

Table 30.67 Layer 2/Layer 3 Update rule learn registers (1 of 2)

Register name	Field name/corresponding field in l23u table	Field explanation
FWL23URL0.L23URNL	Routing number Learn	Address that will be used to learn an entry in the Layer 2/Layer 3 Update rule table
FWL23URL0.L23URPVL	L23U.RPV	See section 30.5.2.7.1. Rule Format .
FWL23URL1.L23UTTUL	L23U.TTLU	See section 30.5.2.7.1. Rule Format .

Table 30.67 Layer 2/Layer 3 Update rule learn registers (2 of 2)

Register name	Field name/corresponding field in I23u table	Field explanation
FWL23URL1.L23UMDAUL	L23U.MDAU	See section 30.5.2.7.1. Rule Format .
FWL23URL1.L23UMSAUL	L23U.MSAU	See section 30.5.2.7.1. Rule Format .
FWL23URL1.L23UCVIDUL	L23U.CVIDU	See section 30.5.2.7.1. Rule Format .
FWL23URL1.L23UCPCPUL	L23U.CPCPU	See section 30.5.2.7.1. Rule Format .
FWL23URL1.L23UCDEIUL	L23U.CDEIU	See section 30.5.2.7.1. Rule Format .
FWL23URL1.L23USVIDUL	L23U.SVIDU	See section 30.5.2.7.1. Rule Format .
FWL23URL1.L23USPCPUL	L23U.SPCPU	See section 30.5.2.7.1. Rule Format .
FWL23URL1.L23USDEIUL	L23U.SDEIU	See section 30.5.2.7.1. Rule Format .
FWL23URL1.L23URTUL	L23U.RTU	See section 30.5.2.7.1. Rule Format .
FWL23URL1.L23UMDALP0/ FWL23URL2.L23UMDALP1	L23U.MDA	See section 30.5.2.7.1. Rule Format .
FWL23URL3.L23UCVIDL	L23U.CVID	See section 30.5.2.7.1. Rule Format .
FWL23URL3.L23UCPCPL	L23U.CPCP	See section 30.5.2.7.1. Rule Format .
FWL23URL3.L23UCDEIL	L23U.CDEI	See section 30.5.2.7.1. Rule Format .
FWL23URL3.L23USVIDL	L23U.SVID	See section 30.5.2.7.1. Rule Format .
FWL23URL3.L23USPCPL	L23U.SPCP	See section 30.5.2.7.1. Rule Format .
FWL23URL3.L23USDEIL	L23U.SDEI	See section 30.5.2.7.1. Rule Format .

Table 30.68 Layer 2/Layer 3 Update rule learn result

Register name	Field name/ corresponding field in I23u table	Field explanation
FWL23URLR.L23ULF	Learning fail	Learning fail with happen in one of the following conditions: The Layer 2/Layer 3 table is not ready (FWL23UTIM.L23UTR is not set)

30.5.2.7.3 SW Reading

Reading is used to read an entry in the Layer 2/Layer 3 Update rule table. [Table 30.69](#) describes register used to read an entry in the Layer 2/Layer 3 Update rule table. [Table 30.70](#) describes the read results.

Table 30.69 Layer 2/Layer 3 Update rule read registers

Register name	Field name/corresponding field in I23u table	Field explanation
FWL23URR.L23RNR	Routing number Read	Address that will be used to read an entry in the Layer 2/Layer 3 Update rule table

Table 30.70 Layer 2/Layer 3 Update rule read result (1 of 2)

Register name	Field name/corresponding field in I23u table	Field explanation
FWL23URRR0.L23URPVL	L23U.RPV	See section 30.5.2.7.1. Rule Format .
FWL23URRR1.L23UTTULR	L23U.TTLU	See section 30.5.2.7.1. Rule Format .
FWL23URRR1.L23UMDAUR	L23U.MDAU	See section 30.5.2.7.1. Rule Format .
FWL23URRR1.L23UMSAUR	L23U.MSAU	See section 30.5.2.7.1. Rule Format .
FWL23URRR1.L23UCVIDUR	L23U.CVIDU	See section 30.5.2.7.1. Rule Format .
FWL23URRR1.L23UCPCPUR	L23U.CPCPU	See section 30.5.2.7.1. Rule Format .
FWL23URRR1.L23UCDEIUR	L23U.CDEIU	See section 30.5.2.7.1. Rule Format .

Table 30.70 Layer 2/Layer 3 Update rule read result (2 of 2)

Register name	Field name/corresponding field in I23u table	Field explanation
FWL23URRR1.L23USVIDUR	L23U.SVIDU	See section 30.5.2.7.1. Rule Format.
FWL23URRR1.L23USPCPUR	L23U.SPCPU	See section 30.5.2.7.1. Rule Format.
FWL23URRR1.L23USDEIUR	L23U.SDEIU	See section 30.5.2.7.1. Rule Format.
FWL23URRR1.L23URTUR	L23U.RTU	See section 30.5.2.7.1. Rule Format.
FWL23URRR1.L23UMDARP0/ FWL23URRR2.L23UMDARP1/	L23U.MDA	See section 30.5.2.7.1. Rule Format.
FWL23URRR3.L23UCVIDR	L23U.CVID	See section 30.5.2.7.1. Rule Format.
FWL23URRR3.L23UCPCPR	L23U.CPCP	See section 30.5.2.7.1. Rule Format.
FWL23URRR3.L23UCDEIR	L23U.CDEI	See section 30.5.2.7.1. Rule Format.
FWL23URRR3.L23USVIDR	L23U.SVID	See section 30.5.2.7.1. Rule Format.
FWL23URRR3.L23USPCPR	L23U.SPCP	See section 30.5.2.7.1. Rule Format.
FWL23URRR3.L23USDEIR	L23U.SDEI	See section 30.5.2.7.1. Rule Format.

30.5.2.7.4 HW Reading

Hardware reading is used by agents [GWCA][ETHA] to fetch the Layer 2/Layer 3 Update rule corresponding to a frame and update the frame. The read result is given to agents to perform Layer 2/Layer 3 update along with L23U.ERR which is used to notify to agents that the corresponding data should be rejected.

30.5.2.7.5 Routing Number Remapping Function

The routing number remapping function is used to change the routing number for a specific port using FWL23URMCi register. That way, different ports can use different Routing Rules for the same frame. It can be used for FRER during replication ([section 30.5.3.2.1. Replication](#)) to send a duplicated frame through several ports with a different stream ID.

Functions:

- When FWL23URMCi.RME is set, a frame sent to port FWL23URMCi.RMDPN with FDESCR.RV set and FDESCR.RN equal to FWL23URMCi.RMRN will be routed with rule number FWL23URMCi.RMNRN.

30.5.3 Filtering

Forwarding engine allows frame filtering and shaping using the following functions:

- PSFP (Per Stream Filtering and Policing) [802.1Qci]
- ATS (Asynchronous Traffic Shaping) [802.1Qcr]
- FRER (Frame Replication and Elimination for Reliability) [802.1CB]

30.5.3.1 PSFP/ATS

PSFP functions is divided in three functions that happen sequential following forwarding flow in [Figure 30.87](#):

- MSDU filter
- Meter filter/ATS

30.5.3.1.1 MSDU Filter

The MSDU filter aims at checking frame size using FWPMFGCi register and can be monitored using FWEIS2 interrupt register. MSDU filtering flow for MSDU filter i is described in [Figure 30.103](#).

Functions:

- FWPMFGCi is used to set MSDU filter number i maximum frame size and mode.
- FWEIS2 interrupt register notifies that an ethernet descriptor has been rejected because of an error.

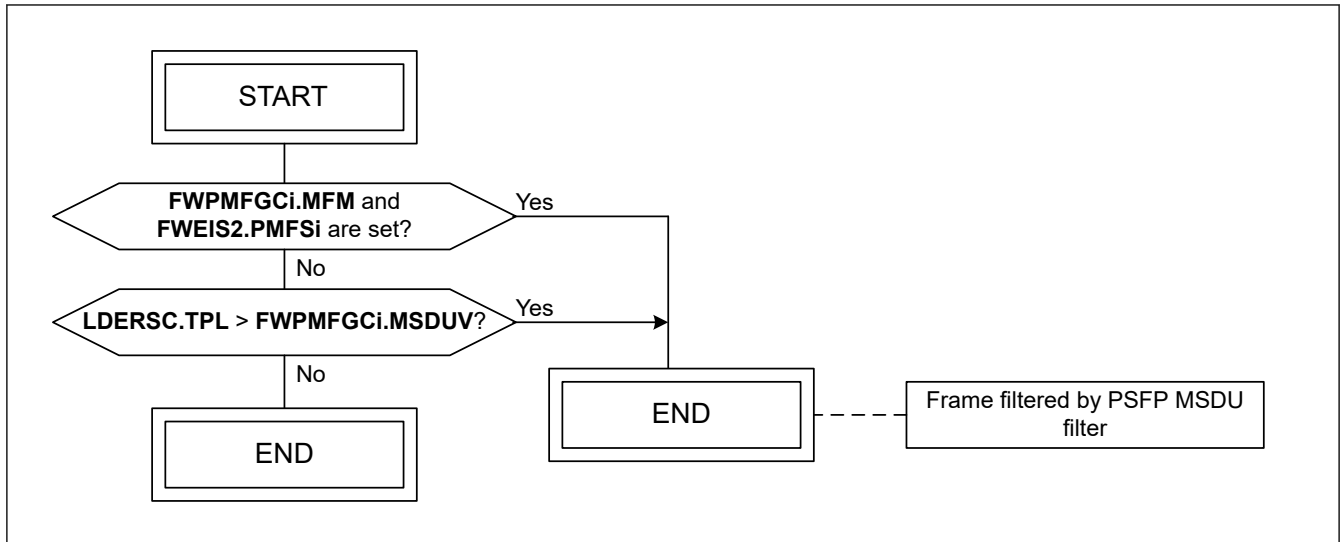


Figure 30.103 MSDU filtering flow for MSDU filter i

30.5.3.1.2 Meter Filter/ATS

The meter filter aims at checking a stream throughput or, in ATS, at controlling a stream throughput using FWPMTRFCi, FWPMTRCBSCi, FWPMTRCIRCi, FWPMTREBSCi, FWPMTREIRCi registers and can be monitored using FWPMTRFMi register and FWEIS5 interrupt registers. Meter filtering flow for meter i is described in [Figure 30.104](#).

Functions:

- FWPMTRFCi register is used to set Meter filter i modes.
- FWPMTRCBSCi and FWPMTRCIRCi registers are used to set Meter filter i green bucket.
- FWPMTREBSCi and FWPMTREIRCi registers are used to set Meter filter i yellow bucket.
- FWPMTRFMi is used to monitor ATS shaper i.
- FWEIS5 interrupt register notifies that an ethernet descriptor has been rejected because of an error.

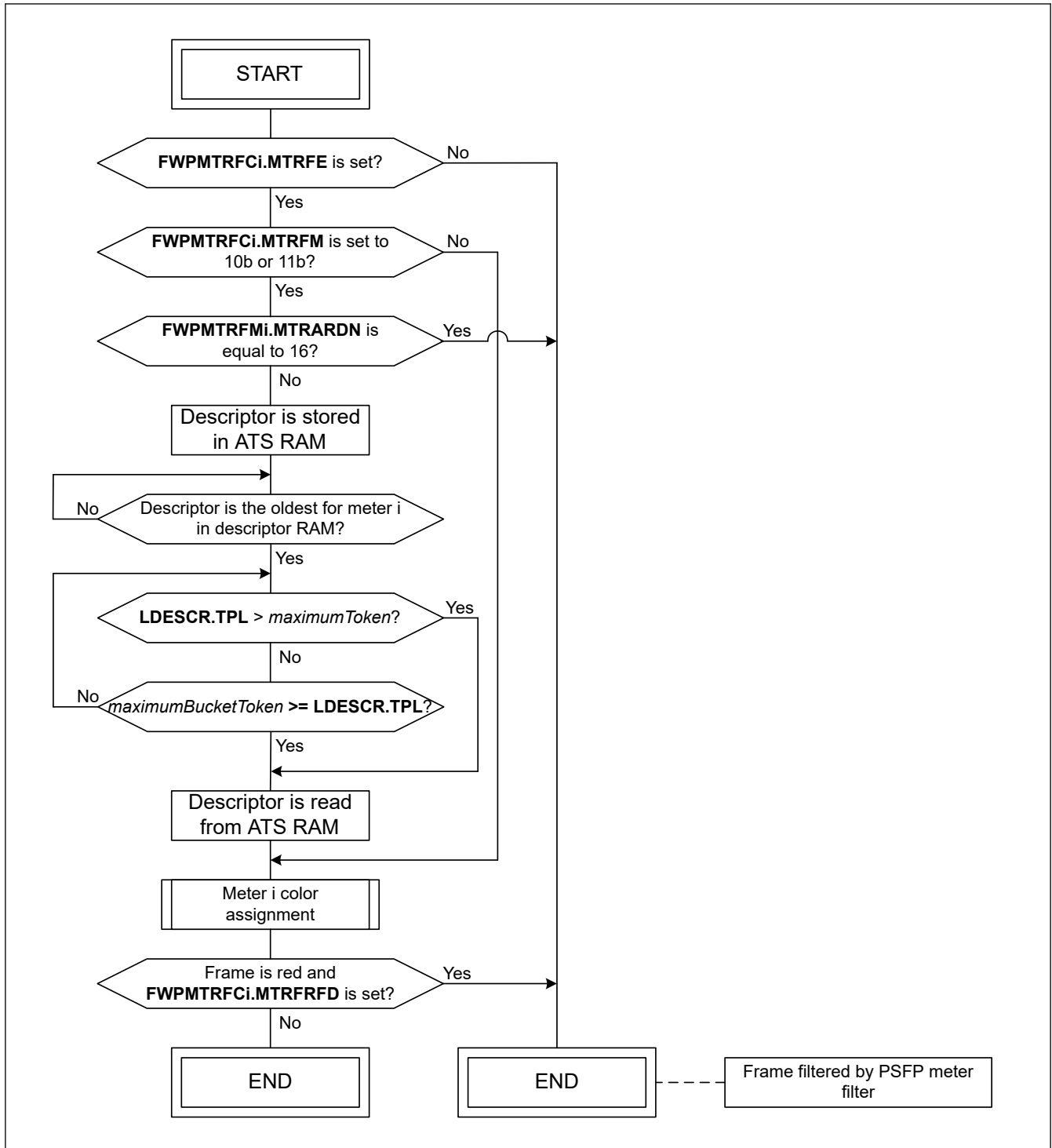


Figure 30.104 Meter filtering flow for meter i

- Note:
- For Meter i color assignment refer to [Figure 30.105](#) and [Figure 30.106](#)
 - In [Figure 30.104](#), *maximumToken* is the maximum number of tokens that can be contained in one meter biggest bucket. For Double bucket meters, it corresponds to the maximum value between FWPMTRCBSCi and FWPMTREBSCi and for Single bucket meters, it corresponds to FWPMTRCBSCi value. Any frame rejected by “LDESCR.TPL [GWCA] > *maximumToken*” condition will be marked as red.
 - In [Figure 30.104](#), *maximumBucketToken* is, for a double bucket the maximum number between the number of tokens actually contained in the yellow bucket and the number of tokens actually contained in the green bucket, for a single bucket the number of tokens actually contained in the green bucket.
 - In [Figure 30.104](#), any frame rejected by “FWPMTRFMi.MTRARDN is equal to 16” conditions (meter corresponding ATS descriptor queue full) will have no color.

Recommendations:

- For every L3 rule mapped to a meter, it is recommended to map also an MSDU filter with a value smaller or equal to the meter corresponding *maximumToken* value. It will be allowed to know that whenever FWEIS5.PMRFSi is set, it is due to an ATS RAM overflow in ATS mode and due to a maximum throughput reached error in other modes and, not due to a frame format error (frame being too big for the corresponding meter).

(1) Double bucket meters

Double bucket meters aim at giving a color to a frame. Meters from meter number 0 to meter number 7 are Double bucket meters.

(a) Double bucket color assignment

Double bucket meter *i* color assignment flow is described in [Figure 30.105](#).

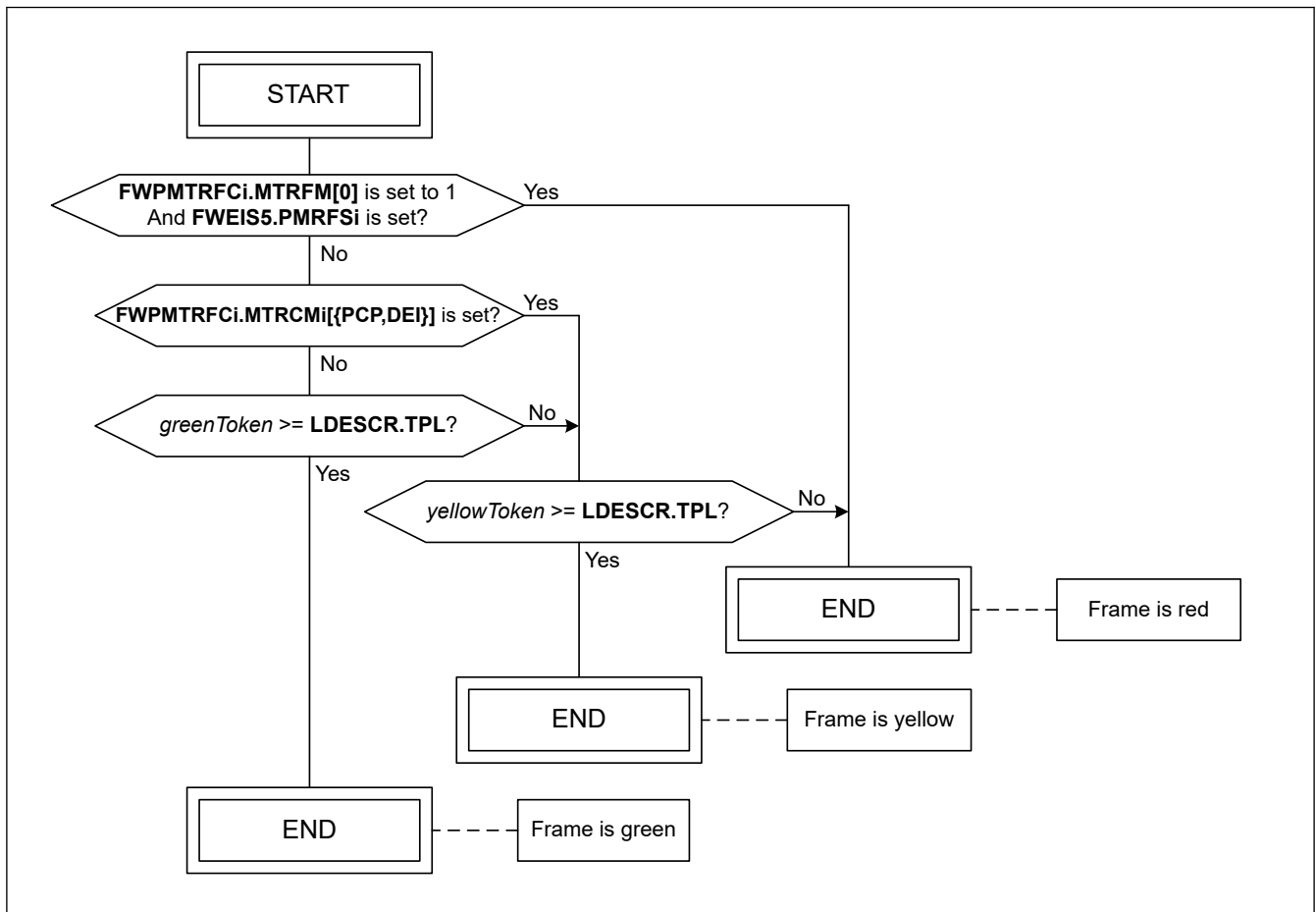


Figure 30.105 Double bucket meter *i* color assignment flow

In [Figure 30.105](#):

- PCP and DEI are respectively the frame incoming S-TAG PCP and DEI values if **FWGC.SVM** is set to 10b, the frame incoming C-TAG PCP and DEI values if **FWGC.SVM** is set to 01b and, 000b and 0 if **FWGC.SVM** is set to 00b.
- greenToken* is the number of tokens contained in the green bucket.
- yellowToken* is the number of tokens contained in the yellow bucket.

(b) Double bucket setting

For double buckets, the green bucket ('Committed' or 'C' bucket) and the yellow bucket ('Excess' or 'E' bucket) needs to be set. For detail explanations concerning coupling flag, color mode and bucket utilization, refer to MEF standard [MEF10.3]. Equation (6) shows how to set information rates in FWPMTRCIRCi.CIR and FWPMTREIRCi.EIR. FWPMTRCBSCi.CBS and FWPMTREBSCi.EBS should be respectively set to the maximum excepted burst size for green and yellow frame in byte.

$$(6) \text{ InformationRate}[\text{byte/cycle}] = \text{MaximumThroughput}[\text{bps}] / (8 * \text{clk_freq}[\text{Hz}])$$

Where:

- *InformationRate* is the value to set to FWPMTRCIRCi.CIR or FWPMTREIRCi.EIR register.
- *MaximumThroughput[bps]* is the expected maximum throughput.
- *clk_freq* is the clock frequency of ESWCLK.

[Example]

If the meter is to be programmed for information rate of 200Mbps and system clock of 100 MHz :

$$\text{InformationRate} = ((200 * 1000000) / (8 * 100 * 1000000)) * 2^{16}$$

$$\text{CIR}[15:0] \text{ or } \text{EIR}[15:0] = (0.25) * 2^{16} = 16384 \text{ (0x4000)}$$

$$\text{CIR}[19:16] \text{ or } \text{EIR}[19:16] = 0 \text{ (Underlined integer)}$$

$(\text{CIR} \times 2^{-16})$ is added to the committed token bucket every system clock.

If the meter is to be programmed for information rate of 200 Mbps and system clock of 320 MHz :

$$\text{InformationRate} = ((1000 * 1000000) / (8 * 320 * 1000000)) \times 2^{16}$$

$$\text{CIR}[15:0] \text{ or } \text{EIR}[15:0] = (0.078125) \times 2^{16} = 5120 \text{ (0x1400)}$$

$$\text{CIR}[19:16] \text{ or } \text{EIR}[19:16] = 0 \text{ (Underlined integer)}$$

$(\text{CIR} \times 2^{-16})$ is added to the committed token bucket every system clock.

Note:

- HW: DropOnYellow is not available in PSFP. To achieve drop on yellow function, the yellow bucket *InformationRate* FWPMTREIRCi.EIR and FWPMTREBSCi.EBS should be set to 0. As a result, all yellow frames will be marked as red and dropped.

(2) Single bucket meters

Single bucket meters aim at giving a color to a frame. Meters from meter number 8 to meter number 31 are Single bucket meters.

(a) Single bucket color assignment

Single bucket meter i color assignment flow is described in [Figure 30.106](#).

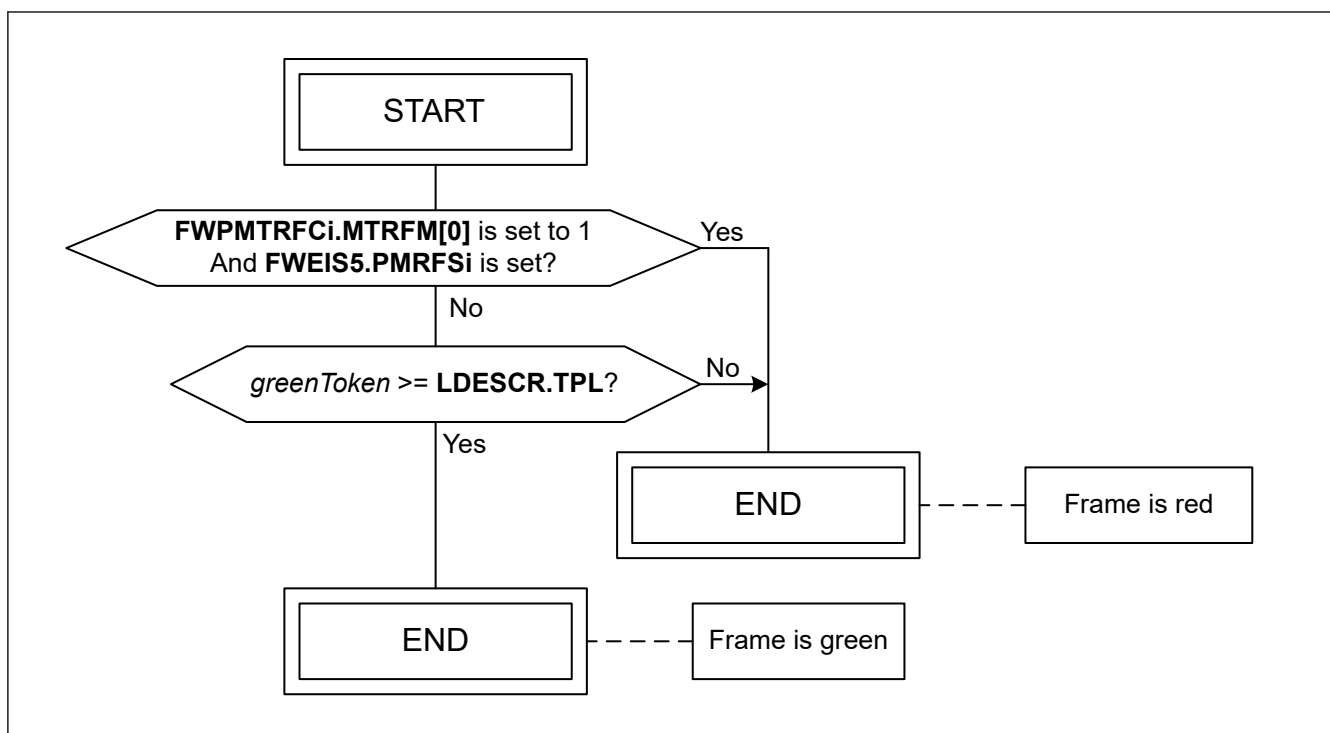


Figure 30.106 Single bucket meter i color assignment flow

In Figure 30.106:

- *greenToken* is the number of tokens contained in the green bucket.

(b) Single bucket setting

The single bucket should be set as the double bucket green bucket, refer to section (b) Double bucket setting.

30.5.3.2 FRER

FRER functions is divided in three functions that happen independently (they can be used simultaneously):

- Replication
- Sequence number generation
- Elimination

Note: • SW: Latent error detection is not implemented in HW and should be implemented in SW using FWFRPPCNI and FWFRDPCNI counters.

30.5.3.2.1 Replication

FRER replication function is used to send a same frame to different ports with different streams. To do so, Layer 3 forwarding/routing/filtering (section [section 30.5.2.4. Layer 3 Forwarding/Routing/Filtering](#)) should be used along with Layer 2/Layer 3 Update function (section [section 30.5.2.7. Layer 2/Layer 3 Update](#)).

Layer 3 forwarding/routing/filtering is used to multicast the incoming FRER stream to several ports and to map the FRER stream to a routing number. [Table 30.71](#) shows an example where Layer 3 table is used to multicast a frame coming from port 2 to port 0 and port 1 and to map the corresponding stream to routing number 5.

Table 30.71 Layer 3 table setting for replication (1 of 2)

Field name L3.	Values	Explanation
SLV	100b	Not mandatory, ensure that the corresponding stream only comes from port 2.
DV	011b	Send the frame to port 0 and port 1
RV	1	Enable routing

Table 30.71 Layer 3 table setting for replication (2 of 2)

Field name L3.	Values	Explanation
RN	5	Map the stream to Layer 2/Layer 3 Update rule number 5

Layer 2/Layer 3 rule remapping is used to remap Routing number L3.RN for one of the ports to send the frame with different stream IDs and allow to learn the rules and the new fields for stream update. [Table 30.72](#) shows an example where port 1 Layer 2/Layer 3 Update rule number is remapped to rule number 8 with 8.2.8.5 Routing number remapping rule number 4. [Table 30.73](#) shown an example for Layer 2/Layer 3 Update rule setting for FRER.

Table 30.72 Layer 2/Layer 3 Update rule number remapping

Register name	Values	Explanation
FWL23URMC4.RME	1	Enable rule number remapping
FWL23URMC4.RMRN	5	Remap rule number 5
FWL23URMC4.RMDPN	1	Remap when routing access comes from port 1
FWL23URMC4.RMNRN	8	Remap rule to rule number 8

Table 30.73 Layer 2/Layer 3 Update rule settings

Field name L23U.	Values for rule 5	Values for rule 8	Explanation
RPV	001b	001b	Each rule is used for its corresponding port
TTLU	0	0	Do not update TimeToLive (user define)
MDAU	1	1	Update destination MAC address (user define)
MSAU	0	0	Do not update source MAC address (user define)
CVIDU	1	1	Update C-TAG (user define)
CPCPU	1	1	Update C-TAG (user define)
CDEIU	1	1	Update C-TAG (user define)
SVIDU	0	0	Do not update S-TAG (user define)
SPCPU	0	0	Do not update S-TAG (user define)
SDEIU	0	0	Do not update S-TAG (user define)
RTU	00b or 01b	00b or 01b	Set to 00b if the replicated stream contains an R-TAG at ingress Set to 01b if the replicated stream does not contain R-TAG at ingress (section 30.5.3.2.2. Sequence Number Generation)
MDA	MAC0	MAC1	Set the new MAC address per each rule (user define)
CVID	VLAN0	VLAN1	Set the new VLAN ID per each rule (user define)
CPCP	PCP0	PCP1	Set the new PCP per each rule (user define)
CDEI	DEI0	DEI1	Set the new DEI per each rule (user define)
SVID	0x0	0x0	Do not update S-TAG (user define)
SPCP	000b	000b	Do not update S-TAG (user define)
SDEI	0	0	Do not update S-TAG (user define)

30.5.3.2.2 Sequence Number Generation

FRER sequence number generation is used during replication to generate a sequential sequence number to insert in stream corresponding frames (refer to section [30.5.3.2.1. Replication](#) for replication) using FWSEQNGCi and FWSEQNRC and can be monitored using FWSEQNGMi.

Functions:

- FWSEQNGCi register is used to enable sequence generation. While enabled, any frame forwarded with FWSEQNGCi.SEQNGRN as routing number will see its sequence number (FDESCR.SEQN) set to FWSEQNGMi.SEQN.
- FWSEQNGMi register is used to monitor the next sequence number to be used by sequence number generation rule i.

- FWSEQNRC.SEQNR is used to reset to 16'b0 the next sequence number to be used by sequence number generation rule i.

Note:

- SW: Sequence number generation should not be used along with elimination.
- A frame rejected by PSFP will not see its sequence number updated by sequence number generation.

30.5.3.2.3 Elimination

FRER elimination aims at identifying the FRER streams (Layer 3 forwarding/routing/filtering, section [30.5.2.4. Layer 3 Forwarding/Routing/Filtering](#)), verifying the integrity of a stream (FRER individual recovery, section [\(e\) Individual recovery algorithms](#)), at eliminating duplicated frames (FRER sequence recovery, [\(f\) Sequence recovery algorithms](#)), at recovering the stream ID (Layer 2/Layer 3 Update function (section [30.5.2.7. Layer 2/Layer 3 Update](#))) and at checking that either the individual recovery or sequence recovery algorithm are stuck (Timeout function, section [\(3\) Timeout](#)). FRER elimination uses FWFTL0/1/R, FWFTOC and FWFTOPC registers and can be monitored using FWFTIM and FWFTR/RR0/RR1/RR2 registers and FWEIS60/61/62/63, FWEIS70/71/72/73 and FWEIS80/81/82/83 interrupt registers.

Functions:

- FWFTL0/1/R registers are used to set/reset an entry in FRER table.
- FWFTOC and FWFTOPC registers are used to set timeout function.
- FWFTIM register is used to reset the FRER table before utilization.
- FWFTR/RR0/RR1/RR2 registers are used to monitor FRER entries in FRER table.
- FWEIS60/61/62/63, FWEIS70/71/72/73 and FWEIS80/81/82/83 interrupt registers notify that an error happened during FRER elimination.

Note:

- SW: Sequence number generation should not be used along with elimination.

(1) Stream identification, FRER recovery, initial stream recovery

In this section, example will be given to explain the different function dependencies and settings. In this example two streams, 0 and 1, are respectively entering the switch from port 0 and 1 that separately should pass through individual recovery algorithm for integrity check and through the same sequence recovery algorithm for duplicate elimination. The eliminated stream should see its stream ID updated to new one and should be forwarded to port 2.

Stream identification is done by Layer 3 forwarding/routing/filtering function, refer to section [30.5.2.4. Layer 3 Forwarding/Routing/Filtering](#). L3 table is used to map each stream to its corresponding FRER individual recovery individual recovery algorithm and also used to map stream to a Layer 2/Layer 3 update rule for stream ID recovery. In [Table 30.74](#) example, stream 0 and stream 1 are respectively mapped to FRER entry number 15 and 23 for individual recovery and both streams are mapped to routing rule number 5 for stream ID recovery.

Table 30.74 Layer 3 table setting for elimination

Field name L3.	Values for stream 0	Values for stream 1	Explanation
SLV	001b	010b	Not mandatory, ensure that each stream comes from its corresponding source port.
DV	100b	100b	All streams for the same sequence recovery algorithm are targeting the port
FRERV	1	1	FRER individual recovery
FRERN	15	23	All streams are targeting a different individual recovery algorithm
RV	1	1	Enable routing
RN	5	5	All streams for the same sequence recovery algorithm are targeting the same Layer 2/Layer 3 update rule for stream ID recovery

FRER table is then used to map stream 0 and 1 individual recovery algorithms to a sequence recovery algorithm and used to set all the recovery algorithms. [Table 30.75](#) shows an example where individual recovery algorithm are mapped to FRER entry number 67 for sequence recovery.

Table 30.75 FRER table setting for sequence recovery

Field name FRER.	Values for FRER entry number 15	Values for FRER entry number 23	Values for FRER entry number 67	Explanation
SRPV	1	1	0	Enable sequence recovery for an individual recovery algorithm
SRP	67	67	All0	Sequence recovery entry number

Layer 2/Layer 3 Update functions is used to recover the output stream ID in the frame. [Table 30.76](#) shown an example for Layer 2/Layer 3 Update rule setting for FRER.

Table 30.76 Layer 2/Layer 3 Update rule settings

Field name L23U.	Values for rule 5	Explanation
RPV	100b	Set to the target port
TTLU	0	Do not update TimeToLive (user define)
MDAU	1	Update destination MAC address (user define)
MSAU	0	Do not update source MAC address (user define)
CVIDU	1	Update C-TAG (user define)
CPCPU	1	Update C-TAG (user define)
CDEIU	1	Update C-TAG (user define)
SVIDU	0	Do not update S-TAG (user define)
SPCPU	0	Do not update S-TAG (user define)
SDEIU	0	Do not update S-TAG (user define)
RTU	00b or 01b	Set to 00b if the output stream should contain the ingress R-TAG Set to 10b if the output stream should be sent without R-TAG
MDA	MAC0	Set the new MAC address per each rule (user define)
CVID	VLAN0	Set the new VLAN ID per each rule (user define)
CPCP	PCP0	Set the new PCP per each rule (user define)
CDEI	DEI0	Set the new DEI per each rule (user define)
SVID	0x0	Do not update S-TAG (user define)
SPCP	000b	Do not update S-TAG (user define)
SDEI	0	Do not update S-TAG (user define)

In this section only an example has been developed, by using the same kind of settings it is possible to:

- Have only one ingress stream and one output stream with the same stream ID and only apply individual recovery.
- Have more than 2 input streams for elimination.
- Have more than 1 output stream after elimination (by applying FRER replication, [section 30.5.3.2.1. Replication](#))

(2) FRER table

FRER table is used to store setting and information used for recovery algorithms.

FRER table is divided in the following items:

- Rule format: Determines what information will be store in the FRER table
- SW Learning: Function to set an FRER entry in the FRER table.
- SW Reading: Function to read an FRER entry in the FRER table.

(a) Rule format

The FRER table is used to store FRER entries. A FRER entry contains information recovery algorithms. All the fields in this table, if quoted, will be written FRER.{Field name}. [Table 30.77](#) describes fields contained in a FRER entry.

Table 30.77 FRER entry format

Field name	Field size (bit)	Field explanation
SHL	4	Sequence History Length Values: 0: Recovery match algorithm - i different from 0: Recovery vector algorithm with history length of i+1
SH	15	Sequence History: Functions: Used to save sequence history for recovery algorithms
RSN	16	Recovery Sequence Number Functions: Used to save last highest sequence number (modulo 2^{16}) for recovery algorithms
TNS	1	Take No Sequence Values: 0: Frames received without R-TAG (LDESCR.RTGI [GWCA] not set) are rejected 1: Frames received without R-TAG (LDESCR.RTGI [GWCA] not set) are forwarded
SRPV	1	Sequence Recovery Pointer Valid Values: 0: After using this entry for individual recovery, sequence recovery will not happen 1: After using this entry for individual recovery, sequence recovery will happen using FRER entry number FRER.SRP
SRP	7	Sequence Recovery Pointer Functions: Set the number of the FRER entry that should be used for sequence recovery Note: Any entry used for sequence recovery cannot be used for individual recovery. As a result, when a Recovery number is set in FRER.SRT when FRER.SRPV is set, the same number cannot be set in L3.FRERN when L3.FRERV is set.
SRRT	10	Set Recovery Remaining Tick Functions: - Time in tick (one tick is FWFTOC.TOC) before timeout. - Setting this register to All0 disable timeout function for the corresponding entry
RRT	10	Recovery Remaining Tick Functions: Used by timeout function to detect timeout error.

(b) SW Learning

Learning is used to set/reset an entry in the FRER table. [Table 30.78](#) describes register used to learn an entry in the FRER table. [Table 30.79](#) describes the learn results. To reset an entry, software should re-learn the same entry again.

Table 30.78 FRER entry learn registers

Register name	Field name/corresponding field in I23u table	Field explanation
FWFTL0.FEAL	Entry Learn address	Corresponding the entry number that should be learnt in FRER table
FWFTL1.FSHLL	FRER.SHL	See (a) Rule format .
No register Set to All0 While learning (TakeAny)	FRER.SH	See (a) Rule format .
No register Set to All1 While learning	FRER.RSN	See (a) Rule format .
FWFTL1.FTNSL	FRER.TNS	See (a) Rule format .
FWFTL1.FSRPVL	FRER.SRPV	See (a) Rule format .
FWFTL0.FSRPL	FRER.SRP	See (a) Rule format .
FWFTL1.FSRRTL	FRER.SRRT	See (a) Rule format .
No register Set to All0 While learning	FRER.RRT	See (a) Rule format .

Table 30.79 FRER entry learn result

Register name	Field name/corresponding field in FRER table	Field explanation
FWFTLR.FLF	Learning fail	Learning fail with happen in one of the following conditions: The FRER table is not ready (FWFTIM.FTR is not set)

(c) SW Reading

Learning is used to read an entry in the FRER table. [Table 30.80](#) describes register used to read an entry in the FRER table. [Table 30.81](#) describes the read results.

Table 30.80 FRER entry read registers

Register name	Field name/corresponding field in I23u table	Field explanation
FWFTR.FEAR	Entry Read address	Corresponding the entry number that should be read from FRER table

Table 30.81 FRER entry learn result

Register name	Field name/corresponding field in I23u table	Field explanation
FWFTRR0.FSHLR	FRER.SHL	See (a) Rule format.
FWFTRR1.FSHR	FRER.SH	See (a) Rule format.
FWFTRR2.FRSNR	FRER.RSN	See (a) Rule format.
FWFTRR0.FTNSR	FRER.TNS	See (a) Rule format.
FWFTRR0.FSRPVR	FRER.SRPV	See (a) Rule format.
FWFTRR0.FSRPR	FRER.SRP	See (a) Rule format.
FWFTRR0.FSRRT	FRER.SRRT	See (a) Rule format.
FWFTRR2.FRRTR	FRER.RRT	See (a) Rule format.

(d) Recovery algorithms

Individual recovery or sequence recovery algorithm can both be performed using either a match recovery algorithm or a vector recovery algorithm. For an entry, match recovery algorithm can be selected by setting All0 to its corresponding FRER.SHL field. If any other value is set vector recovery algorithm will be selected.

Match recovery algorithm is only used to detect if a stream is stuck (two frames are received consecutively with the same sequence number).

Vector recovery algorithm is used to check a stream sequence by accepting all packets within a given range ($\pm$ FRER.SHL) from the last accepted highest sequence recovery number. This algorithm also ensures than within this range no sequence number is received more than on time.

(e) Individual recovery algorithms

Individual recovery match algorithm and vector algorithm are respectively described in [Figure 30.107](#) and [Figure 30.108](#). The variable *Delta* present in these figures is the distance from the ingress frame sequence number to FRER.RSN (This number should be considering overflow and should be contained in [-32767:32768] range).

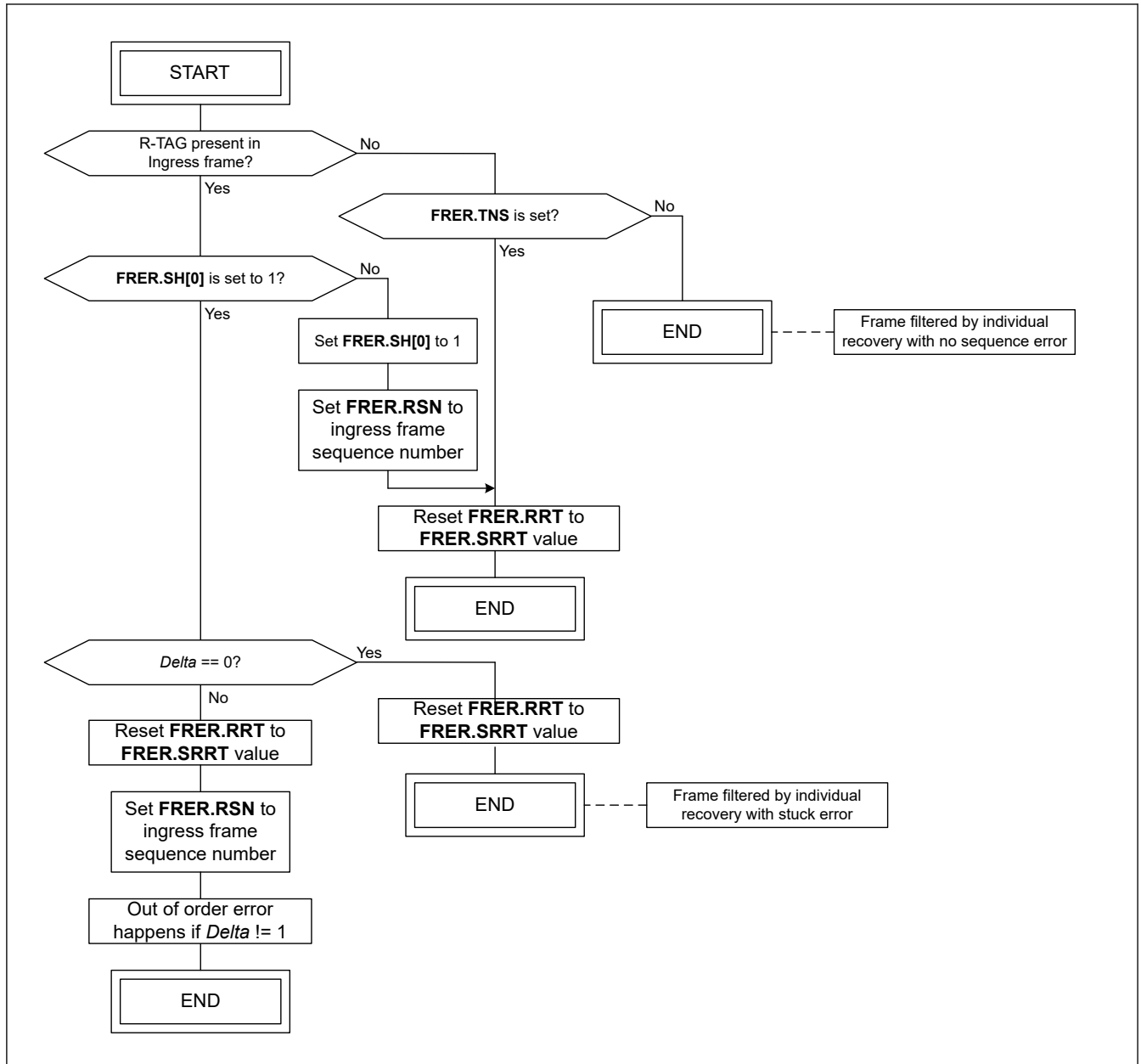


Figure 30.107 Individual recovery match algorithm

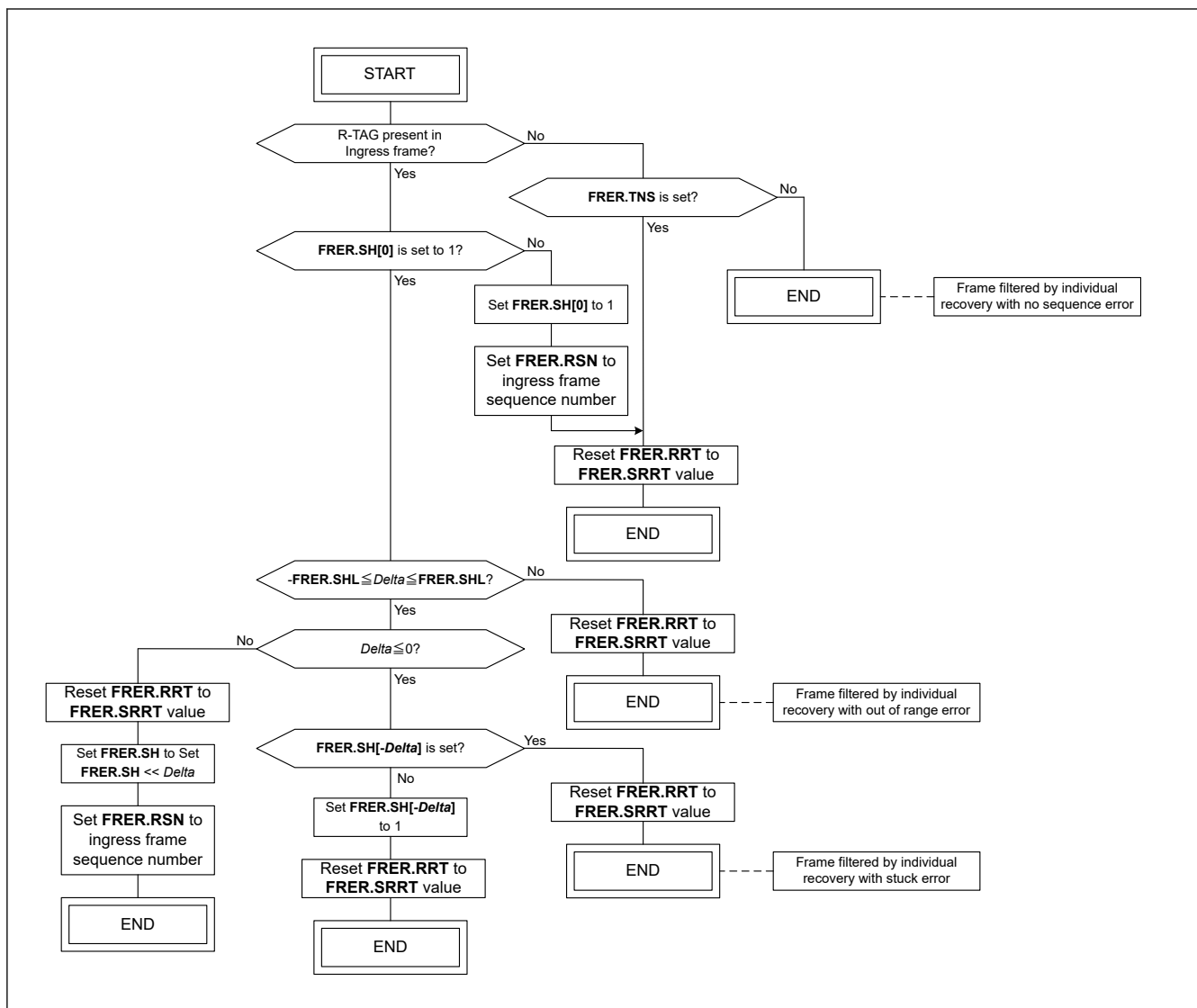


Figure 30.108 Individual recovery vector algorithm

Note: Even if the recovery length is smaller than 14, the sequence history saved in the FRER table will contain the history until 14. Vector recovery algorithm will ignore the useless bits.

(f) Sequence recovery algorithms

Sequence recovery match algorithm and vector algorithm are respectively described in [Figure 30.109](#) and [Figure 30.110](#). The variable *Delta* present in these figures is the distance from the ingress frame sequence number to FRER.RSN (This number should be considering overflow and should be contained in [-32767:32768] range).

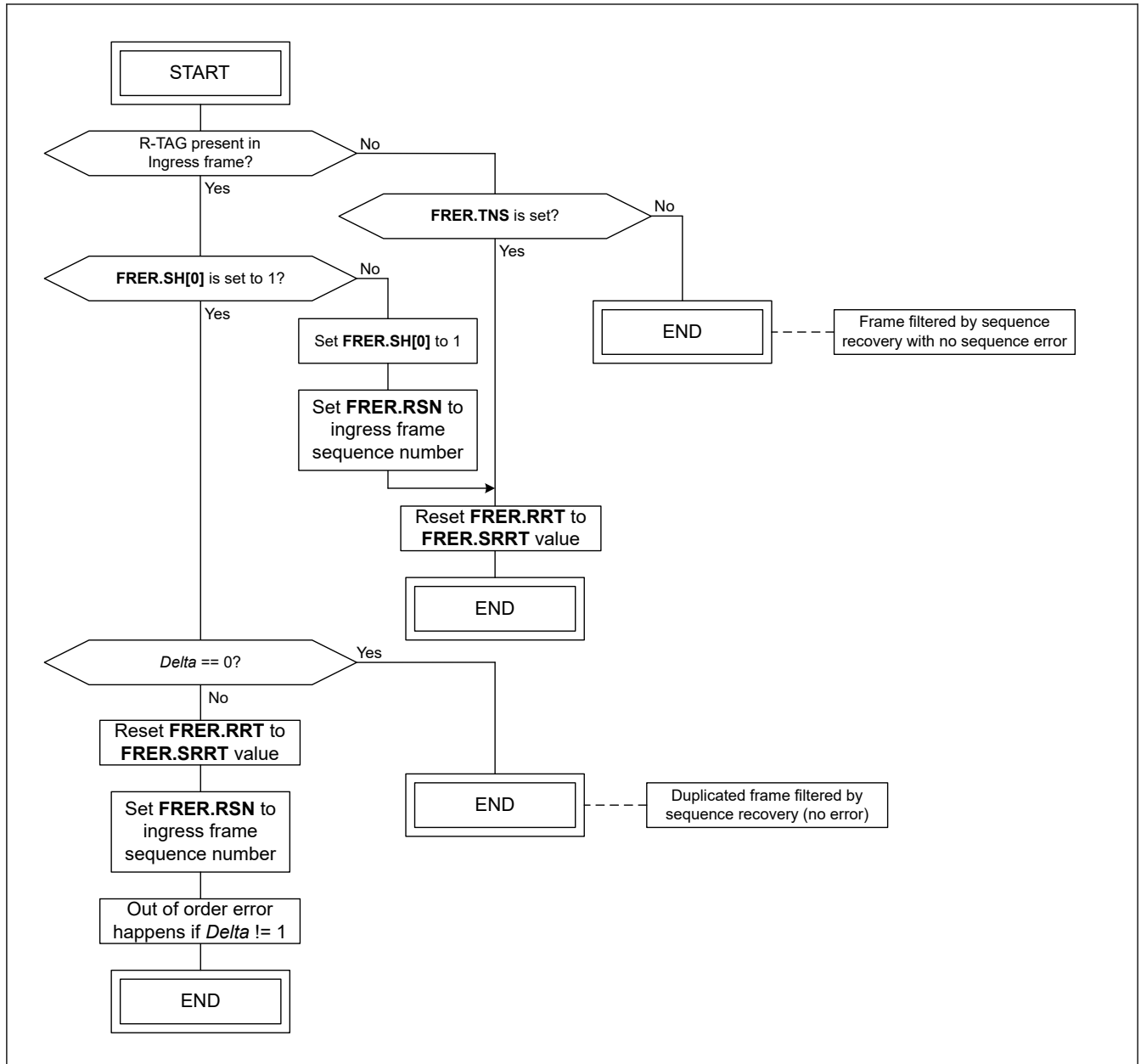


Figure 30.109 Sequence recovery match algorithm

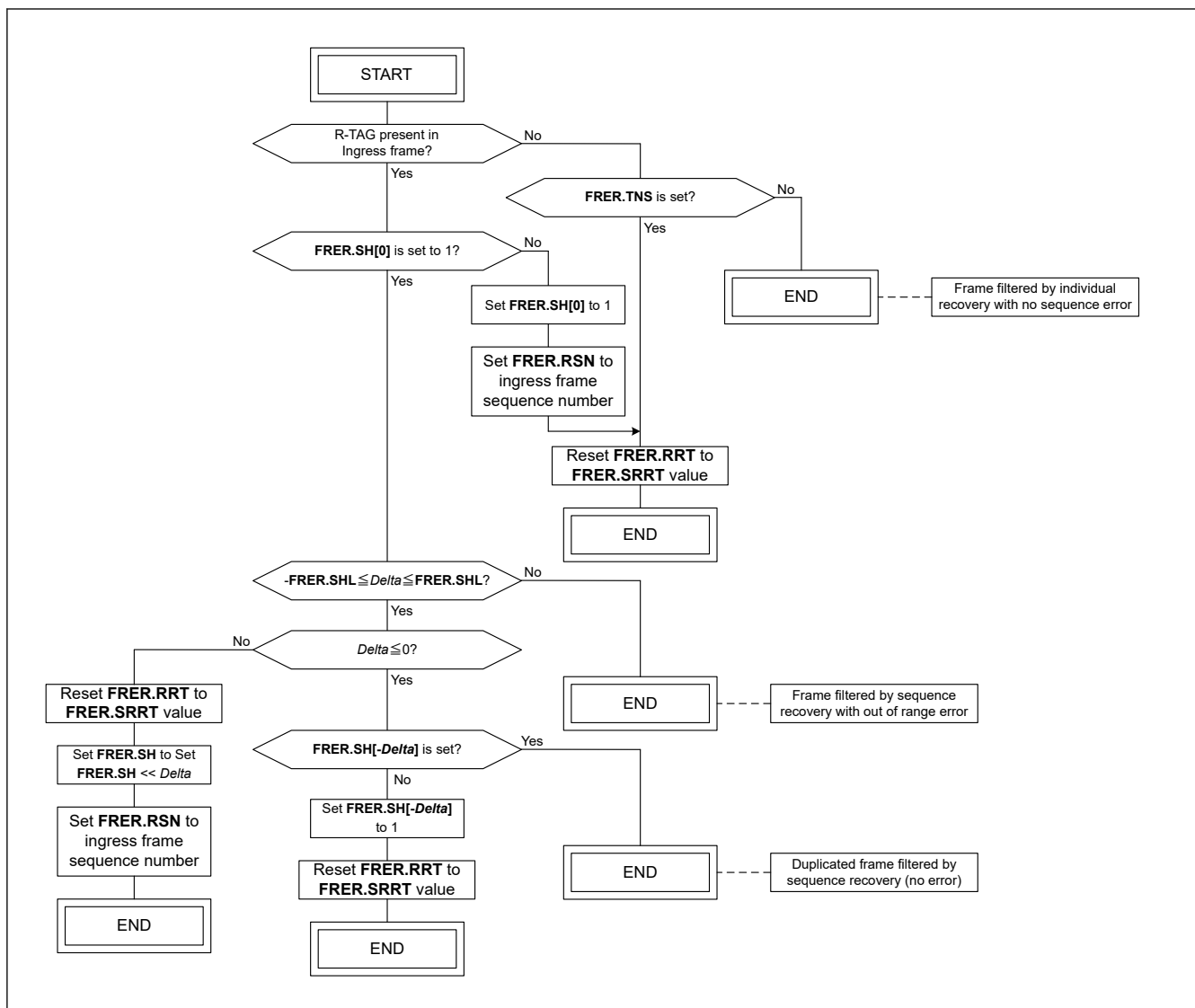


Figure 30.110 Sequence recovery vector algorithm

Note: Even if the recovery length is smaller than 14, the sequence history saved in the FRER table will contain the history until 14. Vector recovery algorithm will ignore the useless bits.

(3) Timeout

The timeout function aims at detecting when a recovery algorithm had no passing frame since a certain time and should be reset because it could be stuck. The timeout functional flow is described in for when timeout functions is enabled (FWFTOC.TOC set to 1). This flow happens for all the FRER entries.

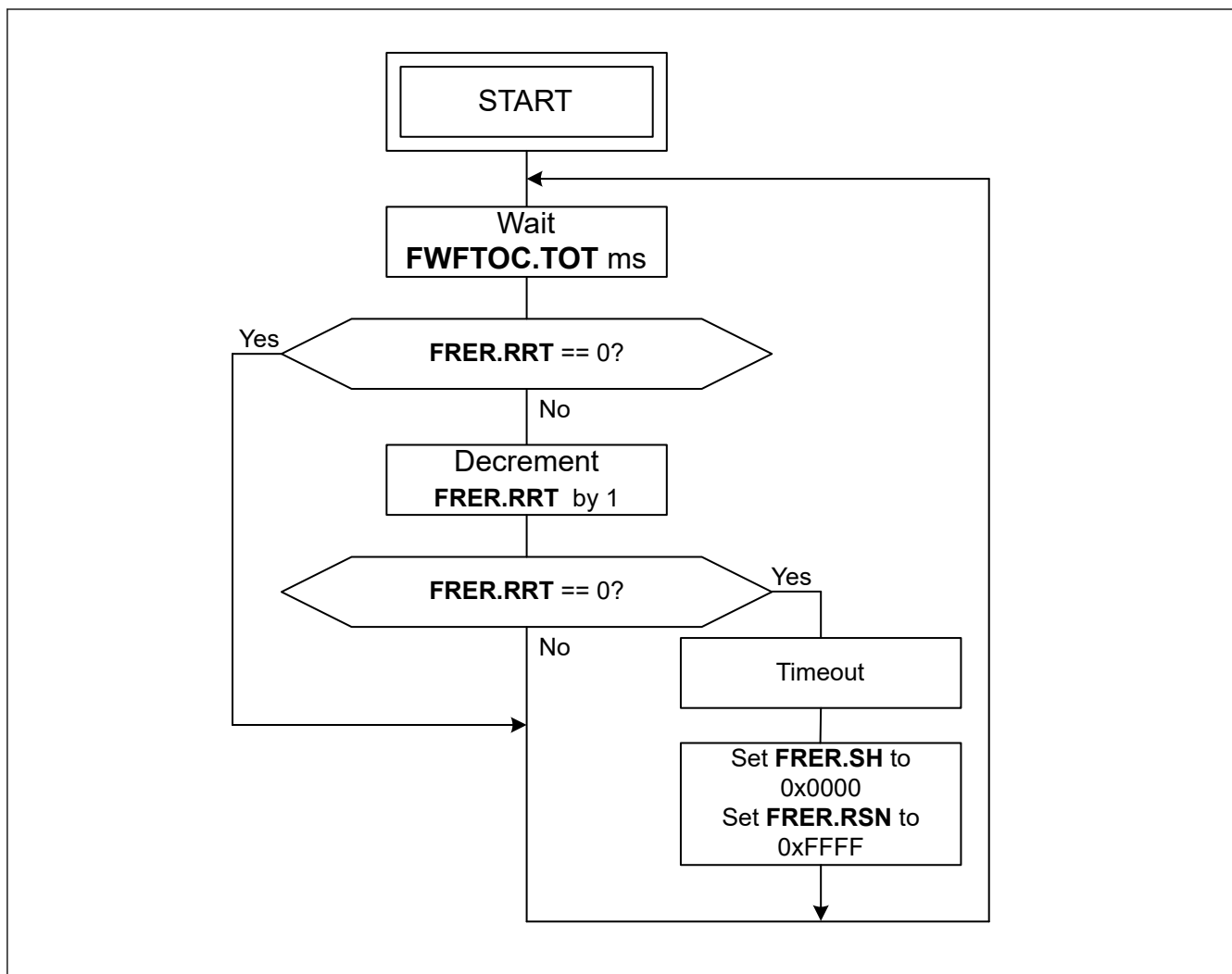


Figure 30.111 Timeout functional flow

30.5.4 Fabric (MFAB)

Fabric Bus (MFAB) is an internal bus used to exchange data between Ethernet Agent [ETHA], Ethernet Common Agent [COMA], Ethernet CPU Agent [GWCA], local RAM, TAG RAM, pointer RAM, and Ethernet Message Forwarding Engine [MFWD].

[Table 30.82](#) lists Fabric features.

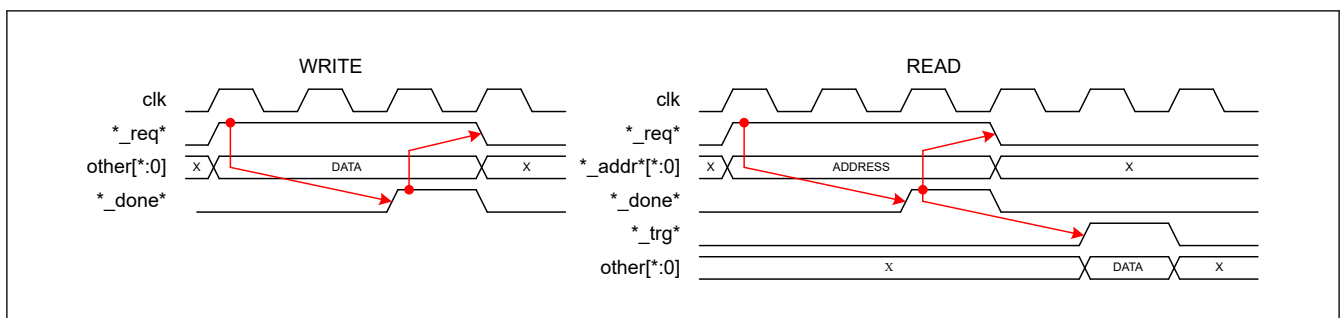
Table 30.82 Fabric feature list (1 of 2)

Function		Details
Overall function	Data transfer	Function to transfer data and access to RAMs
Data provision		- Data for Fabric - Initial frame data + Intermediate data + Last frame data + Frame Descriptor - Data width: 128 + 48 + 9 bits

Table 30.82 Fabric feature list (2 of 2)

Function		Details
Control function	Time arbiter	Arbitration between Time-critical agents [ETHA] for data bus. Arbitration scheme depends on each Time critical agents [ETHA] PHY link/Internal speed.
	Arbitration	- LRU for Time critical-agents [ETHA], Slow agents [GWCA] and Ethernet Common Agent [COMA] error read requests arbitration. - LRU between Slow agents [GWCA] for read/write bus arbitration. - Strict priority like arbitration between time arbiter, Error LRU, and slow LRU. - Priority: time arbiter > slow LRU > Error LRU
	RAM access	Function that writes and reads to RAM
	Transfer write request to Forwarding Engine	Function to transfer write request sent from ETHA Agents [ETHA] or GWCA's [GWCA] to Forwarding Engine.

Figure 30.112 shows Fabric interface protocol.

**Figure 30.112 Fabric interface protocol**

30.5.4.1 Fabric Time Arbiter

Time arbiter controls done signal of Ethernet Agent [ETHA] Write and Read buses depending on link speed.

The faster a link is the more the corresponding agent will be able access to the local RAM.

Agents Done Signal Distribution

Table 30.83 lists done signal distribution depending on PHY link speed possible combinations. Any disabled agent has a doneSignalDistributionValue equal to 0.

Table 30.83 Done signal output ratio by combination of PHY link speed

PHY link speeds of time-critical agents in OPERATION mode [ETHA]	doneSignalDistributionValue [cycle]
Only one Agent enabled	1
PHY link speed of all Agents is the same	1
100 Mbps : 1 Gbps	1 : 10

Done Signal Patterns

Figure 30.113 shows a done signal pattern example where time-critical port 0 has a 100 Mbps link and time-critical port 1 has a 1 Gbps link. When a request signals is not asserted at the timing when the corresponding done signal should have been asserted the done signals is not asserted and the handshake does not happen. In Figure 30.113 the request signals are always asserted to make the done pattern visible but, in a real system, request signals can be asserted anytime.

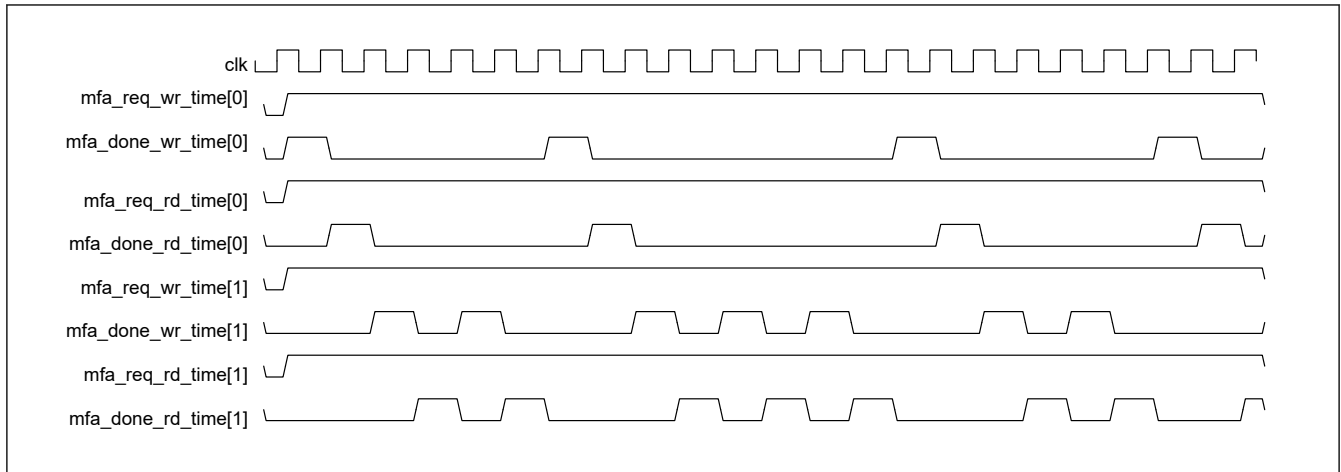


Figure 30.113 Done signal pattern example

The done signal pattern has inherent properties that are always respected:

- For a static configuration the done pattern is cyclic with a constant cycle time *doneCycleClk* described in equation (4).
- During a period of *doneCycleClk* clocks starting from anywhere, a time-critical agent receives for both read and write buses a number of done signals equal to its associated done signal distribution value (Table 30.83).

$$(4) \text{ doneCycleClk [cycle]} = (\Sigma_2 \text{ doneSignalDistributionValue [cycle]}) \times 2$$

Where:

- $\Sigma_2 \text{ doneSignalDistributionValue}$ is the sum of all time-critical agents in OPERATION mode *doneSignalDistributionValue* (Table 30.83).

Restriction:

- HW: Done signals are only given to ethernet agents in OPERATION mode [ETHA].
- HW: If an unknown link speed is set for time-critical port *i* [ETHA] (*mfa_xmii_speed*[$3 \times (i + 1) - 1:3 \times i$] > 101b [RMAC]), Fabric will consider port *i* as disabled and port *i* will not be able to access the time arbiter.

Fabric Jitter

When an Ethernet Agent reads data using fabric time arbiter, some jitter is induced in the reading process by Fabric and it should be taken in account while using TAS [802.1Qbv] or PSFP [802.1Qci]. The fabric jitter can be calculated for an agent using equation (6).

$$(6) \text{ fabricJitter [ns]} = (\lceil \text{doneCycleClk[cycle]} / \text{doneSignalDistributionValue[cycle]} \rceil + (\text{PortTimeOperation[number]} - 1) \times 2) \times \text{clk_period [ns]}$$

Where:

- *doneCycleClk* is detailed in [Done Signal Patterns](#).
- *doneSignalDistributionValue* is corresponding agent value detailed in [Agents Done Signal Distribution](#).
- *PortTimeOperation* is the number of Ethernet Agents in OPERATION mode.
- *clk_period* is the clock *clk* period.

Restrictions:

- SW: The fabricJitter equation takes in account the number of Ethernet Agents [ETHA] in OPERATION mode and also *doneCycleClk*. In fact, enabling/disabling an agent or changing an agent PHY speed will indeed affect the fabric jitter value. However, reconfiguration TAS [802.1Qbv] or PSFP [802.1Qci] jitter in OPERATION is not a recommended option, so the maximum number of Ethernet Agents in OPERATION mode and their corresponding maximum PHY speed should be determined to calculate the worst fabricJitter per agent and it should be kept to the same value all the time.

Fabric Minimum Latency

When an Ethernet Agent reads data using fabric time arbiter, some latency is induced in the reading process by Fabric and it should be taken in account while using TAS [802.1Qbv] or PSFP [802.1Qci]. The fabric minimum latency can be calculated for an agent using equation (7).

$$(7) \text{ fabricLatency [ns]} = (\lfloor \text{doneCycleClk [cycle]} / \text{doneSignalDistributionValue [cycle]} \rfloor - 1) \times \text{clk_period [ns]}$$

Where:

- doneCycleClk is detailed in [Done Signal Patterns](#).
- doneSignalDistributionValue is corresponding agent value detailed in [Agents Done Signal Distribution](#).
- clk_period is the clock clk period.

Restrictions:

- SW: The fabricLatency equation takes in account doneCycleClk. In fact, enabling/disabling an agent or changing an agent PHY speed will indeed affect the fabric latency value. However, reconfiguration TAS [802.1Qbv] or PSFP [802.1Qci] latency in OPERATION is not a recommended option, so if ethernet agent are disabled or PHY speed is changed during OPERATION, it is recommended to set the fabricLatency to $2 \times \text{clk_period [ns]}$ and add the fabricLatency calculated value in the worst case to fabricJitter.
- SW: When preemption is enabled for a port, fabricLatency should be set to 0ns and fabricLatency calculated value when all agents are enabled should be added to fabricJitter.

30.5.4.2 Source Port Generation

Fabric gives automatically a source port number to each Agent depending on the bus they are accessing to. [Table 30.84](#) shows the relation between the Bus accessed by an agent and its source port number.

Table 30.84 Source port number assignment

Bus from which agent accesses the fabric	Source port number
ETHA0	0
ETHA1	1
GWCA0	2